

A Systems Engineering Based Assessment of The Greater New Orleans Hurricane Surge Defense System Using the Multiple Lines-of-Defense Framework



By

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Disclaimer

The authors are solely responsible for the content and conclusions of this report. The workshop participants listed above have contributed valuable insights. However, they have not necessarily endorsed the content and/or conclusions of this report.

1) Introduction and Problem Statement

This report seeks to address a major lesson learned primarily from the catastrophic 2005 flooding of 2005 by applying existing methods from systems engineering and risk and safety assessments of offshore platforms to the greater New Orleans (East Bank) Hurricane Surge Defense System (HSDS). While the surge defense system has been largely designed and constructed in a piecemeal manner, experience in the aerospace, defense, offshore, telecommunications, and other industries has shown that it is possible to design large, complex, safety-critical engineering efforts from a system's perspective which takes into account issues related to maintenance through the project's life cycle, interactions between the system elements, resiliency, and the human and operational factors of concern. In addition, this report moves beyond previous studies on the surge defense system, where risk assessments are based purely on property damage, to include human health and safety outcomes.

Toward addressing this lesson learned, this report will further specify and elaborate on the Multiple Lines of Defense Strategy, introduced by Lopez (2006, 2009). As **Error! Not a valid bookmark self-reference.** indicates, this strategy explicitly couples *structural surge* defenses, such as levees, floodgates, and pump stations, with coastal and community lines of defense. The *coastal lines of defense* consist of features of Louisiana's coastal landscape, such as barrier islands and natural ridges, that mitigate storm surge in front of the structural defenses. The *community lines of defense* are the policies and practices that communities and individuals undertake to reduce their flood risk. These include house elevation, evacuation plans, and insurance. As this report illustrates, the flood risk for greater New Orleans is best reduced when these different lines of defense are integrated into a functioning system of engineered structures, coastal landscape features, and community practices.

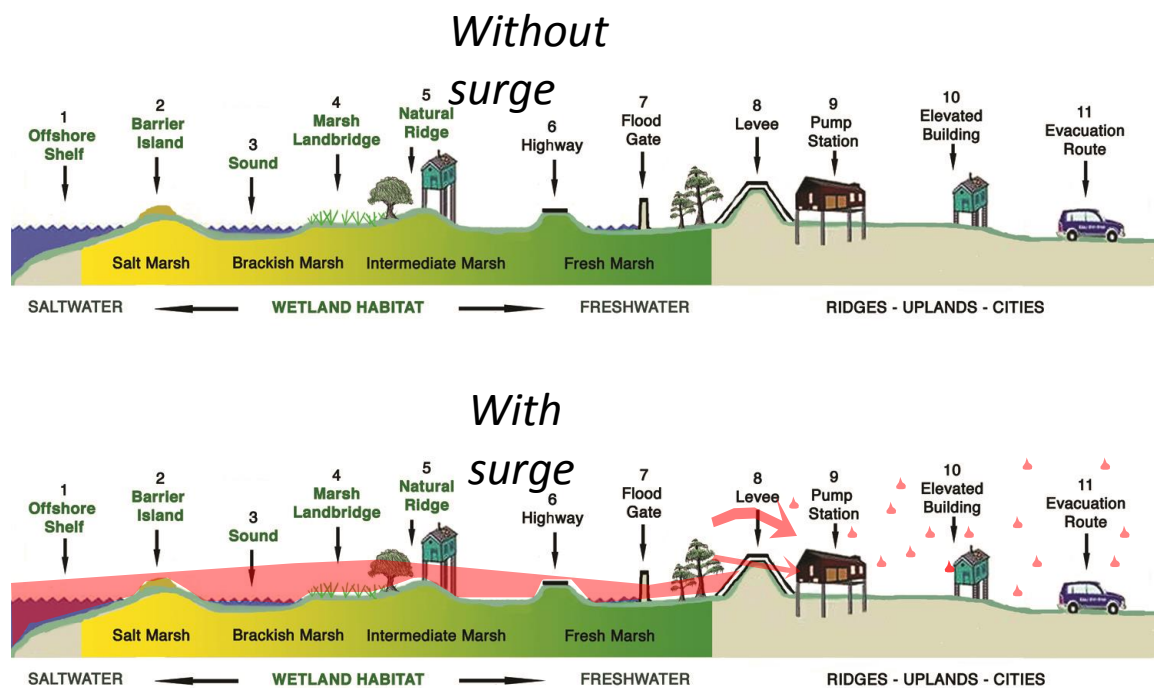


Figure 1: Diagram showing the 12 lines of defense of MLODS and how they impact storm surges. An important aspect of the strategy is to maintain the continuum in habitat (from saltwater to freshwater) outside of the perimeter protection.

Overview and Lessons Learned

Hurricane Katrina was a powerful storm that revealed and exploited a key vulnerability in the hurricane protection system for greater New Orleans (GNO): that “the system” was designed and constructed in a piecemeal fashion, which failed to integrate the disparate elements into a functioning system that achieved its defined objective. While the storm itself was capable of widespread destruction, the overwhelming majority of the damage endured in Southeast Louisiana is attributed to the design and construction failures in the set of levees, floodwalls, and floodgates meant to contain the type of storm surge kicked up by a hurricane of Hurricane Katrina’s strength and magnitude. Additionally, the decline of critical coastal landscape features along with complacency within the community contributed to the magnitude and severity of the flooding. Simply put, a region that required multiple lines of defense for flood protection, really only had one line of defense and that one proved to be faulty.

In the time since Hurricane Katrina, the specific failure mechanisms of the levees, floodwalls, and pump stations have been studied and analyzed and, in most cases, remediated through repairs and upgrades to the structural protection system. Additional design shortcomings have also been addressed, and major “gaps in the system” have been plugged. However, in many ways, the overall vulnerability of this system – that it has not been designed, built, maintained, or operated as an integrated system – largely remains as major source of uncertainty and risk. It has been nearly nine years since the Federal government’s official investigation declared “the system was a system in name only” (Interagency Performance Evaluation Task Force (IPET) 2009) and the Corps pledged to adopt “a ‘systems approach’ to project design, planning, implementation, and maintenance” (USACE 2009a). Billions of dollars have been spent on improving infrastructure, housing, and emergency plans, and the State of Louisiana has approved two Coastal Master Plans. Now is the right time to assess whether these investments have addressed the overriding vulnerability of the system.

The “System in Name Only” vulnerability had diverse and extreme implications for the impacted population of greater New Orleans. In the absence of substantial coastal lines of defense, Hurricane Katrina’s storm surge toppled floodwalls and breached earthen levees, resulting in deep and often fast moving floodwaters engulfing a major urban area, which had been home to nearly 500,000 people (Heerden, Kemp et al. 2008; Jonkman, Maaskant et al. 2009; Boyd 2011a). The damage included flood impacts to homes, businesses, community centers, and infrastructure. Some structures, particularly those near breaches, suffered extensive structural damages. Some homes along the Inner Harbor Navigational Canal (IHNC) in the Lower Ninth Ward were reduced to unrecognizable piles of debris after the floodwalls there toppled and flood waters, pushed by the Mississippi River-Gulf Outlet (MRGO) storm surge funnel, rushed into this residential neighborhood (Heerden, Kemp et al. 2008; Schleifstein 2013). In all over 100,000 homes in GNO (Boyd 2011a) suffered flood damage.

While it could not mitigate damage to homes and structures, a largely successful pre-storm evacuation did reduce the loss of life. Still, lives were lost, a reflection of vulnerabilities in the community lines of defense. In total, approximately 1,575 fatalities (Boyd 2011a) have been associated with the catastrophic impacts of the Hurricane Katrina disaster in southeast Louisiana. Of these, 600 to 700 victims are estimated to have died from exposure to flood waters (Boyd

2011a). Most of these victims are individuals who drowned in their homes, but it also includes other causes of death associated with flood exposure. Another 284 victims (Boyd 2011a) are believed to have died due to causes related to the widespread emergency circumstances that resulted from the flood damage, while approximately 631 victims (Boyd 2011a) died due to evacuation or extended displacement. Additionally, a study based on newspaper obituaries estimated over 2,000 excess deaths during a period of six months that started in January 2006 (Stephens, Grew et al. 2007).

The lesson from these statistics is that a failure in the surge defense system does not just cause deaths due to drowning. It also creates conditions, such as hospitals without power and chronically ill individuals displaced from their medical providers, that led to additional morbidity and mortality. In numerous cases, disaster health impacts were observed hundreds of miles from the flooded areas. Failures in the surge defense system cascade into failures of other important public health and safety systems, and citizens suffer adverse impacts during each step.

Five engineering studies have examined the specific design and engineering causes of the numerous failures of the structural components of the flood defense system. The three major investigations are:

- The Interagency Performance Evaluation Team (IPET) investigation commissioned by the U.S. Army Corps of Engineers (USACE)
- The Team Louisiana investigation commissioned by the State of Louisiana Department of Transportation and Development,
- The Independent Levee Investigation Team (ILIT) team which was funded by the National Science Foundation (NSF) and conducted by a team of University of California, Berkeley engineers.

Two additional groups also provided independent reviews of the IPET study:

- The External Review Panel (ERP) from the American Society of Civil Engineers (ASCE)
- A special committee formed by the National Academy of Engineering (NAE) within the National Research Council (NRC),

To a large degree, these investigations focused on the specific design aspects of individual elements of the structural perimeter protection of the system (particularly levees and floodwalls that breached). However, as shown below, each of them also provided a consistent assessment of the overriding vulnerabilities of a “system” that lacked system’s planning and engineering:

- IPET: “Systems planning and design methods are needed. Planning and design methodologies need to allow for examination of system-wide performance. It is obvious from the IPET analysis that the piecemeal development of the New Orleans Hurricane Protection System provided a system in name only.” (Interagency Performance Evaluation Task Force (IPET) 2009)
- Team Louisiana: “It is clear that this system while possibly originally conceived as a coherent and integrated system (1965) the NOFDS [New Orleans Flood Defense System] was not designed, constructed, nor maintained as such. This system was riddled with major and systemic flaws whose source lies primarily in organizational and institutional

dysfunctions. The so-called engineering failures are not the causes of these failures; the causes are firmly rooted in the organizational and institutional dysfunctions.” (Heerden, Kemp et al. 2008)

- ILIT: “This unacceptable performance was to a large degree the result of more global underlying ‘organizational’ and institutional problems associated with the governmental and local organizations jointly responsible for the design, construction, operation, and maintenance of the flood protection system, including provision of timely funding and other critical resources.” (Seed, Bea et al. 2006)
- ASCE: “The hurricane protection system was constructed as individual pieces — not as an interconnected system — with strong portions built adjacent to weak portions, some pump stations that could not withstand the hurricane forces, and many penetrations through the levees for roads, railroads, and utilities.” (ASCE Hurricane Katrina External Review Panel 2007)
- NAE: “Analysis of system-wide condition[s] will also be crucial in enhancing public confidence in the hurricane protection system.” (Committee on New Orleans Regional Hurricane Protection Projects 2006)

These studies consistently point to the importance of systems-level thinking in designing, constructing, maintaining, and operating the HSDS. However, digging deeper into how these studies frame “the system” illustrates the magnitude of the task ahead. Essentially, these studies view the system as the set of perimeter structures (levees, floodwalls, floodgates, and pump stations) meant to impede the flow into and reduce the accumulation of water in the protected areas. The full Hurricane Surge Defense System for Greater New Orleans is more than just these structural components. In fact, to achieve the objectives of the system – reducing hurricane related flood risks for the protected areas of the GNO Eastbank – the structural components of the system must be coupled with the coastal and community components of the system. As **Figure 1** illustrates and as is described in detail below, the multiple lines of defense strategy (MLODS) makes this coupling explicit.

In the traditional (structural) view of the HSDS, the system specification is confined almost entirely to pumps, floodgates, floodwalls and levees (see **Figure 2**). These structural components have been designed to control the flow of water in a manner that is supposed to prevent storm surge up to a given magnitude, termed the “100-yr year surge”, from inundating protected areas. However, this confined view of the system misses key components that either reduces the magnitude of flooding or the severity of flood losses.

In the MLODS view of the system, the structural components are coupled with the coastal and community components that also mitigate flood risk. The coastal components of the HSDS are the landscape features outside the levee system that reduce the height and intensity of surge forces exerted on the perimeter protection. These include barrier islands, sounds, marshes, and ridges. The community components of the HSDS are the policies and practices within the

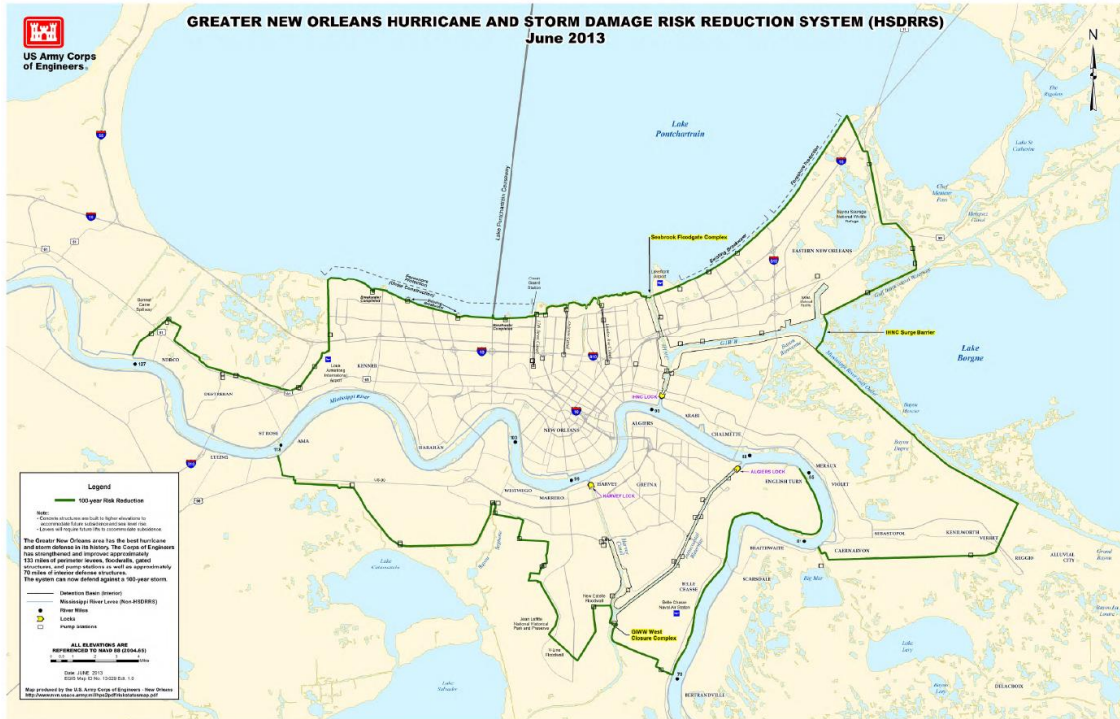


Figure 2: The 2013 Status Map for the GNO Hurricane and Storm Damage Risk Reduction System shows that USACE continues to specify the system in terms of the structural elements of the perimeter protection system only.

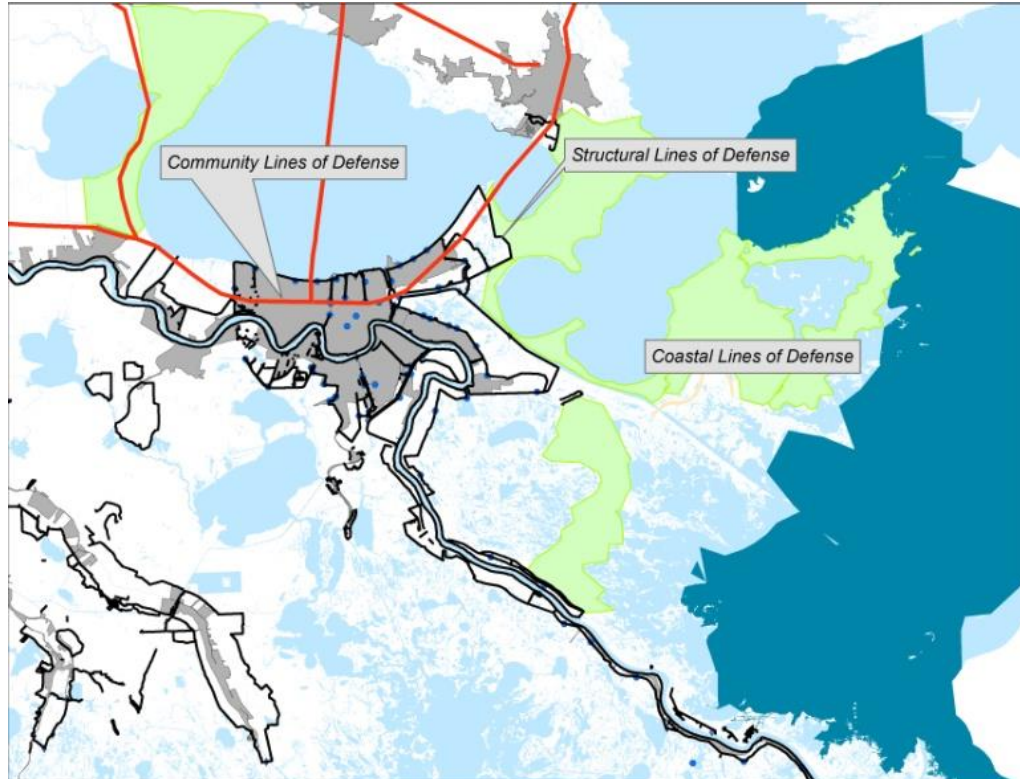


Figure 3: The MLODS specification of the system consists of the coastal, structural, and community lines of defense, all of which span different geographic footprints.

protected communities that reduce flood risks in the event that flood waters impact the protected area. Community components include elevation of structures, evacuation plans, and insurance.

Importantly, MLODS views the Coastal, Structural, and Community lines of defense and their functional integration as necessary elements of a robust and reliable system for mitigating surge risk and supporting a resilient coastal population, economy, and culture. The strategy could be described as “Levees protect our dense community assets, while the coastal landscape protects our levees along with our rural communities and less dense assets without levee protection.”

However, it is important to realize that just adding marsh grasses and evacuation lanes to our list of pieces and parts that make up the HSDS is not sufficient for implementing systems level planning. To fully adopt a systems approach, the human and organizational factors that impact system behavior are just as important as the pieces and parts that make up structure. This central principle of systems engineering is particularly true of the very complex GNO HSDS, which involves many levels of government, crosses local jurisdictions, requires different agencies within each jurisdiction to work in tandem, and impacts a host of local, regional, and national stakeholders. It goes without saying that competing interests and occasional mistrust have impacted the system’s performance. The five major levee investigations along with a host of other studies and investigations make clear that the system failures observed in 2005 were both structural and organizational.

Problem Statement: Specifying the System Based on the Multiple Lines of Defense Strategy

Given that multiple, independent engineering studies consistently cited the lack of systems planning, the current study seeks to make initial steps toward addressing this vulnerability in the HSDS. Toward the goal of applying the best resilient systems engineering practices, we seek to produce an initial specification of the system using the MLODS architecture, and then to utilize this system specification to assess the risks present in the current “as-is” system. Specifying the system includes generating an inventory of system elements (physical parts or organizational processes) along with identifying major system interactions, factors of concern, and scenarios of concern. These steps lay the foundation for a more rigorous probabilistic risk assessment of the system (which is beyond the scope of the current study).

To be clear, the system that is the subject of this study is the Hurricane Surge Defense System for the Eastbank portions of Greater New Orleans (see **Figure 4**). Our study area does not include the Westbank of GNO, though our methods and some of our results likely apply there, and likewise, for other portions of coastal Louisiana. Additionally, this study is limited to tropical system induced flooding (primarily from storm surges with contributions due to rainfall) and we do not assess the system under high river conditions or non-tropical extreme rain events. Nor do we assess the possibility of an extreme tide or levee failure during non-tropical conditions.



Figure 4: Greater New Orleans-East Bank Study area.

Importantly, MLODS (Lopez 2006; Lopez, Snider et al. 2008; Lopez 2009) outlines a system architecture that extends beyond just the engineered structures that provide the perimeter protection. MLODS couples the functionality of structural component with the coastal and community elements that work in tandem with the perimeter protection to provide robust and redundant protection of homes and people.

Essentially the task ahead consists of specifying and then assessing the current “as-is” system based on MLODS. To accomplish this goal, concepts and tools from Systems Engineering, Systems Modeling, and QMAS have been utilized.

Systems engineering is a professional discipline that provides standardized methods for determining whether a set of elements are integrated to function as a system that fulfills a specified objective, meaning it is a system (Technical Board 2004). The discipline provides a number of analytical tools for identifying factors or scenarios that reduce the functionality of the system. Systems modeling, one of the key tools for systems engineering, provides software and methodological tools for specifying a system and examining its performance under different conditions.

QMAS, short for Quality Management Assessment System, is a technique, pioneered in the offshore oil and gas industry, to systematically obtain information on complex systems from the people (termed assessors) who interact daily with system elements. In this case, assessors include levee engineers, coastal scientists, emergency responders, and a host of individuals from related professions. The next section of this report provides more details on System Engineering concepts and practices along with the QMAS methodology.

Brief Overview of the Current “As-Is” Flood Defense System

In terms of critical life-safety systems, there are only a few examples that are as large and complex as the GNO HSDS. To understand the current “as-is” system, it is important to first view it as the result of 300 years of human modifications to a 7,000 yr old, geologically dynamic natural system. However in recent times, the land loss of the Mississippi River deltaic plain has accelerated and has been more heavily influenced by human forces. The human modifications are the numerous adaptations to the landscape that have been undertaken since the arrival of the first European settlers in the early 1700’s (Lopez 2003; Boyd 2011a). Ironically, the human modifications include constructing the structures meant to control the flow of water (both storm surges and high water in the Mississippi River) and man-made canals built to facilitate water borne commerce. Currently, the “natural system” shows signs of large-scale human induced decline along with isolated cases of engineered restoration. This complexity is one reason why we need MLODS and Systems Engineering.

Utilizing MLODS to specify the system has many advantages. First, MLODS advocates landscape-centric design, not design centered on specific engineered structures. MLODS also identifies commonalities between flood protection and habitat management, both of which are key to economic and cultural sustainability. Finally, it couples disparate elements of the HSDS, all of which impact storm surge dynamics and/or impacts, spread across multiple jurisdictions into a functionally integrated system for reducing storm surge risk.

MLODS emerged shortly prior to Hurricane Katrina as an alternative to the organizational-silo approach to flood protection that was dominant at the time (Lopez 2009). During the pre-Katrina period of dysfunctional silos, projects within flood protection agencies were designed and engineered on a project basis, with little thought given to integrating a specific project with other projects that make up the surge defense system. Further, the flood protection agencies were isolated from coastal restoration agencies, and both were isolated from emergency response agencies. On top of that, federal, state, and local agencies all jockeyed amongst each other for funding while also trying to shift responsibility elsewhere. There was also the ongoing conflict between sustaining the coastal population and the powerful navigation and petroleum industries.

Historically, the city has benefited from a robust layer of coastal protection coupled with community practices, such as building elevated homes. Anyone who understands the history of New Orleans knows that levees have been important to the city’s modern development. However, for the most part, these have been river levees meant to prevent flooding during the river’s spring high stage period. Ironically, it was these levees that prevented new wetland growth which contributed to the net loss of coastal protection and partially contributing to the necessity of construction of hurricane protection levees for the city.

High net rates of coastal wetland loss in Louisiana are largely a 20th century phenomena resulting from numerous causes, including the levees along the Mississippi River. These levees stopped the process of annual spring flooding that introduced the freshwater and sediment that sustained the coastal system. Structural hurricane protection commenced in the latter half of the 20th century, spurred by deadly flooding from Hurricane Betsy in 1965. However, wetland loss continued and elevated construction was replaced with slab-on-grade construction.

The period that has been termed “Louisiana’s Crisis of Vulnerability” began during the last decade of the 20th century as shown in **Figure 5**, which depicts New Orleans’ flood history. During the 19th century, New Orleans’ population largely settled on the higher ground near the natural ridge of the Mississippi River. In addition, the urban settlement was surrounded regionally by a healthy coast, resulting in a high level of protection from hurricane surges (an estimated 400 year return period level of protection). During the 20th century, the shift in housing trends toward low elevation and widespread coastal wetland loss was accompanied by a myopic “Levee Only” solution to storm surge flooding, particularly following Betsy in 1965. This approach set the stage for catastrophic levee failures that occurred as Hurricane Katrina past near the city in 2005.

Despite the engineered network of levees and floodwalls that ringed the city, the region’s overall level of protection continued to decrease during the latter half of the twentieth century. For example, while hurricane levees were built around the city, the Mississippi River Gulf Outlet was constructed as a wholly separate and unrelated project, that happened to pierce the levee system and became a major contributing factor to flooding during Hurricane Katrina (Heerden, Kemp et al. 2008). Coastal wetland loss continued, largely unabated. The sense of security provided by the ring of levees along with a multi-decadal lull in Atlantic hurricane activity contributed to widespread complacency within the community. Slab-on-grade homes continued to be built on low lying land, while evacuation planning struggled to keep pace with the region’s increasing vulnerability. In **Figure 5**, the dashed, red line depicts the decline in net protection level for New Orleans.

Hurricane Katrina struck the region in 2005 and brought the underlying crisis of vulnerability to the forefront. In the immediate aftermath of Hurricane Katrina, the USACE’s Task Force (TF) Guardian implemented emergency repairs to the levees, floodwalls, and pump stations that had been damaged, essentially restoring the system to the pre-Hurricane Katrina level of protection (depicted in **Figure 5**). Following TF Guardian, Task Force Hope was created to upgrade the system to provide a level of protection corresponding to a surge that has a 1% annual chance of occurring (the “100 year level of protection”). This is currently the structural system assessed in this study, along with the full complement of other Lines of Defense. It should also be noted that the pre-Katrina levee project known as the Lake Pontchartrain and Vicinities authorized a protection level specified by a “standard project hurricane” (in essence equivalent to Hurricane Betsy), which is a high level of protection than the currently authorized 100-year level for the post-Hurricane Katrina system. Beyond these efforts, the Louisiana Coastal Protection and Restoration Authority (CPRA) and Louisiana Coastal Protection and Restoration (LACPR) report recommend a 400 year level of protection for the urbanized areas of central New Orleans. CPRA’s Coastal Master Plan is described shortly.

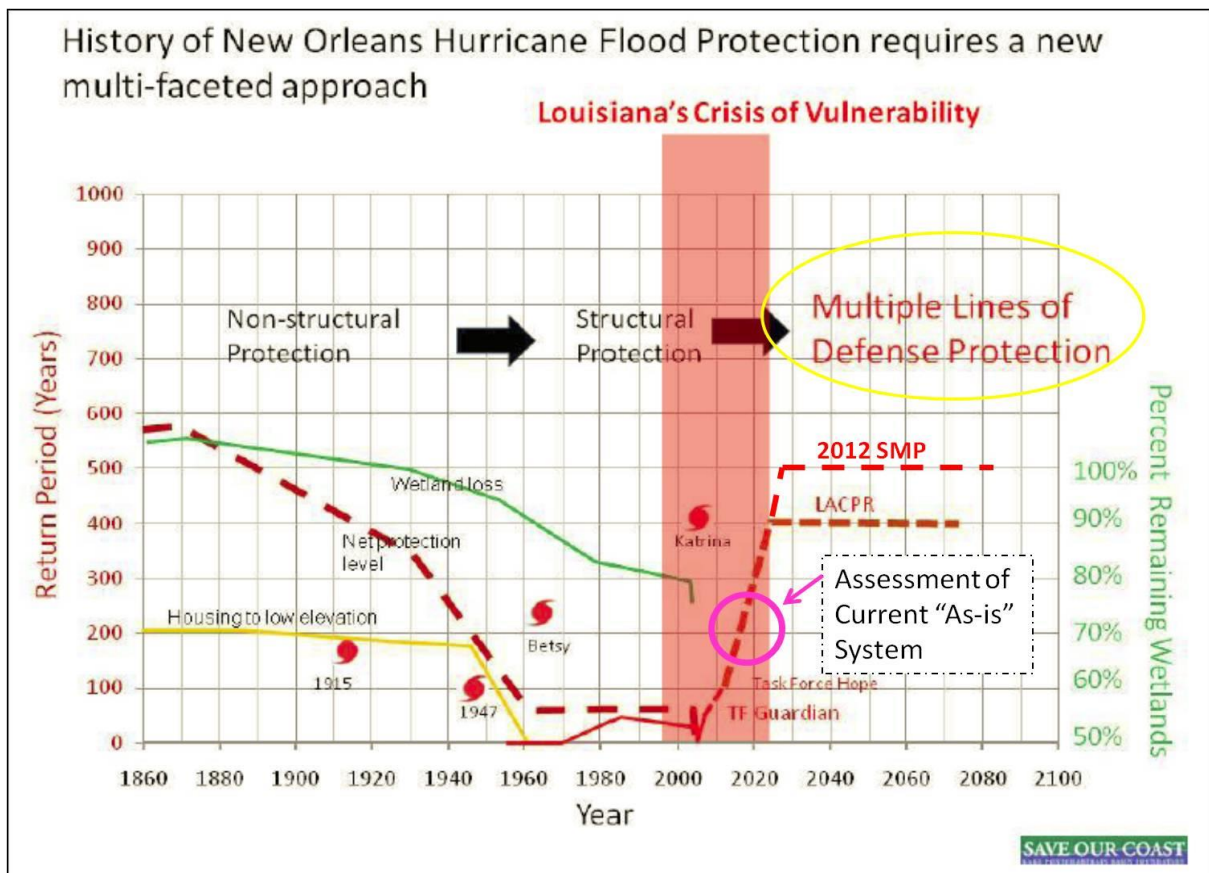


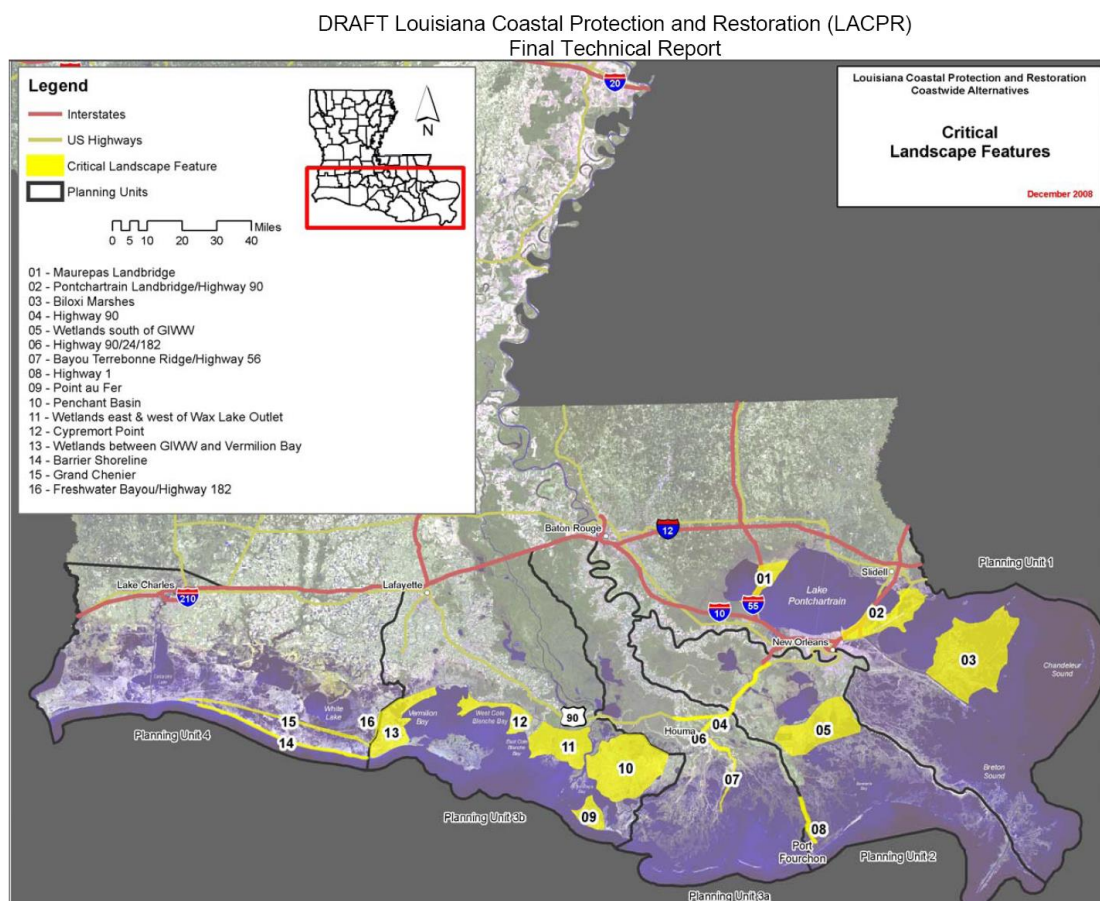
Figure 5: Louisiana’s Crisis of Vulnerability shown by the red area. The dashed red line depicts the decline in the net protection level for New Orleans in the years prior to Hurricane Katrina.

The 19th century city was located on high ground and used elevated housing surrounded regionally by a healthy coast resulting in a high level of protection from hurricane surge (> 400 year return period). The shift in housing, wetland loss, and a myopic “Levee Only” solution all set the stage for catastrophic flooding during Hurricane Katrina.

The years prior to 2005 saw growing awareness of the intimate linkages between flood protection and coastal restoration, at least amongst scientists, engineers, and other technical experts if not necessarily the general public or policy makers. Naturally, the devastation of Hurricane Katrina brought considerable attention to improving the resiliency of coastal Louisiana, and a considerable amount of this attention went to MLODS. In 2007, the Lake Pontchartrain Basin Foundation (LPBF) held a series of public meetings on MLODS, which led to a set of coastwide recommendations published in the 2008 report. Since then MLODS has received numerous endorsements, including local governments and levee boards. In their 2011 strategic plan, the Southeast Louisiana Flood Protection Authority-E (SLFPA-E) asserts that “a ‘multiple lines of defense’ strategy should be adopted to reduce flood risks” (Dupray, Tourment et al. 2010). The 2007 Louisiana State Master Plan stated that “plans for hurricane protection must rely on multiple lines of defense” (Coastal Protection and Restoration Authority of Louisiana 2007), while the newer 2012 State Master Plan draws extensively on MLODS (Louisiana Coastal Protection and Restoration Authority 2012). The USACE’s endorsement of

MLODS came with the 2009 Louisiana Coastal Protection and Restoration (LACPR) “Final Technical Report,” which states explicitly “the best strategy [for storm surge risk reduction] is to rely on multiple lines of defense” (USACE 2009b). **Figure 6** below, from the LACPR report, identifies “critical landscape features” which are essentially synonymous with coastal lines of defense.

Of these, the 2012 Louisiana State Master Plan (SMP) is the most recent, detailed, and ambitious, so it is worthwhile to examine how this plan incorporates MLODS. The SMP is largely consistent with the MLODS approach. Both plans utilize a similar set of projects (levees, diversions, marsh creation, etc.) While there is some overlap in the location of specific projects, there are also notable differences in location and extent (see **Figure 7**). Additionally, MLODS proposes much more expansive marsh restoration than the SMP, and the SMP excludes restoration of the Chandeleur Islands (a key line of defense for the GNO East Bank). These differences likely reflect of the SMP’s modeling based project selection process as compared to the MLODS process which relied on collaborative consultations with experts, citizens, stakeholders, and scientists with detailed local knowledge. However, it should be stressed that these differences in the details should not overshadow the fact that the SMP largely follows MLODS and allocates funding across all the major components of MLODS (**Figure 8**).



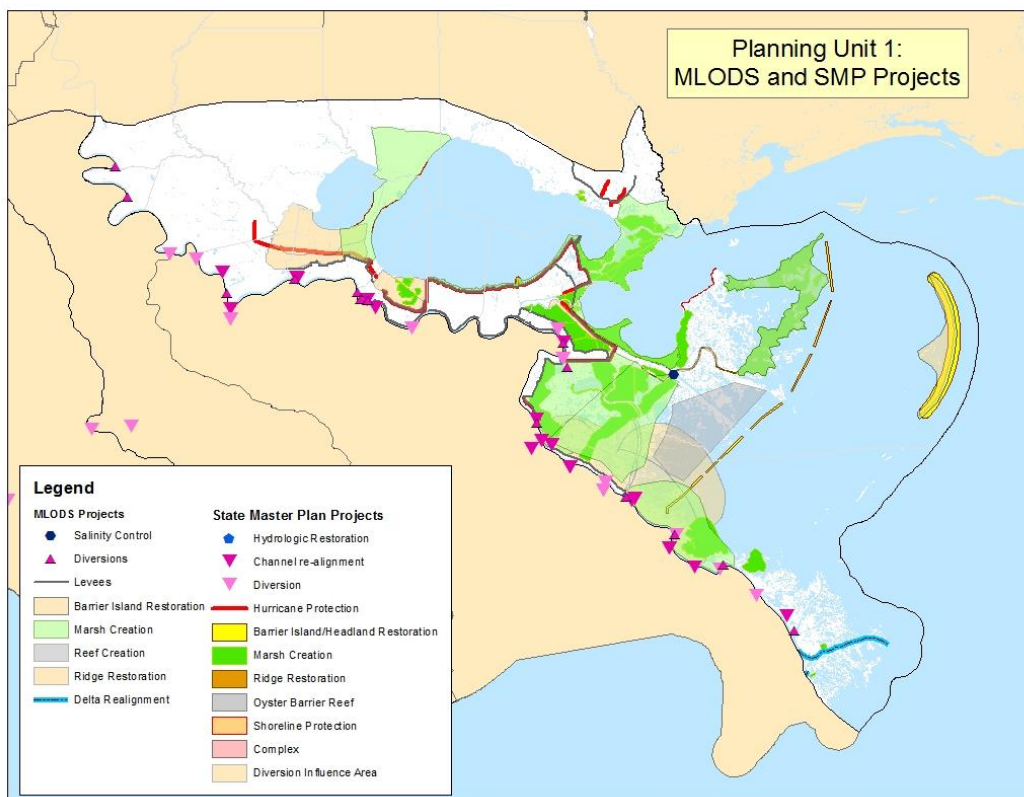


Figure 7: A comparison of coastal restoration projects proposed for the Pontchartrain Basin in the 2008 MLODS report with projects considered for the 2012 State Master Plan.

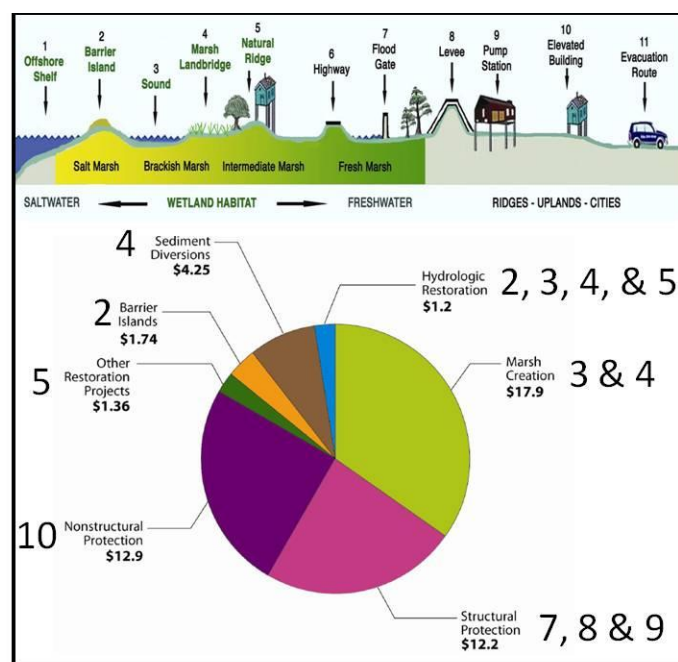


Figure 8: Comparing the 2012 Louisiana State Master Plan budget recommendations (bottom) to LPBF's Multiple Lines of Defense Strategy (top).

Essentially, this study examines three related questions:

- Are these components integrated?
- Do they accomplish a defined objective?
- How can it be assessed and modeled as a system?

In the current, “as-is” system, a number of factors limit MLODS’s ability to function as an integrated system. By and large, the coastal lines of defense have suffered from decades of decline, mostly due to mechanical land removal, subsidence and erosion. Recently, major efforts to reverse this trend have gained steam, though considerable effort and resources are still required before the decline will be stopped or reversed. While the structural lines of defense have benefitted from billions of dollars in improvements (The Times-Picayune 2011) that supposedly bring it up to 100-yr level of protection (USACE 2012), these improvements may be undermined by cost cutting measures, such as sheetpiling that lacks rust protection, and organizational constraints, particularly limited local funding for operations and maintenance. Without a doubt, the community lines of defense have benefited from ongoing progress, such as thousands of homes raised and the most comprehensive evacuation plan the region has ever seen, but many barriers, particularly economic, social, and cultural remain. Most importantly, these improvements have largely followed the traditional organizational silo approach with very little considerations of the coupling between these lines of defense during the planning, design, and implementation stages.

It is important to note that while the HSDS is designed to function during extreme, tropical weather conditions, it faces threats both during extreme weather and during fair weather. Levee breaches occur most often during extreme tides and waves. But, they could also occur during fair weather conditions, due to a variety of causes such a colliding barge or train. Likewise, storm surges and waves erode marsh land bridges, but during fair weather industrial activity also removes land and erodes shorelines. Additionally, for many elements serve dual objectives with flood risk reduction being secondary. For example, while interstates and highways serve as evacuation routes during hurricane preparations, the primary purpose is to facilitate the daily flow of traffic during fair weather days.

Multi-level description of the Hurricane Surge Defense System

To better understand MLODS, it is important to consider how this system spans geographic scales, from landscape scale features to individual parts, and how the system performance is impacted by the interaction of various elements across different scales. This theme will be expanded on throughout this report, but consider the example of how the dynamics of the barrier islands impacts the functionality of evacuation routes.

During an evacuation, timing is everything. Over 1 million residents of GNO must utilize a limited number of evacuation routes during the brief window between the release of official warnings and the onset of hazard conditions which typically spans 30 – 50 hours (Wolshon 2006; Wolshon 2008). One such route is I-10 East, which crosses Lake Pontchartrain before passing through Slidell, LA. While much of I-10 in GNO is within the levee system and while the newly constructed I-10 bridge over Lake Pontchartrain is 30-ft above sea level, the foot of the bridge on

the New Orleans side is an unprotected, low-lying (6 – 8 ft. above sea level) stretch of highway (see **Figure 10**). Once a storm surge rises to the elevation of the interstate at that location, this crucial evacuation route becomes blocked, an illustration of how the capacity of this line of defense depends on the timing of a storm surge's rise at that location. Before the still water level reaches the elevation of the interstate, waves and debris create a transportation hazard.

While it initially might be hard to imagine how the Chandeleur Islands, located 60 miles southeast of I-10, could influence evacuation capacity, the state of the Chandeleur Islands are in fact a major factor limiting the performance of this evacuation route. Like most of Louisiana's coastal landscape, the Chandeleur Islands have been in a state of decline for many decades now (Penland, Beall et al. 2002; Irish, Song et al. 2011; Louisiana Coastal Protection and Restoration Authority 2012). Whereas in the past healthy barrier islands provided a robust line of defense against storm surges, the islands have lost significant land area and have become fractured. Studies (Grzegorzewski, Cialone et al. 2008; Wamsley, Cialone et al. 2009; Grzegorzewski, Cialone et al. 2011) have shown that as the Chandeleur Islands continue to decline, storm surges will reach higher levels and, importantly, move inland quicker (see **Figure 11**). This decline implies that future storms will flood the low spot along I-10 sooner than they have in the past, thus reducing the capacity to evacuate people through this route. According to Wolshon (Wolshon 2008), the observed maximum capacity of this evacuation route is 1,900 vehicles per hour, which implies that nearly 5,000 people could be prevented from evacuating if this route floods 1 hour earlier (assuming 2.5 persons per vehicle). In systems engineering terminology, this is an example of a system interaction where one element of the system influences the performance of another element, and in this complex, expansive system interactions occur over many miles. Furthermore, these two elements also cross a wide gap in professional disciplines.

Given the large footprint of the HSDS and the fact that system interactions occur across geographic scales, now is an important time to briefly sketch the structure of the HSDS across geographic scales. Complex, large scale systems can be packaged according to various scales or resolutions. As one moves from the high-level (10,000 ft) view to the detailed view, more and more system detail is revealed. One of the major challenges in specifying the system is distinguishing high-level (general) system elements from low-level (high detail) system elements. For this study, six levels of detail were developed to help facilitate discussions and development of the system specification (**Figure 12**). The scale steps are logical decomposition groupings specific to the system under consideration. The structures of the HSDS are described at the landscape, regional, local, site, and part level. In **Figure 13**, LIDAR elevation data is used to depict the various landscape features at some of the spatial levels.

At the landscape level, the primary components of the system (the lines of defense) are apparent. At this level, the HSDS spans from the continental shelf in the Gulf of Mexico to the evacuation routes that convey people outside of the coastal zone. Located in the middle, Louisiana's coastal zone is the focus of the MLODS strategy. At this level, general trends in elevation are apparent but not the structure of individual landscape features.

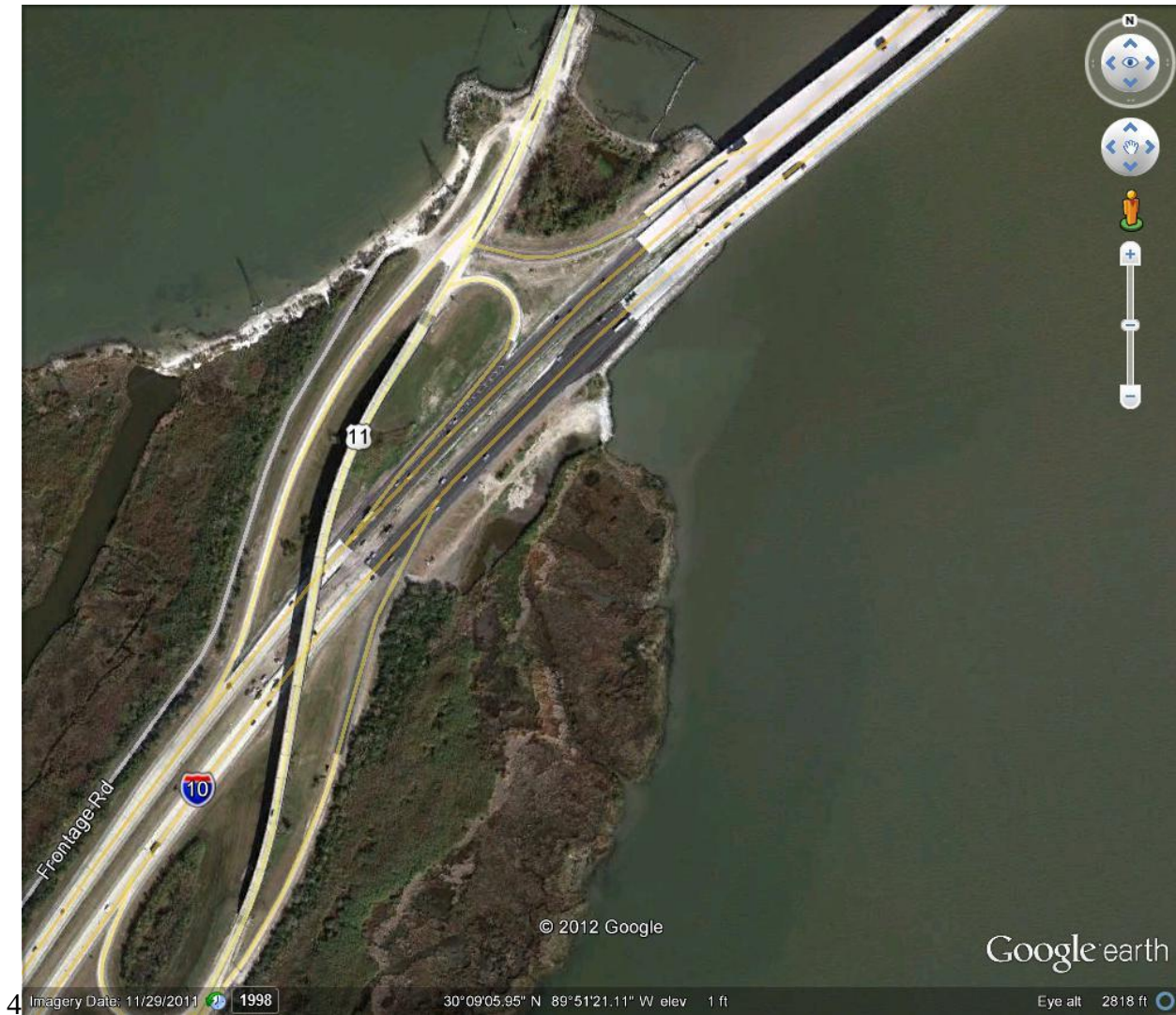


Figure 10: Low spot of Interstate 10, a major evacuation route, at the foot of the newly rebuilt and elevated “Twins Spins” bridge over Lake Pontchartrain. While this route has a maximum capacity of nearly 2,000 vehicles per hour, the section of “black top” before the bridge is just a few feet above sea level and prone to flooding during storm surges.

Zooming to the Regional (state/parish) level of the HSDS, particular aspects of MLODS become more prominent than others. For example, the perimeter protection crosses parishes, while marsh land bridges are major features of the regional landscape.

At the Local (city/town/neighborhood) level, differences between cities and towns become apparent. This is most notable with the interior drainage systems, a major infrastructure system maintained by local governments with Federal assistance. In a similar manner, local officials are the key players in evacuation procedures while housing elevation requirements are approved and enforced at the local level.

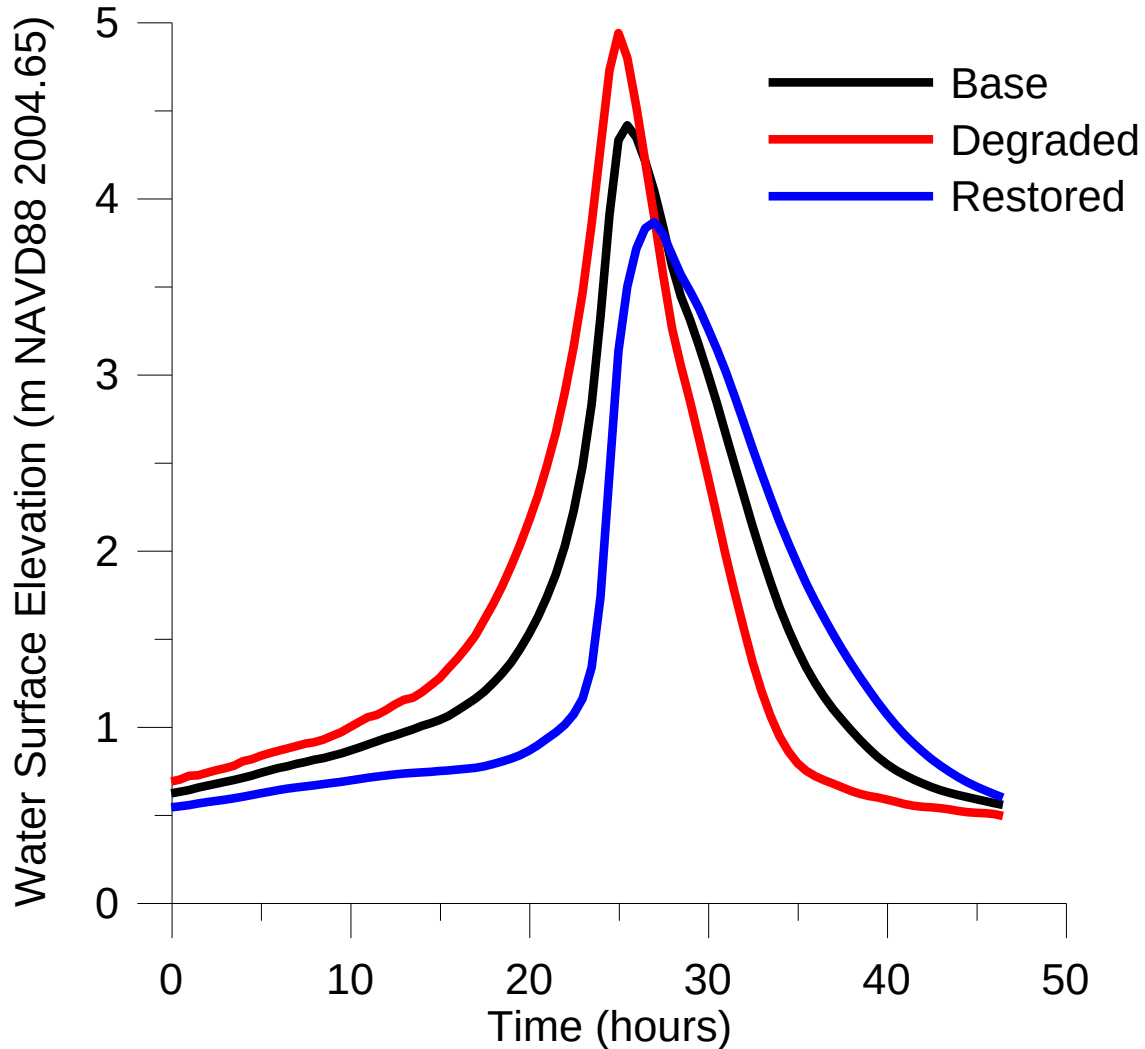


Figure 11: Simulated hydrographs in Chandeleur Sound for hypothetical storm surges based on three scenarios for the Chandeleur Islands. If the Chandeleur Islands degrade (red line) from the current base conditions (black line), then storm surges will peak 0.5 m higher and will arrive 1 hr earlier. Likewise, restoring the Chandeleur Islands, could reduce surge levels nearly 0.75 m and delay the peak by approximately 1 hr (Grzegorzewski, Cialone et al. 2008). The timing of the surge constrains the evacuation procedures, an example system interaction between the barrier islands line of defense and the evacuation route line of defense.

At the Site level, the diversity and complexity of the HSDS becomes apparent. For example, pump stations are site level features. Each one contains complex engineering and millions of parts, but each one has a unique design. Systems elements at the site level assessment include particular ridges and marsh land bridges, specific floodgates and related structures, and thousands of homes that have been elevated.

System Scale (Resolution)

- Level I (Landscape, Federal/State)
- Level II (Regional, State/Parish)
- Level III (Town/City/Neighborhood)
- Level IV (Site)
- Level V (Site Components)
- Level VI (Detailed Parts)



Figure 12: The hierarchy of spatial scales used to order the various elements and components of the MLODS system.

The Component level gets inside the major structural elements of each site. For example, pump stations contain pumps, a crucial component. They also include intake and discharge pipes, grates for clearing debris, buildings to house the pumps, mainline power, and backup power.

At the Part level, accounting for the HSDS almost gets untenable. The archetypical example of a complex system, the Boeing 747 system contains approximately 6 million parts. It is not hard to imagine that a single pump station (which is of similar size to a 747) contains a similar of parts. But, unlike Boeing 747's which are mass produced from nearly identical designs, each of the HSDS approximately 70 pump stations are unique units with site specific design considerations. Then there are parts involved in other lines of defense. At some point, a precise accounting for each system part becomes untenable. For example, large limestone rocks, a major part for levees and floodwalls are counted by the truck load, not individually, implying a complete accounting of every part is not possible or even beneficial.

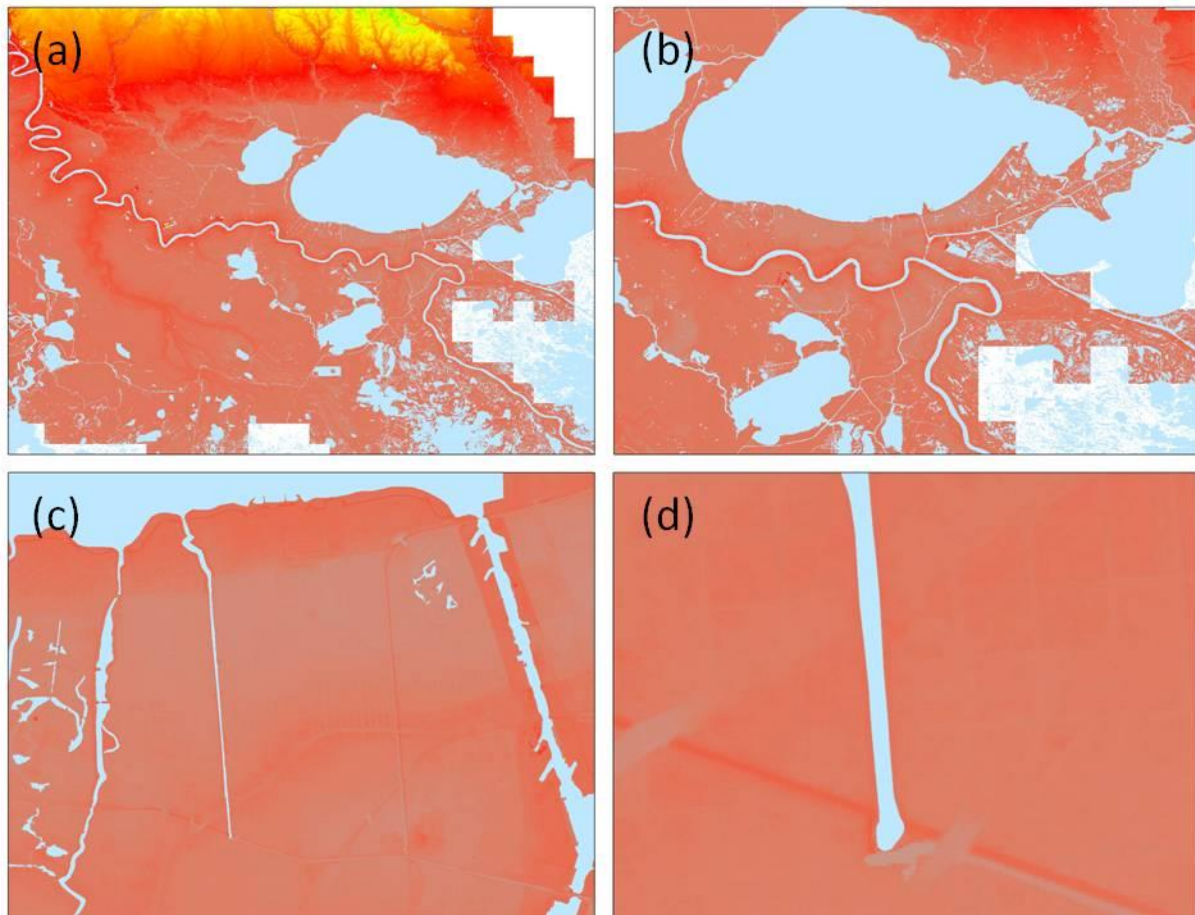


Figure 13: Series of LIDAR elevation maps at different scales, showing how different landscape features are prominent at the different spatial scales. At the landscape level (a), the primary feature is the transition to the deltaic plain, while at regional level (b) many of the natural ridges are visible, though subtle. In the neighborhood level (c) the Gentilly Ridge and the central New Orleans polder are visible. Finally, at the location (d) the catch basin for a pump station is visible, along with a railroad track embankment along two highway underpasses through the embankment.

The discussion of scale issues with the HSDS would not be complete without covering the numerous regulatory jurisdictions. It goes without saying that numerous agencies exert influence over the HSDS. **Figure 14** maps the boundaries of some of the agencies involved in the HSDS, and illustrates the complicated and often overlapping governance of this system. The map shows individual local levee districts along with the larger Southeast Louisiana Flood Protection Authority-East Bank. It also shows the boundaries of local governments, various state agencies tasked with coastal management and restoration functions, the Louisiana Department of Transportation Districts, the Louisiana State Police Districts, and USACE districts. Again, since many of these organizations exist for primary purposes other than flood risk reduction, the elements under their authority are not optimized objectives related to flood risk reduction. More

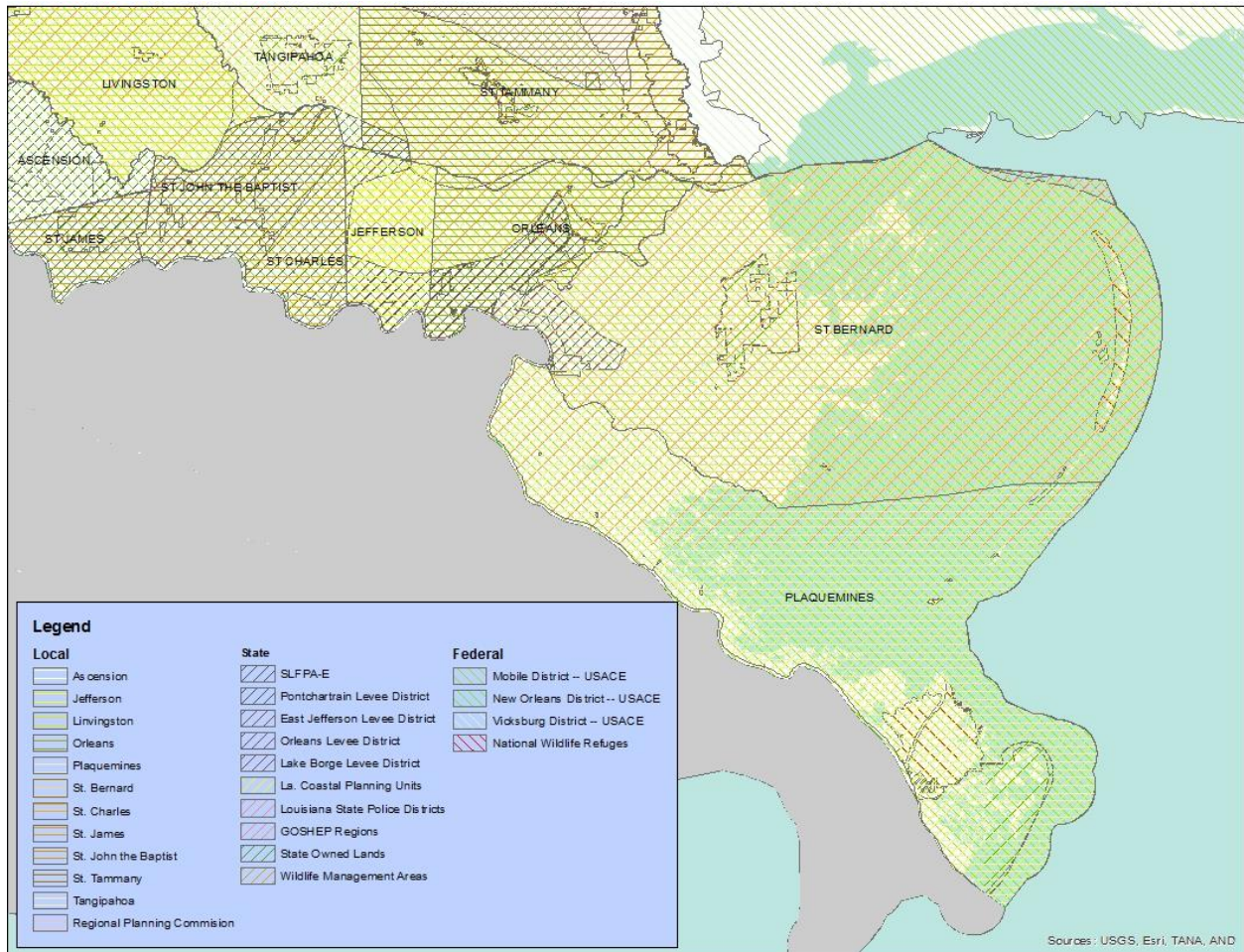


Figure 14: A sample of jurisdictions that are involved in the GNO HSDS.

importantly, the map's limited sample of agencies show a complex set of crisscrossing administrative jurisdictions with authority over the HSDS and influence over the dynamics of an incoming storm surge, thus illustrating the complicated governance of the HSDS.

2) Background and Methodology

Post-Hurricane Katrina Levee Investigations

As mentioned in the previous section of this report, the catastrophic damage that resulted from the widespread failures of the Hurricane Surge Defense System (HSDS) during Hurricane Katrina sparked a major reassessment of flood risk management for greater New Orleans. This section starts with a more in-depth summary of the post-Hurricane Katrina levee investigations, including IPET, ILIT, Team Louisiana, ASCE, and NAE.

IPET (Interagency Performance Evaluation Task Force (IPET) 2009) was a comprehensive study involving numerous teams across various disciplines. Because it was funded by the USACE, major questions about impartiality of the interpretation offered by IPET have been raised (Irish, Resio et al. 2011). Still, this report remains a reliable source for basic data and analysis, including the storm surge and the performance of levees, floodwalls, and pumps. IPET likewise acknowledges that multiple failures resulted from poor design and construction and that the system failed to function as an integrated system.

The Team Louisiana investigation was funded by the Louisiana Department of Transportation and Development and completed by LSU coastal scientists in collaboration with independent engineers (Heerden, Kemp et al. 2008). Its stated goal was to ensure that the state and local perspectives on the causes of the flooding were adequately documented. The report addresses six key questions on the design and construction of the structural components of the HSDS along with the MRGO navigation canal. Among its major findings: the levee system was not designed to fulfill its Congressional mandate, crown elevation were built lower than design specifications, the USACE did not follow existing engineering practice or even its own guidance, the MRGO/GIWW compromised system performance, and shortcomings in maintenance and operations further reduced the level of protection provided by the system.

The Independent Levee Investigation Team (ILIT) team provided the independent study from unbiased investigators with no affiliation with any of the implicated parties (Seed, Bea et al. 2006). This study was funded by the National Science Foundation and completed by engineers and scientists from UC Berkeley. A central theme of the ILIT report is that the system was “Organized for failure.” In terms of specific conclusions, ILIT lists three main causes of the disaster: “(1) a major natural disaster (the Hurricane itself), (2) the poor performance of the flood protection system, due to localized engineering failures, questionable judgments, errors, etc. involved in the detailed design, construction, operation and maintenance of the system, and (3) more global “organizational” and institutional problems associated with the governmental and local organizations responsible for the design, construction, operation, maintenance and funding of the overall flood protection system.”

As part of the IPET process, the American Society of Civil Engineers (ASCE) was asked by the USACE to provide an independent review of the IPET study (ASCE Hurricane Katrina External Review Panel 2007). This group was termed the Hurricane Katrina External Review Panel (ERP). The ERP largely concurs with IPET results in regards to specific engineering causes of

the levee failures and it moves beyond that to address contributing factors, such as flaws in the piecemeal funding and construction process and poor risk assessments and education for the region. The ERP concluded that “the ASCE Hurricane Katrina External Review Panel believes strongly that if lives and public safety are to be protected, significant changes will be required in the way hurricane and flood protection systems are funded, designed, managed, and maintained.”

As a second review of the IPET, the National Academy of Engineering and the National Research Council launched an investigative committee which released five reports on the New Orleans region flood protection projects (Committee on New Orleans Regional Hurricane Protection Projects 2006; Committee on New Orleans Regional Hurricane Protection Projects 2008; Committee on New Orleans Regional Hurricane Protection Projects 2009). The NAE report largely summarizes the IPET conclusions, most of which it endorses with only additional comments on how to better present the findings to the public.

Importantly, these reports agree on many of the key failure modes that were observed in 2005. Many of the failures, such as along the outfall canals and the IHNC, are attributed to sheet piling in floodwalls not being driven deep enough underground. At the 17th Street Canal, sheet piling was driven less than 20 ft. underground, into a layer of soft clays which lacked the soil strength to support the floodwall during surge conditions. A similar situation existed in the London Ave. Canal, except that the layer of soft clays was not present at this location and the sheeting piling was driven in a layer of sand, an even weaker base. At the 17th Street Canal, this design resulted in the floodwall leaning toward the protected side during Hurricane Katrina’s storm surge, which led to a water filled gap that eventually undermined the base of the floodwall. At the London Ave. location, the surge water seeped through the sand layer and underneath the sheet pile support for the floodwall. Again, this flow eventually undermined the base of the floodwall, resulting in catastrophic failure. While poor design allowed water flowing underneath the ground to undermine the floodwalls at these two locations, the IHNC levees were compromised by water that flowed over the top of the floodwall, causing scour and erosion on the protected side that eventually undermined the floodwall’s support. The investigations revealed that design errors, most importantly a datum mix-up, resulted in the “as built” structure being 1-2 ft below the design specification. In all four failures, poorly sampled and largely unknown sub-surface soil conditions were a major contributing factor.

While forensic engineering highlighted these specific failure mechanisms, the major studies also found additional underlying problems with the design and construction process. In designing the system, the USACE utilized an outdated method to determine the characteristics of the “standard project hurricane” specification. However, as the design and construction process dragged on over decades, the USACE failed to update its definition of the standard project hurricane using updated climatological analysis from NOAA (Heerden, Kemp et al. 2008). This shortcoming was followed by two additional design shortcomings. First, engineers over-estimated the soil strengths at the levee locations, resulting in less robust design (particularly sheet piling not being driven deep enough). Second, an elevation datum mix up equated the water level based vertical datum used in the design and modeling of the system to the geodetic (land-based) vertical datum used in construction.

While the factors above set the stage for disaster, additional factors contributed the magnitude of flooding. Importantly, local settling and regional subsidence were not considered in the design of levees and floodwalls, and some floodwalls, particularly along the IHNC, actually had sunk by nearly 1 ft. since construction. Additionally, design and engineering did not account for the loss of coastal landscape features and the related reduction in their ability to reduce surge. Investigators also cited the lack of inter-agency coordination and the absence of a single person or agency in charge of the system. This factor meant that important conflicts of interest between agencies, most notably the dispute of blocking the outfall canals at Lake Pontchartrain (Heerden, Kemp et al. 2008), were not managed from a perspective grounded in maintaining system performance. Furthermore, the system was funded on a project basis over many years without a rigorous external review of project designs. Finally, interior drainage pump stations, while a crucial part of the hurricane protection system, were not designed or built to be part of the surge defense system. Most of the pumps were designed with a capacity to handle a 10-yr rainfall event and many of the pump stations were not engineered to withstand hurricane or storm surge conditions. Some of the pumps even contained parts that deteriorated when exposed to salt water.

System Upgrades Since Hurricane Katrina

The recovery period following the flood disasters of 2005 has seen a number of repairs and improvements in the GNO HSDS. As mentioned previously, the USACE undertook two major initiatives. First, Task Force Hope repaired the damaged sections of levees and then Task Force Guardian upgraded the structural system to the 100 yr. level of performance. Notably, there have been attempts to address the systems-level issues (Coastal Protection and Restoration Authority of Louisiana 2007; Louisiana Coastal Protection and Restoration Authority 2012). However, these attempts, while indicative of progress, still fall short of the goal of fully implementing systems thinking and design in the GNO HSDS.

One of the most notable and concrete changes since 2005 has been the ongoing repairs and improvements to levees, floodwalls, and pump stations. In the early days after the storm, the USACE's Task Force (TF) Hope had a simple, but important mission: repair levees and floodwalls damaged by Hurricane Katrina. Following TF Hope, the USACE launched TF Guardian with a broader mission: provide protection against a 100 yr level storm. Through these two task forces, the USACE has implemented crucial repairs and major improvements in the structural protection component of the HSDS. These improvements include replacing damaged I-wall type sheetpiling with the more reliable T-wall sheetpiling, adding concrete scour and erosion protection on the protected side of floodwalls, hardening transitions between earthen levees and concrete floodwalls, building temporary gates and pump stations at the three major outfall canals of central New Orleans, and storm proofing municipal pump stations. With these upgrades, the USACE claims that the system is built to 100 yr. surge heights and that the structures will be resilient against overtopping and waves from a 500 yr. surge event once planned resiliency measures are constructed (Schleifstein 2013).

Coincident with these improvements, the USACE has also begun efforts to correct problems resulting from the Mississippi River Gulf Outlet (MRGO), which has been described as “a massive human-induced environmental, economic, and storm disaster” (Shaffer, Day et al. 2009)

and a cause of the lethal flood conditions observed in the Lower Ninth Ward of New Orleans (Boyd 2011b). The MRGO was de-authorized in 2008. The first step involved plugging the MRGO channel by constructing a 950 ft. rock dam just south of the Bayou La Loutre Ridge, a natural ridge (and coastal line of defense) in St. Bernard Parish that was breached by the channel. Additional steps at addressing the vulnerability from the MRGO, include constructing the Inner Harbor Navigation Canal-Lake Borgne Surge Barrier (hereafter referred to as the Surge Barrier), replacing the earthen levee along the MRGO channel with a concrete floodwall, and building floodgates at the lake end of the IHNC. The additional, thusfar uncompleted step is implementing the MRGO Restoration Plan. The MRGO Ecosystem Restoration Plan has been approved by the White House's Office of Management and Budget and sent to Congress in 2013. \$1.325 billion of the \$3 billion plan was determined to be cost-effective, environmentally acceptable, and technically feasible and is approved for construction if they get a non-federal sponsor. At this time, no Federal funding has been appropriated for further study or construction regarding the MRGO.

This period has also seen the development and utilizations of new technical knowledge related to assessing the risk posed by storm surge flooding. One such example has been considerable improvement in the ADvanced CIRCulation (ADCIRC) hydrodynamic code for simulating storm surge. While utilized prior and during Hurricane Katrina, this numerical code was considered experimental at the time (Dean, Powell et al. 2004). Since Hurricane Katrina, the core code has been further refined and validated and at the same time more widely adopted (FEMA 2008; Ebersole, Westerink et al. 2010; Lynett, Melby et al. 2010; U.S. Coast Guard 2010; Louisiana Department of Transportation and Development 2013; Port of New Orleans 2013). A similar technical advance has been the development of Joint Probability Method-Optimized Sampling (JPM-OS) for determining storm surge height-frequency relationships, which was used in the IPET's Risk and Reliability Report (Resio 2007; Interagency Performance Evaluation Task Force (IPET) 2009; Levinson, Vickery et al. 2010; Toro, Resio et al. 2010). Additionally, climate scientists at LSU have compiled the most comprehensive database on historical surge events for the Gulf coast region (Needham 2010).

These developments in the quantitative analysis of surge risk fit within a larger trend of growing emphasis on comprehensive water management for greater New Orleans, for the Mississippi River, and for the Gulf of Mexico. Initiatives at the local level include the Master plans for New Orleans and St. Bernard parishes along with revamping and modernizing the respective agencies involved in water management. Other initiatives include efforts to investigate comprehensive, basin wide management of the Mississippi River, including the LCA's Mississippi River Hydrodynamic and Delta Management Study, and similar Gulf-wide efforts, including the newly launched Water Institute of the Gulf of Mexico in Baton Rouge, LA. Capitalizing on this trend, LPBF has greatly expanded its Coastal Sustainability Program and continues to actively monitor and analyze the development and implementation of MLODS.

These research and management initiatives are complemented by new governance structures. At the state level, a 2006 amendment to the Louisiana constitution explicitly links flood protection and coastal restoration, while a new state agency, the Coastal Protection and Restoration Authority (CPRA), pursues this integrated mission. In a similar manner, governance over the GNO levees has been consolidated into two regional authorities. The Southeast Louisiana Flood

Protection Authority-East Bank (SLFPA-E) merges the previously independent Orleans Parish, Jefferson Parish, and Lake Borgne (St. Bernard Parish) levee boards that operated on the East Bank of GNO, while a corresponding SLFPA-W does the same for the West Bank. Of crucial importance to specifying our system, these changes in governance reflect an emerging spirit of civic engagement by the public in the post-Hurricane Katrina era. However, it also important to point out that no major reforms have been undertaken at the USACE and (as mentioned in one of the QMAS workshops) they remain the only branch of the military that has not implemented systems engineering.

The above discussion provides an indication of the momentum that has developed toward modernizing views on managing the relationship between the people and the landscape of Southeast Louisiana, and this momentum culminated in the State of Louisiana's recently released 2012 State Master Plan (SMP) (Louisiana Coastal Protection and Restoration Authority 2012). Without a doubt, systems level thinking was at the heart of the developers of the master plan, though there are areas where they fell short of this goal. As a key principle, the SMP posits: "Systems approach: The Master Plan will be evaluated using a systems approach to flood risk reduction and restoration, whereby benefits of actions and trade-offs among potential solutions will be identified and considered." The SMP process employed both a wide variety of technical experts and engaged stakeholders through planning forums and public comment sessions. The system specification utilized by the SMP is shown in **Figure 15**, which shows the models used by the SMP and linkages between the models.

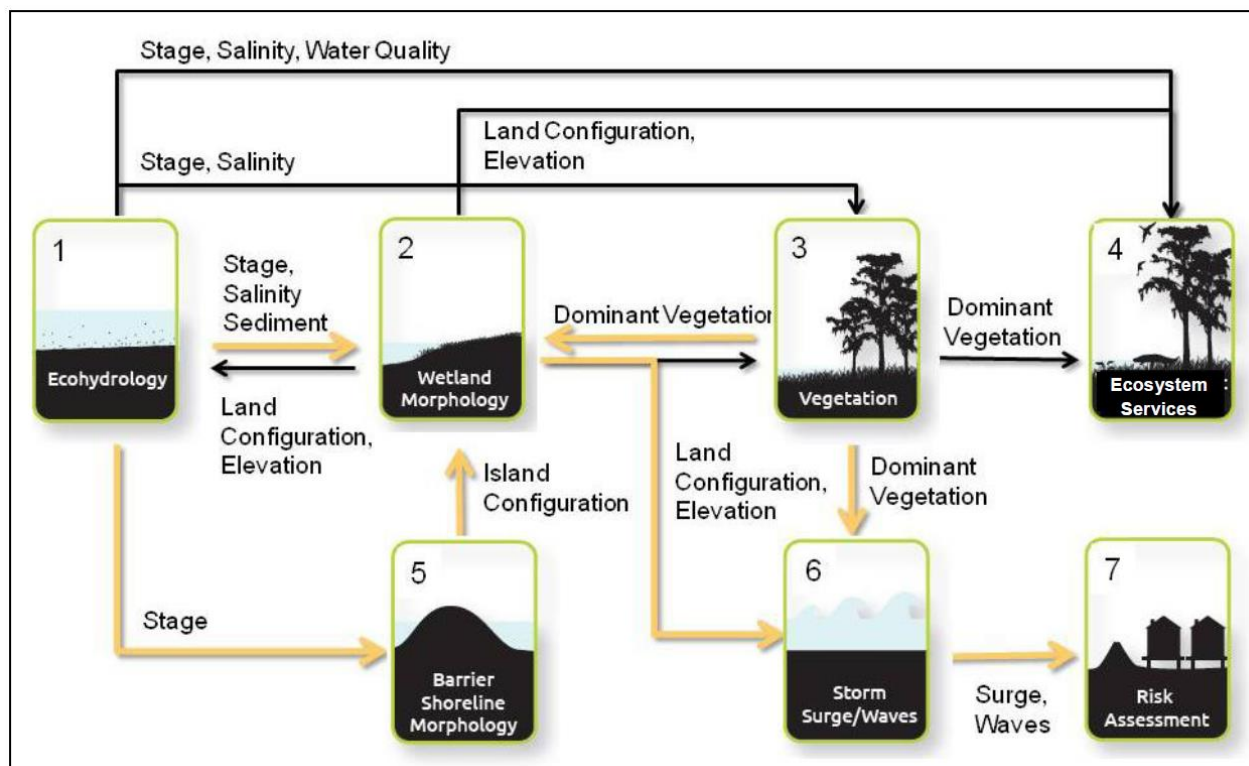


Figure 15: The SMP specification of system, as a set of coupled models that simulate various elements of the system (Louisiana Coastal Protection and Restoration Authority 2012).

Since 2005, there have also been improvements in the community lines of defense. The rebuilt I-10 bridge over Lake Pontchartrain improves both the capacity and reliability of this evacuation route. Additionally, local governments have complemented the personal vehicle based evacuation with assisted evacuation plans and procedures for special needs residents with impaired mobility. Likewise, all of the coastal parishes of Louisiana now have nursing home sheltering agreements with inland communities (Lopez 2003). At the same time, many parishes have adopted new building elevation requirements based on the National Flood Insurance Program's guidelines (City of New Orleans Office of Homeland Security Emergency Preparedness 2011) and thousands of homes have been elevated (Taffaro 2012), some of them up to 15 ft. above ground level. Additional codes, based on standards published by the International Code Council (International Code Council 2013), have been adopted statewide to mitigate wind damage.

Bringing all of this together is continued improvements in public risk communication and education. IPET's Risk and Reliability contained the first ever publically available maps depicting flood risk for GNO. The 2012 SMP has depicted land change and estimates of flood depths due to a 100-yr event over the next 50 years if nothing is done. It also contains a map that explicitly delineates which areas should reasonable expect to have structural protection (if the SMP is fully implemented) and areas that will be dependent on coastal and community protection. These maps certainly help citizens understand flood risk and make risk informed decisions.

Limits to the Utilization of Systems Planning

Unfortunately, given the HSDS's vast footprint, complexity, and large number of diverse stakeholders, it should not be surprising that there have been limits to the full utilization of systems planning. Many of these limitations were identified by participants in 3 QMAS workshops that were held during 2012. Engineers who participated in the workshops observed that TF Guardian and TF Hope largely ignored the IPET recommendations on adopting systems planning and engineering. They considered landscape a baseline, not part of the system. While TF Hope looked at the functionality of its projects, they adopted a limited view of system. Also, the repairs and improvements that have been undertaken largely ignored life cycle considerations. Examples include the pumps on the new pump stations for the Orleans parish outfall canals, which are considered inefficient and expensive to maintain and the decision by the USACE to go without rust blocking protective coating on sheetpiling, which limits the useful lifetime of the sheetpiling. Similarly, the USACE's hurricane flood protection projects remain in organizational silos and separated from other large USACE projects, such as the SELA Urban Flood Control Project. This massive USACE project addresses rainfall flooding by improving interior drainage canals and pump stations. The goal is to provide a level of protection against 10-yr rainfall events, a much different level of protection than that specified for the hurricane flood protection. These and many related issues are elaborated on in later sections of this report.

While the SMP's effort to adopt a systems approach to flood risk is notable, there are shortcomings that cannot be ignored. More specifically, the SMP process utilized an incomplete system specification that did not include all of the lines of defense. Evacuation routes were ignored while coastal transportation embankments and the offshore shelf were only implicitly

included in the terrain dataset used in modeling. Additionally, while the SMP utilized a comprehensive suite of coupled models, the approach was still modular and did not fully capture interactions between system elements. Finally, the SMP projects were evaluated based on economic criteria while health and safety outcomes were not considered.

Methods for “Integrated Systems” Definition: Where do we go from here?

While the efforts to implement systems planning and engineering are commendable, it is also true that these efforts have fallen short of this goal and the current system in 2012 largely remains a system in name only. Clearly, there are many legitimate and staggering obstacles that must be overcome before the disparate elements of the GNO HSDS are fully integrated into a functional system that achieves a defined objective. The past 300 years of human modifications to a 7,000 year old deltaic landscape (Wamsley, Cialone et al. 2009) have produced system elements that are spread over a large footprint and not easily integrated. This complex structure is exacerbated by equally complicated organizational relationships.

Where do we go from here? How do we take this complicated and diverse set of elements and begin to conceptualize them as an integrated system? What is needed to see an integrated system architecture realized on the ground? The bulk of this study focuses on utilizing three tools that have proven useful in designing and building complex systems. They are (i) systems engineering, (ii) systems modeling, and (iii) the Quality Management Assessment System. Each of these tools is described below.

Systems engineering is a field of engineering that draws upon various disciplines to ensure that complex projects are designed and built as integrated systems (Technical Board 2004). It involves coordinating designs of multiple individual elements to enable complete integration, clearly defined objectives based on stakeholder requirements, management strategies over the full life cycles of the system, and incorporating operational behaviors into design considerations. To a large extent, systems engineering is based on the central premise that the properties of a system as a whole differ considerably from the sum of the parts' properties (Oliver, Kelliher et al. 1997; Technical Board 2004). As just one example, even if individual levee and floodwall elements were designed properly, transition failures between these elements can result in a failure of the system to fully function. It boils down to managing the complexity of systems that emerges in large scale systems, where designing elements in isolation can lead to failures when the elements are integrated into the system.

Systems engineering has early origins in electronic design and defense systems. As one of the earliest applications of the discipline, Bell Labs utilized early system engineering principles and practices when it developed one of the world's first complex systems in the 1940s: the national telephone network. Boeing adopted it for early designs of its signature 747 model aircraft (which contains 6 million parts) and NASA has used systems engineering in designing the Saturn V rockets, the Space Shuttle, modules for the international space station, and numerous other spacecraft. For most of the last 6 decades, the US Department of Defense has utilized systems engineering in a variety of projects, such as missile defense systems, fighter jets, and naval vessels. Perhaps because its civil engineering projects are too complicated and must fit

competing interests and objectives, the USACE is exempted from DOD requirements on using systems engineering.

In 1990, the National Council on Systems Engineering was formed to improve and standardize systems engineering practices. In 1995, the name was changed to the International Council on Systems Engineering (INCOSE) and the group emerged as the source of accepted professional standards and certification in systems engineering. Today INCOSE has over 8,000 members and 60 local chapters spread across the globe. It is the publisher of the *INCOSE Systems Engineering Handbook*, the *Journal of Systems Engineering*, and other authoritative publications. It has been heavily involved in enacting professional standards for systems engineering, hosts international conferences on the topic, and provides professional certification for system engineering specialists.

INCOSE, particularly the *INCOSE Systems Engineering Handbook* (Technical Board 2004) stresses that systems are the combination of structural elements and behavioral elements. As such, the analytical tools developed and promoted by INCOSE cover both of these arenas. Structural tools, such as the system hierarchy and block diagrams, inventory and describe the physical objects that make up a system. Behavioral tools, such as activity diagrams and use case diagrams, focus on the behavioral and organizational elements of the system.

The goal of systems engineering is to ensure that the various disparate elements of a system are designed and built in a way that integrates them into a coherent and functioning system (Oliver, Kelliher et al. 1997; Technical Board 2004). To achieve this goal, systems modeling is used extensively to derive executable models for predicting system behavior and performance (Technical Board 2004). Without a doubt, systems modeling has become an important tool in systems engineering, and new modeling languages have been developed just for this purpose. One such modeling language is SysML (Systems Modeling Language), a derivative of the Universal Modeling Language (UML). In SysML, the user creates explicit models of the system's structure and behavior that accounts for system elements along with their attributes, relationships, requirements, and activities. Essentially a set of diagrams based on block diagram notation, each "block" in a SysML model represents a system element. The block contains properties and attributes that describe the system element and its characteristics. Once the system specification has been completed in SysML, the diagrams and modeling tests provide checks that all the required properties, attributes, and functionality are present within the system.

To our knowledge, past practice in system engineering has been mostly limited to systems such as aircraft, automobiles, electronic equipment, and other such moveable objects. While these systems may contain millions of parts, these are still isolated products with a small footprint and largely functionally independent of the landscape. Telecommunications and air traffic control systems are two well known applications of systems engineering that involve a system spread over a large geographical footprint. Still both of these systems consist of a relatively limited set of electronic and mechanical parts that function largely isolated from landscape. To our knowledge, there have not been any attempts to design or assess a major, landscape scale infrastructure project using systems engineering. Likewise, we are not aware of any attempt to apply the discipline when designing a system that involves as many governance structures as are involved with the GNO HSDS. In other words, while systems engineering is widely utilized in

designing products that involve degrees of complexity, there are no precedents for applying the discipline to a system that resembles the scale and complexity of the GNO HSDS.

The only known exception to this observation is a recent assessment of the flood protection levees for Sherman Island in California's Sacramento Delta (de Corn 2011). While not labeled as a systems engineering analysis, this study followed a process called Quality Management Assessment System (QMAS) that parallels many of the principles of systems engineering (see **Figure 16**). QMAS is essentially a risk assessment instrument that utilizes system "assessors" who possess knowledge and experience with system elements. The QMAS process consists of a facilitated discussion whereby assessors share their experiences and the facilitators formalize this knowledge into a system specification. Once this information has been collected, it can be used in quantitative estimates of the probability of system failure. Pioneered for offshore platforms, QMAS has undergone limited testing in a flood infrastructure context, such as the Sherman Island exercise.

Most of the systems engineering guidance and examples are geared toward industrial production and consider the entire process from initial planning to design and then to manufacturing and maintenance. In this regard, it is important to note the differences between designing a system to be built and assessing a system as it has been built. For this reason, many aspects of systems engineers are not applicable to the current study. We simply cannot go back in time and request that the original French settlers conduct a thorough Requirements Analysis and Systems Engineering Management Plan before they start building the first levee along the Mississippi River. What we can do is use the concepts and tools of systems engineering to assess the current "as-is" HSDS. In this respect, QMAS, an assessment tool, greatly compliments the published systems engineering guidance.

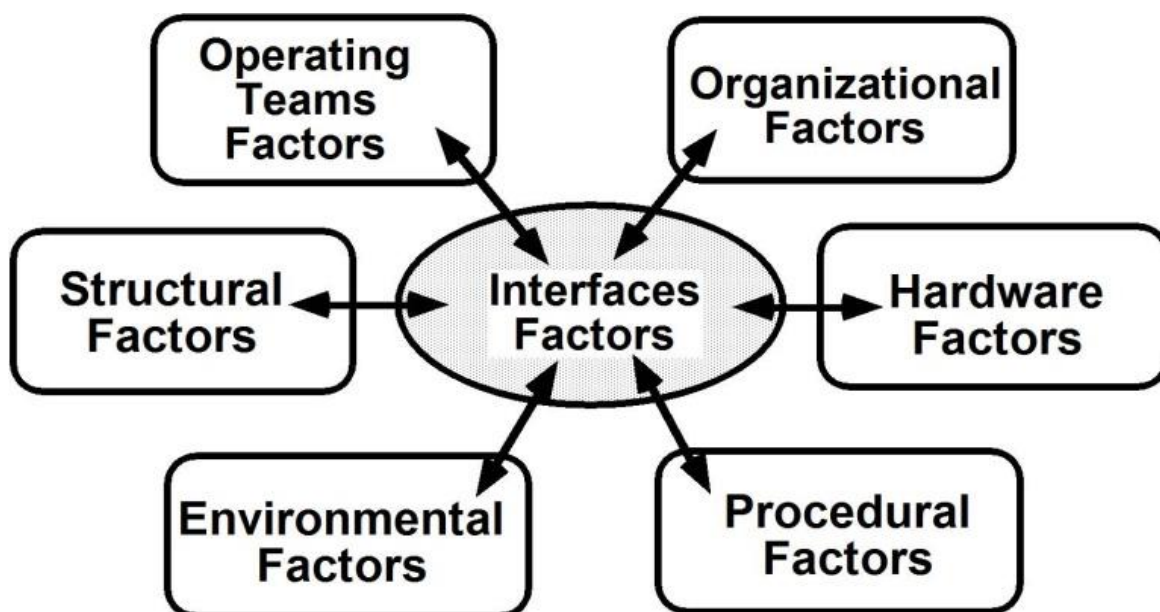


Figure 16: Quality Management Assessment System (QMAS) Components.

3) Preliminary System Specification

A system is defined by INCOSE as “an integrated set of elements, segments and/or subsystems that accomplish a defined objective” (Technical Board 2004). INCOSE further notes that “the growing complexity in all areas of systems development has increased the need for Systems Engineers” (Technical Board 2004). This is certainly the case with the GNO HSDS, a very complex system with a large geographic footprint, interactions over a range of spatial scales, and subject to unknown environmental stressors.

The bulk of this study centers on three QMAS style workshops held over the fall and summer of 2012. These workshops are summarized in the next section, while Appendix A provides a detailed recounting of the workshops. This section covers research undertaken to prepare for and complement these workshops. To prepare for the workshops, we also conducted extensive background research on the GNO HSDS. Key tasks include background research on systems engineering and related tools, compiling and reviewing the relevant literature on the GNO HSDS, and an initial system specification to frame the workshop discussions. We also compiled a geodatabase that includes data layers related to terrain, the lines of defense, surge magnitudes, and organizational jurisdictions.

Risk: Why we are doing this report

As a risk based assessment of the GNO HSDS, risk is a basic concept that underlies this study. Based on a complicated, future looking concept, the emergent science of risk analysis has yet to acquire a universal definition and standardized measurement techniques of the core concept of risk. There exists great agreement that risk involves some measure of uncertainty (probability) of an outcome along with negative consequences of that outcome. Mathematically rigorous definitions of risk define the concept as the integral of the probability density function (PDF) times a consequence function. However, this rigorous definition is of limited practical use for many reasons, including unknown PDF's and consequence functions along with the practical limits of integrating over all possible scenarios. A number of analysts use the much simpler definition of risk as probability multiplied by consequences. However, this definition only calculates the risk of a single possible event, not the overall risk of the many events that are possible.

The INCOSE handbook defines risk as “measuring of the uncertainty of attaining a goal, objective, or requirement pertaining to technical performance, cost, and schedule” (Technical Board 2004). While this definition is of limited use in quantitative risk assessments, it gets straight to the heart of the issue: uncertainty and goals. In this particular case, the goal consists of protecting the people, communities, and infrastructure of GNO from storm surge induced flooding, a necessary condition toward achieving the much larger goal of a sustainable city given predicted sea level rise and ongoing coastal land loss. However, tremendous uncertainty exists over achieving that goal: uncertainty related to possible storm surge conditions, uncertainty in the future state of coastal lines of defense and the performance of the structural defense, and uncertainty in the effectiveness in community practices, such as evacuation and house elevation.

In many ways, this study seeks to identify these uncertainties and to assess how these uncertainties impact the goal of preventing flood damages.

While quantitative risk assessments based on a rigorous definition have many benefits, this type of analysis is far beyond the scope of this study. There are simply too many unknowns to plug meaningful numbers into the risk equations. Others, most notably the IPET Risk and Reliability Analysis along with LACPR and the SMP, have made meaningful attempts at quantitative flood risk analysis for GNO. However, even the most thorough analysis, IPET's Risk and Reliability Analysis (Interagency Performance Evaluation Task Force (IPET) 2009), only assesses three characteristic scenarios: the 50 yr., 100 yr., and the 500 yr. storm surge. This fact illustrates the limits of the state of the art in applying risk analysis to such a complex system.

Instead of adding a few decimal places to the results provided by IPET, this study is an attempt to start from scratch with a clean slate. INCOSE writes that "risks generally needs to be assessed subjectively because adequate statistical data are rarely available" (Technical Board 2004), a statement that is particularly true about the GNO HSDS. In particular, we seek to more accurately specify the flood defense system and to identify the major uncertainties related to system performance. In doing so, we hope to provide a body of baseline information that can be used in future modeling that is systems based.

Systems Engineering Concepts and Tools

Systems engineering provides many of the tools needed to deal with the uncertainties spread across various scales. It provides a systematic method for inventorying the elements of a complex system and identifying potential factors that impact system performance. The next paragraphs describe these tools, while most of the rest of this report applies these tools to the HSDS.

As one of the first tasks, the system specification establishes the functional baseline to ensure user requirements are met. Once the system is specified, the functional baseline is then decomposed and allocated to lower level elements of the system, which comprise the allocated baseline (meaning the functions have been allocated to specific system elements). This then leads to the product baseline, which includes engineering drawings and configurations which describe the system as it should be built. Stressing the importance of getting it right at the beginning, INCOSE notes that "the most common source of development problems is flaws in the system specification process" (Technical Board 2004).

When describing complex engineered systems, Oliver, Kelliher, and Keegan (1997) stress "the importance of separating structure from behavior." They write that "structure is *how it is built*," but "behavior is the *what it does* part of the system configuration." That is to say that an inventory of just the parts needed to build a system without describing what those parts do and how they interact with system operators does not completely specify a system.

INCOSE points out that one of the key first tasks in engineering a complex system is to "establish nomenclature and terminology that support clear, unambiguous communication and

definition of the system, its functions, components, operations, and associated processes” (Technical Board 2004). The first step for this task is to develop a system hierarchy, that is an ordered inventory of system elements (both parts and processes). The system hierarchy establishes a standard nomenclature for communicating between team members from different disciplines.

INCOSE defines the following levels of the system hierarchy:

System	An integrated set of elements, segments and/or subsystems that accomplish a defined objective, such as an air transportation system.
Segment	A major product, service, or facility of the system, e.g., the aircraft element of an air transportation system (commonly used, but subsystems can be used instead of element/segments).
Subsystem	An integrated set of assemblies, components, and parts which performs a cleanly and clearly separated function, involving similar technical skills, or a separate supplier. Examples are an aircraft on-board communications subsystem or an airport control tower as a subsystem of the air transportation system.
Assembly	An integrated set of components and/or subassemblies that comprise a defined part of a subsystem, e.g., the pilot’s radar display console or the fuel injection assembly of the aircraft propulsion subsystem.
Subassembly	An integrated set of components and/or parts that comprise a well-defined portion of an assembly, e.g., a video display with its related integrated circuitry or a pilot’s radio headset.
Component	Comprised of multiple parts; a cleanly identified item, e.g., a cathode ray tube or the ear-piece of the pilot’s radio headset.
Part	The lowest level of separately identifiable items, e.g., a bolt to hold a console in place.

(Source: Technical Board 2004)

They also provide the following generic system hierarchy diagram (**Figure 17**).

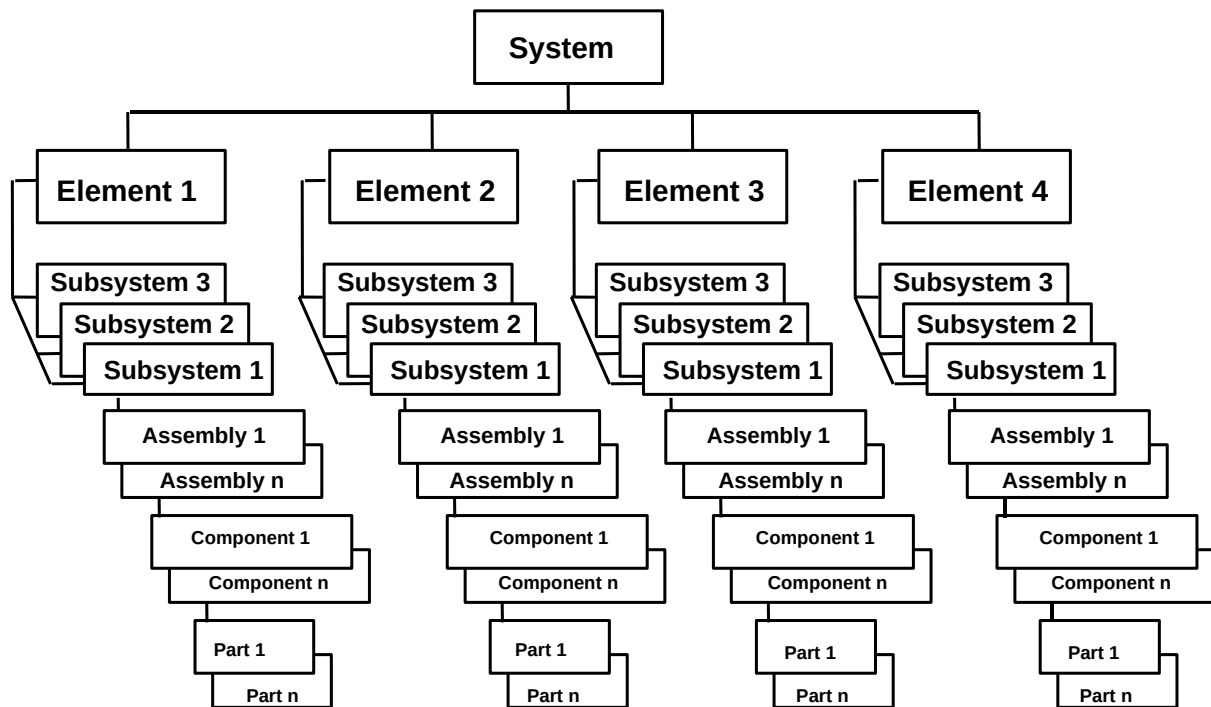


Figure 17: Generic system hierarchy diagram (Source: INCOSE 2004).

Echoing the Systems Engineering standards, risk analysts tasked with assessing the storm surge risk for the study area have noted the need for a complete, standardized system inventory. IPET writes that “the reliability analysis starts with a detailed inventory of the engineering characteristics of every section of the hurricane protection system” (Interagency Performance Evaluation Task Force (IPET) 2009). In turn, Ayyub, et al. (2009) hint at the need to inventory all features (not just the engineered elements) that impact surge and wave dynamics: “Defining the physical features of the system required an accurate inventory of all components that provide protection against storm surge and waves” (Ayyub, Foster et al. 2009a). However, both of these studies adopted a structural view of the system, and their inventories are largely confined to just levees, floodwalls, floodgates, and pump stations.

Once the system is specified and the system hierarchy is completed, the systems engineer has the information needed to begin systematically identifying the factors and situations that could inhibit system performance.

This step involves identifying the system interactions, factors of concern, and scenarios of concern. System interactions are functional connections between two system elements. For example, in the GNO HSDS, levees, floodwalls, floodgates, and pump stations are all separate system elements but their interaction creates the perimeter structural protection component of the HSDS. Likewise, barrier islands, sounds, and marsh are all elements of the coastal protection component of the HSDS. While individual elements, the exchange of tidal waters, freshwater, sediments, nutrients, and animal species between them is crucial to sustaining them and ensuring

that they perform in an integrated manner. With a properly specified system, systems modeling can be used to help identify the system interactions along with factors and scenarios of concern.

The goal of systems modeling, as described by Oliver, Kelliher, and Keegan (1997) is to produce “rigorous, executable models of behavior (what things do) and structure (how things are built)” of the system and its performance under a variety of representative use cases. Essentially, IPET, LACPR, and the SMP all completed this step. However, to fully fulfill this goal requires overcoming what they call the gap “between needs expressed in informal, natural language and component specifications described in the multiple engineering notations.” To overcome this gap, the authors suggest that systems engineers “capture... system requirements and specifications in models that are computer executable and unambiguous,” a step that these studies only partially fulfilled.

Systems modeling languages, such as SysML, are key tools in capturing requirements and specifications and in modeling structure and behavior. In SysML, the system elements are represented by “blocks.” Blocks contain labels for the elements along with properties to describe the attributes and characteristics of the element. Blocks can then be joined in different types of diagrams, such as the block definition diagram (BDD) which describes the structure of the system. Other SysML diagrams that represent structure are the internal block diagram, the parametric diagram, and the package diagram. Additional SysML diagrams are used to represent behavior. These include the activity diagram, the sequence diagram, the state machine diagram, and the use case diagram.

Of note, systems engineering and SysML contain many more concepts and tools that are important for ensuring system performance but beyond the scope of this current study. Such additional concepts include: Requirements validation, Functional Analysis/Allocation, and the Systems Engineering Management Plan.

Multiple Lines of Defense Strategy in a Systems Engineering Context

MLODS has been described by its creators as “a planning methodology to coordinate and prioritize conventional restoration methods and projects for coastal habitats and flood protection” (Lopez, Snider et al. 2008). However, beyond just a methodology or strategy, MLODS also represents an initial, high level system specification for the GNO HSDS. The well known MLODS diagram provides a list of components in the system hierarchy (see **Figure 18**). The rest of this section deals with our efforts prior to the workshops to fill in additional information in the system hierarchy.

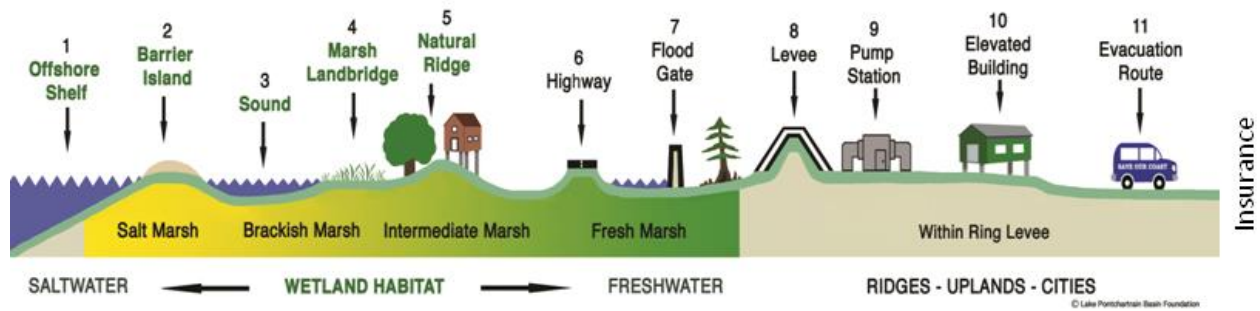


Figure 18: The Multiple Lines of Defense (MLODS) diagram.

Numerous documents provide plain text descriptions of the GNO HSDS, such as the MLODS reports and papers published by LPBF (Lopez 2006; Lopez, Snider et al. 2008; Lopez 2009). Other plain text descriptions of the system include Louisiana’s 2007 and 2012 State Master Plans (Coastal Protection and Restoration Authority of Louisiana 2007; Louisiana Coastal Protection and Restoration Authority 2012), local master plans, local infrastructure improvement plans, state and local hazard mitigation plans, and the major post-Hurricane Katrina levee investigations. Media reports, particularly in regards to coastal restoration efforts and the housing elevation program, also provide useful plain text descriptions of the system components. As part of the initial system specification, these documents were reviewed to identify the relevant pieces, parts, and processes that comprise the GNO HSDS.

A number of deliverables resulted from this work, including a comprehensive bibliography of plain text descriptions of the system, a geodatabase that contains relevant spatial data layers (including layers for each of the system components listed in **Figure 18** above), a requirements table based on numerous regulations and plans from Federal, state and local governments, and a summary of lessons-learned drawn from the levee investigations. The next few sections detail how these resources were used to fill in the system hierarchy and other initial system specification diagrams.

Literature Review

To help specify this system, we reviewed a number of documents related to this and similar systems. Planning documents, such as the State Master Plan, the 2008 MLODS report, various local master plans and mitigation plans, and evacuation guides helped us compile a comprehensive inventory of system elements along with illustrating how these elements integrate into the system. Also useful were various operation and design manuals from the USACE and local agencies. The scientific literature covered a number of important topics related to our system. Evacuation was one such topic that is prevalent in the literature. Likewise, literature on ADCIRC helped illustrate how coastal landscape features impact storm surge and various studies in the natural sciences related to the coastal lines of defense.

In addition to a comprehensive bibliography on the HSDS, two additional major deliverables were derived from our literature review: a) an initial requirements table, and b) an initial system

inventory and hierarchy. Both of these products are presented later in this report, while a few general illustrative comments are provided here.

In systems engineering, the requirements table lists the functional requirements necessary to meet the system objective. Once listed, the requirements can then be allocated to specific system elements. For example, to fulfill the system objective of a resilient city, the City of New Orleans' Master Plan calls for 1-in-400 year protection where appropriate. This requirement can be linked with specific system elements, specifically the structural protection element. Likewise, the evacuation and house elevation lines of defense are used to fulfill the requirement of minimizing losses to hurricane induced flooding.

The system hierarchy is an ordered inventory of system pieces, parts, and procedures. The INCOSE System's Engineering handbook provides a generic system hierarchy. To fill in the hierarchy for this specific system, the literature database was reviewed to indentify most elements of the system. For example, the literature on levees and floodwalls lists clays, grass, sheet piling, cement, rocks, and mortar as parts. Likewise, the evacuation literature includes highway and interstate lanes, contraflow crossovers, traffic control personal, temporary signage, NHC warnings, and informational pamphlets. Importantly, many attributes or characteristics of the various elements are also listed in the literature. For example, levee sections have specified heights and evacuation routes have specified traffic capacities. As described later, SysML allows the system analysts to include this information in the system specification.

Geodatabase

Compilation of spatial datasets related to the GNO Hurricane Surge Defense System is an ongoing endeavor. During the course of the project, we have compiled a basic set of spatial datasets. However, the spatial datasets still need to be populated with important attribute data.

At the most high level view, our geodatabase includes shapefile layers for most of the LODs. We have shapefiles, which depict the LOD's footprint at the 10,000 ft. level, covering the barrier islands, sounds, marsh land bridges, ridges, transportation embankments, floodgates, levees, pump stations, and evacuation routes. We have not acquired any spatially referenced dataset on housing elevation or insurance. Our geodatabase also includes supporting datasets such as land loss layers, predicted surge levels, climatology, elevation, population, and imagery.

Case Studies

Two tropical events have tested the system during the course of this study: Tropical Storm Lee and Hurricane Isaac. The paragraphs that follow provide a brief overview of the performance of the system during these two case studies.

Tropical Storm Lee

Tropical Storm Lee in September 2011 is one tropical event that occurred during the course of this study and after most of the post-Hurricane Katrina upgrades had been completed. This

relatively mild storm was still very wet. Most of the region experienced over 8" of rainfall, while some areas got up to 15". As the storm crossed land the winds of the storm tilted Lake Pontchartrain in various directions. But the relatively weak offshore winds only partially filled it before hand. Tide levels reached about 4 - 6 ft. across Lake Pontchartrain (**Figure 20**).

Unprotected coastal areas flooded due to the storm surge, including US Highway 90, an evacuation route for the eastern sections of GNO (see **Figure 19**). Without structural protection, these communities depended upon coastal and community lines of defense. Protected areas experienced only isolated, street level rainfall flooding. No major flood impacts occurred within protected areas. Outside the protected areas approximately 275 homes flooded: 40 in St. Tammany, 25 in Lafitte, 10 in Terrebonne and Lafourche parishes, and 200 in Livingston Parish. Region wide five shelters opened, hosting approximately 50 people total. Emergency responders conducted 30 assisted evacuations from flooded areas in Lafitte. There was one near drowning in Slidell, and one reported fatality related to Tropical Storm Lee, a drowning in northeastern Mississippi.



Figure 19: Photo of U.S. Highway 90, an important evacuation route along the Gentilly Ridge, closed due to storm surge flooding outside the levee protection system during T.S. Lee, September 2011.

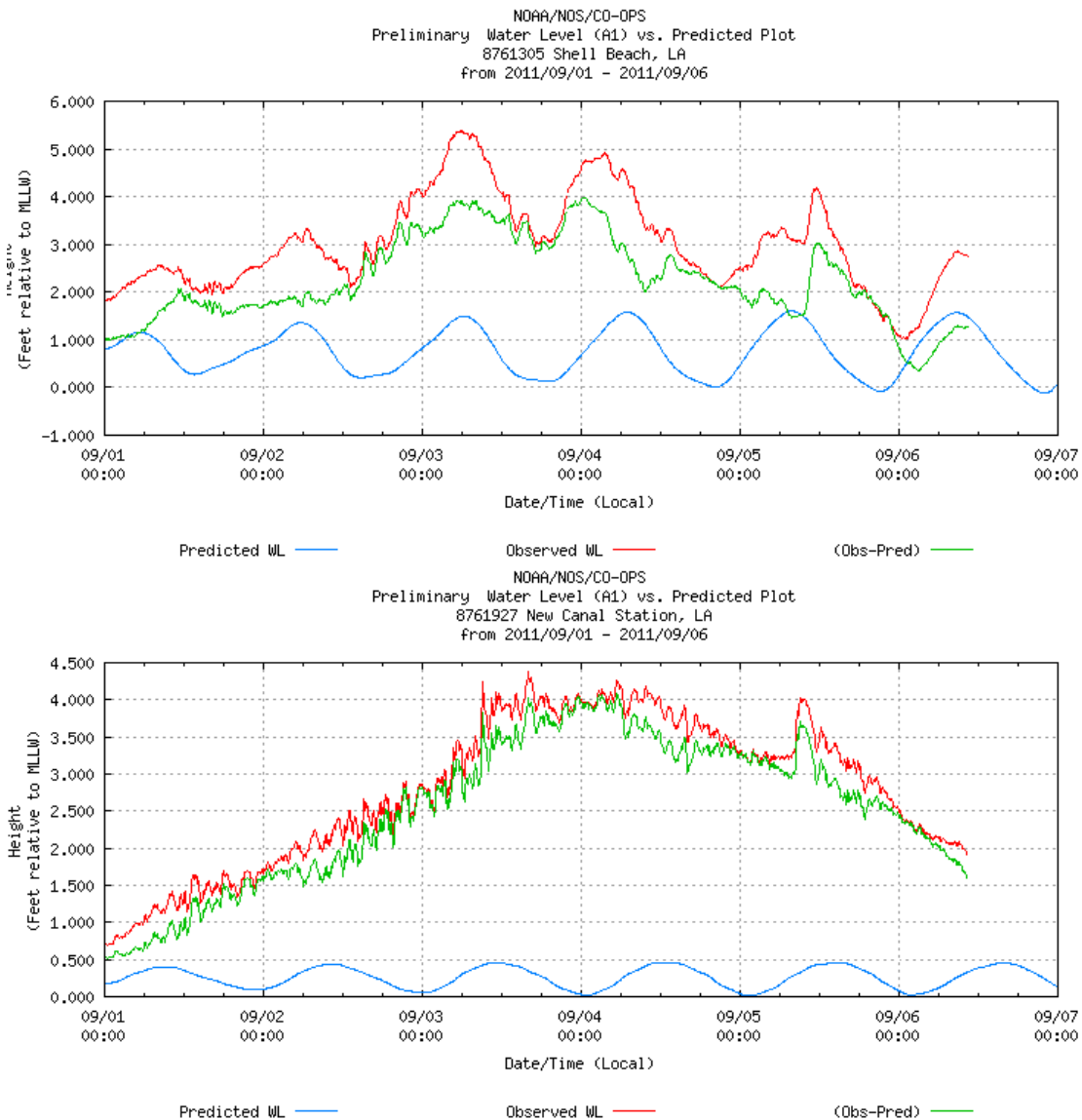


Figure 20: Tropical Storm Lee hydrographs from (a) Shell Beach near Lake Borgne and (b) New Canal Station on Lake Pontchartrain’s south shore.

Hurricane Isaac

The first major test of the GNO HSDS, Hurricane Isaac in August 2012 brought a significant surge into the region. Slowly skirting Louisiana’s southeast coast before turning north to pass near Baton Rouge, this category 1 storm slowly followed a “worst case” path that meant offshore winds bore down on the region for two days. In Lake Borgne, water levels reached 13.6 ft. against the Surge Barrier. The storm’s slow trek westward along the coastline filled Lake Pontchartrain and then tilted the Lake in many directions as the storm’s winds shifted during its trek across land. **Figure 21** shows the Hurricane Isaac’s storm surge in Southeast Louisiana. Of

note, Hurricane Isaac occurred prior to QMAS workshop 3, and the performance of the HSDS during Isaac was discussed then (see the next section and the Appendix A).

Facing its first event, closure operations at the Lake Borgne Surge Barrier were successful, keeping the Lake Borgne surge away from protected areas. Most other operations worked successfully, including the closure of all the necessary floodgates. There were some malfunctions that interfered with pump operations, including a failure of automatic controls to start a pump and a fire at a pump station. Additionally, power outages and the lack of backup power meant that rainfall accumulated when pumps stopped working (see **Figure 24**). Still no major flood impacts occurred within the levee protected areas. At the community level, the population within protected areas complied with emergency recommendations and sheltered in place for the duration of the storm. Outside the levee system, housing elevation was praised as reducing flooding impacts. Of note, the community demonstrated significant resiliency against two days of heavy rain and strong winds followed by a week of widespread debris and power outages.

Outside of the levee system, flooding severely impacted communities. On the East Bank of GNO, two communities were particularly hard hit by the largely unmitigated storm surge. A 14 ft. surge flooded Braithwaite to White Ditch in Plaquemines Parish. A total of 500 structures flooded, including 350 homes. Around 150 people were rescued, and two drowning deaths occurred. A 9 ft. surge took the Laplace subdivisions by surprise. Over 3,500 people were rescued, but no deaths occurred. Both flood events and regional flood patterns in general point to the weaknesses in all three major components of the system for communities outside the urban center. Both areas lacked sufficient structural protection and houses were largely slab-on-grade. Of particular concern, both areas are adjacent to significant land loss, which meant that Isaac's Category 1 winds were able to push water inland with reduced resistance from the coastal lines of defense.

Also, of particular concern, Hurricane Isaac's storm surge managed to flood, and hence close, I-10 in both directions. Not only did the aforementioned low-lying section of I-10 before the Twin Spans bridge flood, but also a low-lying section west of the city near LaPlace, La. also flooded.

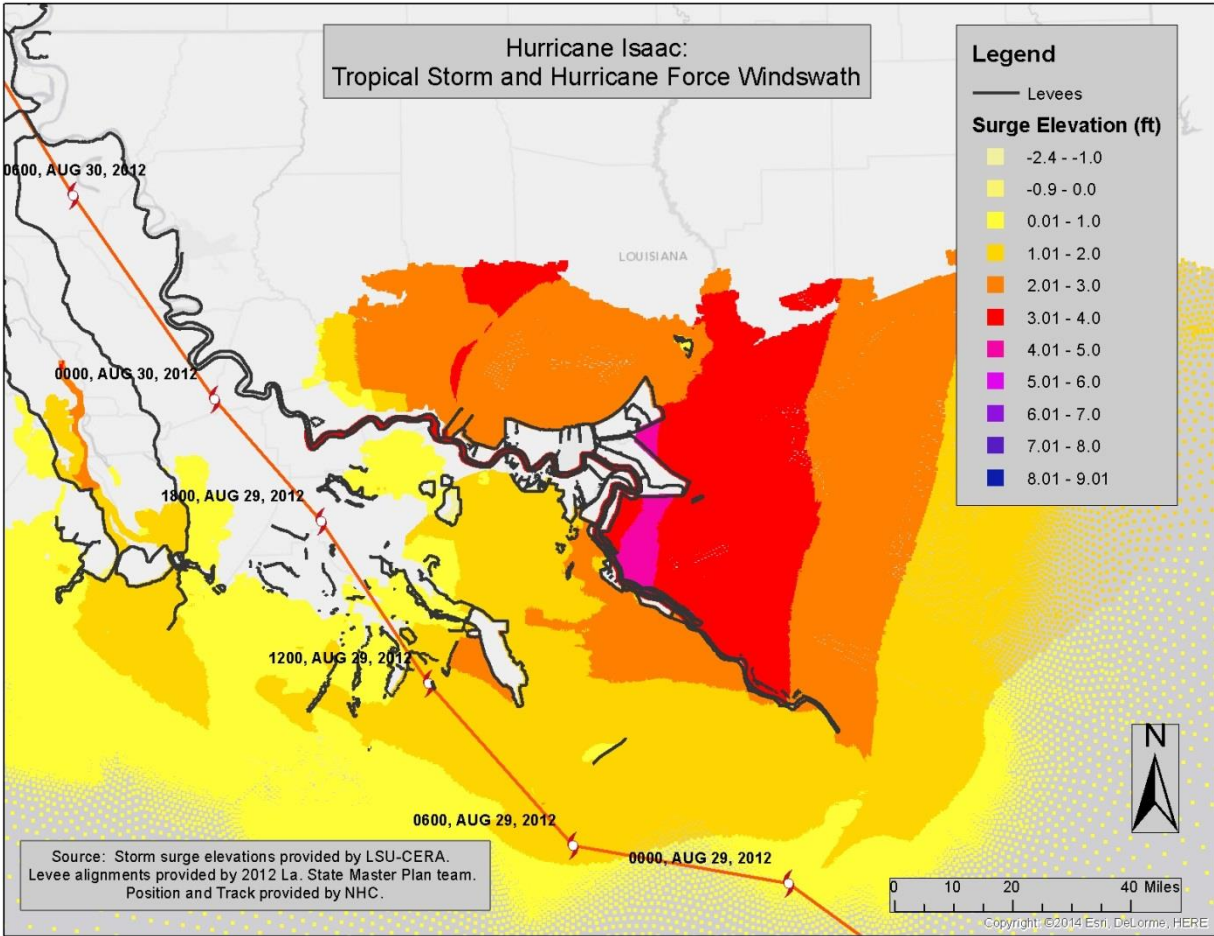


Figure 21: Hurricane Isaac storm surge heights based on the ADCIRC simulation.

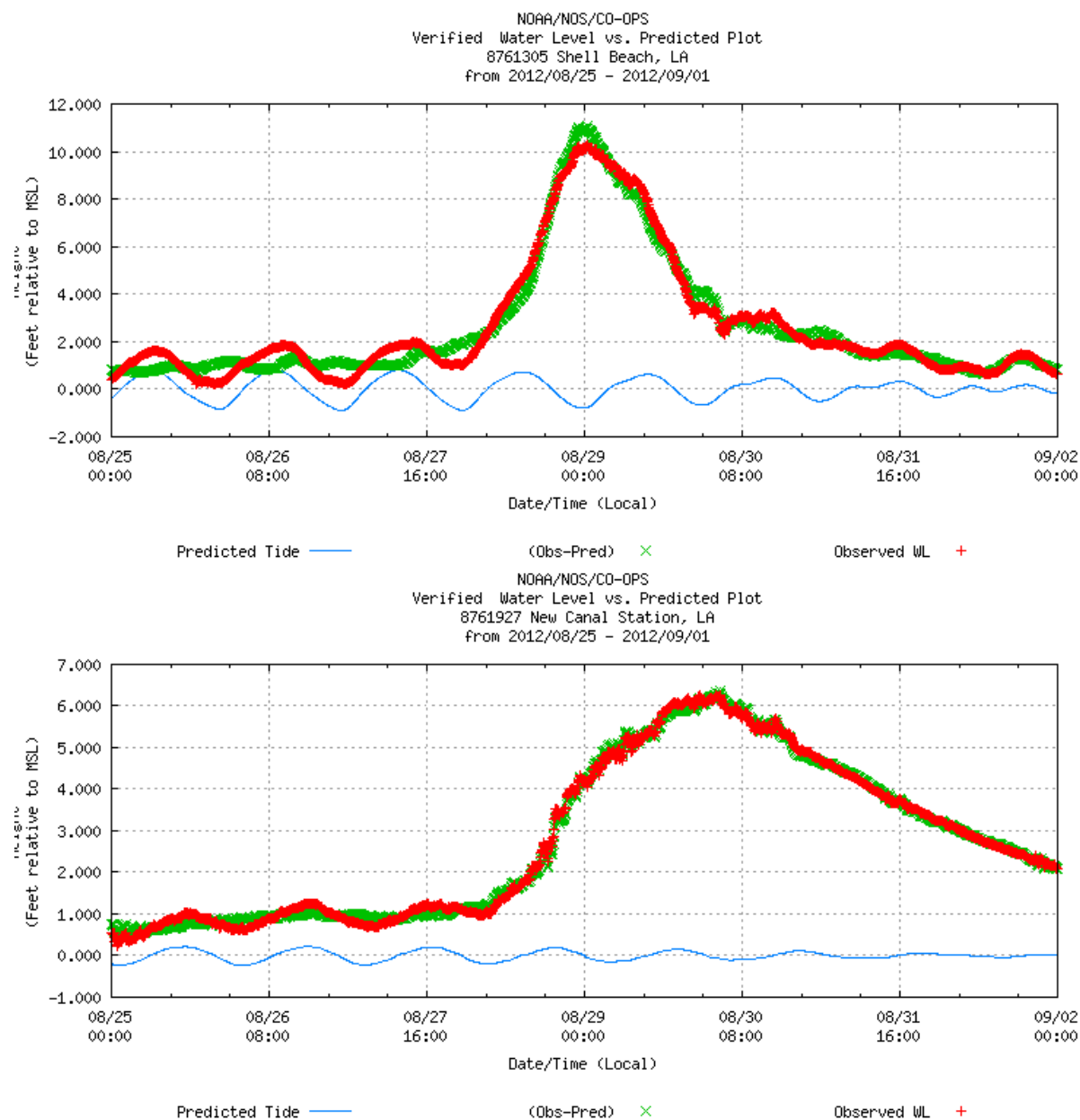


Figure 22: Hurricane Isaac hydrographs from (a) Shell Beach near Lake Borgne where water levels reached 10 ft. above sea level and (b) New Canal Station on Lake Pontchartrain's south shore where water levels reached 6 ft. above mean sea level.

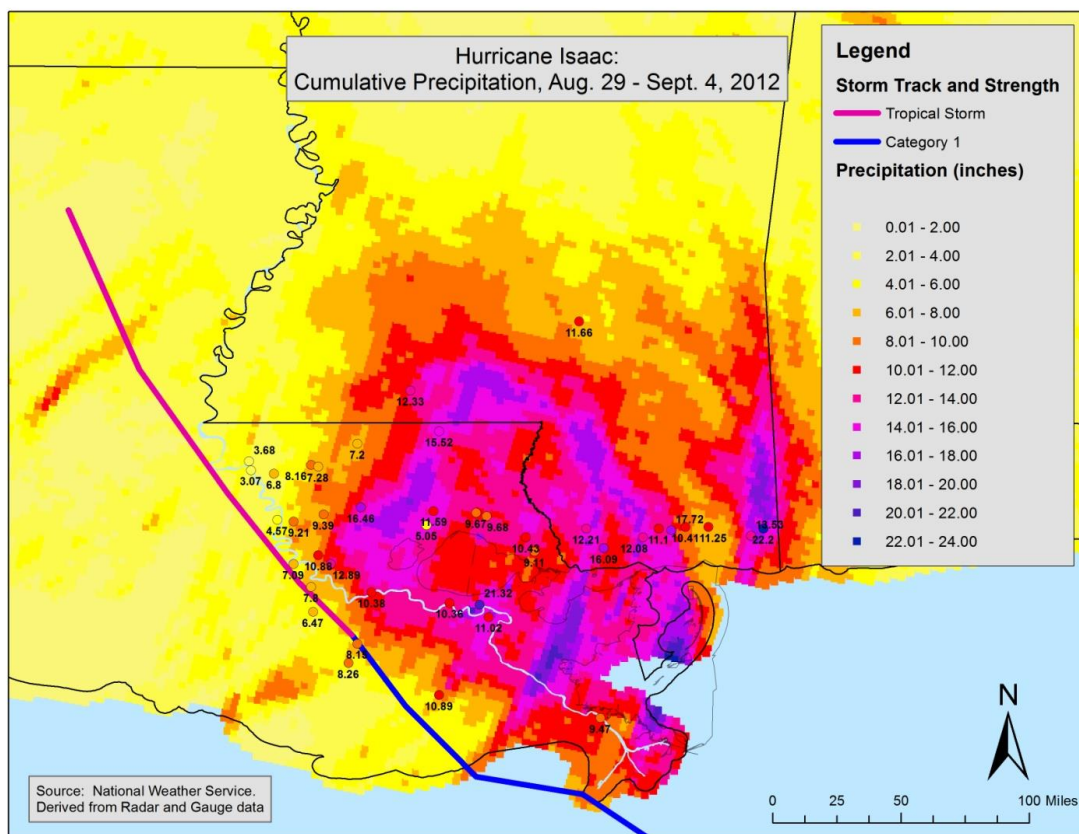


Figure 23: Hurricane Isaac rainfall, from radar and gage data.



Figure 24: Street flooding in the Gentilly neighborhood of New Orleans. The floodwaters first started to accumulate on Wednesday, August 29, 2012 around 10 am, following a power outage. Within the next few hours, pump operations resumed on back up generators and the streets were quickly dried.

Specifying MLODS as a System: High level overview

The preliminary research described above was synthesized into an early system hierarchy which was then used to identify a straw list of system interactions and factors of concern. At the outset, it was decided to group the 12 LODS into three categories: i) Coastal, ii) Structural, and iii) Community. This led to the initial system hierarchy shown in **Figure 25** below.

With this preliminary system hierarchy available, we developed a straw list of system interactions and factors of concern:

- Increased consequences of minor storms due to coastal land loss and sea level rise
- Unprotected low points along evacuation routes
- Central Wetlands Unit: Pump stations provide freshwater that maintain wetlands and wetlands protect levees

While not comprehensive, this list was presented in Workshop #1 and served to exemplify our approach and goals for the workshop participants.

Sea level rise and coastal land loss are certainties in Louisiana's present and future landscape. However, there remains significant uncertainty regarding the magnitude of these factors and their implications. **Figure 26** below shows the lines of defense along with coastal land loss on one map, and major factors of concern impacting system performance are immediately apparent. **Figure 26** shows that nearly all of the lines of defense, including even the perimeter protection

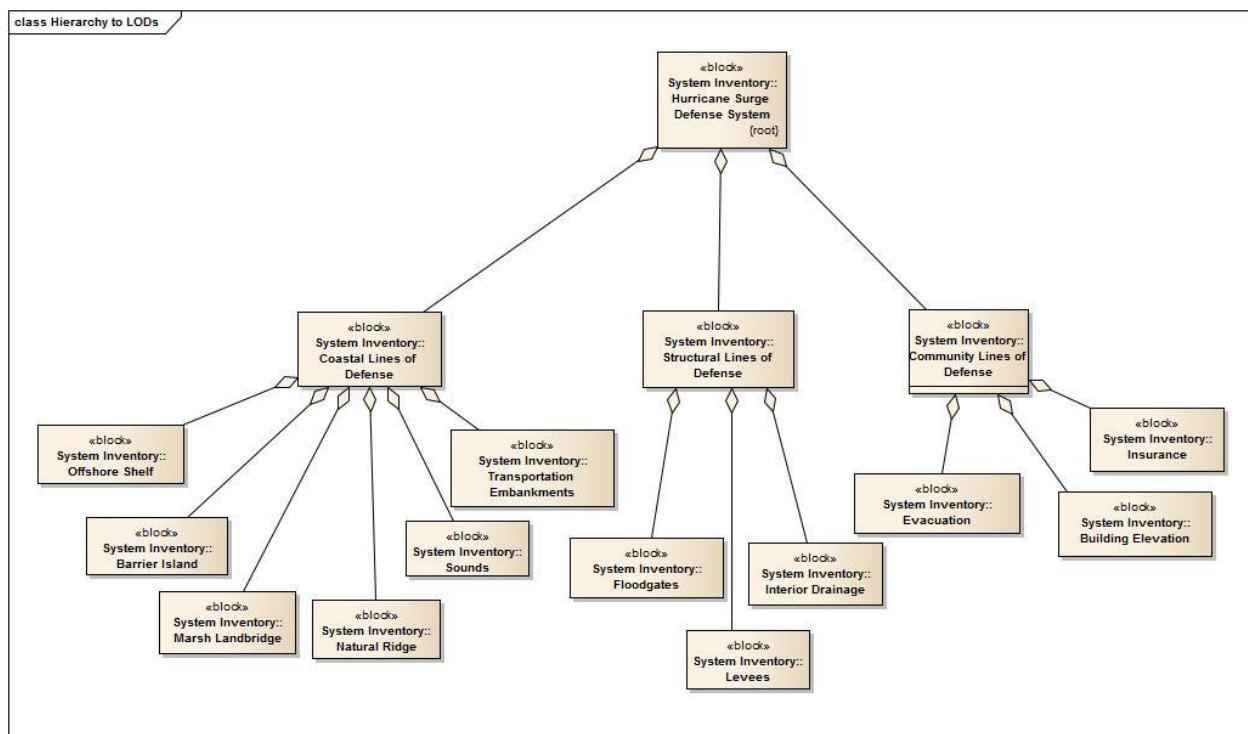


Figure 25: An initial, three level system hierarchy that lists the 12 lines of defense at the component level.

and the major evacuation lanes are impacted by land loss. Zooming in, **Figure 27** and **Figure 28** show that two crucial coastal lines of defense – barrier islands and marsh land bridges – face near term prospects of disintegrating due to land loss.

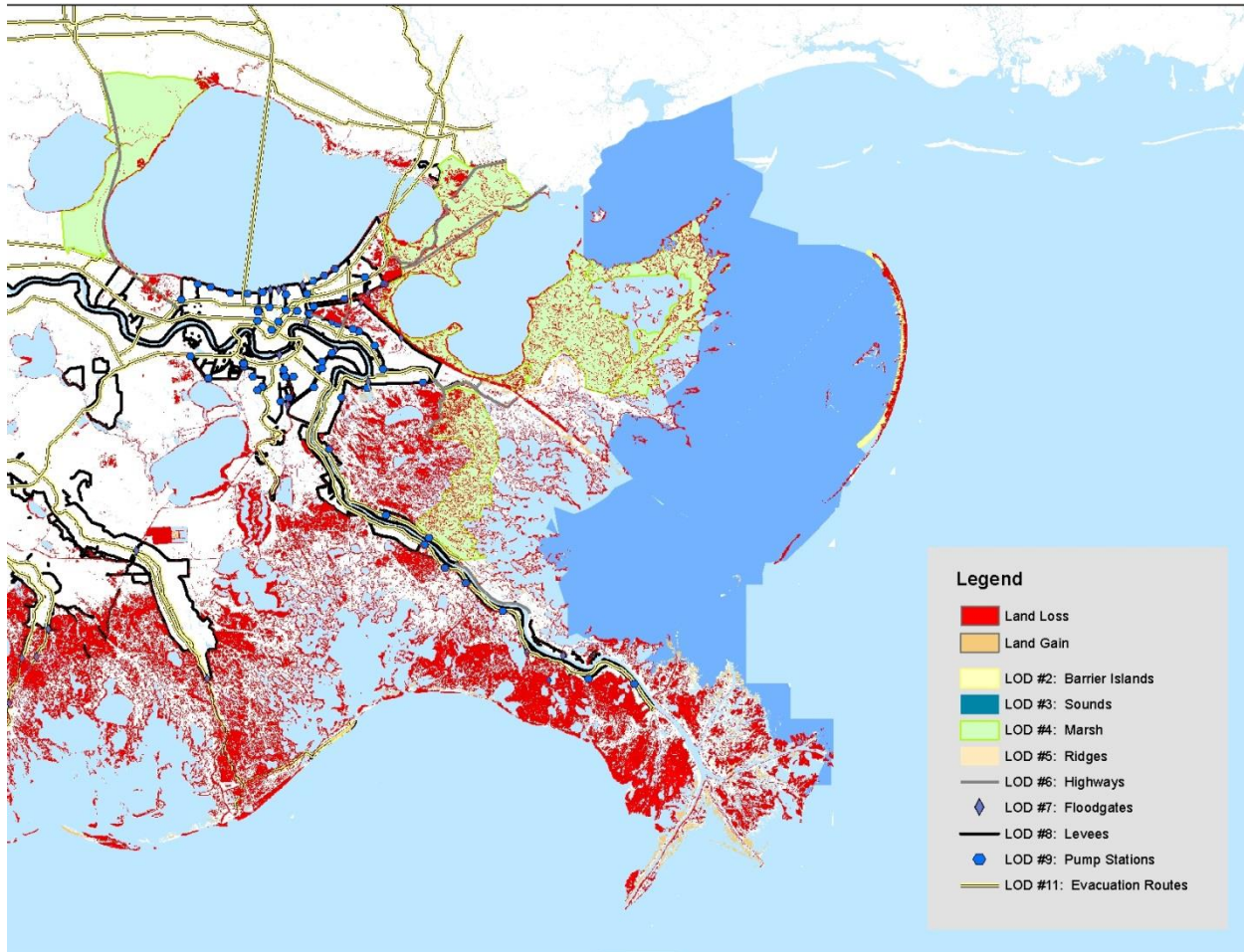


Figure 26: Coastal land loss (USGS 2011) and the lines of defense as part of the GNO HSDS.

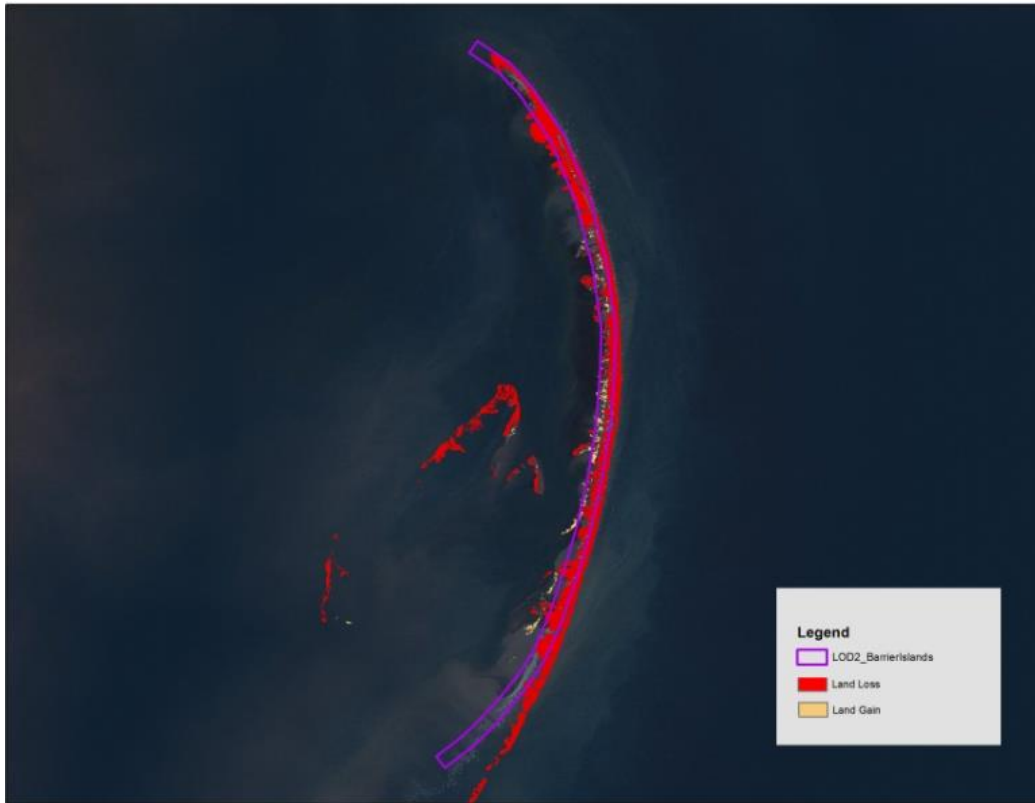


Figure 27: The Chandeleur Islands line of defense and land loss.

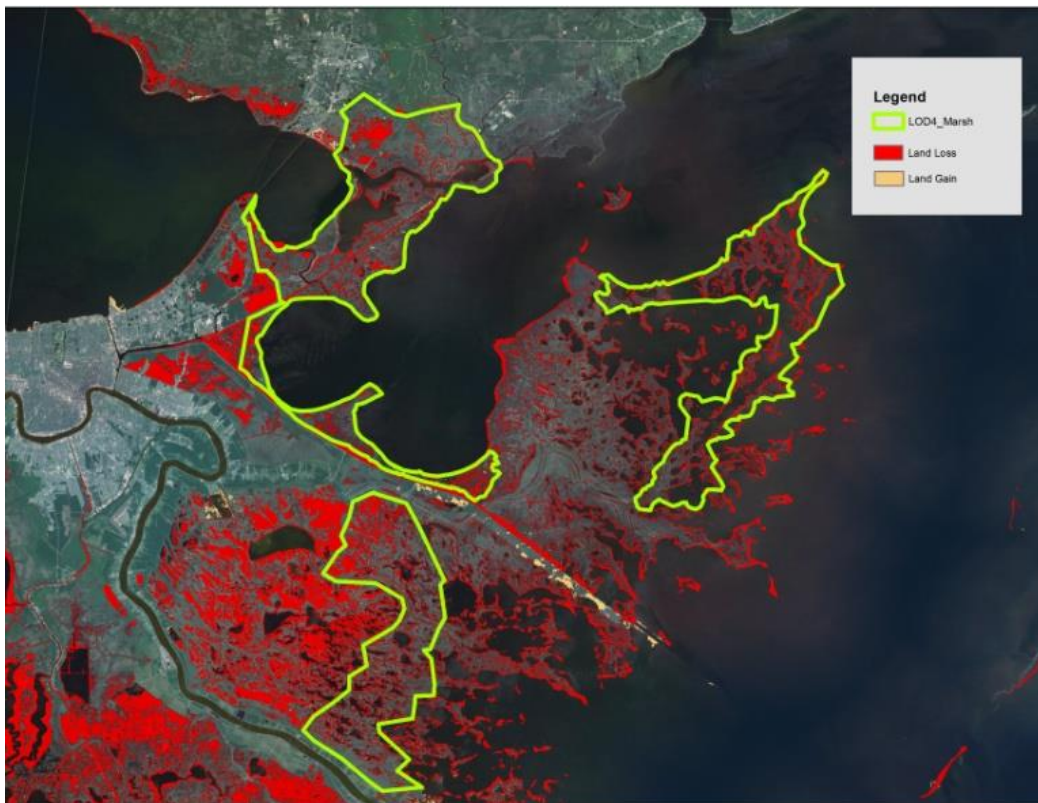


Figure 28: Marsh land bridge lines of defenses and land loss.

While land loss will almost certainly impact evacuation routes in the near future, it is important to note that there are currently significant factors of concern related to this line of defense. **Figure 29** and **Figure 30** shows the I-10 and Hwy 11 bridges crossing Lake Pontchartrain from New Orleans East. Also shown on the map are levees and elevation contours. It can be seen that outside the levee system there exists a nearly 2 mile stretch of interstate leading to these two bridges that crosses a narrow peninsula surrounded by water. Furthermore, the inset shows a 2,000 ft. section of the interstate right at the foot of the bridge that is about 6 - 7 ft. above sea level. Once the still water height at this location reaches greater than 7 ft., this section of interstate will flood, thus blocking this evacuation route. Even before that stage, it is possible that wave action and debris could cause the interstate to be the unpassable. **Figure 31** shows that all of the eastbound evacuation routes have sections that are outside of the levee protection and adjacent to significant coastal erosion.

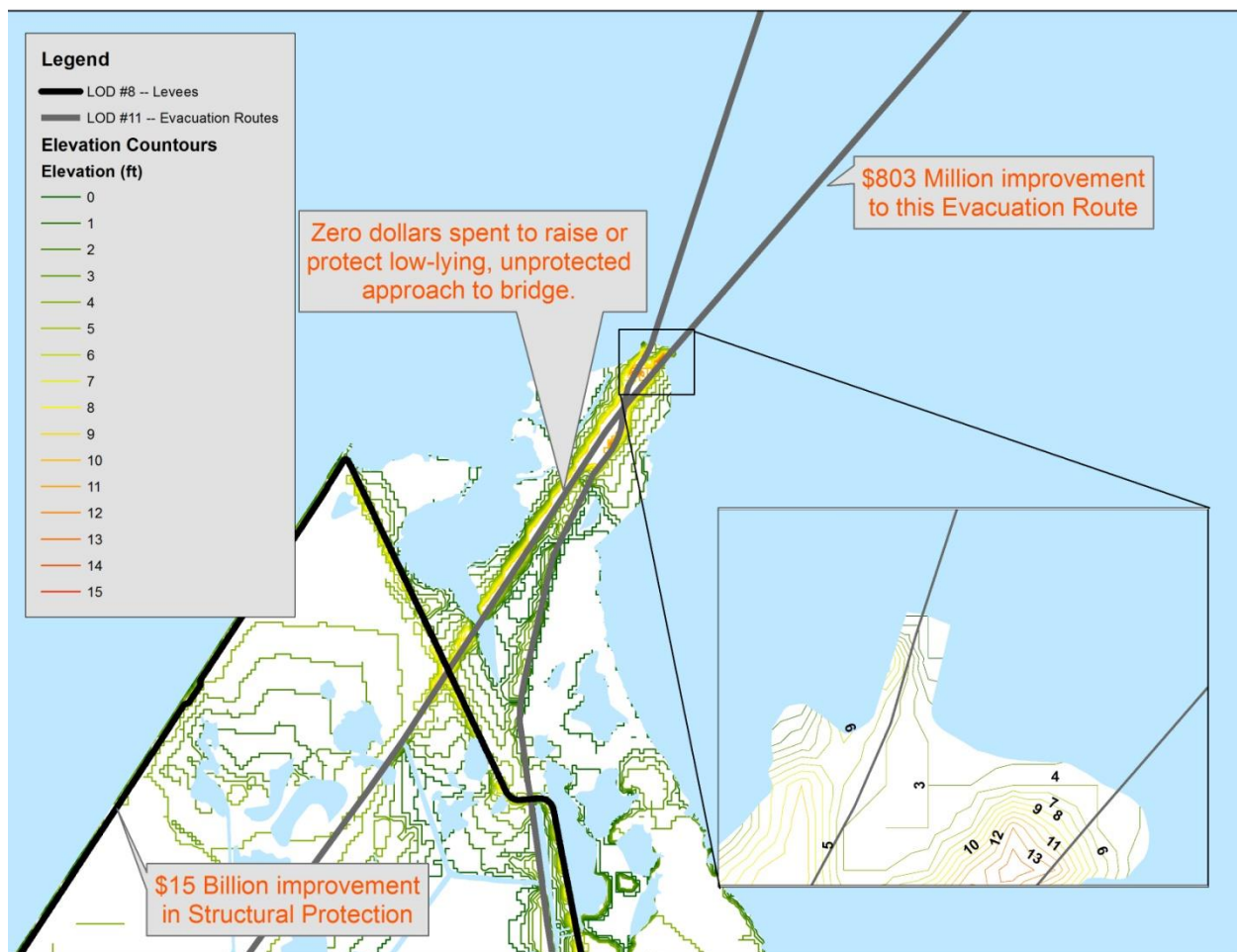


Figure 29: An unprotected low spot just before the I-10 “Twin Spans” bridge, a major eastbound evacuation route, illustrates how system interactions can negatively impact system performance by prematurely blocking a major evacuation route. Elevation contours derived from LiDAR.

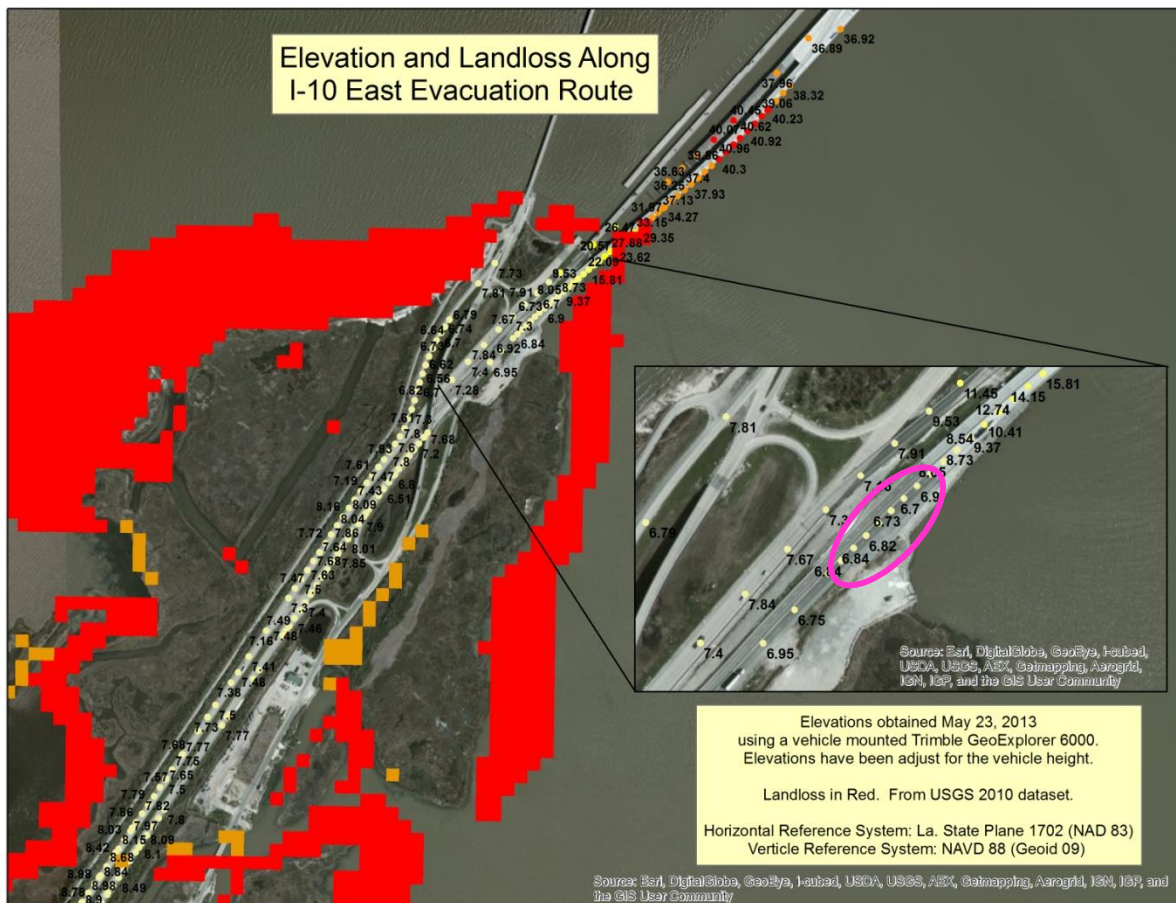


Figure 30: Elevation points along the I-10 East evacuation route, obtained by LPBF scientists using a vehicle mounted Trimble GeoExplorer 6000 along with coastal landloss (red) and gain (brown) from the 2010 USGS landloss dataset. The lowest elevation of 6.7 ft was observed just before the bridge ramp and adjacent to an area of significant shoreline loss. The purple oval identifies the some of the lowest elevation measures that were observed.

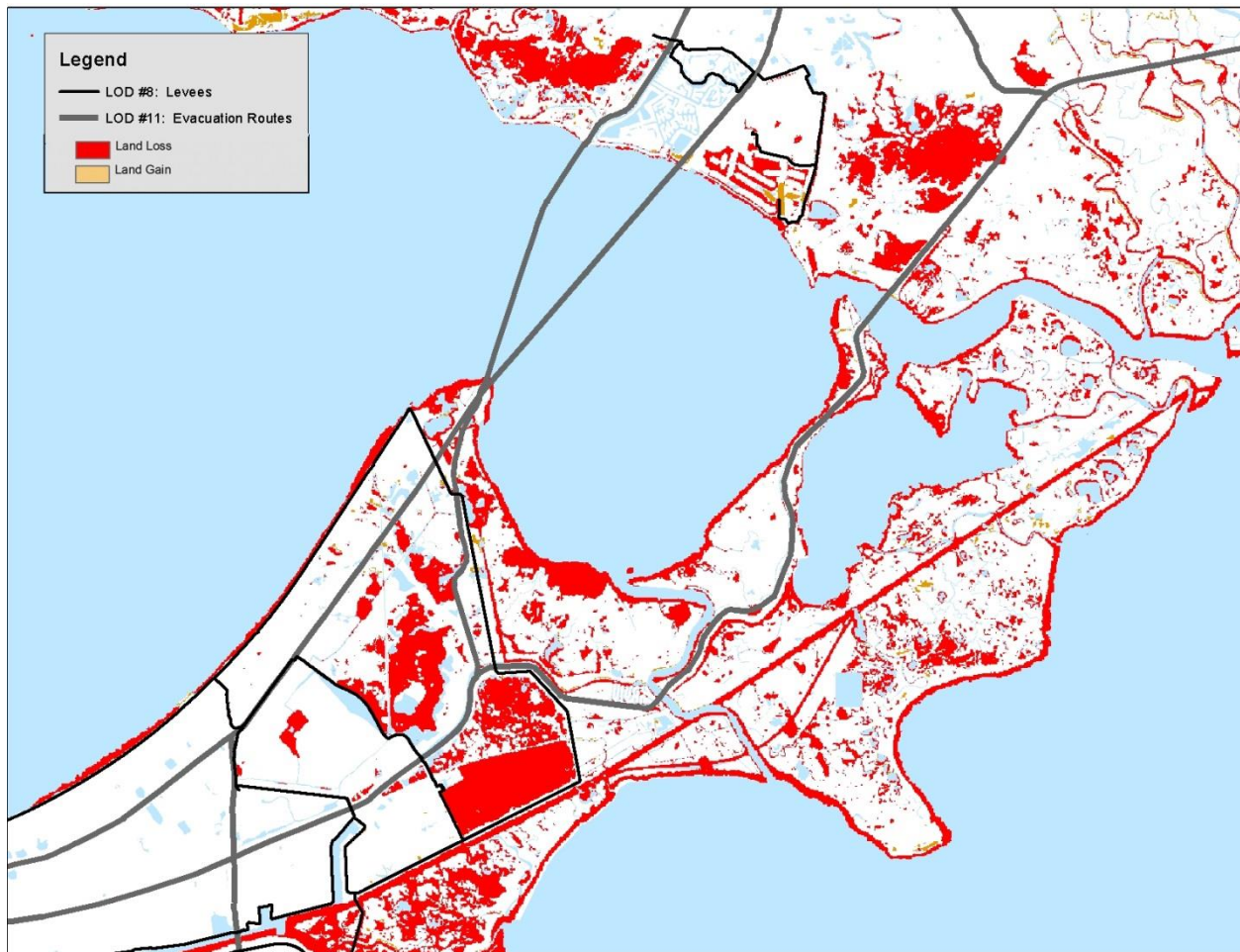


Figure 31: Evacuation routes in the east of New Orleans along with levees and landloss. This figure shows that all three east bound evacuation routes have sections outside of the levee system with nearby shoreline erosion.

Not all systems interactions result in factors of concern. Some interactions potentially enhance performance. Where these can be identified there may be opportunities to replicate and enhance them. One such possible case is the complex interaction between pump stations, wetlands, and levees in the Central Wetland Unit (CWU) in St. Bernard parish, see **Figure 32** below. Previous research has speculated that the pump stations, by serving as a source of freshwater, help maintain arcs of more resilient wetlands in the CWU (Day, Shaffer et al. 2011). This effect is a classic system interaction between two elements of HSDS, but it is also hypothesized here that there possibly exists an additional beneficial interaction. **Figure 32** also shows breaches in the 40 Arpent levee (along the southwest boundary of the CWU) that occurred during Hurricane Katrina in 2005. It is seen that levee breaches occurred in areas away from the pump stations and that no major breaches occurred adjacent to the pump stations and their associated resilient wetlands. While not conclusive, this observed spatial association suggests an interaction between pump stations, wetlands, and levees whereby the pumps stations maintain resilient wetlands and the resilient wetlands enhance the resiliency of the levees. If confirmed, this

beneficial interaction could certainly be replicated elsewhere and those areas could be targeted for strategic cypress tree planting. In this respect, systems engineering provides a framework and methodology to investigate this and other possible interactions that enhance system performance.

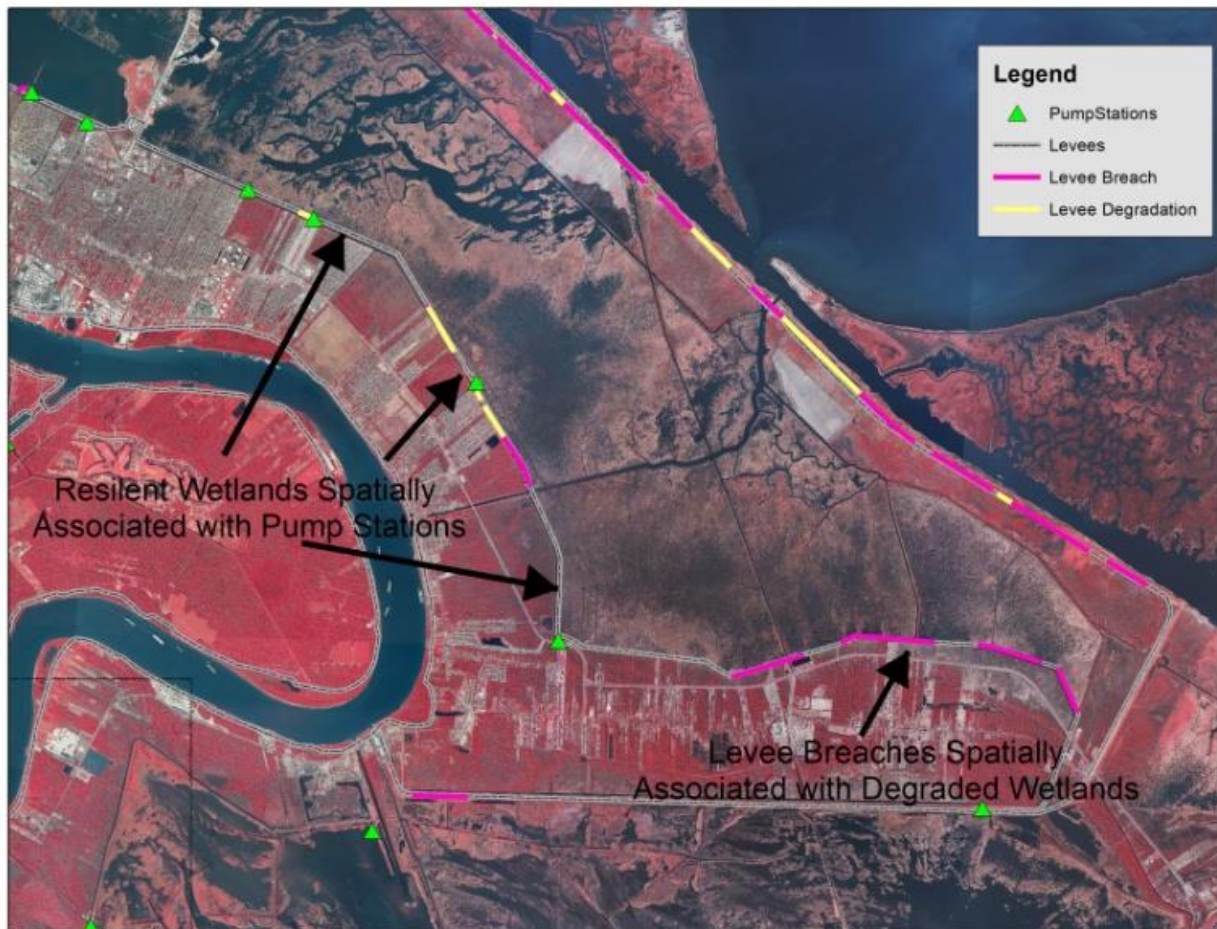


Figure 32: The hypothesized beneficial system interaction between pump stations, which feed freshwater and thus maintain wetlands in front of the 40 Arpent levee in St. Bernard Parish, illustrates that system interactions can also increase the performance and resiliency of the system.

4) Summary and Analysis of QMAS Workshops

This section provides a synopsis of the major observations from the QMAS workshops. The workshops were very productive events and participants provided a variety of observations representing numerous disciplines and perspectives. Appendix A provides a detailed recounting of the workshops. This section begins with a concise summary of the salient points of the workshops, along with any follow up information that we compiled. This section also includes a sub-section that provides a detailed narrative of closure operations for the flood controls structures along the IHNC-GIWW, a major factor of concern identified and described in the QMAS workshops. The findings along with the background research described in the previous section are then used to develop the major outputs of this study (and the subject of the next two sections): an inventory of system elements, a list of system interactions, factors of concern, and scenarios of concern, and our initial attempt to develop a SysML model of the hurricane surge defense system for the greater New Orleans East Bank.

Description of Workshops

As part of the HSDS study, LPBF hosted three QMAS workshops during 2012, on April 6, June 29, and October 26. All three workshops were held at LPBF offices in New Orleans. In organizing the workshops, we started with a large list of potential invitees that represented the broad set of professional disciplines associated with the HSDS, as specified by the Multiple Lines of Defense Strategy. This means that we sought to include civil engineers, coastal scientists, evacuation officials, emergency management professionals, building industry representatives, and others.

Table 1 below lists the actual workshop attendees along with their areas of expertise. It can be seen that while we did achieve broad based representation, engineering related fields were most prevalent among our set of system assessors. Out of 26 total participants, 12 (46%) had expertise in engineering. Combining environmental sciences, environmental management, and water management, the workshops included 6 (23%) assessors with expertise in these areas. Other assessors represented disaster response, evacuation management, climatology, GIS, and systems engineering. It is also shown in the table that assessors represented a variety of organizations, including government agencies, levee boards, non-profit groups, academic institutions, and consultants.

Table 1: List of workshop participants along with agency and areas of expertise.

Name	Agency	Expertise	Workshop Attendance
Ezra Boyd	LPBF	Hazard Geography, Risk Analysis	Facilitator
Rune Storesund	Storesund Consulting	Engineering, Risk Analysis	Facilitator
John Lopez	LPBF	Engineering, Environmental Management	Facilitator
Bob Turner	SLFPAE	Engineering	1, 2, 3
Stradford Goins	Formerly SLFPA-E	Engineering	1, 2
Joe Suhayda	LSU	Engineering	1, 2, 3
Officer Ron Tardo	LSP	Evacuation, Public Safety	1
Bruce Cuber	Red Cross	Disaster Response	1
Scott Eustis	Gulf Restoration Network	Environmental Management	1, 2
Hal Needham	LSU	Climatology	1, 2
Elissa Miller	USACE	Engineering	1, 2
Mark Kulp	UNO	Geology, Environmental Science	1
HJ Bothsworth	Levees.org	Engineering	1, 2
Stevan Spencer	SLFPAE	Engineering	2, 3
Barry Keim	LSU Climate Center	Climatology	2
Robert Hoover	East Jefferson Levee District	Engineering	2
Earthea Nance	University of New Orleans	Engineering, Urban Planning	2
Roger Cowell	SLFPAE	GIS	2, 3
Bob Jacobsen	SLFPAE/BJPE	Engineering	2, 3
Steve Picou	LSU Ag Center	Water Management	2, 3
Grasshopper Mendoza	Horizon Water Committee	Water Management	2, 3
Sheila Grissett	SLFPAE	Public Communication	2, 3
Dinah Maygarden	UNO	Environmental Science	2
Carlton Dufrechou	GNOEC	Environmental Management	1, 2, 3
Harold Daigle	CPRA	Environmental Management	2
KC King	Consultant	Systems Engineering	3

The QMAS workshops have been designed as a facilitated discussion among individuals with direct and hands-on experience with the system being assessed (de Corn 2011). It largely serves as a qualitative data gathering exercise, where facilitators record the experience and knowledge shared by assessors. By encouraging an open and honest conversation between system assessors, it is hoped that they provide insights into the system processes that might not otherwise be discussed in a discipline-centric work environment. Key to making the exercise a success is facilitating an open, but focused discussion amongst the assessors. Each participant was assured that their comments and observations would be recorded anonymously, so that they could speak openly about the strengths and weaknesses of the HSDS.

Toward achieving this goal, each workshop was organized around specific themes or goals. The first workshop focused on the system definition, the second workshop focused on factors of concern, and the third workshop focused on scenario development. For each workshop, we developed introductory presentations along with handouts designed to steer the discussion toward achieving the three goals. Such materials include partially completed system hierarchies meant to elucidate missing system elements from the assessors, a preliminary listing of factors of concern for the assessors to critique and add to, and summaries of recent storm surge events meant to help assessors analyze the system performance across all aspects. Of note, the third workshop occurred following Hurricane Isaac, which has been considered the first test of the redesigned and rebuilt structural protection system. Naturally, this event was a prominent topic during this workshop.

The following lists some of more illustrative points that were provided by the assessment team:

- “We can’t go back to what it was” which was followed by the “system has changed so much that we are forever dependent on structural protection”
- “Every flood protection structure shifts risk” and that there is “no risk neutral flood structure”
- “We bought into the systems approach, but money knows jurisdictional boundaries while water does not.”
- The stated level of protection is based on hydrology. “Does adding behavior reduce the level of protection?”
- Design-Build means “Built to a fixed budget for design and construction” so cost overruns during design come out of construction.
- Local governments “will not be able to raise enough tax revenue to cover operations and maintenance”
- “Will \$15 billion spent on protection lead people to think New Orleans is safe?”
- “People look for reasons not to evacuate” and “We are obligated to realize” some people will not evacuate.
- “Evacuation authority is fragmented and exercised on a case-by-case basis.”
- “The IHNC is the Achilles Heel of the system”
- When asked “What are our vulnerabilities?” an assessor with intimate knowledge of the HSDS replied “Besides the IHNC, operations”
- “Understanding the failure modes is only as good as our knowledge of the sub-surface.”
- “Pinnacle of the systems engineering approach is the shared vision of the system.”
- Detailed information on the HSDS and flood risk for the region is not publically

available. For example, the Corps has released flood risk maps, but not the details of the models or assumptions used in the analysis.

- When communicating risks to the public: people of authority “need to explain that a catastrophic storm may have these characteristics... and you list them all... then explain that we aren’t smart enough to know which ones will pop up with a given event at a given time.”

Brief Synopsis of Workshop Observations

Throughout the course of the workshops a few points consistently emerged as particularly salient in regards to specifying the system and identifying factors of concern. One important point relates to the validity of the system requirements. In particular, is the hydrologic modeling of surge events over a 100 yr or 500 yr time span accurate? Related to this issue, it was also consistently observed that the stated level of protection is based solely on hydrologic modeling and does not account for behavior and operations.

Assessors listed a number of concerns related to system behavior: funding, governance, life cycle considerations, and a design-build approach under a fixed budget and an arbitrary deadline. Specific examples include the lack of rust coating on steel sheetpiling, a sandcore levee in St. Bernard parish, the cost of future levee lifts, and the High Performance Geotextile Material (HPGM) used as erosion protection for earthen levees. These points are particularly salient given that the system exists in an ever changing landscape. Additionally, assessors noted that poorly secured objects in the IHNC/GIWW navigational corridor created an impact induced failure risk because the design specifications assumed that the USCG would require that these vessels be evacuated though they do not have the necessary authority and capability to compel operators and owners to evacuate their vessels.

Also related to the question of system requirements, assessors consistently pointed out that the state and Federal risk assessments (and subsequent mitigation plans) calculate benefits based exclusively on property damage. These assessments exclude public health and safety benefits, such as reducing storm induced mortality and preventing hazardous releases. This accounting leads to undervaluing certain measures, such as increasing evacuation capacity or providing structural protection to an industrial facility with hazardous materials in a sparsely populated area, when analyzing the cost and benefits of different policy options.

Beyond system requirements and behavior, concerns related to evacuation were pervasive throughout the workshops. The state’s evacuation plan rests on a 50 hour window to evacuate, but recent experiences have demonstrated that unpredictable storms can provide a much shorter window. Furthermore, this timeline is based on the onset of hazardous winds, and not the arrival of the storm surge. The issues of timing the evacuation is further complicated by the fact that the decision to call an evacuation is inherently political, with the potential for political considerations to trump meteorological considerations. Furthermore, once the decision has been made, implementation requires a high level of coordination across jurisdictions and levels of government. Finally, assessors noted that even after an official and coordinated evacuation call has been made, it is ultimately a personal decision that is highly emotional and reflects

conflicting interests. One assessor noted that the personal costs of evacuating (gas, shelter, food), while another pointed out that residents are keenly aware that they risk losing personal property to looting while evacuated. In the end, one assessor bluntly stated that “we are obligated to realize that not everyone will evacuate,” a bold statement that goes against the basic policy assumption that 100% evacuation compliance is possible.

Concerns about interior drainage were also salient throughout the workshops. It was pointed out that the networks of drains, pipes, canals, and pumps that make up the interior drainage system are designed with sufficient capacity for a 10-yr rainfall event. In contrast, a typical hurricane will produce much more significant rainfall. In this regard, it was noted that when compared to the volume of water from overtopping, rainfall is a more significant cause of inundation within protected areas. Operating the pumps requires a high level of coordination. Losing capacity from just one pump would cause the entire system to back up. Likewise, poor communications or negligent operations could cause water levels in the drainage canals to exceed the safe water levels, creating a risk of floodwall failure.

Finally, it was pointed out that the public along with independent scientists and engineers lack access to detailed information on the system, a salient point of the workshops that impacts all aspects of the system.

Table 2 lists the major factors of concern identified in the QMAS workshops, while the next sections provides additional details from the workshops.

Table 2: Factors of Concern Summary.

SYSTEM PERFORMANCE

- * Basis for storm surge - the data behind the development of the storm surge was questioned. Significant concern was expressed that not 'starting from the beginning' would lead to a situation where improper storm surges are used to perform more refined system configurations.
- *IHNC/GIWW impact loads - USACE developed design criteria for IHNC floodwalls based on the presumption that the US Coast Guard (USCG) would require removal of barges from IHNC. In some instances, this requirement has been waived by USCG, putting the IHNC/GIWW floodwalls at risk.

CONSEQUENCES

- *State plan focuses on property, not mortality. Also need to account for hazardous materials. Evacuation is not just people.

TRANSPORTATION

- *Need for widening of roadways and bridges to expedite evacuation egress and ingress.
- *Deep-draft shipping/navigation channels and impact on land loss.

EVACUATION

- *Tiered evacuation with evacuation guidance issued by individual parishes. The actual evacuation decision is personal and emotional, implying the sequence can vary and result in chokepoints, both OUTGOING and INCOMING (after the storm).
- *Wind as evacuation trigger - there is a separation in sequence between surge arrival (elevated water levels) and increased wind speeds. Surge arrives first, before residents have 'wind' warning.
- *Coordination in timelines between towns/cities; parishes; and state are very important and requires improvement.
- *Enough time to evacuate? 38 hrs can be political.
- *Looting - major factor for residents to NOT evacuate.
- *Costs associated with temporary evacuation (travel, lodging, food).

SYSTEM FUNDING

- *Funding - current funding distributions are based on jurisdiction, not system functionality.
- *Federal funding assurance a large concern.

DRAINAGE

- *Storm drains are configured for the 10-yr rainfall event. Not necessarily representative of actual hurricane rainfall.
- *Rainfall is a much larger inundation contributor than overtopping.
- *Unwatering operations following inundation is not well coordinated between the levee district and sewerage and water board.
- *For major storms, need all pumps running at once, if one goes down, pumping operations back up.
- *If operator 'falls asleep at the switch,' need an overflow system so the water elevation does not exceed the maximum, otherwise, potential for catastrophic failure.

FLOOD PROTECTION

- *Levee armoring - there is a current push for levee armoring with grass. There is significant concern regarding the performance of this armoring strategy.
- *Operations and maintenance requirements associated with turf (grass) armoring unknown, sensitivity of turf condition unknown.
- *Future levee lifts – settlement is a chronic problem that results in lower crest elevations. The available funding (appropriations) for such work has not been identified, so there is concern about maintaining the target protection in the future.
- *Impact loading from: pleasure crafts, RVs, railroad cars, fuel tanks, buoyant houses all located in IHNC (no evacuation or securing requirements)

PUBLIC COMMUNICATION

- *Accessibility of information to the public about the system is LACKING.

Analysis of Workshop Findings

The previous section provided a brief synopsis of the observations raised by system assessors during the three QMAS workshops. This section takes the next step and provides a more thorough summary along with an initial analysis of the workshop findings. A first step of this process was to categorize the observations into a complete and consistent set of categories based on what assessors said and how the multiple lines of defense frames the HSDS. Some of QMAS findings fit into categories based on specific lines of defense (such as interior drainage), others fall into categories based on system engineering topics (such as system requirements). The sub-sections that follow are based on these categories. For each category, the sub-section starts with a table listing most of the recorded QMAS observations that fit within the category and then follows that with a 1 - 3 paragraph narrative summarizing and analyzing the most important observations. Unlike the previous sub-section and Appendix A, the narrative below moves beyond strictly reciting what was stated in the workshops and draws on additional knowledge, research, and analysis.

System Issues

Category	Comment
System	System lacks a clear functional objective
Issues	Arbitrary timelines mandated from Congress meant that life cycle considerations were sacrificed
	System implies boundary, but elements extend outside boundary
	Residents outside Federal protection system
	Surge height-frequency relationships varies spatially, within events, and are non-stationary
	River levees breaching during surge. (River flood protection is a different system from the HSDS).
	Nearby infrastructure not considered part of system
	Post-K design and construction process driven by congressional deadlines
	Lack of a thorough, independent investigation of all projects completed by USACE

The first category is a catch-all category for observations related to general issues on the overall system. One of the most important observations, which was raised repeatedly, is that the HSDS lacks a clear functional objective. It was pointed out that the objectives which are available are either ambiguous or not explicit to the complete system. For example, the State Master Plan lists as a key requirement: “sustainable long term solution for coastal restoration and for comprehensive risk reduction from storm surge and waves” (Louisiana Coastal Protection and Restoration Authority 2012). However, the term “sustainable” is undefined in the plan and the requirement does not list a specific criteria for measuring sustainability. On the other hand, while the National Flood Insurance Program lists the specific criteria that levee systems protect against the 100-yr flood threat (FEMA 2002), this requirement only relates to the structural lines of defense and does not apply to the coastal or community lines of defense.

System Requirements

Category	Comment
System Requirements	"Are the requirements right?" Validity of storm surge height-frequency relationships based just on hydrologic modeling Risk analysis does not account for behavior, both maintenance and operations, that affects the level of protection FEMA and USACE requirements and standards are different Interaction between swamps/marsh and levees impacts freeboard calculation USACE design based on acceptable overtopping rate, but FEMA uses freeboard for NFIP. Freeboard calculation does not consider inundation of swamp and how that impacts wave amplitude and periods. This factor influences flood risk for areas with swamps/marsh directly in front of levee, where 100 yr flood depths are possible from a 85 yr storm River levee requirements based on 800 yr risk level Drainage and pump stations requirements based on 10 yr risk level Need a complete failure tree analysis No elevation requirements for GNO in SMP because it calls for 400-yr structural protection Current 3 ft BFE's creates institutional momentum Need clear, realistic evacuation requirements

The system assessors raised a number of issues that fall into the system requirements category. As the most basic level, an assessor asked early during the first workshop: "Are the [system] requirements right?" Based on the assessor's follow-up comments, his concern related to the validity of the USACE's method, termed the Joint Probability Method-Optimal Sampling (Resio 2007) to develop storm surge height-frequency relationships based on statistical analysis of hydrologic modeling results. In short, the 100-yr surge elevation, which determines the design height of levee reaches, was calculated by first simulating a sample of storms that was believed to be representative of a long term storm record, then plugging these storm characteristics into the coupled ADCIRC+SWAN storm surge and wave model, and finally statistically analyzing the computed surge heights to calculate the height-frequency relationship. While the assessor's particular concern questioned whether or not the simulated sample of storms consisted of a representative sample, there are other concerns related to method. These include the accuracy of the hurricane model, the accuracy of the ADCIRC+SWAN model, the accuracy of the terrain grid used for the storm surge model, and the lack of external validation based on direct comparison of the simulation results with observational data from tidal gage and other datasets on extreme tides.

Also related to system requirements and how the level of protection is calculated, assessors consistently raised concerns regarding system behavior. Assessors described a system that involves a number of moving parts that require human operation along with ongoing maintenance, both of which depend on adequate technical expertise and funding. However, the risk analysis performed by USACE only considered hydrodynamics. It did not include human

dynamics. In this regard, one assessor asked: “What happens if someone falls asleep at the switch?” While this specific question was addressed to maintaining water levels below the safe water level in Orleans parish’s three discharge canals, the question applies to any system element that requires human operation. Furthermore, beyond just operating system elements during a storm event, there are also related concerns regarding ongoing maintenance of system elements, particularly related to the availability of funding and expertise within the local sponsor. (Reflecting their salience throughout the workshops, similar behavioral concerns are also described in sub-sections below.)

Reflecting the previously mentioned lack of a clear functional objective for the system, another important system requirements concern reflects the lack of consistent system requirements across different elements. While the levee and floodwall elements of the HSDS are designed based on the 100-yr risk level, other elements of the HSDS are based on other risk levels. Assessors pointed out that the levees along the river are designed for the 800-yr river stage, but pump stations are designed based on the estimated 10-yr level of precipitation. Beyond just the concerns expressed during the QMAS workshops, it is worth pointing out that housing elevation requirements for most of the urban areas of GNO are arbitrarily set at 3 ft. above grade, regardless of the 100-yr flood level of the location, while the evacuation requirements for local and state officials are ambiguous and not explicitly tied to predicted storm surge levels.

Stakeholders

Category	Comment
Stakeholders	Stakeholders include organizations that design, build, maintain, and operate system Stakeholders: Actors who gain value from system; Critical Stakeholders: sub-group who require more complete system information Stakeholders who are users that live within the protected areas Residents who live outside the perimeter protection system are also stakeholders.

Stakeholders is an important systems issue that was discussed during the QMAS workshops. In the most general sense, stakeholders are actors who derive value from the system. Naturally, this definition includes users that reside within the protected areas along with other citizens who work or own property within the protected areas. Likewise, stakeholders include the organizations and personnel that design, build, maintain, and operate the system or specific system elements. Simply put, their career success is determined by the performance of the system. Finally, in the third workshop that followed Hurricane Isaac, assessors pointed out that stakeholders also includes residents who reside outside the perimeter protection system, since their level of risk is influenced both directly and indirectly by the configuration of the perimeter protection system.

Availability of Information on the System

Category	Comment
Availability of Information on System	"Pinnacle of a systems approach is a shared vision of the system" which requires that critical stakeholders have full access information on the system Public, independent experts need fuller access to design and technical information on system SLFPA-E and CPRA cannot access the Corps ProjectWise database which has detailed information about the perimeter protection system The SLFPA-E Levee Information System (LIMS) only covers New Orleans East Document-centric databases are characteristic of a project perspective What is needed is a system perspective that models structure and behavior What are the strengths and weaknesses of the system? The answer guides decisions on evacuation, settlement, elevation, land use, emergency planning and response Need to complete the failure tree analysis

Followed by stakeholders, the availability of information for stakeholders was an important topic that was raised throughout the three workshops. In a nutshell, assessors pointed out that crucial stakeholders lack access to crucial information on the system. As just one example, neither SLFPA-E nor CPRA can access the Corps ProjectWise database which contains detailed technical design and “as-built” documents on the perimeter protection system. One database, the Levee Information System (LIMS), has been made publically available by SLFPA-E, but it only covers New Orleans East. Generally speaking, the public (i.e. the systems users) have not been presented with concise and reliable information on their level of risks. As one assessor pointed out, without knowing the strengths and weaknesses of the system, system users lack the guidance that they need to make key decisions on evacuation, settlement, land use, building elevation, and preparedness.

Behavior: Operations, Maintenance, Jurisdictions, and Funding

Category	Comment
Behavior: Operations, Maintenance, Jurisdictions, and Funding	"Money is spent based on jurisdictions not on system specifications" Federal government funds projects piecemeal Independent project reviews are not standard State encourages personal preparedness, not community level planning. Need to encourage neighborhood groups to identify and track elderly, children, and other special needs residents Unique system with a lot of natural elements along with

- fabricated elements
- No complete organizational chart showing who has authority over system elements
- Funding and expertise for maintenance and operations lacking at the state and local level
- Grass maintenance on earthen levees will be expensive, requires equipment, fertilizer, herbicides
- Concerns regarding floodgate maintenance and operations when turned over to local/state control

Elaborating on the previous observations related to system requirements, the assessors voiced a number of concerns regarding the behavior of the system. Generally speaking, these concerns relate to operations, maintenance, funding, and jurisdictions. One observer noted that while water crosses parish lines, funding does not. He further elaborated that funding of projects based on jurisdictional lines constitutes a major impediment to a systems approach.

It was also pointed out that the HSDS under the MLODS framework consists of a unique combination of natural elements and engineered elements, something that complicates behavioral aspects of the system. For example, maintenance of levees and floodwalls is officially mandated for the community to stay in compliance with NFIP requirements. However, maintenance of the coastal lines of defense does not fall under NFIP requirements, despite the fact that they impact the flood risk for the region and the actuarial calculations of the NFIP.

Levee and Floodwall

Category	Comment
Levee and Floodwall Issues	"Achilles heel of the entire system is the IHNC/GIWW... We have a limited time to close the [Lake Borgne Surge Barrier] gate.... We need divers to check sill because of debris... if debris is there it takes time to get equipment and manpower in place to move debris... if the barrier is not closed, the walls in the IHNC are still fragile" Most elements of the system are hybrid floodwall and levee Unsecured objects in the IHNC/GIWW could impact levee, causing a breach. IHNC/GIWW is a Regulated Navigational Area, but if the Coast Guard does not enforce it then the level of protection is reduced. L-Walls are impact resistant, but not widely used Surge barrier design allows overtopping at rate of $0.1\text{ft}^3/\text{s}/\text{linear ft}$ Foreshore protection, grass, armoring, and HPGM are all crucial to preventing erosion during a storm HPGM: Scars are common (particularly mower blades that dig in) and can cause erosion 8-mile section of T-wall with relict sand core levee in St. Bernard Parish

- I-walls still present, though not used widely in the perimeter protection system
- Lack of rust coating on sheet piles used in newly constructed floodwalls in St. Bernard Parish
- USACE granted "categorical exclusions" from armoring certain levee segments
- No investigation of scour on the flood side
- Temporary barriers prone to failure
- Erosion due to overtopping
- Erosion before overtopping is a big unknown
- Sub-surface geotechnical issues have not changed since Katrina
- Higher factor of safety in rebuilt elements, particularly T-Walls
- Breach repairs could be better coordinated
- CPRA (local partner) lacks funding for lifts. Some will be required as early 2017.

Assessors expressed considerable and wide ranging concerns related to levee and floodwall issues throughout all three workshops. Most notable, issues surrounding the floodwalls along IHNC/GIWW corridor led one assessor to describe this navigational corridor as the “Achilles Heel of the system.” Prior to Hurricane Katrina, this region was an open system with direct storm surge exchanges with Lakes Borgne and Pontchartrain. During Hurricane Katrina, the navigational corridor funneled Katrina’s surge from Lake Borgne toward the Lower Ninth Ward of New Orleans, where the surge toppled two large sections of floodwall and then destroyed all but one home over a 10 square block area. Of note, while the breached sections of floodwalls have been rebuilt to stronger standards, most of the other floodwalls have not been upgraded, beyond just the addition of a concrete scour pad on the protected side.

There are now newly constructed floodgates that block storm surge from entering the corridor from either lake, thus making it enclosed retention area. However, the closed retention area still receives water from rainfall, pump stations that discharge into the waterway, and possible overtopping from the Lake Borgne Surge Barrier (which was designed to allow limited overtopping.) Furthermore, unsecured objects, such as barges, RV’s, and large capacity propane tanks pose an impact risk to the floodwalls when they become buoyant. Technically, the area is a Regulated Navigation Area (RNA) giving the Coast Guard the legal authority to compel vessels to evacuate the IHNC/GIWW. However, as a practical matter assessors expressed concern that the Coast Guard does not have the resources to enforce a mandatory vessel evacuation of the RNA. Furthermore, neither the Coast Guard nor any other agency has the authority to force the other objects, including RV’s and a large storage capacity propane tank at an RV park and recreational vessels at a shipyard, to evacuate the RNA ahead of a storm.

Beyond the IHNC/GIWW, assessors described a number of concerns related to levees and floodwalls. Erosion and scour was a major theme. Assessors noted that there has been no investigation of scour on the flood side of levees and that the USACE granted “categorical exclusions” from armoring certain levee segments. There were considerable questions about the reliability of the methods for foreshore protection, particularly the widespread use of High Performance Geotextile Material (HPGM.) One assessor stated that it was prone to tearing by grass cutting equipment and another worried about the costs associated with removing and

resetting the HPGM for scheduled levee lifts. With floodwalls, assessors worried that the USACE's timesaving decision to replace rust coating on sheetpiling by adding to the thickness of the sheetpiling was inadequate and would undermine the floodwall's integrity over the system element's life cycle. In addition, an assessor noted that a newly rebuilt 8-mile section of floodwall in St. Bernard parish consisted of T-Wall sheetpiling driven into a relict sand core levee. He further worried that if the sand core eroded, the T-Wall would not have sufficient support and would breach. He lamented that while the USACE claims the T-Wall would not fail if the sand core eroded away, they have not released the results of the analysis that supports that claim.

When asked about how the current levee and floodwall system compares to the one in place during Hurricane Katrina, one assessor noted that the sub-surface geotechnical issues have not changed since then but another responded that the newly rebuilt elements have a higher factor of safety, particular those where T-walls have replaced I-walls.

While there is no denying the levees and floodwalls were important topics during the QMAS process, it should be noted that it is not clear if the prevalence of levee and floodwall issues during the workshops reflected their prominence as a source of risk in the system or simply reflected the prevalence of civil engineers in the QMAS workshops.

Floodgates

Category	Comment
Floodgates	"Gates are a big vulnerability" Designs of different floodgates based on different standards and criteria New floodgates are very complex and state/local agencies are not properly trained to maintain and operate them Not all gates have backup generators and safe houses

One assessor described floodgates as "a big vulnerability." Assessors observed that various floodgates, a prominent feature of the perimeter protection system, have been constructed at different times using different standards and criteria. For example, some floodgates have backup generators and safe houses, while others do not. The Lake Borgne Storm Surge Barrier illustrates the lack of standardization as well. The Surge Barrier actually includes three gate mechanisms. One of the floodgates blocks Bayou Bienvenue when closed; it is a lift gate. In addition, two gates are used to block GIWW; one is a sector gate and the other is a barge gate. The barge gate takes approximately 7 hours to close, but before that can begin the sector gate must be closed so that crews can access the sector gate. In a follow up conversation, it was reported that closing just the Surge Barrier complex costs approximately \$11,000. When one considers that these three gates were all constructed at the same time as part of the same project, it is easy to appreciate the concerns related to the various other floodgates which have been constructed at different times by different agencies. Later in this chapter, a case study delves into the complex operational sequence for closing the Surge Barrier.

Floodgates also create a number of concerns related to long term operations, maintenance, and funding. Most acutely, state and local agencies expressed concern that they did not possess the proper expertise or funding to maintain and operate the gates at the Surge Barrier, which are highly complex and contain many moving parts.

Interior Drainage

Category	Comment
Interior Drainage	"A raindrop that falls [within the protected areas] has many masters" "Pump stations in a flat city add energy for water to drain to the lake" Different drainage systems within and across parishes. Five separate systems within the study area: Jefferson East Bank, Central New Orleans, New Orleans East, Lower 9th Ward, and St. Bernard Parish Failure of the drainage subsystem is a failure of the entire system Pumps and drainage are not well integrated into the perimeter system "No way to disconnect the Central Wetlands Unit from the system" Unwatering not coordinated Central New Orleans will always require 2 - 3 pumps. Capacity concerns with older pump stations and new pump stations along a single drainage path Designs of different pump stations based on different standards and criteria Hydraulic pumps at the new pump stations for the outfall canals are very inefficient and maintenance intensive Newly formed retention areas Natural retention outside levee system versus engineered retention inside levee system 100 yr floodplain defined by rain, not surge Outfall canal levees remain a problem No full analysis of pumping options for the outfall canals

Like levees and floodgates, interior drainage was considered a major source of concern for the assessors. The third and final element of the structural lines-of-defense, interior drainage, and particularly pump stations, are crucial in preventing the accumulation of water that overtops levees, flows through breaches, or falls within the protected areas. One assessor observed that a failure of the drainage subsystem is a failure of the entire system. Reflecting the different designs used across the floodgate subsystem, interior drainage consists of a complicated mix of pump stations and other elements. Like floodgates and levees, these have been built at different time periods by different agencies and according to different criteria and standards. The lack of standardization is even more acute for the drainage subsystem because of the coupling of historic and modern pump stations along the same drainage canal. The new pump stations must be designed to accommodate the pumping capacity of historic stations. However, because of different design criteria, the parts, expertise, and operational procedures are not interchangeable between them. Furthermore, currently in central Orleans, the New Orleans Sewerage and Water

Board operates the historic pumps that discharge into the canals, while the USACE operates the modern pumps that remove water from the canals. Beyond just different pumping capacities, different lines of command and institutional objectives make coordinating operations between the local and Federal agencies difficult and complicated. Despite the problems associated with this coupling, one assessor pointed out that double pumping (and even triple pumping in one case) is necessary because “pump stations in a flat city add energy for water to drain to the lake.”

Indeed, across the GNO area there are five separate interior drainage systems, based largely on the different polders. In addition to the Central New Orleans system described above, the East Bank of Jefferson Parish and New Orleans East both have separate, compartmentalized interior drainage systems. In contrast to the multiple pumping system of Central New Orleans, the interior drainage systems in Jefferson and New Orleans East both consist of drainage canals that connect to pump stations embedded in the perimeter protection levees and only require single pumping. Also, while St. Bernard Parish and the Lower Ninth Ward are both located within the same levee ring, the drainage systems are run by two different parishes and should be considered different systems. For example, in the Lower Ninth Ward stormwater is conveyed to the pump stations via underground pipes, while St. Bernard largely utilizes open canals.

Beyond pump stations and conveyance mechanisms, differences in the discharge locations create another layer of contrast between the different systems. In Jefferson Parish and New Orleans East, pumps discharge directly into Lake Pontchartrain, which has nearly unlimited capacity as long as a storm surge does not fill the lake to near the levee crowns. In Central New Orleans, the lake is also the ultimate discharge basin, but the newly enclosed drainage canals serve as an intermediary storage basin with limited storage capacity defined by the safe water levels. Drainage discharge in St. Bernard and the Lower Ninth Ward is the most complicated and differs substantially from the other polders. Here the pumps discharge into the Central Wetlands Unit, an enclosed area of limited storage capacity when floodgates along the former MRGO are closed. Furthermore, the storage capacity of the CWU depends on when and if the floodgates along its perimeter are closed.

Assessors expressed considerable concern related to the creation and subsequent management of the new retention areas created by floodgates constructed following Hurricane Katrina. In addition to the discharge canals in Orleans parish that have been converted into closed retention areas, the Lake Borgne Surge Barrier and the Seabrook Floodgate have likewise converted the IHNC/GIWW into a closed retention area. The issues associated with the new IHNC/GIWW retention area were described above in the levee and floodwall section. In Central New Orleans outfall canals, floodwall breaches were a major source of flooding during the Hurricane Katrina disaster. In lieu of repairing miles of floodwalls along each of the three canals, the USACE chose to construct floodgates and pump stations at the end of the canals adjacent to Lake Pontchartrain. When the floodgates are closed to block storm surge, they also block rainfall drainage from the canals. However, the unrepaired floodwalls still pose a breach risk if the canals fill with pumped water. The storage capacity of the new retention areas is characterized by the safe water levels.

Also on the issue of design and pump capacity, it was pointed out that many of the pump stations were built according to a design criteria based on a 10-year rainfall event. Despite the fact that

the pump stations are an integral part of the perimeter protection system, the design criteria for this element of the HSDS contrasts greatly with the criteria used to determine the heights of levees, floodwalls, and floodgates.

Chandeleur Islands

Category	Comment
Chandeleur Islands	Loss of islands increases surge height, quickens arrival, shortens evacuation window Created and sustained by accretion during fair weather Subsidence and sea level rise are fair weather failure modes Loss of shoal, islands creates risk to HSDS Implications of including Chandeleur Islands in Federal system: Regular monitoring, FEMA obligated to pay for damage, Ecological issues must be balanced with flood protection, impacts flood insurance certification

The Chandeleur Islands are an important coastal line of defense for the GNO area. This barrier island and its associated shoal slows the propagation of surge inland and breaks the large amplitude waves of the open gulf. However, assessors questioned if the island chain would be able to provide this surge protection function in the future.

It was noted that the islands were created and are sustained by accretion during fair weather. In turn, subsidence and sea level rise are two fair weather failure modes that led assessors to question the island's long-term sustainability. Likewise, storm generated surge and waves also erode the islands, an extreme weather failure mode that has been observed during recent storms including Hurricanes Georges, Ivan, and Katrina. One participant noted a recent study that examined how deterioration of the islands would both increase surge height and quicken its arrival at inland locations. However, during a follow up conversation, the person pointed out that material eroded off the island just winds up in the shoal, where it still provides resistance to surge and wave propagation and further research was needed to investigate the net effect of sand transport from the island to the shoal. It was suggested that the interaction of the storm surge with an emergent island is not very different compared to a slightly ephemerally emergent shoal. The ecologic value is very different but not necessarily the storm surge dampening.

Of interest, assessors voiced a number of perspectives when asked if the Chandeleur Islands should be formally considered part of the Federal flood protection system and thus tied to the NFIP. It was noted that this would compel regular monitoring of the islands, similar to what is required for levees and floodwalls and that FEMA would be obliged to fund repairs for any damage induced during the Federally declared disaster. On the other hand, assessors expressed concern about tying flood insurance certification to a landscape feature with dismal prospects for long term sustainability and worried that it would be difficult to balance flood protection goals with ecological objectives.

New Orleans East Land Bridge

Category	Comment
New Orleans East Land Bridge	Brackish marsh with remnant Pine Island Ridge Shoreline retreat: ~ 4 ft/ yr on both Lake Pontchartrain and Lake Borgne sides Katrina induced weak spot could breach Navigational dredging: Ongoing and possible future Need to limit oil and gas activities, but CPRA only reviews permits that fall within its project areas Potential USACE Navigational Projects that could harm the marsh: IHNC lock replacement, dredging GIWW for Panama Canal expansion "Filling the Lake" vs. "Tilting the lake": Barrier plan will reduce filling, but not impact tilting Most vulnerable parts of the land bridge: Katrina erosion spot, GIWW, Passes are eroding, Grain Coin Pocket (small bay that is about to breach to the Rigolets), Narrow part of ridge/land bridge along Hwy 90, foot of I-10

The New Orleans Land bridge consists of a brackish marsh adjacent to the Pine Island Ridge. It snakes east of New Orleans toward Slidell, LA, and it provides a barrier to storm surge and waves propagating into Lake Pontchartrain. Two passes, Pass Rigolets and Chef Pass, pierce the barrier and provide a storm surge conveyance route from Lake Borgne into Lake Pontchartrain. The SLFPA-E is currently evaluating options to enhance the surge protection provided by the land bridge through either building a levee along the land bridge, building gates at the passes, or both. However, one assessor noted the distinction between “filling the lake” with storm surge via offshore winds and “tilting the lake” when winds change direction over the lake after a storm makes landfall. He stated that the flood barriers along the land bridge would only reduce filling and would not reduce tilting. LPBF has released a comprehensive framework evaluation of flood gates on the passes, and describes potential pitfalls of such a project (Lopez, Davis et al. 2011).

Assessors noted that the land bridge is subject to a high rate of shoreline retreat, estimated at 4 ft. per year, on both the Lake Pontchartrain and Lake Borgne sides. This fair weather failure mode has been compounded by erosion during storms and one participant described a weak spot in the land bridge created by Hurricane Katrina that could breach into Pass Rigolets. Such a development would then increase the flow through the pass and likely increase the associated erosion. Assessors also noted industry related fair weather failure modes, including oil and gas activities and possible dredging of the GIWW (which runs along the southern shoreline of the land bridge) to accommodate high capacity cargo ships stemming from the IHNC lock expansion and the Panama Canal expansion.

Evacuation

Category	Comment
Evacuation	"a catastrophic storm has these characteristics... and you list them all.... Then explain that we aren't smart enough to know which ones will pop up with a given storm at a given time" Public education and risk awareness are important parts of the system Timing of the coordinated, regional plan is based on winds, not surge Storm and storm surge prediction capabilities need to be considered as part of the system No official, credible storm surge prediction capabilities SMP looks at property protection, not public health benefits of projects SMP neglects risk reduction benefits related to evacuation and related infrastructure improvements Because it just assesses projects on property protection, SMP emphasizes capital improvements over public health and safety Unprotected low spots on evacuation routes Numerous floodgates cross evacuation routes. Closure must be coordinated. Levee design process does not account for fast moving storms and early flooding of evacuation routes Loss of Chandeleur Islands, Biloxi Marsh, and Orleans land bridge would reduce available evacuation window Not everyone assumed the worst for Isaac and evacuated when they should have Coordination between agencies is difficult Even with excellent coordination, evacuation is still a personal decision based on perceived risk. Timing of individual decisions and actions can vary and cause chokepoints Warnings need to be precise and avoid false alarms "People look for reasons not to evacuate"; we are obligated to acknowledge that and plan for it Need to avoid the "bridge scenario" again where post-storm pedestrian evacuation is forcefully blocked by neighboring parish authorities

Evacuation is the last line of defense for protecting lives from hurricane induced flooding. The state and parishes have developed a phased, regional evacuation plan, and assessors expressed general confidence in the plan. Because of the importance of evacuation, assessors generally agreed that public education and risk awareness were important parts of the system. One assessor pointed out that warnings need to be precise and avoid false alarms.

Assessors also described a number of reasons to be concerned. Communication and coordination were major concerns. Many assessors emphasized that while people must evacuate because of the surge hazard, timing the evacuation process is based on the arrival of tropical storm force winds. It was also observed that the surge can arrive before the winds. Having noted that there

is no official, credible source of storm surge predictions, assessors described how coordination between parishes can cause problems with evacuations. They also pointed out that traffic management agencies tasked with evacuation management must coordinate with levee agencies tasked with closing floodgates and with bridge maintenance personnel tasked with protecting those assets from wind and surge. Finally, they pointed out that even with proper coordination and communication, evacuation is inherently a personal decision based on emotional considerations, financial constraints, family commitments, and prior evacuation experience. One assessor stated bluntly: “People look for reasons not to evacuate.”

When prompted about the exclusion of evacuation from the State Master Plan, assessors had a number of interesting comments. One pointed out that the SMP was based on economic criteria and flood protection projects in the SMP were evaluated via property protection. The SMP did not assess the public health benefits of projects. Because of this, the SMP emphasizes capital improvements over health. Probably the best example of the implications of this criterion was the previously described low points and choke points along evacuation routes. In terms of direct economic benefits, addressing these road construction projects provided very little if any direct economic benefits and the SMP did not consider the public health and safety benefits of improving the capacity of evacuation routes.

Pollution Controls

Category	Comment
Pollution Controls	"Pollution controls are important to system" Hazardous releases cause direct damage and health impacts Hazardous materials containers can damage landscape and HSDS Hazardous cleanup operations cause traffic problems, hinder economic recovery

Finally, pollution controls are not an explicit line of defense nor were they a formal topic when planning the QMAS workshops. However, they were a part of the discussion and assessors generally agreed that they are an important element of the HSDS. It was noted that hazardous releases pose a risk of direct damage along with health impacts for the system stakeholders. Furthermore, such releases damage the landscape, including elements of the HSDS. Finally, cleanup operations following a major release can result in traffic restrictions that slow the delivery of relief supplies and hinder economic recovery.

This topic was particularly pertinent at the third workshop, which followed Hurricane Isaac and the associated large scale hazardous release during the Stolthaven incident. The Stolthaven transfer facility is located near Braithewaite, LA, in Plaquemines parish. During Hurricane Isaac, a number of hazardous chemicals were stored there, either in large storage tanks or on rail cars. After the storm surge overwhelmed the local hurricane protection levee, the facility was flooded. The fast moving surge waters damaged the facilities and rail cars, and resulted in a large chemical release. After mixing with the flood waters, the receding surge then dispersed the hazardous substances through the adjacent marsh. Hazardous cleanup operations there lasted for

weeks and caused an extended closure of the only highway that connects with flooded communities further downriver.

Lake Borgne Surge Barrier Closure Operations and IHNC/GIWW Evacuation Activities

During the three workshops, the Lake Borgne Surge Barrier Closure operations and related activities within the IHNC/GIWW were repeatedly discussed as a complex sequence of coordinated tasks which involves numerous stakeholders and various moving parts. As such, it was decided that this sequence of tasks deserved particular emphasis and analysis. The remainder of this section describes this complicated set of procedures, while the next section presents a SysML model of this activity.

This closure sequence involves the closure of the three navigational floodgates and one navigational lock located within the IHNC/GIWW corridors: the IHNC Navigation Lock at the Mississippi River, the Seabrook floodgate at Lake Pontchartrain, the GIWW floodgate at the Lake Borgne Surge Barrier (which actually consists of two gates), and the Bayou Bienvenue floodgate at the Lake Borgne Surge Barrier. The locations of the four gates are shown in **Figure 33**. The limits of the IHNC/GIWW navigational corridor along with the gates and lock are shaded in yellow on **Figure 33**.

During fair weather conditions, the floodgates stay in the open position to allow both tidal exchange and waterborne commerce. If the three gates were left open during a hurricane, the storm surge would enter the IHNC/GIWW, raise water levels there, and exert hydraulic pressure on the walls. Additionally, if the surge is high enough, it could overtop the walls and flood the protected neighborhoods. Even without overtopping, having the storm surge enter the IHNC/GIWW would increase the risk of the floodwalls failing. Other than repairs to the sections that breached during Hurricane Katrina and the addition of concrete scour pads on the protected side, these floodwalls have not been substantially improved or raised.

On the other hand, the IHNC Navigation Lock (**Figure 34**) serves as a daily barrier to regulate water levels between the Mississippi River and the IHNC. During high water season, the Mississippi River water level is typically 8 to 14 feet above sea level; the IHNC is situated near sea level. This navigation lock remains closed to maintain separate water levels between the Mississippi River and IHNC. The IHNC Lock is operated by the U.S. Army Corps of Engineers and it is only opened when needed for navigational traffic.

The GIWW and Bayou Bienvenue Floodgates (**Figure 36**) are part of the newly constructed Lake Borgne Surge Barrier that is designed to prevent storm surges in Lake Borgne from entering the IHNC/GIWW. During fair weather, the gates allow maritime and recreational boating traffic along with aquatic organisms to traverse across the barrier.

The Seabrook Floodgate structure (**Figure 35**), which was completed in early 2012, allows the closure of the IHNC from Lake Pontchartrain. With the Surge Barrier blocking the eastern end of the IHNC/GIWW, storm surges could still enter Lake Pontchartrain through the Pass Rigolets and Chef Menteur Pass, then propagate into the IHNC/GIWW if the Seabrook structure were not closed.

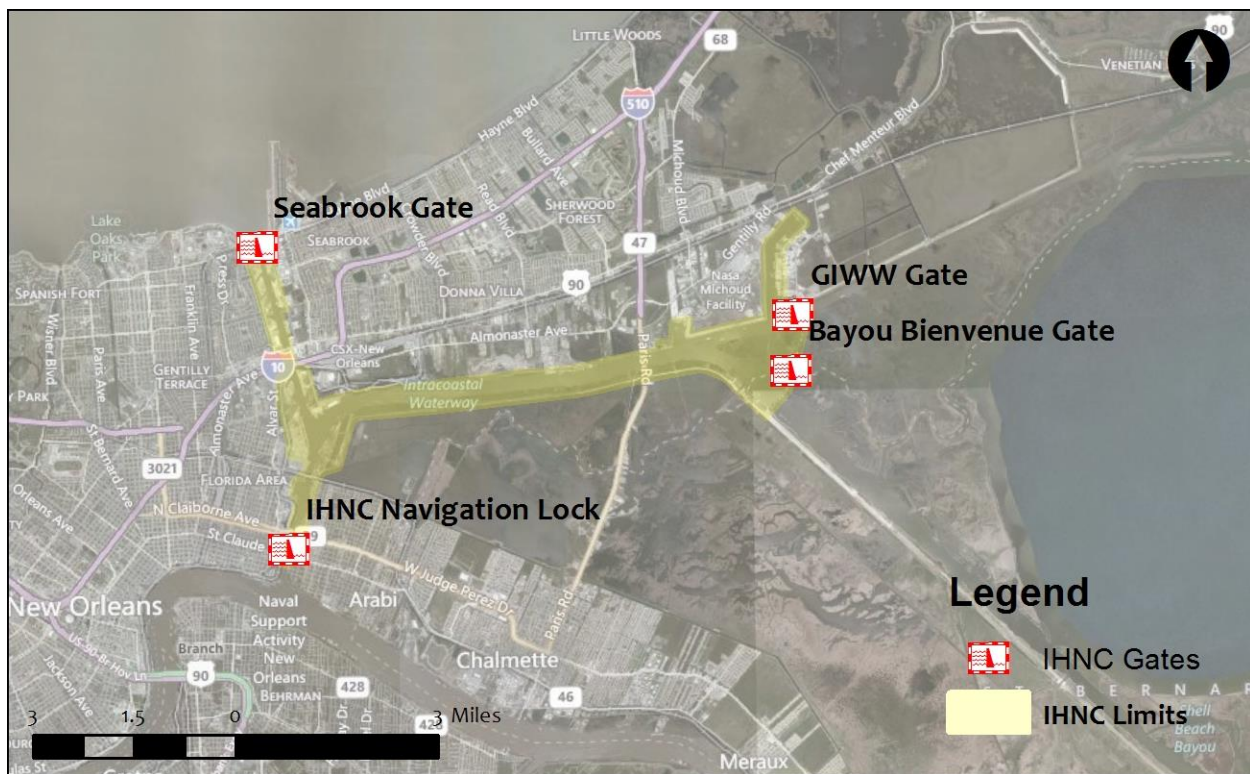


Figure 33: Limits of the IHNC and the four gates that are part of the perimeter Hurricane Defense System.



Figure 34: IHNC Navigation Lock, viewing from IHNC towards Mississippi River (Maples 2000).



Figure 35: Overview of new Seabrook Floodgate Structure (USACE 2009c).



Figure 36: Construction of the IHNC closure structure and new Bayou Bienvenue and GIWW Floodgate Structures (USACE 2009c)

Upon the approach of a storm that would generate a storm surge of concern, there is a procedure by which the three floodgates are closed so that maritime interests within the IHNC/GIWW can exit this corridor. These procedures also require coordination with the IHNC Navigational Lock and with agencies that manage a handful of draw bridges that cross the IHNC. Importantly, these activities must be coordinated to allow evacuation of vessels from the IHNC/GIWW prior

to closing of the gates. Further, these activities must also have a minimal impact on the concurrent vehicular evacuation of the general population.

According to USACE operational guidelines, the criteria for commencing IHNC gate closure operations are:

- Water elevations of 3 ft. or greater in Lake Borgne and/or
- A storm is predicted to make landfall in the 'area' within three days.

The general sequence of activities is as follows:

1. Closure of the Seabrook Gate by USACE (planned to be turned over to CPRA). Gate closure takes approximately 20 minutes and is necessary to reduce currents through the IHNC/GIWW which would imperil closure of the barge gate.
2. Closure of the Barge Gate at the Surge Barrier by the USACE (planned to be turned over to CPRA). Takes about 7 hours to complete and must be completed during calm conditions.
3. Closure of the Bayou Bienvenue Gate by USACE (planned to be turned over to CPRA). Gate closure takes approximately 20 minutes.
4. Closure of the GIWW Sector Gate by USACE (planned to be turned over to CPRA). Gate closure takes approximately 2 hours (sector gates).
5. Closure of the IHNC Navigation Lock by USACE.
6. Storm occurs and then passes.
7. GIWW Sector gate opened first (when maximum water elevation differential is ~3 ft) by USACE (planned to be turned over to CPRA). Gate opening takes 2 hrs.
8. Bayou Bienvenue Gate is opened by USACE (planned to be turned over to CPRA). Gate opening takes 2 hrs.
9. Seabrook Gate opened by USACE (planned to be turned over to CPRA). Gate opening takes 2 hrs. Not opened until Lake Pontchartrain drains.

Impacts to navigation associated with gate closures are coordinated with the U.S. Coast Guard. Once the decision to close the gate has been made, the gate operators notify the U.S. Coast Guard, who then passes the notification on to navigation interests.

Marine vessels within the IHNC are in a U.S. Coast Guard Regulated Navigation Area and operators are legally required to evacuate their vessels. Waivers can be granted if a pre-approved plan for anchoring or sinking the vessel is submitted to the USCG (U.S. Coast Guard 2010). However, currently there are no written criteria for the anchoring of vessel plans. Additionally, the USCG has reported being short on personnel to inspect anchored vessels and issue penalties for non-compliance.

While the vessels are evacuating the IHNC/GIWW, bridge operations must be coordinated with the closure of flood gates in order to allow the larger marine vessels and sailboats with large masts to exit the IHNC/GIWW. There are a number of low-clearance draw bridges that need to be raised to allow vessel evacuation. But when in the raised position, these bridges interrupt vehicular and rail transit, so it is not possible to leave them in the 'up' position. The bridges are operated by the Port of New Orleans and the Louisiana Department of Transportation. The Port of New Orleans operates the following bridges: Seabrook Railroad Bridge, Almonaster Avenue

Bridge, Florida Avenue Bridge, and St. Claude Avenue Bridge (Port of New Orleans 2013). The Claiborne Avenue Bridge (Judge William Seeber Bridge) is operated by the Louisiana Department of Transportation (Louisiana Department of Transportation and Development 2013).



Figure 37: Location of bridges within IHNC that require manual operation.

The Port of New Orleans’ Hurricane Preparation Plan (Port of New Orleans 2013) details the bridge operations and is copied below. (We were unable to locate operational guidelines for the Claiborne Avenue Bridge.)

The Bridge Operations Manager shall notify each of the four bridge operators on duty of the “hurricane watch” advisory. The Bridge Operations Manager will also notify each off-duty operator of the advisory and place them on alert. As the storm approaches, normal operation of each bridge will continue until the wind velocity or water level in the Inner Harbor - Navigation Canal prohibits safe operation, or until the Orleans Levee District closes floodgates leading to the bridge, as follows.

a. Seabrook Railroad Bridge

When wind velocity exceeds 40 MPH, operation of the Seabrook Railroad Bridge will cease and the bridge will remain locked in the “fully lowered” position. When such conditions exist, the bridge may be raised only at the discretion of the Bridge Operations Manager, with approval of a Division Director.

When the water level of the IHNC reaches +5.0 feet NVGD, the Seabrook Railroad Bridge will be closed to marine traffic and the bridge will remain locked in the “fully lowered” position (this is the elevation of the locking bar electrical conduit).

When the Orleans Levee District closes floodgates at the railroad approaches of the Seabrook Railroad Bridge (W39 & E15), the Bridge Operations Manager will advise Norfolk Southern Railroad, the U. S. Coast Guard, and the IHNC Lockmaster that “the bridge will be locked in the ‘fully lowered’ position, closed to all rail and marine traffic, and bridge operations will be suspended”. The Bridge Operator will then be allowed to leave the bridge unless the U. S. Coast Guard informs the Bridge Operations Manager that one or more vessels require transit through the bridge prior to suspension of operations, upon which the Bridge Operator will remain until these vessels have passed the bridge.

Prior to vacating the Seabrook Railroad Bridge, the Bridge Operator must lock the bridge in the “fully lowered” position and make the proper notation in the bridge log book. Then all power to the bridge will be shut off, all doors and gates to the Bridge Operator’s House and Machinery House will be locked, and barricades will be placed across the two access roadways (near France Road at the west side of the IHNC, and on the protected side of Floodgate E14 at the east side of the IHNC).

When floodgates at the railroad approaches of the Seabrook Railroad Bridge are re-opened, weather conditions permit, and the Bridge Operator’s House is accessible, the bridge will be inspected and normal operations will resume if approved by bridge maintenance personnel and the Bridge Operations Manager. Upon completion of the inspection, the Bridge Operations Manager will advise Norfolk Southern Railroad, the U. S. Coast Guard, and the IHNC Lockmaster of the status of bridge operations.

b. Almonaster Avenue Bridge

When wind velocity exceeds 40 MPH, operation of the Almonaster Avenue Bridge will cease and the bridge will remain locked in the “fully lowered” position. When such conditions exist, the bridge may be raised only at the discretion of the Bridge Operations Manager, with approval of a Division Director.

When the water level of the IHNC reaches +5.0 feet NVGD, the Almonaster Avenue Bridge will be closed to marine traffic and the bridge will remain locked in the “fully lowered” position (this is the elevation of the locking bar electrical conduit).

When the Orleans Levee District closes floodgates at the vehicular approaches (W31 & E9) and railroad approaches (W30 & E8) of the Almonaster Avenue Bridge, the Bridge Operations Manager will advise CSX Transportation, Inc., New Orleans Public Belt Railroad, the U. S. Coast Guard, and the IHNC Lockmaster that “the bridge will be locked in the ‘fully lowered’ position, closed to all rail and marine traffic, and bridge operations will be suspended”. The Bridge Operator will then be allowed to leave the bridge unless the U. S. Coast Guard informs the Bridge Operations Manager that one or more vessels require transit through the bridge prior to suspension of operations, upon which the Bridge Operator will remain until these vessels have passed the bridge.

Prior to vacating the Almonaster Avenue Bridge, the Bridge Operator must lock the bridge in the “fully lowered” position and make the proper notation in the bridge log book. Then all vehicular

traffic gates and barriers will be lowered, all power to the bridge will be shut off, all doors and gates to the Bridge Operator's House and Machinery House will be locked, and barricades will be placed across the east-bound and west-bound roadways (near France Road and Jourdan Road).

When floodgates at the vehicular approaches and/or railroad approaches of the Almonaster Avenue Bridge are re-opened, weather conditions permit, and the Bridge Operator's House is accessible, the bridge will be inspected and normal operations will resume if approved by bridge maintenance personnel and the Bridge Operations Manager. Upon completion of the inspection, the Bridge Operations Manager will advise CSX Transportation, Inc., New Orleans Public Belt Railroad, the U. S. Coast Guard, and the IHNC Lockmaster of the status of bridge operations.

c. Florida Avenue Bridge

When wind velocity exceeds 40 MPH, the Florida Avenue Bridge will not be raised higher than 41 feet above its "fully lowered" position (at this height, the elevation of the bottom of the lift span is at +45 feet NVGD, but vertical clearance for marine vessels will vary with the water level in the IHNC). When such conditions exist, the bridge may be raised above 41 feet only at the discretion of the Bridge Operations Manager, with approval of a Division Director.

When the water level of the IHNC reaches +5.0 feet NVGD, the Florida Avenue Bridge will be closed to all rail, vehicular, and pedestrian traffic (at this level, the water/waves of the IHNC produce a "lifting" force on the span and the bridge cannot be fully seated and locked in position).

When the Orleans Levee District closes floodgates at the vehicular approaches (W20 & E1) and railroad approaches (W21 & E2) of the Florida Avenue Bridge, the Bridge Operations Manager will advise Norfolk Southern Railroad, the U. S. Coast Guard and the IHNC Lockmaster that "the bridge will be closed to rail traffic, raised and locked at a height of 41 feet above its 'fully lowered' position, and bridge operations will be suspended". The bridge will then be raised to 41 feet and the Bridge Operator will be allowed to leave the bridge unless the U. S. Coast Guard has informed the Bridge Operations Manager that one or more vessels requiring additional vertical clearance need to pass through the bridge prior to suspension of operations, upon which the Bridge Operator will remain until these vessels have passed the bridge.

Prior to vacating the Florida Avenue Bridge, the Bridge Operator must lower all vehicular traffic gates and barriers, raise the bridge to a height of 41 feet above its "fully lowered" position, ensure that all brakes are set, and make the proper notation in the bridge log book. Then all power to the bridge will be shut off, all doors and gates to the Bridge Operator's House and Bridge Towers will be locked, and barricades will be placed across the roadway (outside Floodgates W20 & E1, near France Road, and near Jourdan Road).

When floodgates at the vehicular approaches and/or railroad approaches of the Florida Avenue Bridge are re-opened, weather conditions permit, and the Bridge Operator's House is accessible, the bridge will be inspected and normal operations will resume if approved by bridge maintenance personnel and the Bridge Operations Manager. Upon completion of the inspection,

the Bridge Operations Manager will advise Norfolk Southern Railroad, the U. S. Coast Guard, and the IHNC Lockmaster of the status of bridge operations.

d. St. Claude Avenue Bridge

When wind velocity exceeds 40 MPH, operation of the St. Claude Avenue Bridge will cease and the bridge will remain locked in the “fully lowered” position. When such conditions exist, the bridge may be raised only at the discretion of the Bridge Operations Manager, with approval of a Division Director. In addition, when the Corps of Engineers ceases operation of the IHNC lock, for any reason, including too high a water level in the IHNC, the St. Claude Bridge will cease operations and the bridge will remain locked in the “fully lowered” position.

At no time before, during, or after a tropical storm event will the St. Claude Avenue Bridge be closed to vehicular or pedestrian traffic. However, such traffic will be temporarily halted to allow passage of marine vessels in advance of an approaching storm and after the storm has passed.

Based on these operational procedures for floodgate and bridge closures, the organizations serving as Stakeholders for the IHNC/GIWW evacuation sequence include:

- United States Coast Guard
- United States Army Corps of Engineers
- National Weather Service (storm alerts)
- CPRA
- Southeast Louisiana Flood Protection Authority
- Louisiana Department of Transportation
- East Jefferson Levee District
- Orleans Levee District
- Lake Borgne Levee District
- Port of New Orleans
- Railroads (CSX, New Orleans Public Belt Railroad, Norfolk Southern Railroad)
- Maritime Industry (index of tenants can be accessed here:
http://www.portno.com/port_directory.asp)

Figure 38 below brings all of this together in one map, while formal modeling of this case study is presented in the Section 6 of this report.

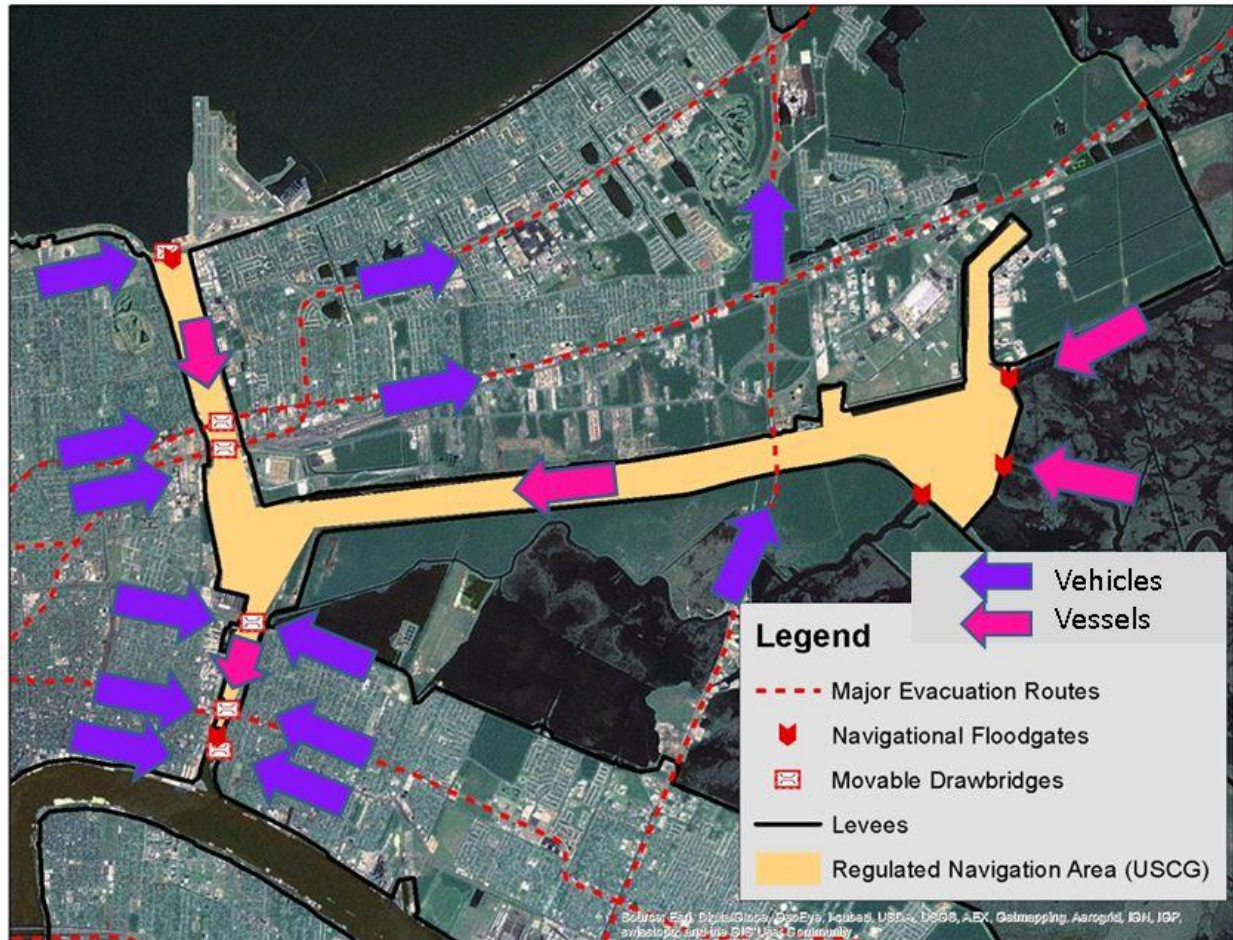


Figure 38: Map depicting the flow of vehicles and vessels along with movable bridges and floodgates for the IHNC/GIWW Regulated Navigation Area.

HSDS in a Systems-of-Systems Context

The IHNC/GIWW case study illustrates the integration of multiple organizations and multiple procedures over an extended spatial area. Likewise, the evacuation route example shows the integration of the national transportation systems (land and marine) and the central Gulf of Mexico estuary ecosystem with the Hurricane Surge Defense System. These cases illustrate the functionality and utility of the Systems Engineering approach that views the HSDS as a system-of-systems, whereby the systems of interest is viewed in conjunction with other overlapping systems.

The Hurricane Surge Defense System in the greater New Orleans area consists of a complex assemblage of organizations, structures, procedures, and environments. The intent of the HSDS is to provide protection to the community and its assets and to provide the foundation for a resilience economy and culture. The protected community consists of a wide range of stakeholders, such as residents, industry, commerce, utilities, and critical organizations (police, fire stations, hospitals). However, the HSDS includes large-scale effects, situated outside the

physical footprint of the protected area, and as such the HSDS interacts with other large scale and equally complex systems. For example, the barrier islands and marsh land bridges that serve as a natural buffer for incoming hurricane storm surges and waves are also part of the larger Gulf of Mexico ecosystem. Similarly, the national transportation and communication systems also enable the flow of information, people, and emergency supplies into and out of the region during a tropical weather event. During evacuations, members of the protected community utilize a national transportation network to temporarily relocate outside of the boundary of the HSDS. In doing so, they exert pressure on host communities, such as increased traffic and sheltering needs. Thus, the Hurricane Defense System can have far-reaching extents, not limited to the physical area being protected by the system.

Likewise, systems external to the HSDS also interact with and impact the HSDS. These larger systems include the navigation system linked to the Mississippi River, railroads, and nearby ports. The Gulf ecosystem including the Pontchartrain Estuary, the global climatic system that impacts hurricane frequency and intensity, the Mississippi river flood protection system, and the US agriculture system which impacts water quality in the Mississippi River and northern Gulf of Mexico. Importantly, there is the government funding system which includes the Federal, state and local funding processes determined largely by our democratic system of governance. Also, ultimately the HSDS protects a system of Louisiana communities engaged in economic and other activities beyond flood risk reduction. In this regard, it cannot be over emphasized that some elements of the HSDS are elements of other systems that are optimized for objectives not related to flood risk reduction. For example, coastal transportation embankments are primarily elements of the transportation system and only secondarily act as a storm surge barrier.

As such, The Hurricane Surge Defense System can be thought of as a “System” that is situated within a larger “Super System-of-Systems” and comprised of a number of “Subsystems” (**Figure 40**). “Super-System” interactions are those large-scale elements that impact the HSDS, but are largely not controllable at the HSDS level. The U.S. Congress, for example, allocates funding to the USACE, which in turn plans, design, constructs, and (in some instances) operates elements within the HSDS. If the U.S. Congress opts not to allocate funding, the local system manager has very limited options to persuade Congress to reconsider. As another example, the Corps’ “upstream” decisions on Mississippi River management impact many aspects of the GNO flood protection, particularly the coastal lines of defense that depend on freshwater and sediment from the river.

Similarly, there are processes within the natural ecosystem and climate system that impact the system. Regional subsidence, sea level rise, and climate change have major impacts on storm frequency, severity, and duration. The physical protection system was designed using historical weather observations as guidance, but changes in environmental conditions can, in turn, impact the ability of the system to provide the desired degree of inundation protection for the protected community and their assets.

In contrast to the external systems, the “Subsystem” (elements of the HSDS) lends itself to be more closely managed and controlled by the system’s critical stakeholders. Subsystems would include components of the system, such as the IHNC/GIWW flood gates and bridges. While

these elements tend to be under the direct control of the organizations that comprise the HSDS, the IHNC/GIWW case study shows that it is not always as straightforward as one would imagine.

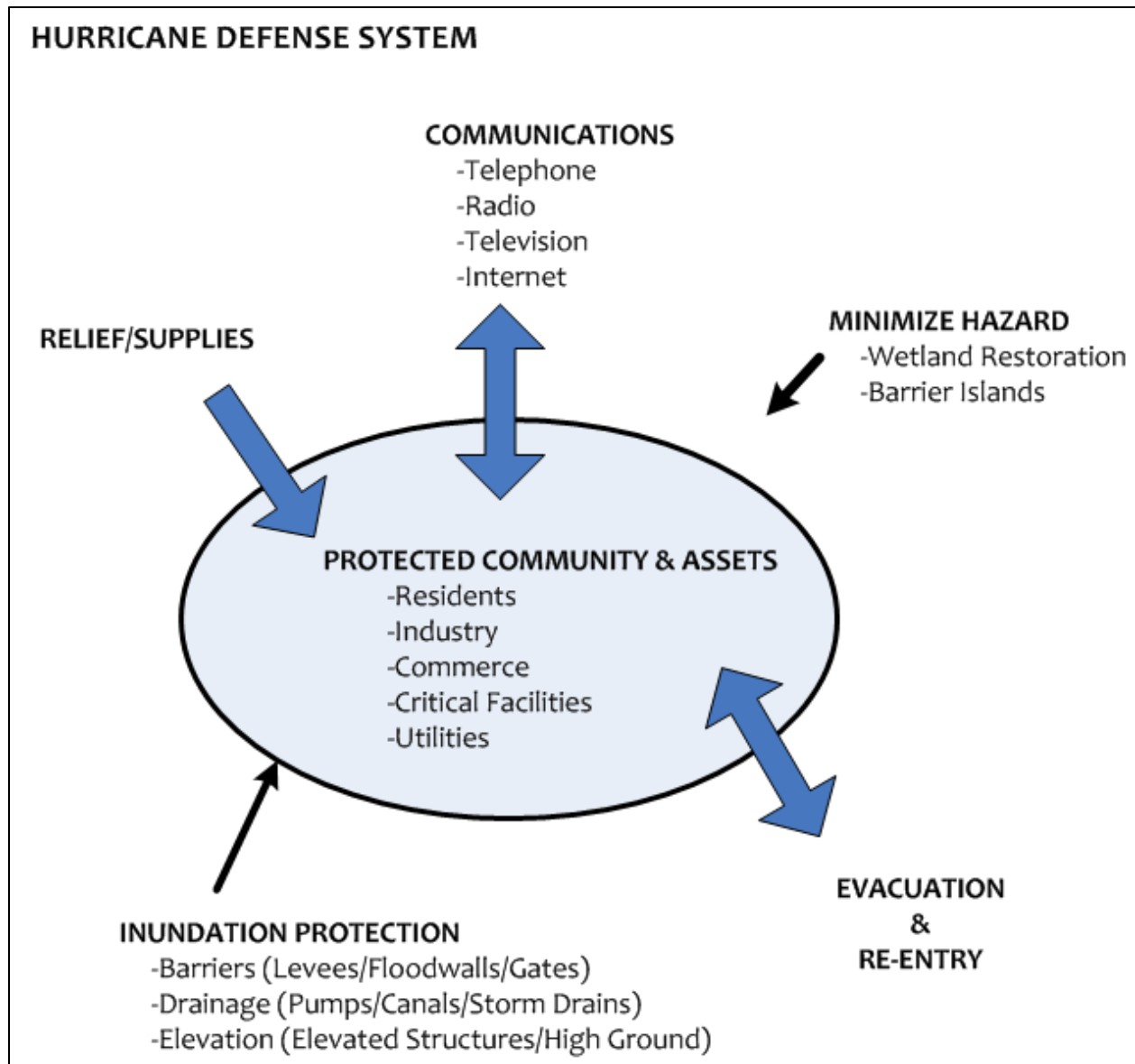


Figure 39: Hurricane Surge Defense System (symbolized by the oval) interacts with other systems.

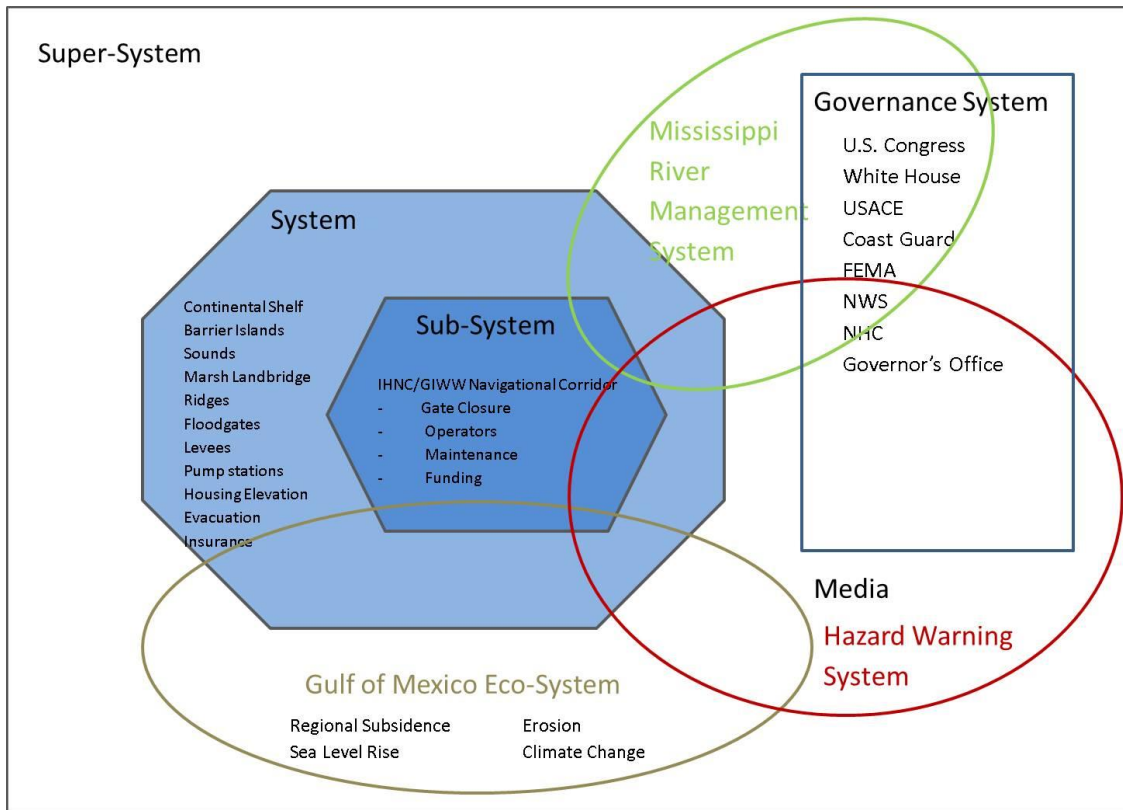


Figure 40: Examples of elements associated with the “Super-System,” the “System,” and the “Subsystem” along with other distinct but overlapping systems.

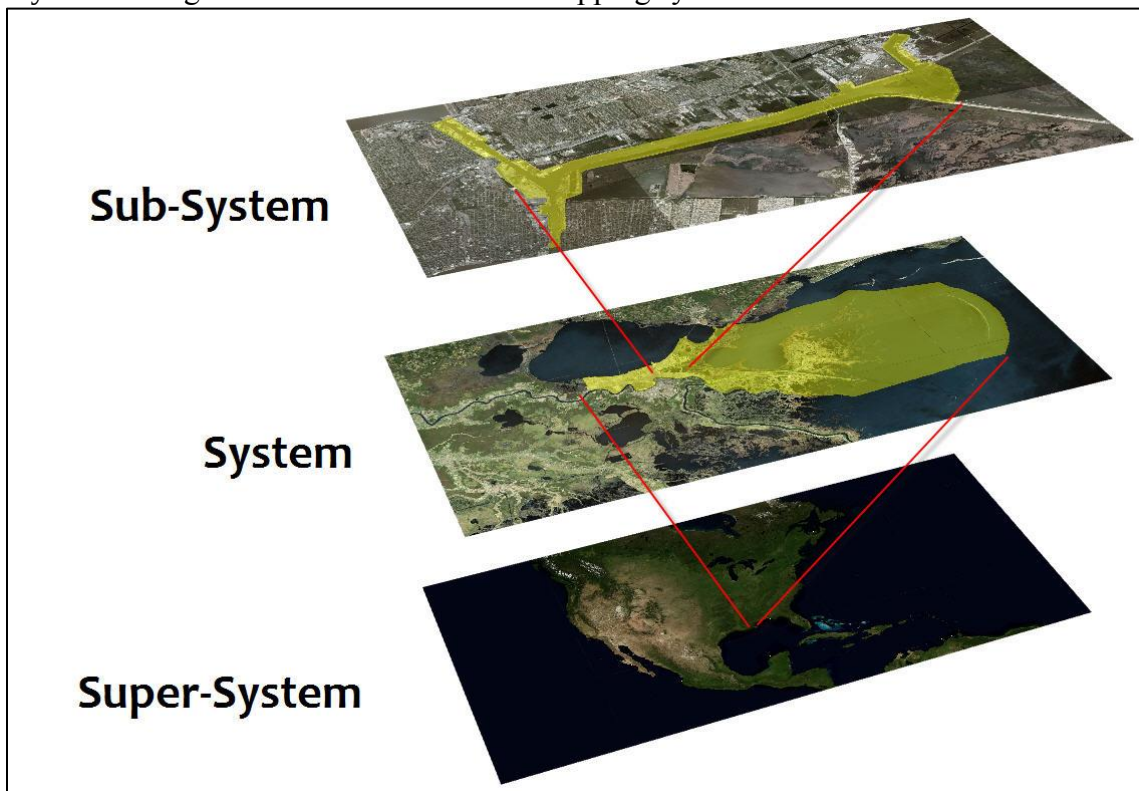


Figure 41: Spatial decomposition of the "Super-System," "System," and "Subsystem."

5) Initial System Specification

This section presents an initial inventory of system elements and a system hierarchy followed by a list of factors and scenarios of concern. This section synthesizes in a narrative format the information that went into our first attempt at creating a high-level SysML model of this system, which is covered in the next section.

In the system specification, the various levels of the system hierarchy largely correspond to different spatial scales of the system elements. At the top level, the Hurricane Surge Defense System, is a system with a regional footprint covering all of Southeast Louisiana. At the next level in the hierarchy, the coastal, structural, and community system segments all take up smaller and sometimes overlapping spatial footprints (see **Figure 3.**)

In our initial generic system hierarchy, system elements are listed down to the location level. Beyond this level, we do not attempt to account for the various parts of a specific system element at a location. For example, individual pump stations all contains parts such as gears, while automobiles involved in evacuation procedures all require engines, brakes, and transmissions. However, this level of detail is beyond our current scope and in some cases counterproductive. For example, in the present context of storm surge risk reduction, automobiles can be considered autonomous elements with varying but statistically predictable reliabilities. Ensuring that the system includes sufficient Motor Assistance Patrols to respond to breakdowns that block evacuation lanes is much more pertinent to the system objective than accounting for the specific part of a specific make and model that might lead to breakdown.

Requirements Table Narrative

As described previously, the literature database was used to create a table listing known requirements for either the system as a whole or for individual elements. The requirements table is a crucial early part of the systems engineering process, and it is used in later steps to ensure that when integrated, the parts create a system that fulfills its objectives. A later stage of the system analysis includes requirement verification and validation, whereby individual requirements are linked to the functionality of specific system elements. Naturally, our initial requirements table was improved with input from the QMAS workshops.

Like the structure of the system, the system requirements can also be organized into a hierarchical diagram with overall system-wide requirements at top and lower level functional requirements listed below. At the top of the requirements table is the overarching requirement of ensuring a resilient and economically viable coastal community given the constraints posed by sea level rise, regional subsidence, coastal land loss, and increasing energy costs. The 2012 State Master Plan expresses this requirement as ensuring a coastal zone "characterized by consistent levels of productivity and resilience (the ability to withstand naturally variable conditions and/or recover from disturbances)." Toward fulfilling this requirement, the HSDS must meet specific performance requirements defined in terms of an acceptable risk level and resiliency level. For

example, the availability of flood insurance is a crucial requirement to maintain economic productivity, and the National Flood Insurance Program requirements include the following:

"To be recognized as providing a 1-percent-annual-chance level of flood protection on the modernized NFIP maps, called Digital Flood Insurance Rate Maps (DFIRMs), levee systems must meet and continue to meet the minimum design, operation, and maintenance standards" (FEMA 2008)

In a similar manner, the USACE requirements augment the 100-yr risk level with a 500-yr resiliency requirement. They specify levee heights must meet the 1% annual probability storm surge height, while the structure must remain resilient given overtopping from a surge with 0.2% annual probability (Schleifstein 2013).

Toward fulfilling these requirements, specific elements of the system must meet more specific requirements. For example, individual levee and floodwall segments must meet a crown elevation requirement determined by calculating the surge height-frequency relationship at that location. Importantly, to meet this requirement over the element's lifespan, there is the additional requirement that the analysis account for sea level rise, local settling, and geologic subsidence. In a similar manner, pump stations must fulfill requirements defined in terms of pumping capacity. In areas with a single set of pump stations, the pumping capacity is defined in terms of precipitation volumes over the drainage area. However, where double pumping is necessary (i.e. the drainage canals within Orleans Parish) the first set of pump stations must meet requirements based on precipitation, while the second set of pump stations must fulfill the requirement of keeping up with the first set of pumps to maintain water levels canals below the safe water level. Of note, QMAS assessors consistently noted that pump station capacities are defined in terms of the rainfall hazard and not in terms of the storm surge hazard and that major parts are not necessarily required to be salt water tolerant.

The non-structural elements of the HSDS must fulfill system requirements. Coastal restoration projects must meet requirements related to subsidence and erosion. Housing elevation requirements are expressed in terms of base flood elevations published by FEMA. They also must fulfill building code requirements related to increased wind loads experienced by elevated structures. Public participation is a crucial requirement for the evacuation line of defense, which in turn requires risk communication with the public including the availability of evacuation maps and emergency preparedness guidelines that are readable by all aspects of the population.

Appendix D consists of a requirements table derived from a review of Federal, state, and local planning documents along with an academic analysis of the appropriate risk level for the GNO area. It is worth noting the academic analysis (Jonkman, et. Al 2009) specifies a system requirement that is an order of magnitude greater than that which is specified in the planning documents.

Generic Subsystem Level System Hierarchy

Our system inventory starts with a three level system hierarchy which includes the 12 lines of defense at the subsystem level. Earlier, we described how spatial scale plays a role in describing a complex system (see **Figure 12**). Later, we introduced the systems engineering based system hierarchy (see **Figure 17**). While the two figures use different terminologies, both refer to essentially the same concept of describing the system with greater detail as you zoom in to smaller spatial scales. Throughout the system specification that follows, we stick with the standard systems engineering terminology. However, it is important to keep in mind that the different levels of the system hierarchy (the system, segment, subsystem, components, etc.) also correspond to different spatial scales (landscape, regional, city, neighborhood, etc.)

At the subsystem level, the spatial footprints of the elements are different, but of similar scale. Each element covers an area that is of the order of 10's of square miles. The one exception is the evacuation routes, which are linear features emanating from the protected region and stretching up to a hundred miles. Below the subsystem level, the LODs are separately broken down into assemblies, components, and parts. This refined level of detail is presented for each LOD in the next sub-sections.

Table 3 presents a generic system hierarchy at the subsystem level. In this table, each row represents a level of the system hierarchy with the higher elements consisting of the aggregate of the element listed below it. So, for example Building Elevation, Evacuation, and Insurance are all elements comprising the Community LODs segment of the system. The sections that follow provide a more detailed inventory for each subsystem. However, at this point it is important to note that is hard to present all of the detailed information to specify this complex system in a clear and concise manner. Some of the tables and figures that follow are difficult to read, not because we have not made our best attempts to simplify them, but because the system is complex and difficult to describe in a clean and simple manner.

Table 3: MLODS based HSDS system hierarchy to the subsystem level. At the top level, the Hurricane Surge Defense System is the system that we are describing. The HSDS is made up three segments – the Coastal, Structural, and Community lines of defense. At the subsystem level, these three components are made up of 12 separate lines of defense (see **Figure 42**.)

System	Hurricane Surge Defense System											
Segment	Coastal LODs						Structural LODs			Community LODs		
Sub-system	Offshore Shelf	Barrier Islands	Sounds	Marsh Land bridge	Natural Ridge	Transportation Embankments	Floodgates	Levees	Pump Stations	Building Elevation	Evacuation	Insurance

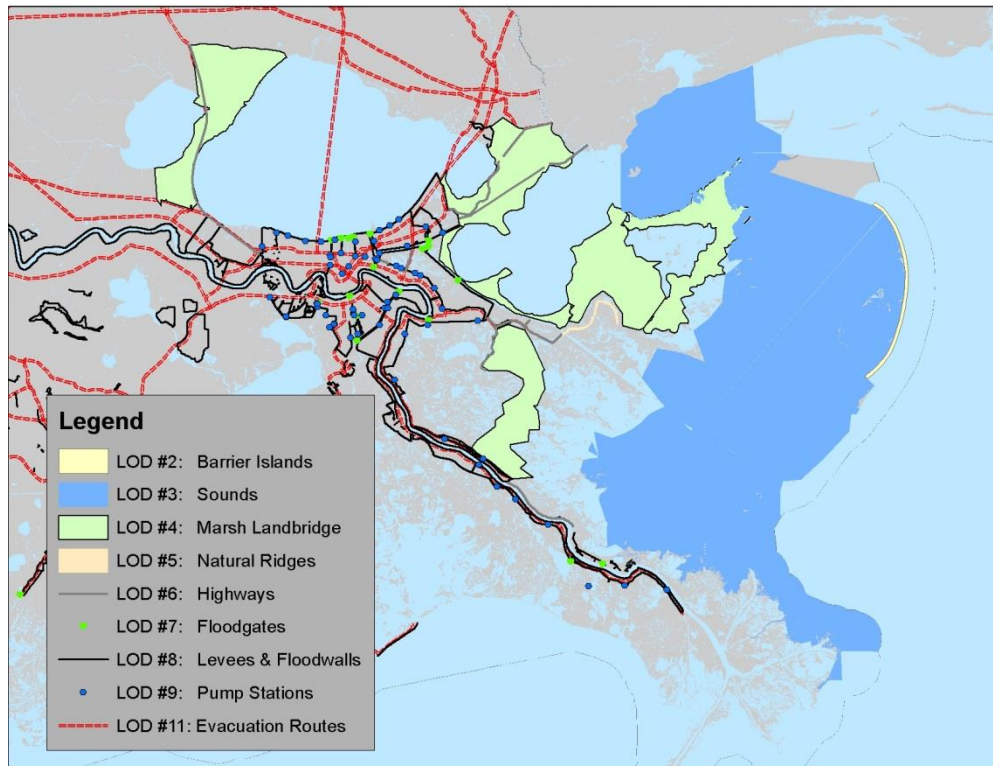


Figure 42: Map of the subsystem elements, a.k.a. the Lines of Defense, listed in the bottom row of the table above. For a fully functioning system, these individual elements must be integrated functionally to fulfill the system objectives and requirements. Note that the continental shelf (LOD #1), housing elevation (LOD #10) and insurance (LOD #12) are not shown on the map.

System Inventory at the Assembly and Component Levels

LOD#1: Offshore Shelf

The Gulf of Mexico's offshore shelf is a geologic feature delineating where the ocean basin meets the continental crust. For a storm surge being pushed inland by a tropical windstorm, water depths jump from deep ocean scale (thousands of feet) to continental scale (hundreds of feet) at the offshore shelf. Moving inland, the shallow shelf greatly reduces the heights and energy of waves from an approaching tropical system. It has also been hypothesized that the reduced water depth reduces the transfer of energy from the ocean to the tropical system, thereby reducing the storm's intensity. On the flip side, because the gentle slope of the shelf as one moves toward the coastline, the shelf can also promote higher surge heights via run-up. While the offshore shelf is largely outside the policy and engineering realm, the shape and geometry of the shelf has a strong influence on the dynamics of an approaching storm surge so it is an important element in the surge risk reduction system.

Table 4 below list elements of the offshore shelf subsystem at the assembly and component levels, while **Figure 43** presents a topographical map and geologic map of the shelf. In terms of assemblies, the offshore shelf primarily consists of the physical terrain and landscape. While

other assemblies are required to fully describe the offshore shelf (such as biologic species and activities, tectonic processes, and petroleum extraction) this is the primary element of the shelf that influences storm surge propagation and risk. Below the assembly level, the offshore shelf consists of a number of geologic components: the oceanic crust, the Texas-Louisiana Shelf, the Upper Central Shelf, the Mississippi Slope, the Gulf of Mexico, and Submerged Sediments of the Mississippi Delta.

Table 4: Assemblies and components for the Offshore Shelf.

System	Flood Defense System					
Segment/Element	Coastal LODs					
Subsystem	Offshore Shelf					
Assembly	Physical Landscape					
Component	Oceanic Crust	Texas-Louisiana Shelf	Upper Central Slope	Mississippi Slope	Gulf of Mexico	Mississippi Delta (Submerged Sediments)

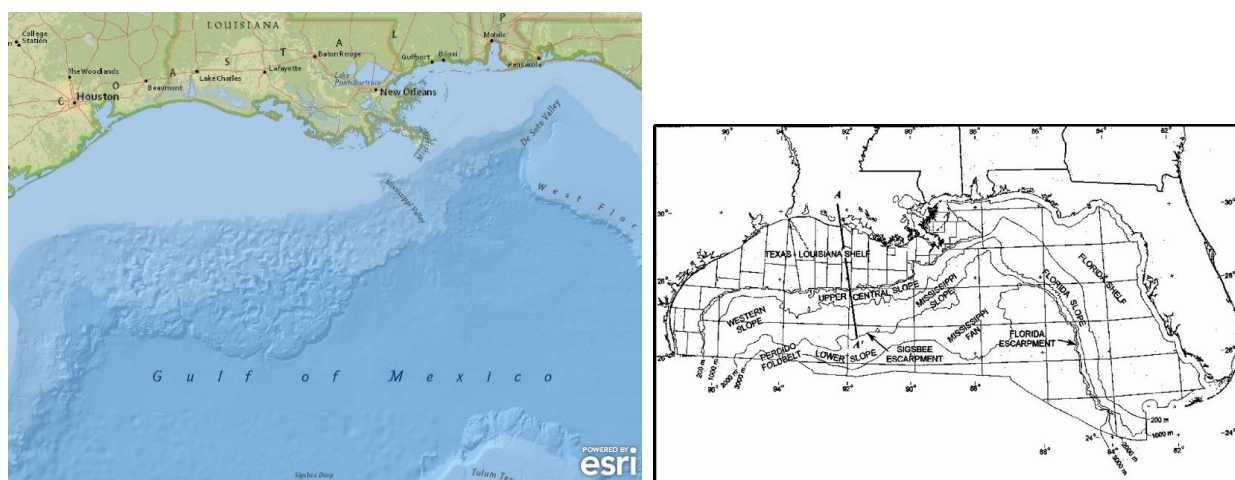


Figure 43: Maps depicting the Offshore Shelf (Sources: Left – ESRI; Right -- Minerals Management Service.)

LOD#2: Barrier Island

The second line of defense, the barrier islands and shoals, provides an important coastal buffer. Despite their low lying and fragmented appearance at the surface, the barrier islands and shoals, the sub-surface materials (i.e. the shoals) also provide storm surge mitigation benefits. These benefits include reducing surge for some hurricane scenarios, as well as reducing the heights of

waves that impact the interior shorelines. Indeed, while many people have expressed great concern about retreat and submergence of the barrier islands, it is important to note that shallow shoals are still part of the system even if below the surface. In fact, one advantage of viewing the loss of barrier islands through a systems approach is that you can better compare how shoals versus barrier islands reduce surge risk.

At the assembly level, the barrier islands consist of the physical landscape, vegetation, habitat, and management. Components of the physical landscape include waves and currents, submerged sand and soils, and dry sand and soils. Vegetation on the barrier islands includes a number of species, and the dominant ones are black mangrove, groundsel bush, wax myrtle, and submerged aquatic vegetation. Naturally, each of these species consists of different types of roots and foliage, attributes which impact the resistance that the vegetation applies to the incoming storm surge. The important habitats of the barrier islands include shellfish reefs and corals, both of which are submerged and comparatively rigid. Finally, the management component of the barrier islands includes the designated conservation areas, such as Breton National Wildlife Refuge, the 2010 emergency sand berm project which introduced new sand into the littoral system, and numerous management and restoration projects that impact the barrier islands.

Table 5: Assemblies and components for the Barrier Islands.

System	Flood Defense System											
Segment/Element	Coastal LODs											
Subsystem	Barrier Islands: Chandeleurs and Breton											
Assembly	Physical Landscape			Vegetation				Habitat		Management		
Component	Waves & Currents	Sand & Soils (Submerged)	Sand & Soils (Dry)	Black Mangrove	Groundsel Bush	Wax Myrtle	Submerged Aquatic Vegetation	Shellfish Reefs	Corals	Breton National Wildlife Refuge	Emergency Sand Berms	Restoration Projects

LOD#3: Sounds

Sounds are large, relatively shallow waterbodies located between the barrier islands and the marsh shoreline. With depths in the 10 – 30 ft range, these relatively shallow waterbodies provide an additional buffer to the deep water currents that drive storm surge dynamics. On the other hand, they also provide a large fetch (20 – 50 miles) that allow storm surge and waves to regenerate before impacting the shoreline.

The sounds comprise four assembly elements: the physical landscape, the vegetation, the habitat, and management. The physical landscape consists of the water, the water bottom, and waves and currents. Since by definition sounds do not have dry land, the only component of the vegetation assembly are the submerged aquatic species. Habitat in the sounds consists of shellfish and corals. Management components consist of dredging, navigation, and restoration projects.

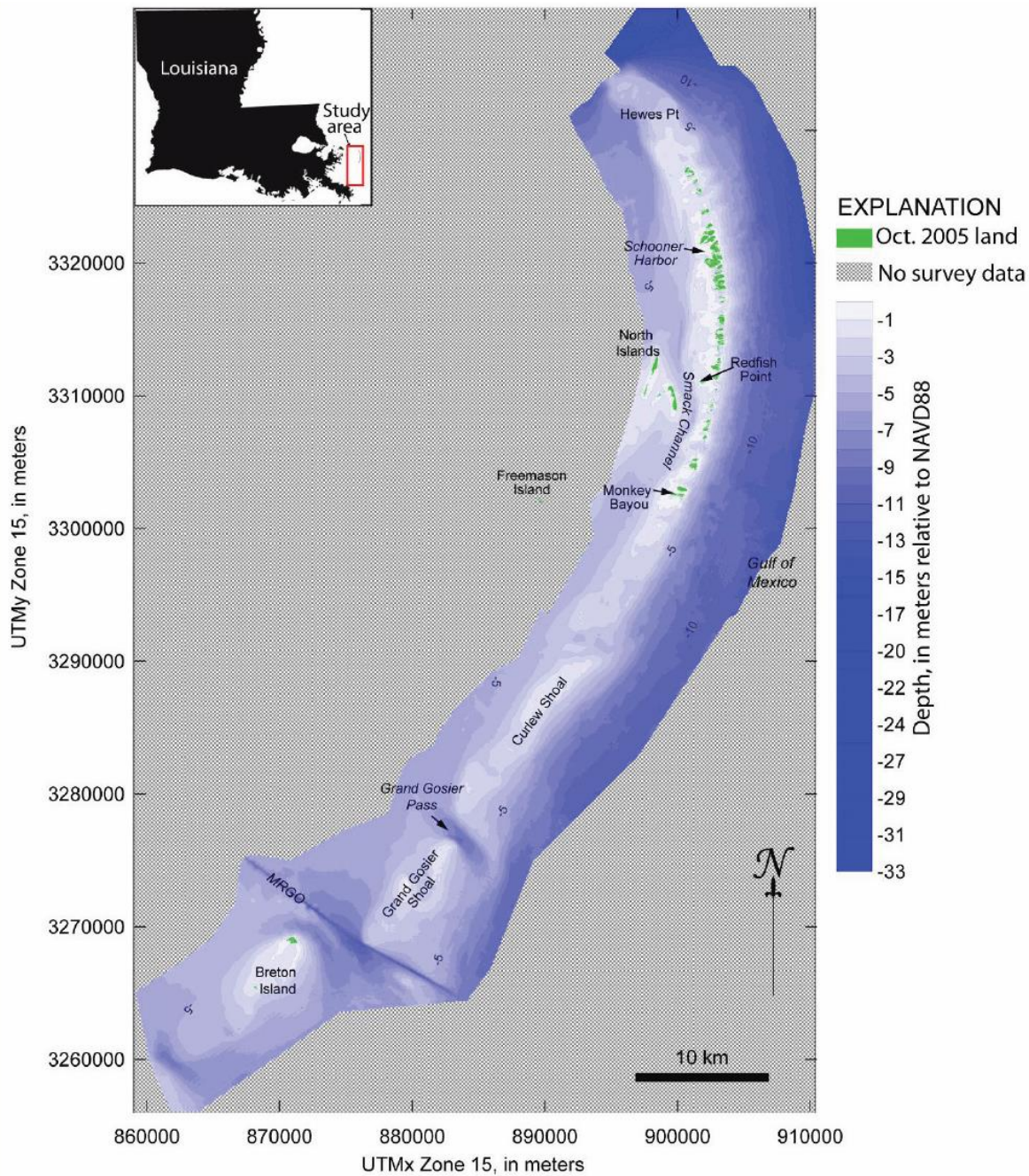


Figure 44: Maps depicting the Chandeleur Barrier Islands and the associated shoals (Miner, Kulp et al. 2009)

Table 6: Assemblies and components for the Sounds.

System	Flood Defense System										
Segment/Element	Coastal LODs										
Subsystem	Sounds: Mississippi, Chandeleur, Breton										
Assembly	Physical Landscape				Vegetation		Habitat		Management		
Component	Water	Water Bottom	Waves & Currents	Sediment Transport	Submerged Aquatic Vegetation	Fringe Marsh	Shellfish	Corals	Dredging	Navigation	Restoration Projects

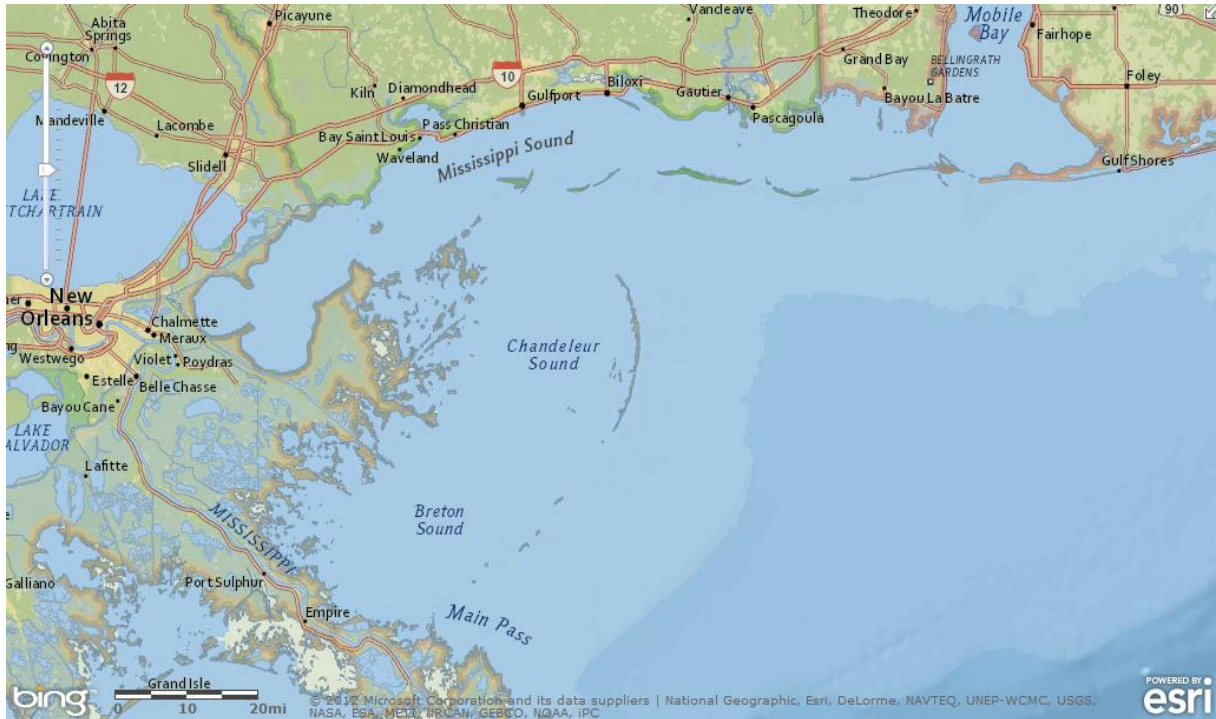


Figure 45: Map depicting the Sounds (Source: ESRI.)

LOD#4: Marsh Land Bridge

Much of southeast Louisiana’s natural shoreline consists of marsh. A marsh land bridge is a mostly continuously and attached mass of marsh soils and vegetation. Many marsh land bridges are extensions of natural ridges, the fifth LOD. While there are certainly some recent exceptions, marsh land bridges have proven resilient to damage during tropical events. As a mostly rigid mass, they provide a physical barrier to storm surges. Even though their low elevation implies that they are typically submerged by moderate surge values, these still reduce the volume of water that can propagate inland. Sounds also reduce wave heights significantly and bottom friction reduces the surge’s forward momentum.

Assemblies of the marsh land bridges include the physical landscape, vegetation, habitat and management. Components of the physical landscape include waves and currents, the nearshore environment, the interior waterways, soils, and geologic anchor of the land bridge (viz. ridge or similar feature). Vegetation components include saline vegetation, intermediate vegetation, and freshwater marsh vegetation. The habitat includes herbivores and fish species. Management includes ownership, diversions and spillways, restoration projects, freshwater pumping, and extractive industries.

Table 7: Assemblies and components for Marsh Land Bridges.

System	Flood Defense System											
Segment	Coastal LODs											
Subsystem	Marsh Landbridge: NO East Landbridge, Biloxi Landbridge, Breton Landbridge											
Assembly	Physical Landscape				Vegetation			Habitat	Management			
Component	Waves & Currents	Nearshore Environment	Interior Waterways	Soils (Dry)	Saline Marsh	Intermediate Marsh	Freshwater Marsh	Herbivores	Ownership	Diversions / Spillways	Restoration Projects	Sediment and Freshwater Pumping

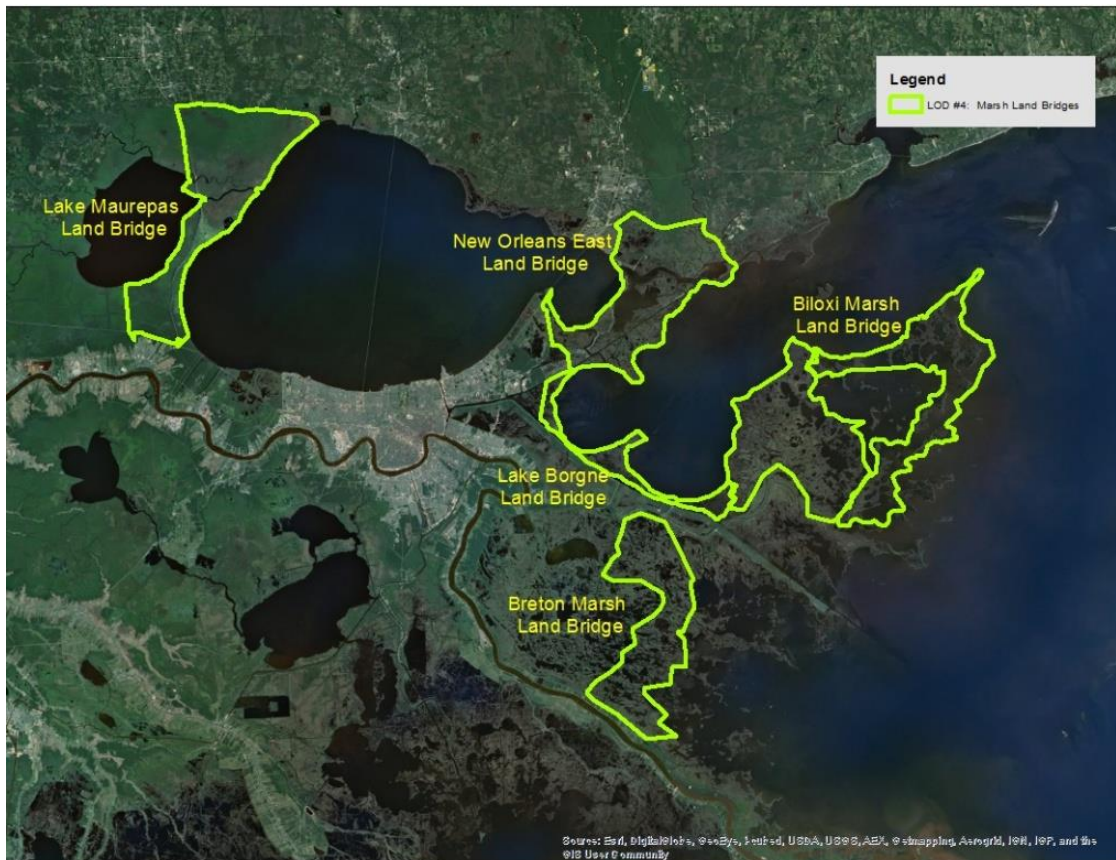


Figure 46: Map depicting the Marsh Land Bridges (Source: ESRI.)

LOD#5: Natural Ridge

Natural ridges are linear, elevated features of the natural landscape that are relicts of the natural levees along abandoned distributaries of the Mississippi River. Built from sediment deposited during overbank flows, ridges consist of compacted heavier soils. With elevations up to 10 ft., these features block surges up to the elevation of the ridge and reduce the volume of water pushed inland by surges that overtop the ridges. While ridges are naturally continuous, human alterations have compromised some ridges, mainly where transportation routes (either highways or canals) have cut through the natural ridge. The cut through LaLoutre Ridge for the MRGO is one such example, though it has recently been filled as part of the MRGO de-authorization process. Where present, these cuts serve as hydrologic leak points that allow both daily tides and storm surges to flow through the ridge. Also, converging ridge geometries can also work as funneling mechanisms that intensify surge and focus waves at the funnel's vertex.

At the assembly level, natural ridges consist of the physical landscape, vegetation, habitat, and management. The physical landscape includes accumulated heavy sediments, the substrate, and faults. Vegetation includes hardwood forest tree species, softwoods, and fringe marsh. Habitat includes forest, swamp, and marsh habitat. Management of ridges includes ownership, land use, and restoration projects.

Table 8: Assembly and components for the Natural Ridges.

System	Flood Defense System						
Segment	Coastal LODs						
Subsystem	Natural Ridge: Gentilly Ridge, LaLoutre Ridge, Delacroix Ridge						
Assembly	Physical Landscape			Vegetation	Habitat	Management	
Component	Accumulated Heavy Sediments	Substrate	Faults	Hardwood Forest		Ownership	Restoration Projects

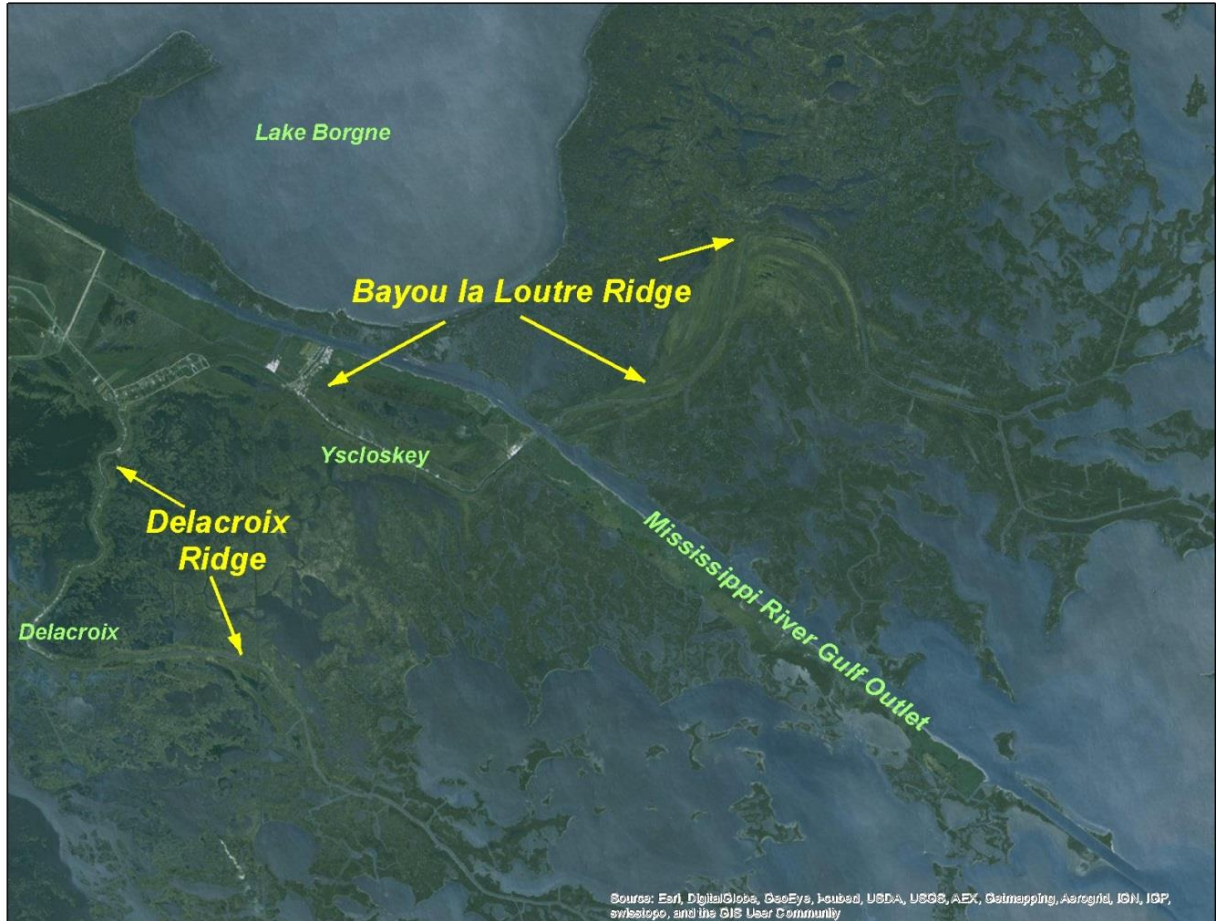


Figure 47: Map depicting the Bayou la Loutre Ridge and Delacroix Ridges in eastern St. Bernard Parish (Source: ESRI.)

LOD#6: Transportation Embankments

The transportation embankments subsystem is the last element within the coastal lines of defense segment of the flood defense system. In many ways they are functionally similar to natural ridges: they act as barriers to surge. In fact, these elements could also be labeled “man-made ridges”. While distinct from natural ridges, these man-made barriers are also not part of the

perimeter protection system. Four primary characteristics distinguish transportation embankments from ridges. Whereas ridges meander and curve, transportation embankments mostly consist of straight segments. Whereas ridges tend to be continuous, transportation embankments have hydrologic leak points, viz. bridges over water bodies, that can convey surge through the feature. Likewise, natural ridges are constructed of natural soils whereas most embankments (except spoil banks) are reinforced with imported limestone and rock. Likewise, highways and railroads are typically under active use and maintenance, while there are no maintenance regimes for ridges. Finally, many of these distinctions reflect the simple fact the transportation embankments are created for objectives related to transportation and commerce.

At the assembly level, transportation embankments include highways, railroads, and spoil banks. Generally speaking, all three assemblies are comprised of the same components: materials, construction and design, and maintenance. However, spoil banks consist of locally dredged materials and are not managed and largely unmaintained.

Table 9: Assemblies and components for the transportation embankments.

System	Flood Defense System								
Segment	Structural LODs								
Subsystem	Transportation Embankments								
Assembly	Highways			Railroads			Spoil Banks		
Component	Materials	Construction	Maintenance	Materials	Construction	Maintenance	Soils	Vegetation	Maintenance

LOD#7: Floodgates

Floodgates are a primary engineered structural element of the perimeter protection system. Whereas levees provide perimeter protection along dry land, floodgates block the gaps in levees where they cross canals, bayous, and other water bodies. Typically, they are built to be open for tidal exchange and navigation during normal weather conditions, but closed to block surge during tropical weather systems or other instances of extreme tide levels. When closed they provide a continuous barrier with a height typically equal to the adjacent levee reaches.

At the assembly level, floodgates consist of the structure, the transitions to levee/floodwall, operations, and maintenance. Components of the structure assembly include gates, the pivot mechanism, power supply, and controls. The transition assembly includes sheetpiles, limestone, and concrete. Operations include the USACE, local agencies, and communications between the agencies. Maintenance largely falls on local agencies, though CPRA is becoming increasingly involved.



Figure 48: Map depicting significant coastal transportation embankments. Minor roads and spoil banks are not shown. (Source: ESRI, US Census.)

Table 10: Assemblies and components for the floodgates.

System	Flood Defense System										
Segment	Structural LODs										
Subsystem	Floodgates										
Assembly	Structure				Transitions			Operations		Maintenance	
Component	Gates	Pivot Mechanism	Power	Controls	Sheetpiles	Limestone	Concrete	Local Agencies	Communications	SLFPA-E	CPRA

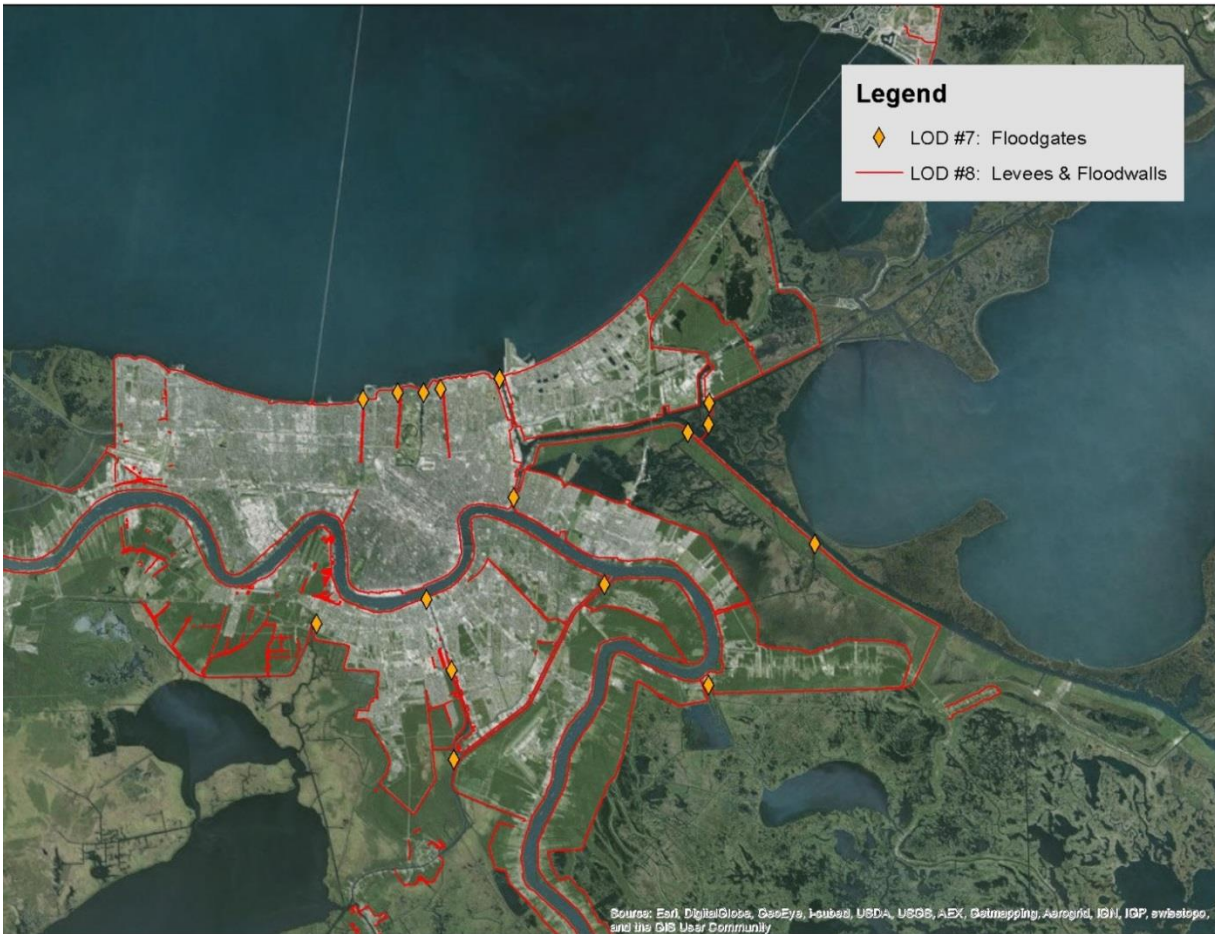


Figure 49: Map showing floodgates and adjacent levee reaches. (Source: Louisiana 2012 SMP, ESRI.)

LOD#8: Levees

Levees are the second segment of the structural LODs. These linear, engineered structures provide a continuous barrier that separates the “flood” side from the “protected” side. Of course, “protected” does not imply never flooded. In fact, it is fairly common for surge to either overtop or breach levees, resulting in flooding on the protected side. In these cases, levees also prevent gravity drainage from the protected side and can lead to flooding when rainfall exceeds the mechanical interior drainage capacities. While traditionally earthen levees are considered distinct from concrete and steel floodwalls, most contemporary perimeter protection structures consists of earthen/floodwall hybrids. Here the term “levee” is used as a general term for all combinations of earthen levee and concrete and steel floodwall. Specific reaches will be distinguished as either an “earthen levee”, a “floodwall”, or a “hybrid”. Naturally, they interact with surge and waves in different manners.

At the assembly level, levees consist of the levee structure, transitions, operations, and maintenance. The structure of the perimeter barrier is made of the following components: earthen levee, concrete and steel floodwall, movable floodgates (non-navigational), overtopping

protection, shoreline protection and stabilization, and temporary barrier such as sandbags, inflatable dams, and HESCO baskets. Operations components include relevant agencies, operations plans, contractors, and communications. Maintenance consists of inspections, grass-cutting, and lifts.

Table 11: Assemblies and components for the levees.

System	Flood Defense System												
Segment	Structural LODs												
Subsystem	Levees												
Assembly	Structure					Transitions			Operations		Maintenance		
Component	Earthen Levee	Concrete & Steel Floodwall	Over Topping Protection	Shoreline Protection and Stabilization	Temporary Barriers	Concrete	Limestone	Grout	Operations Plan	Communications	Inspections	Grass-cutting	Lifts

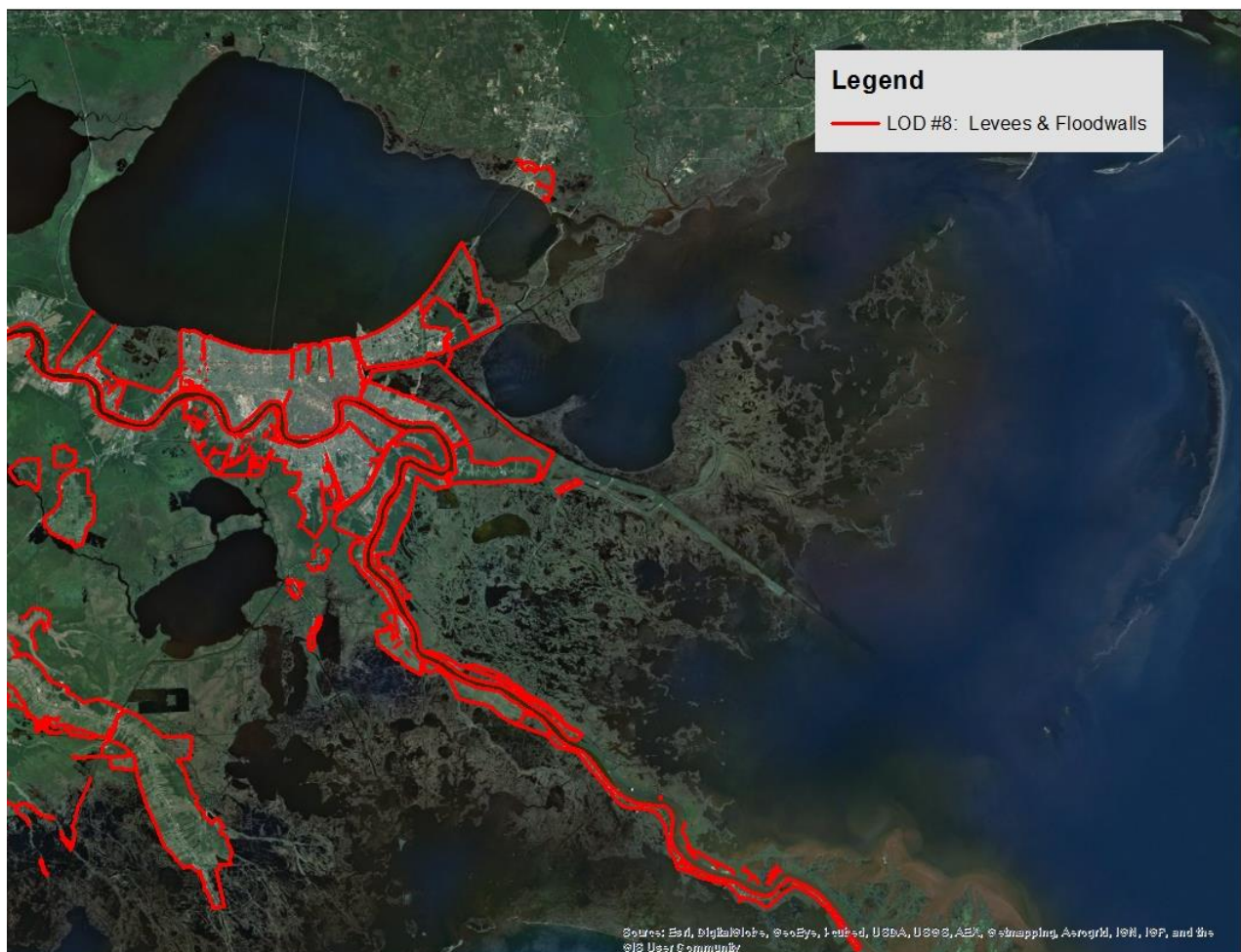


Figure 50: Map depicting the levees. Note that the urban footprint essentially mirrors the levees. (Source: Louisiana 2012 SMP, ESRI.)



Figure 51: The Lake Borgne Surge Barrier, a major post-Hurricane Katrina improvement to the perimeter protection system. (Source: The Times-Picayune | Nola.com.)

LOD#9: Interior Drainage

While levees represent a barrier blocking storm surge from impacting the urban areas, these barriers also block drainage from inside the protected areas. For this reason, an interior, forced drainage system, centered around pump stations, are an important subsystem of the hurricane surge defense system. Without the interior drainage subsystem, rainfall and water from overtopping would accumulate in the protected area with resulting flood impacts.

At the assembly level, the interior drainage system consists of structures, transitions, operations, and maintenance. Components of the assembly include canals and ditches, pump stations, discharge canals, and storage basins. Transitions include transitions to floodwalls and transitions to earthen levees. Operations include power, personnel, safe houses, operations plans, and communications. Maintenance includes maintenance on the pump station, grass cutting along canals, and dredging within canals.

Table 12: Assemblies and components for the interior drainage subsystem.

System	Flood Defense System										
Segment	Structural LODs										
Subsystem	Interior Drainage										
Assembly	Structures			Transitions		Operations					Maintenance
Component	Canals & Ditches	Pump Stations	Discharge Canals	Floodwall Transitions	Earthen Levee Transitions	Power	Personnel	Safe Houses	Operations Plan	Communications	Local Agencies

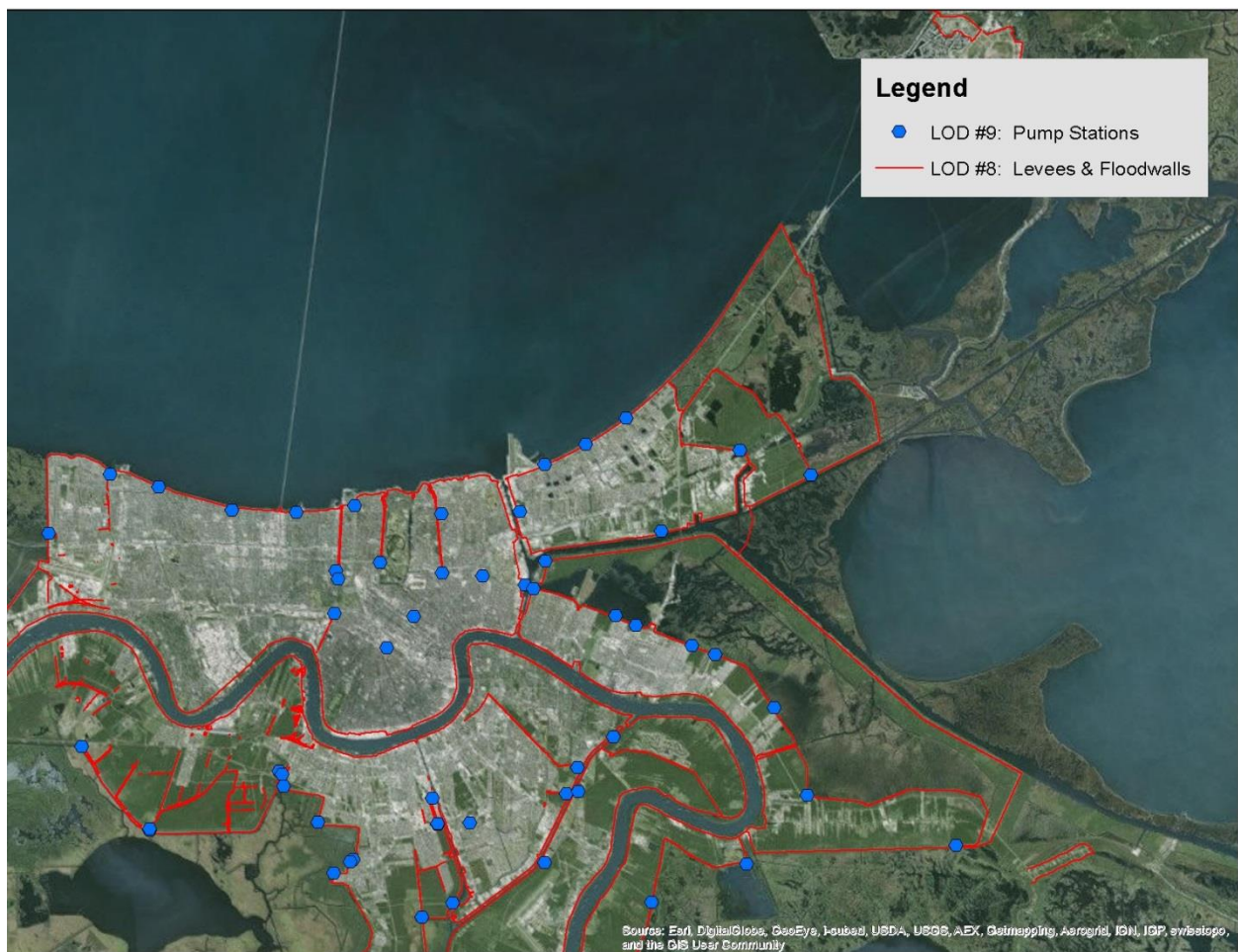


Figure 52: Map depicting pump stations (blue hexagons), the primary component of the interior drainage subsystem. Interior drainage in Greater New Orleans is also partitioned into several drainage basins, such as New Orleans East and Jefferson Parish, and retention areas, such as the Central Wetlands Unit and the IHNC-GIWW corridor, which are bounded by levees (red lines). (Source: Louisiana 2012 SMP, ESRI.)

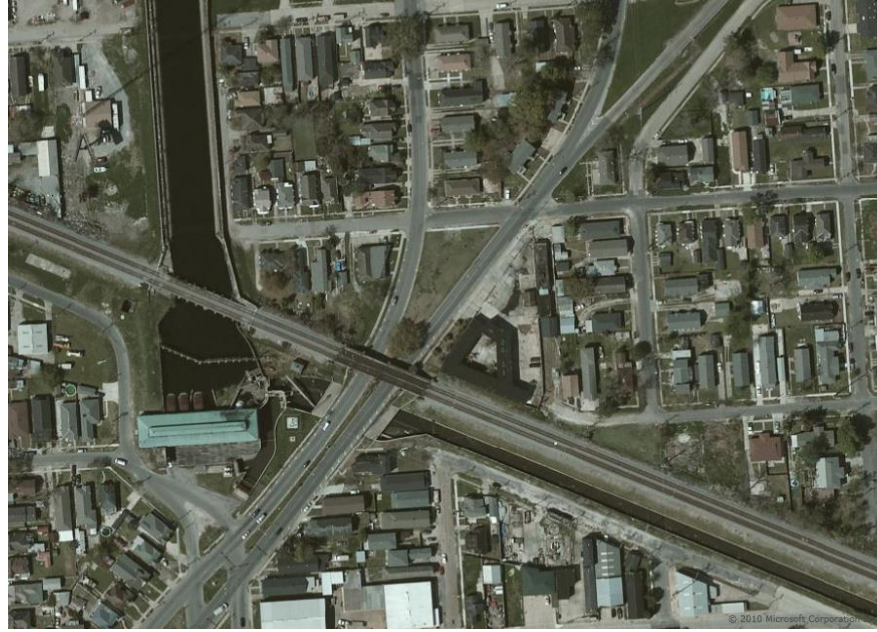


Figure 53: Aerial imagery of a pump station (green metal roof at the head of the canal) linking an interior drainage canal with a discharge (outfall) canal in the Gentilly neighborhood. This is the same location shown in **Figure 13d**. (Source: ESRI.)

LOD#10: Building Elevation

Building elevation is the first community line of defense. This LOD provides a redundant layer of protection since the coastal and structural lines of defense do not and cannot provide complete protection against storm surges and other types of flooding. Simply put, building elevation places immovable assets (homes, businesses, and contents) above potential flood levels, thus reducing the risk of flood damage. However, care must be taken to ensure that raised structures are properly supported and can handle the increased wind exposure.

At the assembly level, this line of defense requires funding, engineering and construction, compliance, and public participation. In terms of funding, there are three Federally funded programs that are components of this assembly: the Hazard Mitigation Grant Program (for Road Home recipients), The Repetitive Flood Loss Program (a FEMA program), and funding programs for buyouts and relocation. While most residents depend on funding from one of these programs, private funding is also possible for some residents. It is also worth noting that local governments are also starting to explore funding mechanisms for house elevation. Engineering and construction consists of piling, tie-downs, flood proofing, wind mitigation, and a workforce. Compliance involves codes and regulations, contractors, and homeowners. Finally, public participation has a voluntary component and potentially a compulsory component (though compulsory buyouts have, and will likely continue to, face community resistance).

Table 13: Assemblies and components for the building elevation subsystem.

System	Flood Defense System												
Segment	Community LODs												
Subsystem	Building Elevation												
Assembly	Funding			Engineering & Construction					Compliance			Public Participation	
Component	Hazard Mitigation Grant Program (Road Home)	Repetative Flood Loss Program (FEMA)	Buyouts & Relocation	Pilings	Tiedowns	Flood Proofing	Wind Mitigation	Workforce	Codes & Regulations	Contractors	Homeowners	Voluntary	Compulsory



Figure 54: Photo of a home being raised (Source: The Times-Picayune | Nola.com).

LOD#11: Evacuation

Personal evacuation involves physically moving persons from an area of potential flood exposure and it is the last line of defense for protecting the health and safety of persons. Furthermore, it is the only subsystem that provides a reasonable level of personal safety for individuals within a region that could be impacted by surge. (We say reasonable level because evacuations can result in health impacts to individuals, particularly vulnerable individuals). While the current assessment is limited to the general population and special needs population, the evacuation of institutionalized populations, such as prisons and hospitals, the evacuation of hazardous materials from potential flood impacts areas, and the evacuation of vessels from navigational water bodies potentially exposed to surge are also relevant to the objective of the HSDS.

Essentially, personal evacuation comprises two assemblies: the personal vehicle based evacuation and the assisted evacuation for special needs persons, including residents with restricted mobility. To a large extent, both assemblies consist of similar components: streets and highways, transportation assets, fuel supply and distribution, traffic and logistics management, compliance, and lodging/sheltering at the destination location. The differences reflect the individual versus group nature of the assemblies. For example, individuals in the general population utilize personal vehicles, while special needs residents utilize public transportation assets. Likewise, management for personal evacuation largely involves traffic management, while management for special needs involves logistic tasks, such as registering and tracking participating individuals. Finally, while the general public has a host of options for sheltering and lodging, the special needs population is constrained to shelters or medical facilities that are equipped to meet their needs.

Table 14: Assemblies and components for the evacuation subsystem.

System	Flood Defense System											
Segment	Community LODs											
Subsystem	Evacuation											
Assembly	Personal Vehicle Based Evacuation						Special Needs Assisted Evacuation					
Component	Streets & Highways	Personal Vehicles	Fuel Supply & Distribution	Traffic Management	Public Compliance	Lodging/ Sheltering	Streets & Highways	Public Transportation Assets - Regional	Public Transportation Assets - Local	Management & Logistics	Public Compliance	Sheltering

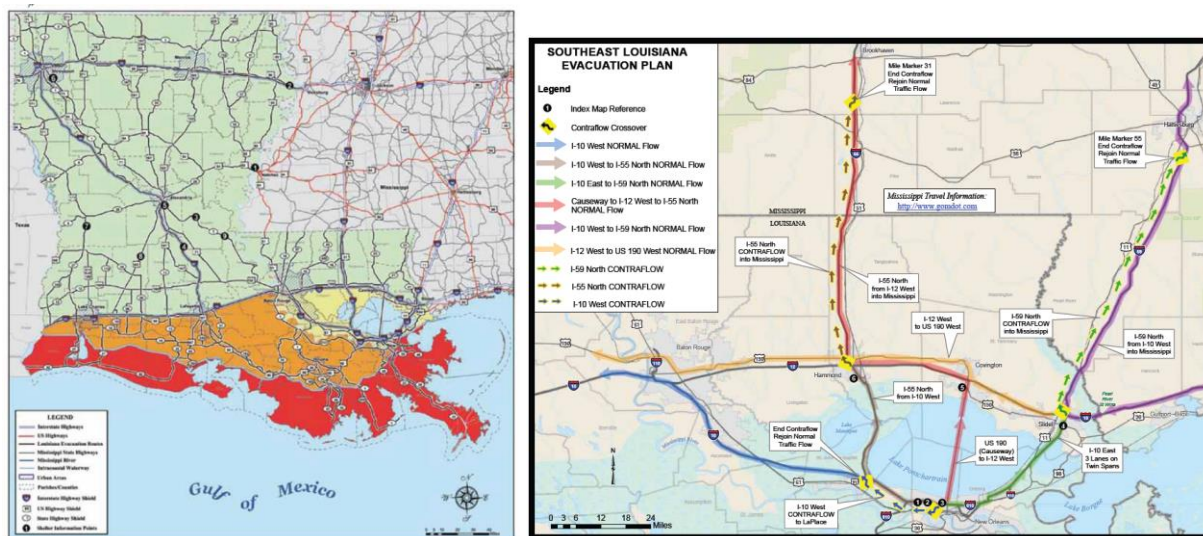


Figure 55: The Southeast Louisiana evacuation maps published by GOHSEP. The left map depicts the three zones of the phased evacuation plan, while the right map shows contraflow lanes for evacuating GNO. (Source: GOHSEP.)

LOD#12: Insurance

Insurance is the last line of defense for communities. While insurance cannot prevent flood damage, it does provide financial resources for recovery following flood losses. Without these resources, recovery would lag and long term impacts would accumulate. These resources provide a level of resiliency that enables a quicker recovery from flood events, along with mitigation incentives.

This subsystem consists of three assemblies: providers, public participation, and oversight. Providers include private insurers, the state based insurer of last resort, and the Federally administered flood insurance program. By in large, participation by homeowners is compelled through mortgage requirements. For homeowners without mortgages, various education and incentive programs can increase participation. Oversight includes the state and Federal regulatory agencies.

Table 15: Assemblies and components for the insurance subsystem.

System	Flood Defense System							
Segment	Community LODs							
Subsystem	Insurance							
Assembly	Providers			Public Participation			Oversight	
Component	Private	State	Federal	Lenders	Voluntary		State	Federal

IMPERIAL FIRE & CASUALTY INSURANCE COMPANY
POLICY NUMBER: 0000000000
AGENCY: DAN BURGHARDT
AGENCY PHONE: (504) 450-7232
INVESTMENT DATE: 10/1/2012 10:01 AM
EXPIRATION DATE: 10/1/2013 10:01 AM
To report a claim, call: 811-254-6819
Endorsement Effective Date: 01/1/2013

DAN J. BURGHARDT INSURANCE AGENCY INC
3008 DAVID DRIVE
METairie, LA 70003

REVISED FLOOD INSURANCE POLICY DECLARATIONS
STANDARD PROGRAM - DWELLING FORM

DELIVERY ADDRESS: [Redacted]
INSURED NAME(S) AND MAILING ADDRESS: [Redacted]

PROPERTY ADDRESS: [Redacted]
RATING INFORMATION:
DATE OF CONSTRUCTION: 12/12/1961
CURRENT FLOOD ZONE: AE
GRANDFATHERED: NO
RATED FLOOD ZONE: AE
COMMUNITY NUMBER: 220303 0005 E REGULAR PROGRAM
COMMUNITY NAME: NEW ORLEANS/ORLEANS PARISH
REPLACEMENT COST: \$77,000
NUMBER OF UNITS: N/A

PROPERTY DESCRIPTION: N/A
BUILDING OCCUPANCY: SINGLE FAMILY
CONDOMINIUM INDICATOR: NOT A CONDO
BUILDING TYPE: ONE FLOOR
FIRST MORTGAGEE INFORMATION: [Redacted]
SECOND MORTGAGEE INFORMATION: N/A

LOAN NUMBER: N/A
DISASTER AGENCY INFORMATION: N/A

LOAN NUMBER: N/A
CASE FILE NUMBER: N/A
DISASTER AGENCY: N/A

PREMIUM CALCULATION -

	COVERAGE	DEDUCTIBLE	BASIC COVERAGE	BASIC RATE	ADOL COVERAGE	ADOL RATE	DEL. DISCOUNT/SURCHARGE	Standard
BUILDING	\$77,000	\$5,000	\$40,000	0.78	\$17,000	0.55	(\$91.00)	\$477.00
CONTENTS	\$0	\$0	\$0	0.58	\$0	1.18	\$0.00	\$0.00
ANNUAL SUBTOTAL:								\$477.00
ENDORSEMENT PREMIUM: \$0.00								
INCREASED COST OF COMPLIANCE:								\$70.00
COMMUNITY RATING DISCOUNT: 10%								(\$15.00)
PROBATION SURCHARGE:								\$0.00
FULL ANNUAL PREMIUM:								\$462.00
FEDERAL POLICY SERVICE FEE:								\$40.00
TOTAL:								\$502.00
PRORATA PREMIUM ADJUSTMENT:								\$0.00
ADJUSTED ANNUAL PREMIUM:								\$502.00
RENEWAL BILLING PAYOR:								INSURED

IN WITNESS WHEREOF, I have signed this policy below and hereby enter into this Insurance Agreement.

[Signature] H. Meno Carter, Jr. / President
[Signature] Dan Boudreaux / Secretary

Zero Balance Due
This Is Not A Bill

This declarations page along with the Standard Flood Insurance Policy Form constitutes your flood insurance policy. Coverage limitations may apply. Refer to your flood insurance policy for details.

Policy issued by IMPERIAL FIRE AND CASUALTY INSURANCE COMPANY

File: 1180088 Page 1 of 2 DocID: 14506243 Printed: 02/07/2013

Figure 56: Example Flood Insurance declarations page.

More Complete Evacuation Hierarchy

The HSDS is a complex system that includes millions of parts spread across a large geographic area. Some elements of this system are manufactured parts, others are part of the natural environment, and others are social in nature. Given the size and complexity of the HSDS, a full hierarchy to the part level is impossible to depict in a clear, readable manner. One strength of depicting the system specification in a hierarchical manner is that the system can be decomposed into individual hierarchies for viewpoints that are more readily understood. The previous sections depicted the system structure to the component level, so that the reader appreciates the various elements needed to comprise the different lines of defense. As a more thorough illustration, this section presents and describes a more complete system hierarchy for the evacuation subsystem to the part level.

Table 16 below shows the evacuation subsystem. This table represents the best known attempt to account for all the pieces and parts needed to implement a regional evacuation due to a hurricane threat. Initially based on the literature database, the QMAS workshop provided additional details. Of note, the table lists the elements necessary for vehicle based evacuation of the general population and the assisted evacuation of the special needs population, but does not include institutionalized populations, navigational vessels, or hazardous materials.

Just filling out the inventory is a worthwhile endeavor that can lead to insights on how to improve system performance. For example, electric power at fuel stations is an important, but often overlooked part of the evacuation subsystem. It is necessary for the stations to pump gasoline for citizen's personal vehicles. Because power outages typically do not occur before the storm, its availability is taken for granted. What happens if a fuel stations lose power during the evacuation stage and before the onset of storm conditions? Portable generators are an available asset within the state's emergency response system and they are typically pre-deployed during the hurricane preparedness phase. However, current plans utilize these assets only after a storm has passed to assist re-entry procedures during widespread power outages. There are currently no plans to deploy them prior to a storm. Through the process of simply listing all the elements needed to implement a large scale evacuation, this shortcoming of planning was identified.

Table 16: More complete system hierarchy for the evacuation subsystem. Red font denotes elements that must be secured prior to the onset of storm conditions

System	Flood Defense System												
Segment/Element	Community LODs												
Subsystem Assembly	Evacuation												
Component	Personal Vehicle Based Evacuation						Special Needs Assisted Evacuation						
	Streets & Highways	Personal Vehicles	Fuel Supply & Distribution	Traffic Management	Public Compliance	Lodging/Sheltering	Streets & Highways	Public Transportation Assets - Regional	Public Transportation Assets - Local	Management & Logistics	Public Compliance	Sheltering	
Part						Destination hotel, shelter, family/friend's residence, campground, or RV							
	Local Streets		Gas Stations	Phased Evacuation Plan	Hazard Awareness Education		Local Streets	Buses	Buses	Pre-Registration System	Hazard Awareness Education	Destination facility	
	Interstates and Highways (Emergency Evacuation Routes)			Intra Government Communication and Coordination		Household Evacuation Plan	Emergency Evacuation Routes (Interstates & Highways)		Bus Stops/Pick Up Points	Event Registration & Monitoring System	Hazard Monitoring (NWS, NHC)	Agreement with Destination Municipality	
			Gasoline		Hazard Monitoring Plan			Trains					Shelter Volunteers
	Secondary Routes		Main Power Supply	Traffic Direction (LSP)	Emergency Warning (NHC)	Family Communications Plan	Interstate Crossovers	Airplanes	Fuel Supply & Distribution	Central Collection Hub	Emergency Warning (NHC)	Public Health Monitoring & Compliance (LDHH)	
	Bridges & Causeways		Backup Generators	Traffic Control Cones, Flares, Barricades, Etc. (LSP, LDOTD)	Hazard Communication - Official	Emergency Supply Kit		Ferries, Barges, Cruiseships		Staff & Volunteers (Local Government/NGO)	Hazard Communication - Official (LDHH)		
	Interstate Crossovers		Personal Cash	Movable Signage (LDOTD)	Hazard Communication - Media	Financial Resources	Permanent Signage	Fuel Supply & Distribution			Hazard Communication - Media		
	Emergency Lanes			Motorist Assistance Patrol Vans (LDOTD)	Hazard Communication-Interpersonal						Hazard Communication-Interpersonal		
	Permanent Signage			Training and Exercises	Law Enforcement Patrol of Evacuated Areas (Police, LANG)		Armoring (Seaward Side)				Law Enforcement Patrol of Evacuated Areas (Police, LANG)		
	Armoring (Seaward Side)				Reentry Plan								

Interior Drainage in Orleans, St. Bernard, and Jefferson Parishes.

The previous sub-sections presented generic system hierarchies for the 12 individual lines-of-defense, which are also termed the subsystems in the system hierarchy. This included a generic system hierarchy for the interior drainage subsystem. The generic hierarchies are meant to provide templates that can be filled with the specific details of actual instances of the subsystems in the HSDS. As an illustrative example, this section completes the interior drainage hierarchy for three specific cases: St. Bernard Parish, the east bank of Jefferson Parish, and the central part of Orleans Parish.

These specific instances of the interior drainage subsystem have a number of commonalities, but also exhibit some significant differences. In terms of commonalities, pump stations are the centerpiece to each instance and each instance requires stormwater conveyance elements and stormwater discharge elements. Generally speaking, the drainage canals and related elements that convey stormwater into pump station are common to each instance, though there are distinctions that are described below. Likewise, each instance requires power, personnel, operational plans, and communications. However the historical differences related to the location of pump station relative to perimeter protection structures between these three instances has resulted in differences in the availability and type of basins where the stormwater is discharged.

In Jefferson Parish's Eastbank (**Figure 57** and **Table 17**) the interior drainage consists of a planned network of drainage canals that connect to a single set of pump stations, most of which are located along the perimeter levees. As such, this instance consists of a single retention area, Lake Pontchartrain and adjoining water bodies, with nearly unlimited capacity. Note that currently, one east Jefferson pump station discharges in the 17th street canal.

In St. Bernard Parish (**Figure 58** and **Table 18**), the instance also consists of a planned network of canals that feed into a single set of pumps along the 40 Arpent Levee that discharge into the Central Wetlands Unit. However, in this case, the CWU is bounded by levees, so the exterior retention element of St. Bernard drainage subsystem has limited capacity under any conditions. The previously noted positive interaction between the pump stations, the marsh in the CWU, and the 40 Arpent levee is one result of the specifics of this instance.

In central New Orleans (**Figure 59** and **Table 19**), the differences in the interior drainage subsystem are most noticeable. Here interior drainage consists of a historical, unplanned mixture of canals, levees, and pump stations. Unlike the other two instances, double and even triple pumping is required here. The previously noted east Jefferson pump station that discharges into the 17th street further complicated coordinated pumping operations. Additionally, the three major drainage canals, formerly directly connected to the Lake, now serve as isolated retention areas with limited capacity. Of note, the capacity of the canals depends on the integrity of the floodwalls along the canals, as specified in the safe water level for each canal. In fact, only the central Orleans interior drainage subsystem has safe water level requirements. A similar situation exists within the IHNC/GIWW, which also receives stormwater discharged by pump stations and is lined with floodwalls of limited integrity. For this instance of the subsystem, operational procedures are complicated by the need to coordinate pumping in a manner that

fulfills this important system requirement. As witnessed during Hurricane Isaac, this can lead to inundation of protected areas when pump stations have to reduce pumping into the outfall canals.

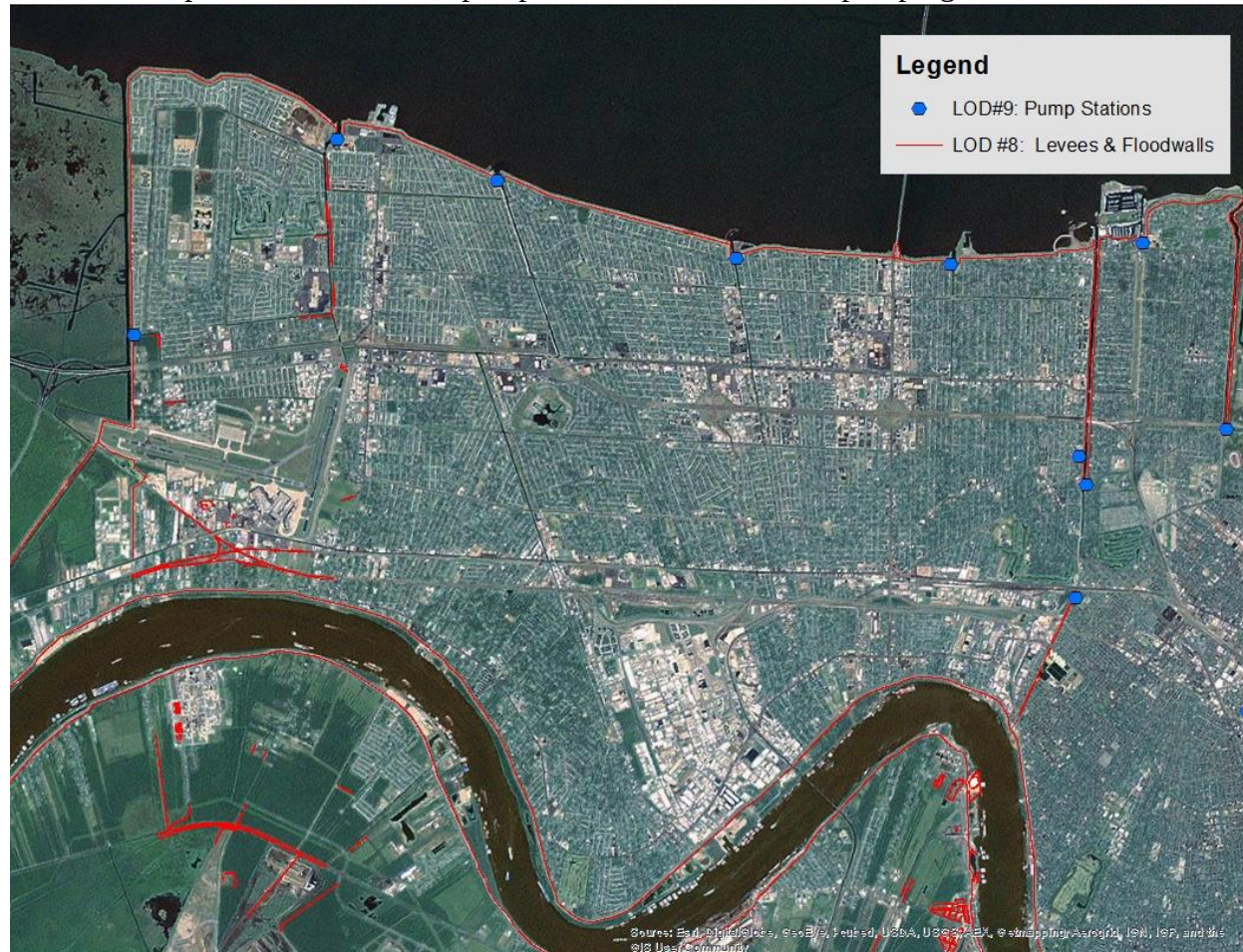


Figure 57: Pump station locations for the East Jefferson instances of the interior drainage subsystem. (Source: Louisiana 2012 SMP, ESRI.)

System	Flood Defense System							
Segment	Structural LODs							
Subsystem	Interior Drainage: Jefferson Parish Eastbank							
Assembly	Structures			Transitions		Operations		
Component	Canals & Ditches	Pump Stations	Discharge Basin	Floodwall Transitions	Earthen Levee Transitions	Power	Personel	Safe Houses
Part	Ditches	Stormproof Building	Lake Pontchartrain			Main Line Power	Jefferson Parish Drainage Department	Elevated, Windproof Structure
	Unpaved Canals	Pumps				Backup Power		Life support systems
	Canals	Intake Pipes						Pump Monitors & Controls
	Underground Pipes	Discharge Pipes						
	Underground Culverts	Intake Gates						

Table 17: Specific system hierarchies for the Jefferson Parish interior drainage subsystem.

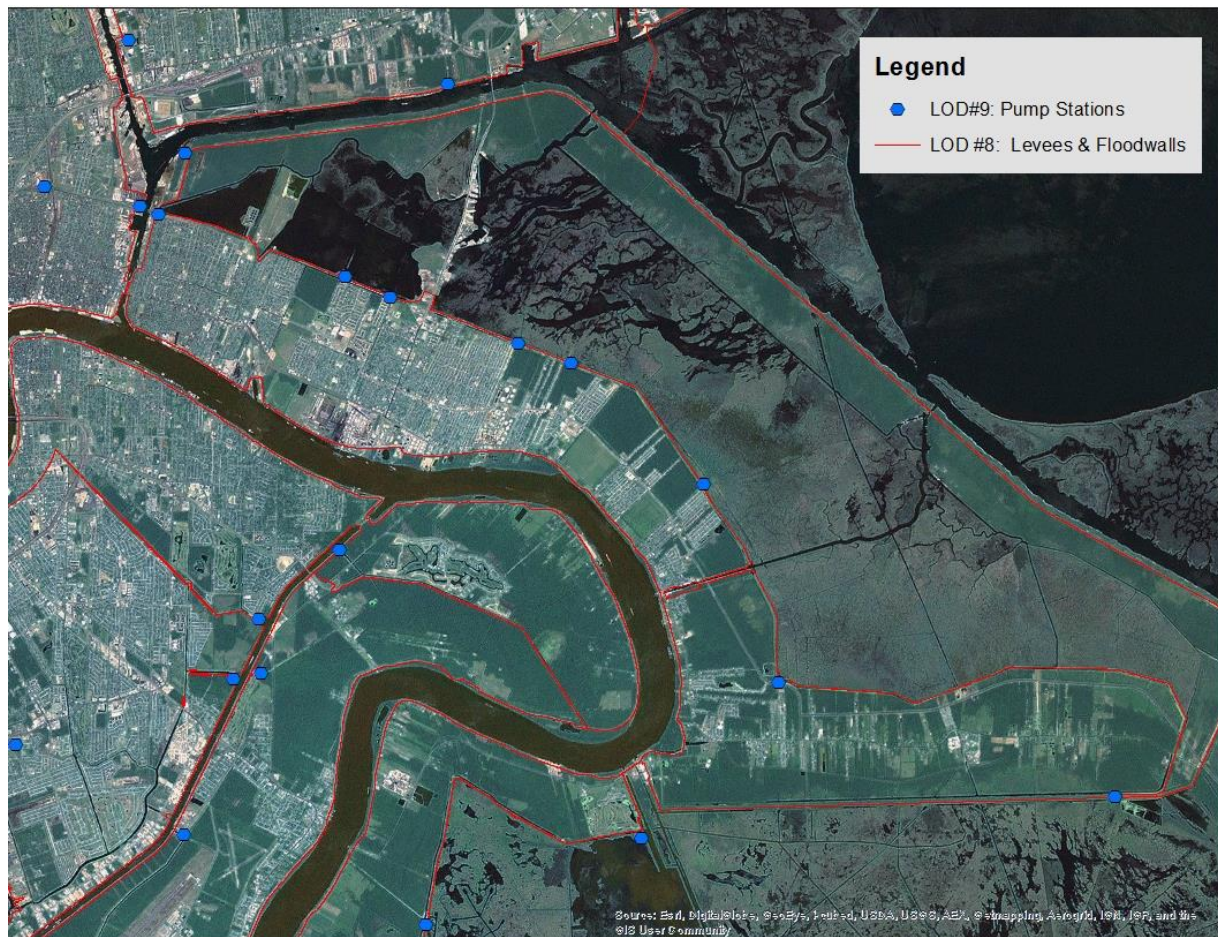


Figure 58: Pump station locations for the St. Bernard Parish instances of the interior drainage subsystem. (Source: Louisiana 2012 SMP, ESRI.)

System	Flood Defense System							
Segment	Structural LODs							
Subsystem	Interior Drainage: St. Bernard Parish							
Assembly	Structures			Transitions		Operations		
Component	Canals & Ditches	Pump Stations	Discharge Basin	Floodwall Transitions	Earthen Levee Transitions	Power	Personel	Safe Houses
Part	Ditches	Stormproof Building	Central Wetlands Unit			Main Line	Lake Borgne	Elevated, Windproof Structure
	Unpaved Canals	Pumps				Backup Power		Life support systems
		Intake Pipes						Pump Monitors & Controls
	Canals	Discharge Pipes						
	Underground Pipes	Intake Gates						
	Underground Culverts							

Table 18: Specific system hierarchies for the St. Bernard Parish interior drainage subsystem.



Figure 59: Pump station locations for the central New Orleans instances of the interior drainage subsystem. (Source: Louisiana 2012 SMP, ESRI.)

System	Flood Defense System							
Segment	Structural LODs							
Subsystem	Interior Drainage: Central New Orleans							
Assembly	Structures			Transitions		Operations		
Component	Canals & Ditches	Pump Stations	Discharge Basin	Floodwall Transitions	Earthern Levee Transitions	Power	Personel	Safe Houses
Part	Ditches	Stormproof Building	Outfall Canals			Main Line Power	NO S&WB	Elevated, Windproof Structure
	Unpaved Canals	Pumps	IHNC/GIWW			NO S&WB 25 Hertz Backup Power	USACE	Life support systems
	Canals	Intake Pipes	Temporary Floodgates/Pump Stations			Onsite Backup Power		Pump Monitors & Controls
	Underground Pipes	Discharge Pipes						
	Underground Culverts	Intake Gates						

Table 19: Specific system hierarchies for the central Orleans Parish interior drainage subsystem.

6) SysML Model of the HSDS

This section presents a preliminary model of the Hurricane Surge Defense System using the Systems Modeling Language (SysML). It incorporates the findings of the three QMAS workshops along with our background research. To be clear, this section describes a work in progress meant to illustrate the utility of using SysML when addressing the “system in name only” problem. A complete and fully specified model of this highly complex system is beyond the scope of this initial assessment.

Our goals include producing a generic SysML inventory of system elements to the location level along with a template for creating specific system hierarchies (such as for taking the Barrier Island template and completing it for the Chandeleur Islands.) Additionally, we use this high level, generic system inventory along with different types of SysML diagrams to illustrate system interactions and scenarios of concern described previously in this report. Following good systems engineering practices, we start with the system requirements.

Requirements Diagram

System requirements are the capabilities, functions, and conditions that the system and its elements must meet to ensure that the system achieves its objectives. Typically, system requirements are represented via text-based declarations and scattered across various design and policy documents. SysML, particularly the “requirements diagram”, provides a platform to compile these various requirements into concise statements which can then be related to specific system elements (blocks) that fulfill the requirement.

As a simple introduction to SysML diagrams, **Figure 60** presents the high level requirements of the HSDS. It is based on a review of technical documents from Federal, state, and local stakeholders that propose the optimal levels of protection for GNO. While the figure presents four high level system requirements, Appendix D includes a much larger list of 45 system requirements that were identified in an initial review of relevant Federal, state, and local documents along with a peer-reviewed, quantitative risk based estimate of the optimal level of protection (Jonkman, et Al. 2009.)

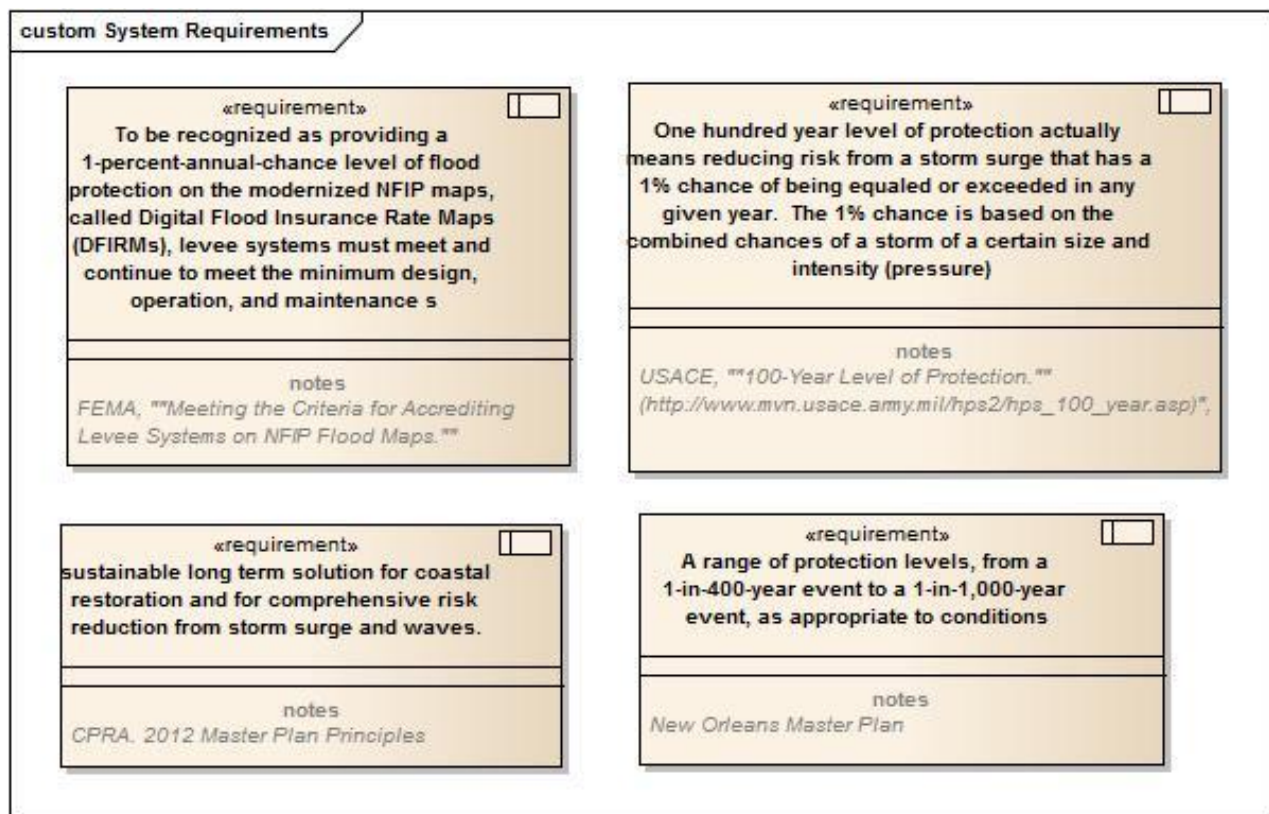


Figure 60: High level system requirements table for the Hurricane Surge Defense System (HSDS). The top row includes four approved requirements from FEMA, USACE, CPRA, and the New Orleans Master Plan. The bottom row contains two proposed requirements based on an analysis by Jonkman, et al. (2009). Note that each requirement block lists the reference document.

The Lines of Defense in Block Notation

In SysML, the block is the primary unit to represent the system's structure. Blocks can represent parts, sites, facilities, operations, and personnel. The system itself can even be represented by a block, as for example in **Figure 61**. Likewise, blocks can be used to represent the 12 lines of defense, as in **Figure 62**. In both figures, the block contains a label at the top. Below the label are the attributes of the block, such as the risk level for the HSDS or the depth and fetch of a sound. The blocks also include the responsibilities of the block, its constraints, and general notes. As elaborated below, blocks form the basis for modeling the structure of the system.

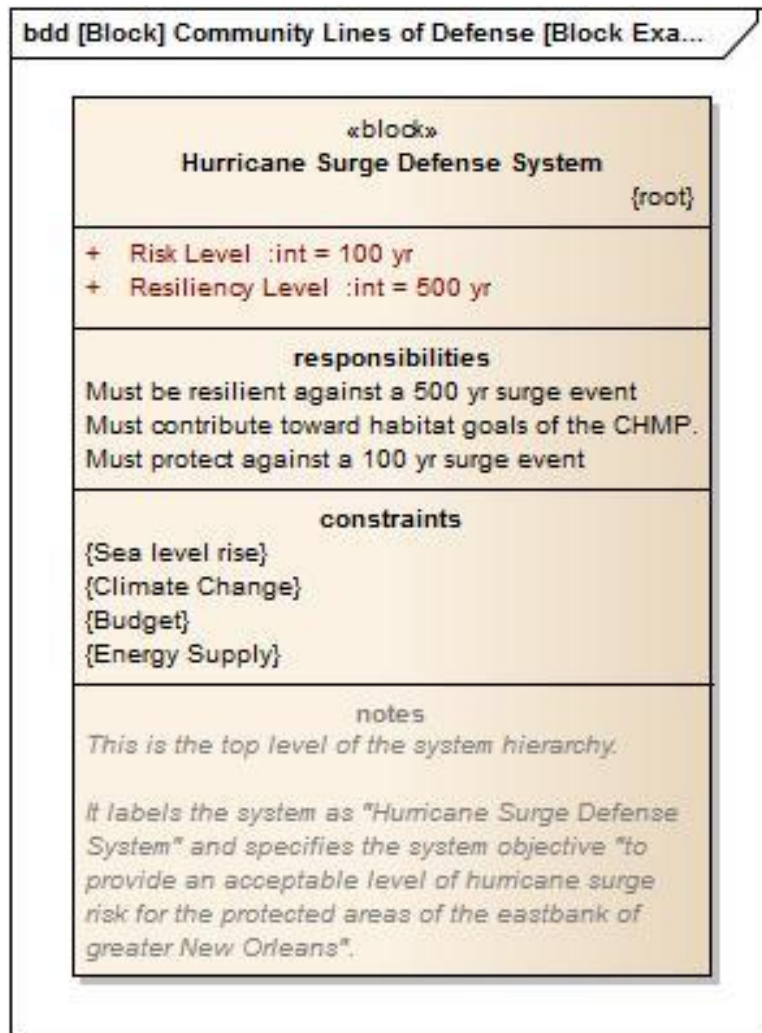


Figure 61: A block for the GNO-East Bank Hurricane Surge Defense System.

System Hierarchy and Inventory

Using blocks as the primary unit, SysML can be used to create various diagrams to represent the system's structure and operations. One such diagram is the block definition diagram (BDD) which can be used to represent the system's structure, including both the physical parts and operator behavior. Essentially, a BDD is a diagram that contains blocks and connections between the blocks. **Figure 63, Figure 64, Figure 65, Figure 66, and Figure 67** show various representations of the HSDS system hierarchy using block definition diagrams. In these diagrams, the blocks represent the elements of the system while lines depict relationships between the elements. The open diamond at the upper end denotes "aggregation", which implies that the parts (blocks) below make up to whole (block) above. Other types of relationships include Association, Composition, Generalization, Allocate, Item Flow, Dependency, Trace, and Information Flow.

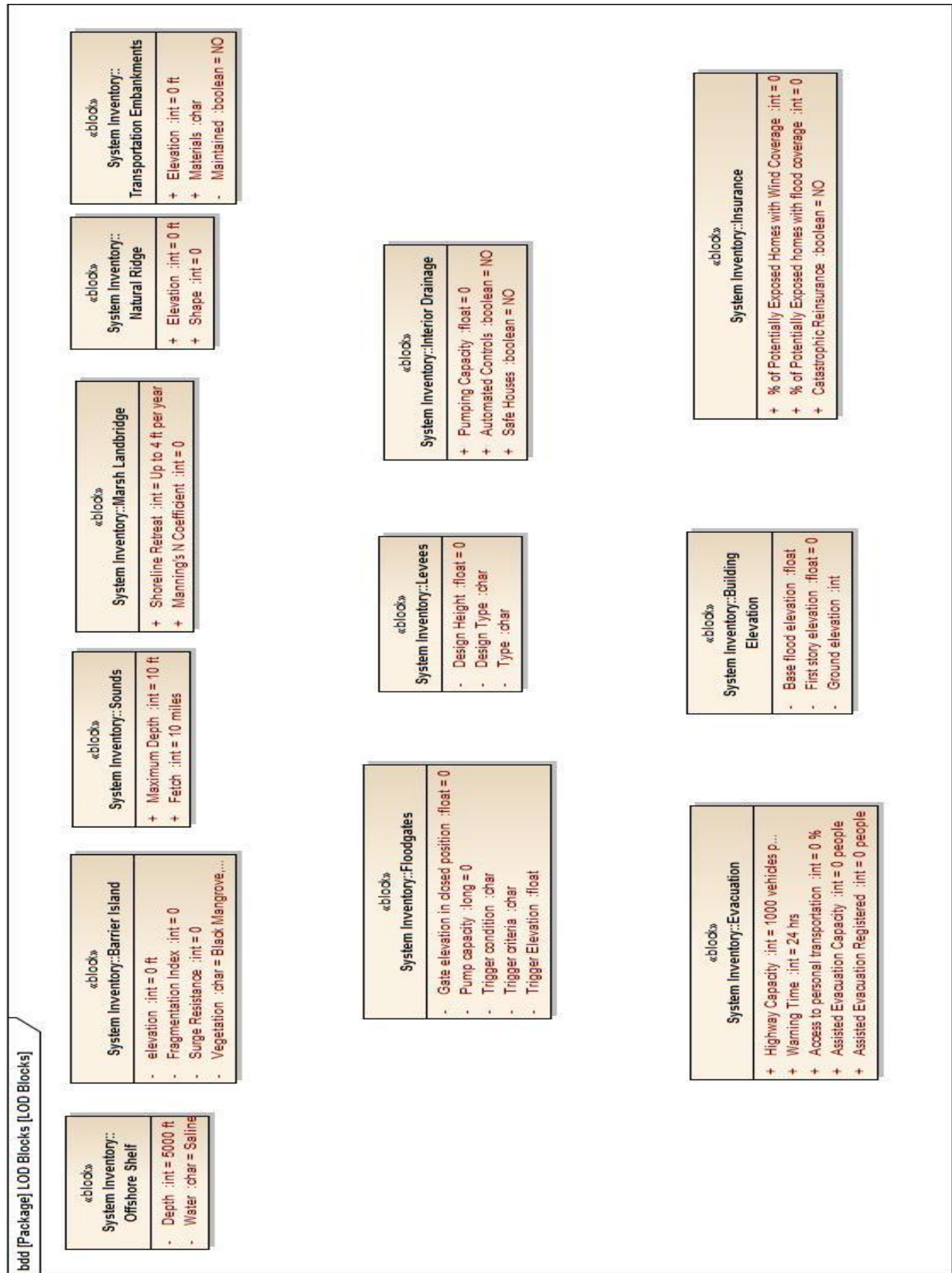


Figure 62: Individual blocks for each Line of Defense. Coastal LODs are listed on the top row, structural LODs are in the middle row, and bottom row contains community LODs.

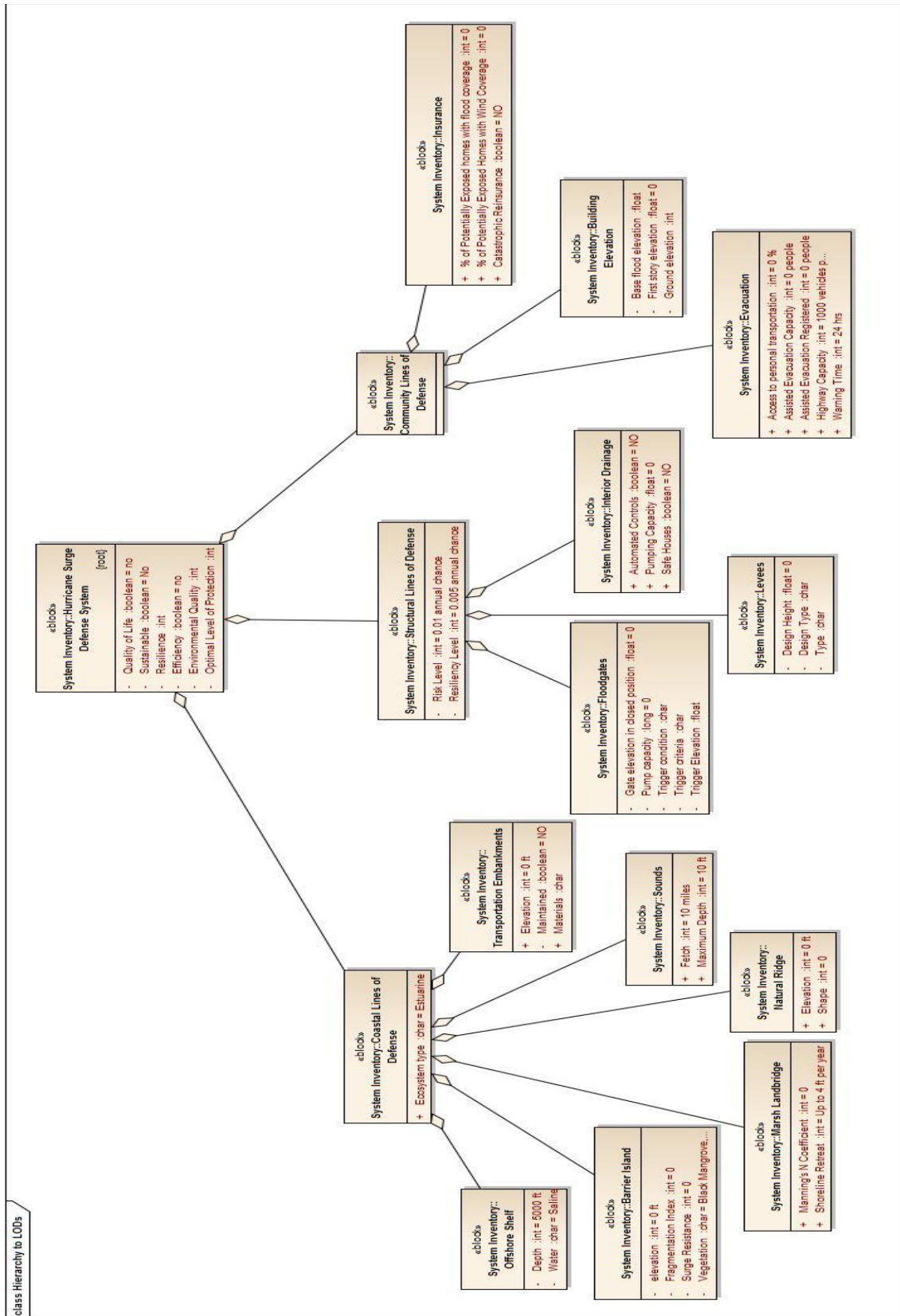


Figure 63: Three level system hierarchy that depicts the HSDS as an aggregate of the 12 lines of defense.

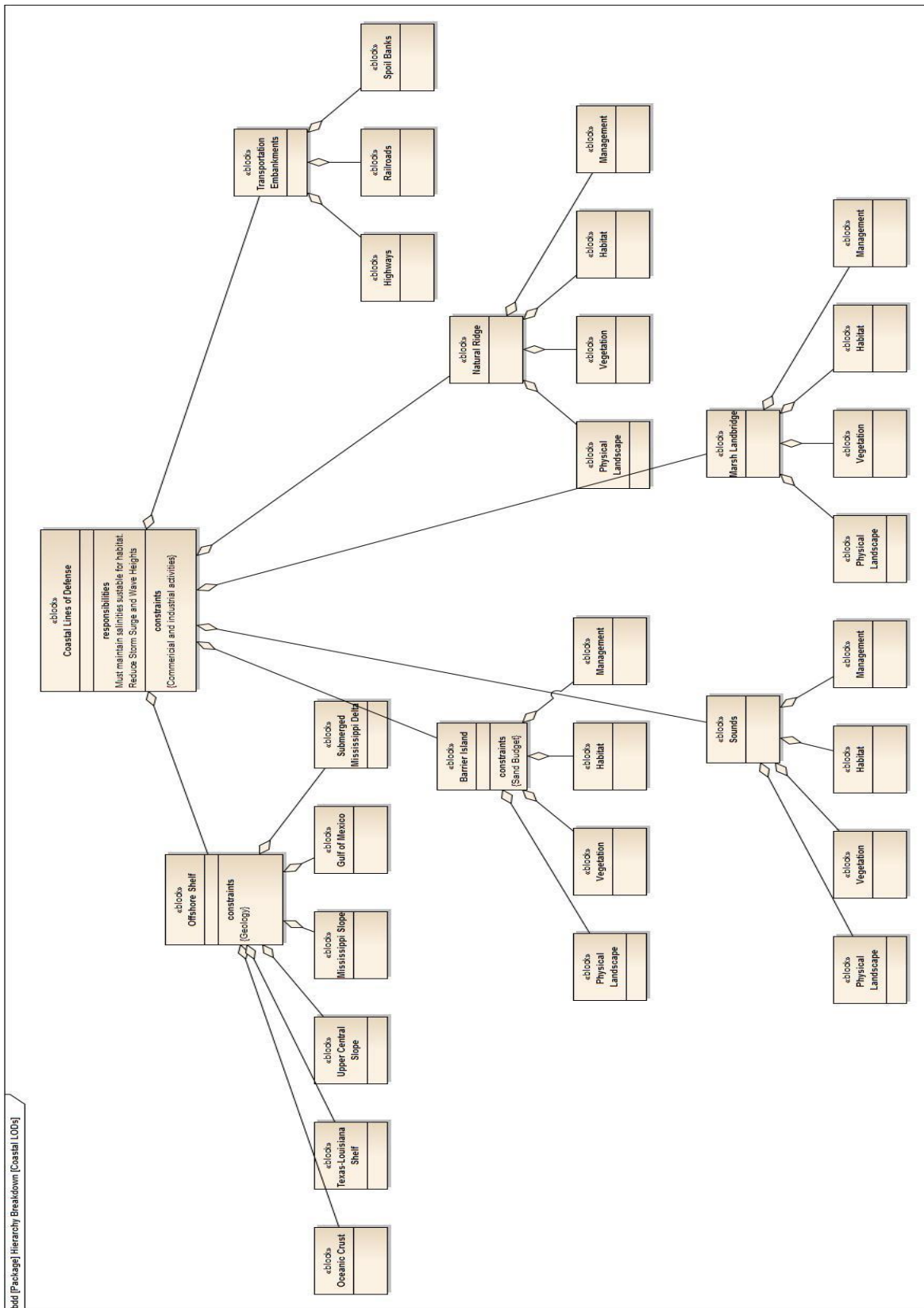


Figure 64: System hierarchy for the coastal lines of defense. This subsystem includes the first six lines of defense and the hierarchy aggregates these to the assembly level.

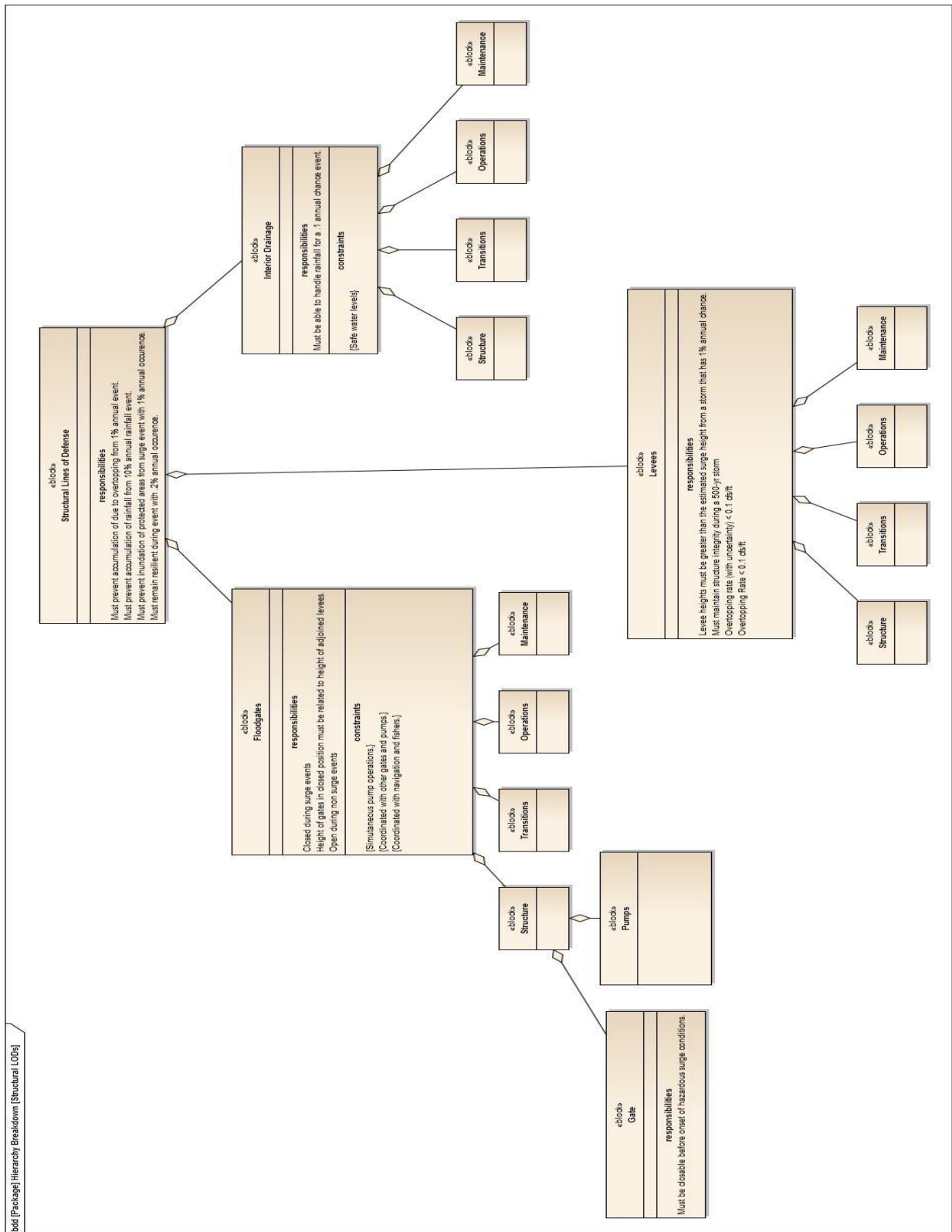


Figure 65: System hierarchy for the structural lines of defense. This subsystem includes the lines of defense that comprise perimeter protection system and the hierarchy aggregates these to the assembly.

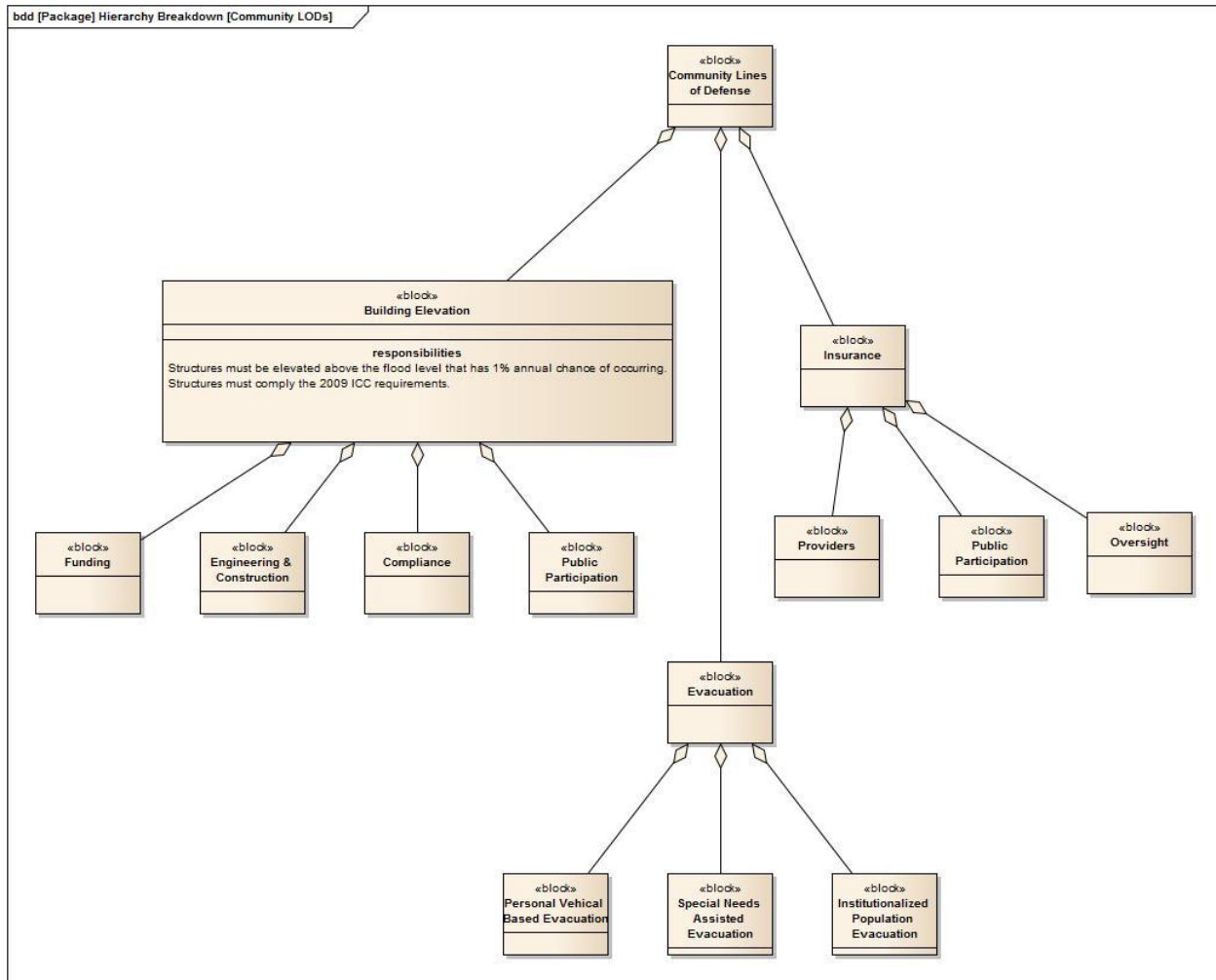


Figure 66: System hierarchy for the community lines of defense. This subsystem includes the last three lines of defense and the hierarchy aggregates these to the assembly level.

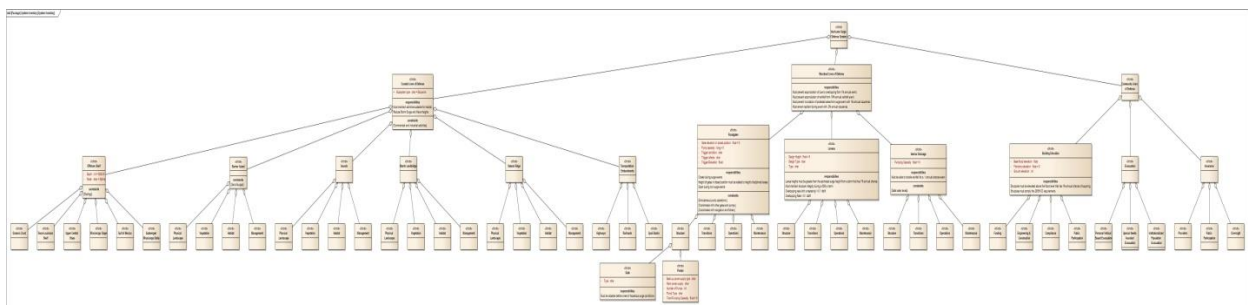


Figure 67: System hierarchy for the HSDS to the assembly level. The difficulty in presenting this diagram illustrates the complexity of this system along the utility of the scalable, hierarchical view that is central to systems engineering and SysML software.

«block» Evacuation

- + Highway Capacity :int = 1000 vehicles p...
- + Warning Time :int = 24 hrs
- + Access to personal transportation :int = 0 %
- + Assisted Evacuation Capacity :int = 0 people
- + Assisted Evacuation Registered :int = 0 people

responsibilities

- Ensure Public Compliance with Evacuation Orders
- Ensure Sufficient Highway Capacity
- Provide Sufficient Warning Time & Guidance to the Public
- Reduce Obstructions to Evacuation Traffic

constraints

- (Warning Time)
- (Vehicle Access)
- (Fuel Supply)
- (Public Compliance)

«block» Personal Vehical Based Evacuation

- «block» Streets & Highways
 - «block» Local Streets
 - «block» Interstates and Highways (Evacuation Routes)
 - «block» Bridges & Caisways
 - «block» Interstate Crossovers
 - «block» Emergency Lanes
 - «block» Permanent Signage
- «block» Personal Vehicles
- «block» Fuel Supply & Distribution
 - «block» Gas Stations
 - «block» Gasoline
 - «block» Main Power Supply
 - «block» Backup Generators
 - «block» Personal Cash
- «block» Traffic Management
 - «block» Planned Evacuation Plan with Distribution
 - «block» Extra Government (Communication and Coordination)
 - «block» Traffic Direction Officers
 - «block» Traffic Control Devices
 - «block» Movable Signage
 - «block» Motorist Assistance Patrol
- «block» Public Compliance
 - «block» Hazard Awareness Education
 - «block» Hazard Monitoring
 - «block» Emergency Warnings
 - «block» Hazard Communications - Official
 - «block» Hazard Communications - Media
 - «block» Hazard Communications - Interpersonal
 - «block» Law Enforcement Patrols of Evacuated Areas
- «block» Lodging/Sheltering
 - «block» Hotel, Shelter, family/friend's residence, campground, or city
 - «block» Household Evacuation Plan
 - «block» Family Communications Plan
 - «block» Emergency Supply Kit
 - «block» Financial Resources

«block» Special Needs Assisted Evacuation

- «block» Streets & Highways
 - «block» Local Streets
 - «block» Interstates & Highways (Evacuation Routes)
 - «block» Interstate Crossovers
 - «block» Emergency Lanes
 - «block» Permanent Signage
- «block» Transportation Assets - Regional
 - «block» Local Buses
 - «block» Coast Buses
 - «block» Trains
 - «block» Airplanes
 - «block» Ferries, Barges, Cruis+ships
 - «block» Fuel Supply & Distribution
- «block» Transportation Assets - Local
 - «block» Local Buses
 - «block» Bus Stops/Pick Up Points
 - «block» Fuel Supply & Distribution
 - «block» Fuel Supply & Distribution
- «block» Management & Logistics
 - «block» Pre-Registration System
 - «block» Event Registration & Monitoring System
 - «block» Central Collection Hub
 - «block» Staff, Volunteers, and Medical
- «block» Public Compliance
 - «block» Hazard Awareness & Education
 - «block» Hazard Monitoring
 - «block» Emergency Warning
 - «block» Hazard Communication - Official
 - «block» Hazard Communication - Media
 - «block» Hazard Communication - Interpersonal
 - «block» Law Enforcement Patrol of Evacuated Areas
- «block» Sheltering
 - «block» Sheltering Facility
 - «block» Adaptation with Dehydration Municipality
 - «block» Shelter Volunteers & Supplies
 - «block» Shelter Medical Staff
 - «block» Public Health Monitoring & Standards Compliance

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Behavior Diagrams: Evacuation Route-Barrier Island Interaction Example

The previous set of diagrams depicted the high level structure of the HSDS. SysML also includes elements and diagrams to depict the system's behavior. This section utilizes the example of the evacuation procedures to depict this behavior and to graphically illustrate the system iteration between the Chandeleur Islands, a coastal line of defense, and the I-10 east evacuation route, a community line of defense. This system interaction was identified in the QMAS workshops, and this section presents an initial attempt to model it using SysML

Use Case diagrams are used in SysML to represent functionality in terms of how actors use the system to accomplish a set of goals. In this case, the actor's goal is to reduce flood risk by transporting him or herself outside of the area potentially impacted by surge conditions. **Figure 69** shows the Use Case diagram for the evacuation subsystem. In Use Case diagrams actors are depicted by stick figures, actions are depicted by ovals, and relationships between actions are depicted by lines connecting the oval. In **Figure 69**, the Evacuation Use Case diagram includes three main actors – the National Hurricane Center (NHC), local officials, and residents. The first activity consists of the NHC monitoring tropical weather conditions. If conditions warrant, the NHC extends this activity by issuing a warning (really a set of warnings) which are consumed by local officials. Local officials can then extend the warning activity by ordering an evacuation. Residents then assess the available guidance and can choose one of three activities: evacuate outside of region, evacuate within the region, or shelter in place.

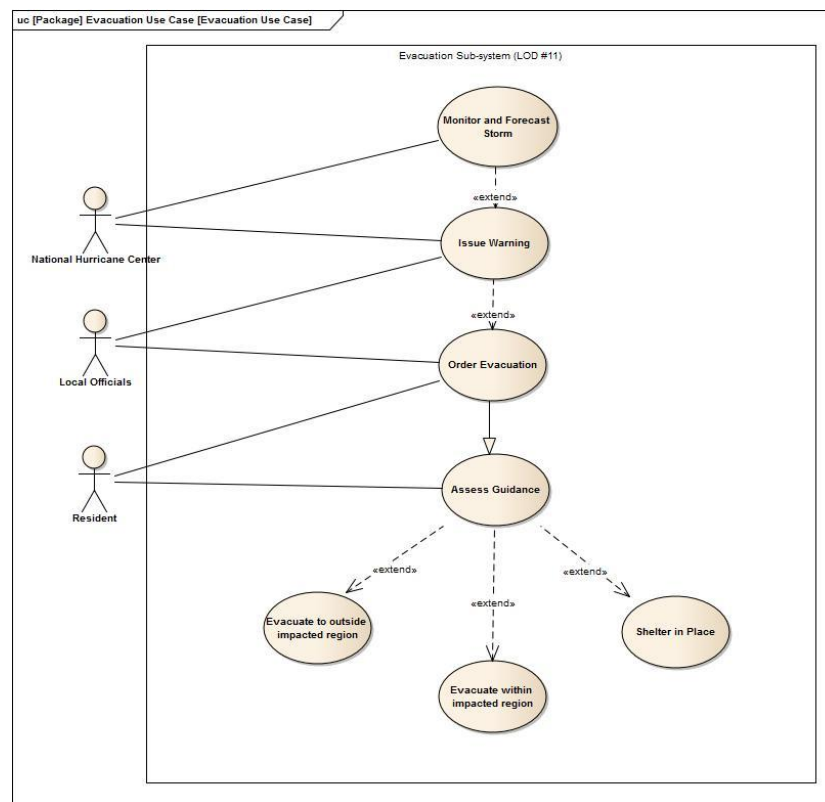


Figure 69: The Evacuation Use Case Diagram shows the process by which individuals arrive at the decision to evacuate, based on a series of warning and assessment activities which are depicted by the ovals.

Building on the Use Case diagram, Activity diagrams are used to represent the system's behavior by depicting an ordering of actions. **Figure 70** depicts the evacuation activity as an ordered progression of three activities. The solid line depicts the actor's initial state – in this case it is labeled “Normal Activity – No Alert”. The first activity in the sequence is that the actor receives an evacuation order from local officials, which then prompts the second activity “assess and confirm flood risk.” Below Activity 2, a diamond depicts a decision node, where one activity can be followed by multiple activities. If the actor then follows Activity 3a and chooses to evacuate, the actor first prepares to evacuate and then evacuates, which takes him or her to the final state of “Safe from flood impacts.” Naturally, the actor could decide to pursue other activities which do not lead to a safe final state.

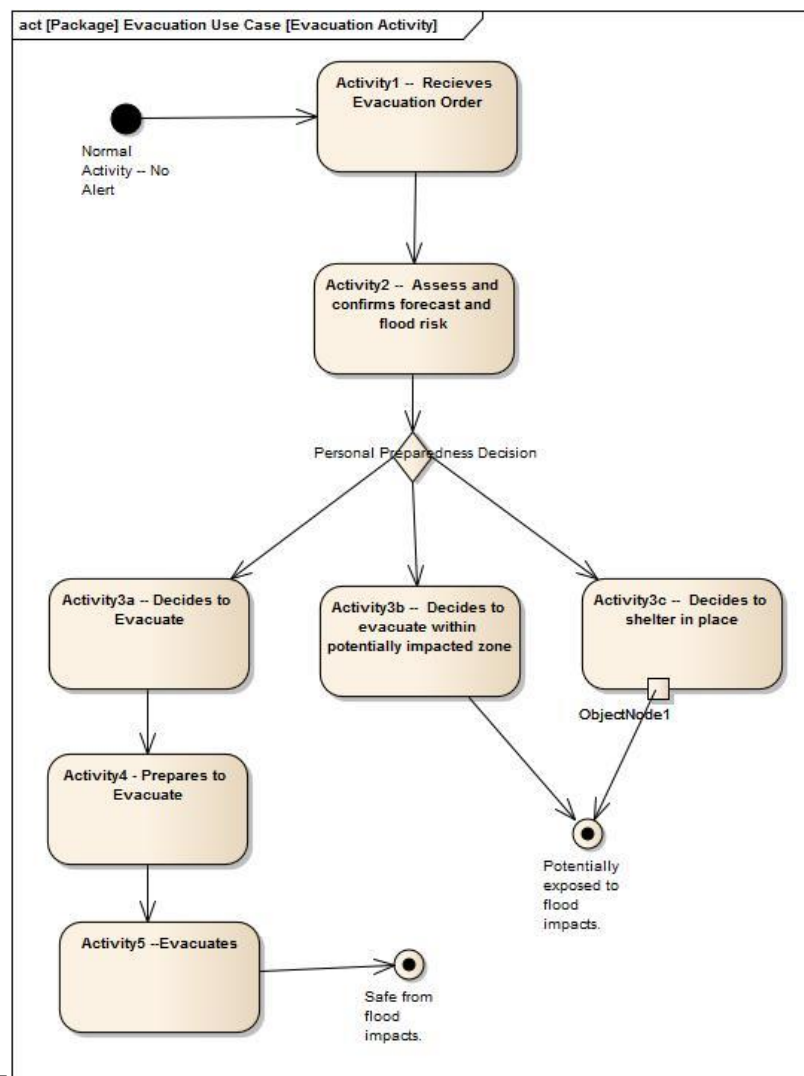


Figure 70: Evacuation activity diagram. The activity begins at an initial state, depicted by the dark circle. Following two activities (depicted by the rectangles), the user then reaches a decision mode, which is depicted by the diamond. Based on the decision and subsequent activities, the user arrives at one of two possible final states.

A third behavior diagram, the State Machine Diagram, is used to represent transitions between states (either actors or system elements) triggered by specific events. For example, an actor using the HSDS may transition between not evacuated to evacuated. In **Figure 71**, the initial state is “normal state – no alert.” The activity of assessing the hazard and warning information then prompts a decision fork, to either evacuate or not evacuate. This decision is based on a risk trigger and if the user’s perceived risk from flooding is greater than the estimated costs to evacuate, the decision is followed by Activity 3a – Decides to Evacuate. Finally, the user’s evacuation activity utilizes LOD#11 – the evacuation subsystem – which leads the actor to the final state – evacuated.

The final behavior diagram, the Sequence Diagram, represents behavior through a sequence of messages exchanged between parts. It is used to depict complex interactions between system elements. **Figure 72** depicts the evacuation sequence along with the interaction between the coastal LOD elements and the evacuation subsystem. In the sequence diagram, time is depicted vertically with higher activities preceding the lower activities. Above the sequence, stick figures and blocks are used to depict the actors and system elements involved – the NHC, local officials, residents, the evacuation subsystem, the coastal lines of defense, and the storm surge. Horizontal arrows depict messages, or information, that flow from one actor/element to another. At the top left, the earliest message is a Hurricane Warning (based on anticipated wind impacts) conveyed from the NHC to local officials. Shortly after that, the NHC also conveys a storm surge prediction, which may prompt local officials to convey an evacuation order message to residents. Residents then monitor the public hazard communication elements of the evacuation subsystem to obtain information about traffic conditions, while the NHC also conveys storm surge predictions directly to the resident. Shortly

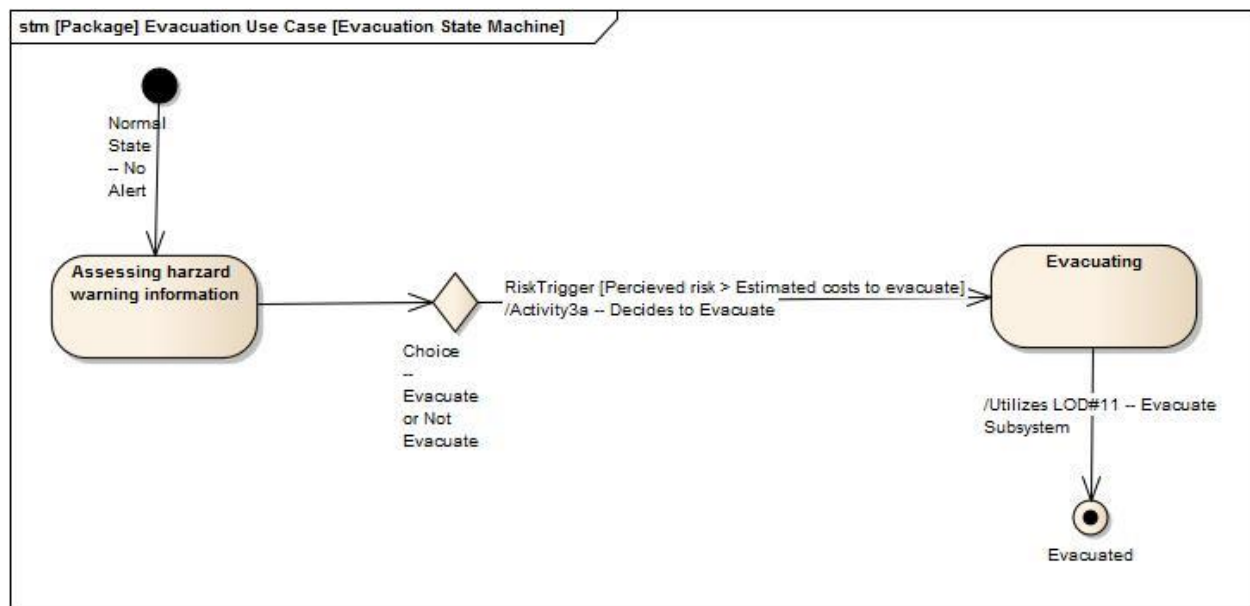


Figure 71: Evacuation state machine diagram represents a user’s transition from a normal state to an evacuated state. Key to the transition is decision fork (depicted by the diamond) based on a risk trigger.

thereafter, the surge interacts with the coastal lines of defense and one can think of the coastal lines of defense as “sending” a “mitigation message” to the surge. However, eventually, when the surge height is greater than the elevation of the evacuation route, the surge floods the evacuation route. Local officials who are monitoring evacuation routes then receive the signal that the evacuation route has flooded and they then convey a “hunker down” message to residents. In the final step of the sequence, the surge causes impacts on residents.

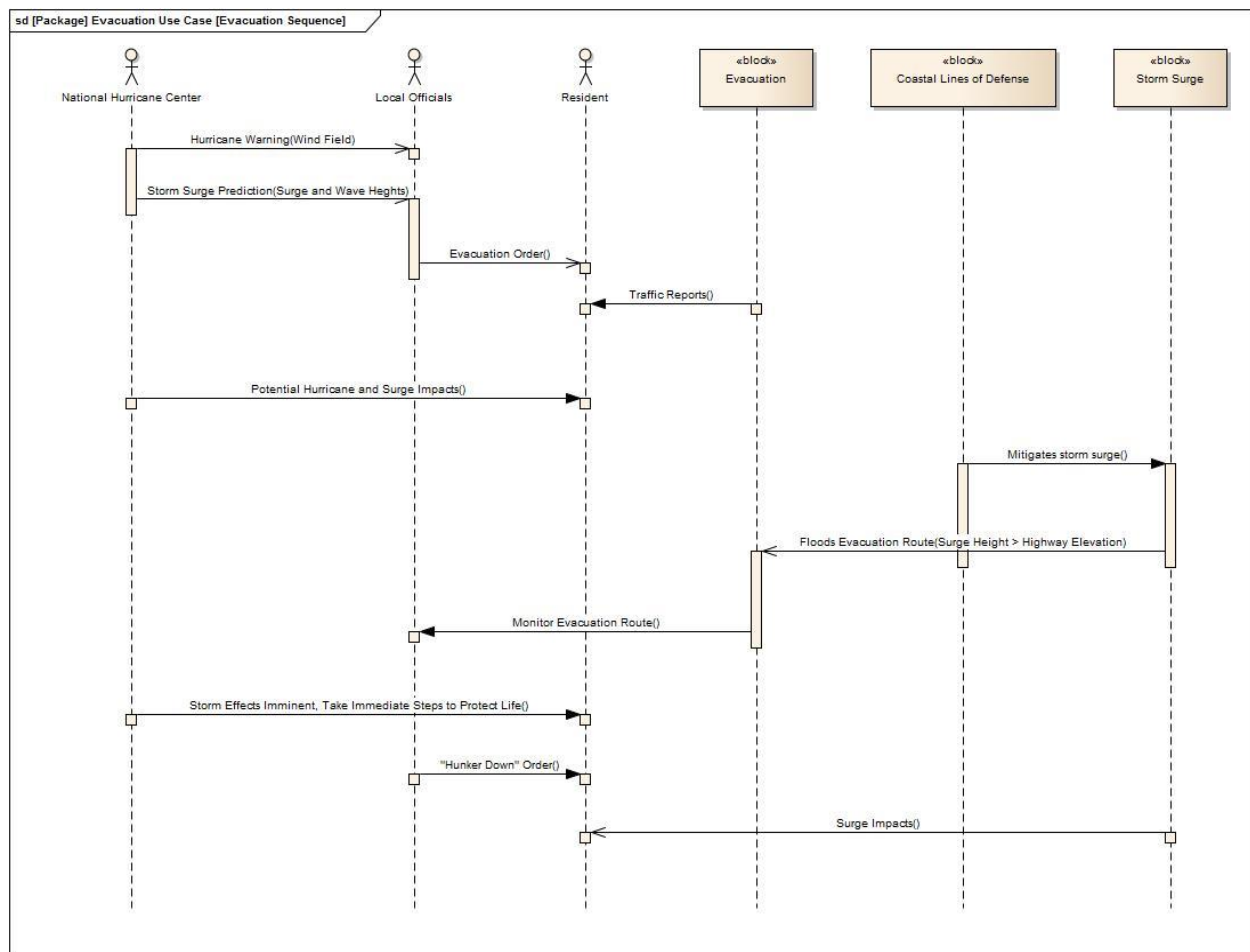


Figure 72: The evacuation sequence diagram which illustrates interaction between the evacuation subsystem, the structural protection subsystem, and the coastal lines-of-defense subsystem. Time is depicted vertically, with higher events preceding lower events. The horizontal arrows depict “messages”, which are information exchanged between elements. They serve as triggers for events. In this example, the sequence begins with the National Hurricane Center sending a Hurricane Warning, which is based on the wind field, to local officials. The sequence of preparedness activities ends with local officials sending a “hunker down” message to residents, which is followed by a storm surge impact message.

Case Study: Surge Barrier Closure Operations SysML Model

The IHNC/GIWW Floodgate Closure Operations model is a sub-set of the larger HSDS model **Figure 73**. The system package elements (use cases, behavior, structure, etc.) will extend from the larger system to the subsystem being analyzed, in our case the IHNC evacuation.

As discussed earlier in the report, the Hurricane Surge Defense System is a domain that relies on other domains (these can either be other “Super-Systems” or “Systems”). This is conceptually shown in **Figure 74**. For example, transportation is a Super-System that includes numerous types of transportation types: air, rail, vehicular, and maritime. All of these transportation systems play a role in the HSDS. For the IHNC/GIWW case study, only a sub-set of that larger transportation system will be included in the model – specifically maritime and railroads, as the maritime industry is impacted by operation of the flood gates and bridges and the railroad industry is impacted by the operation of the bridges.

The requirements for closure of the IHNC/GIWW consist of two categories; bridge closure requirements and flood gate requirements, which are shown in a “requirements diagram” in **Figure 75**. The flood gate closure requirements consist of a water surface elevation trigger (El. + 3ft) and/or storm proximity (landfall within 3 days). The bridge closure requirements are more extensive. The bridges are closed when the wind speed is greater than 40 mph (with an additional requirement that once the bridge is down, it can only be raised with approval of a District Director). The bridge must be secure before the operator vacates and the bridge must be inspected following the hurricane and before it is opened to the public.

System elements can be assigned to groups of requirements (**Figure 76**). In this case, the bridges are assigned to satisfy the bridge closure group of requirements and the flood gates are assigned the flood gate closure requirements. The bridge closure requirements and the flood gate closure requirements are sub-sets of the IHNC closure requirements, as a result, the bridges are associated with the IHNC closure requirements.

Although a simplistic figure, this capability is very powerful when dealing with a complex system (having both many elements as well as many inter-relationships) as it is possible to track the functional extent and origin of requirements. It is then possible to identify conflicts/clashes as well as gaps in system requirements.

For our application, the IHNC/GIWW Closure Use Case is shown in **Figure 77**. The two primary activities are issuance of a hurricane forecast (by the National Weather Service) and the closure of the flood gates and bridges within the IHNC/GIWW retention area. The information sent out by the National Weather Service is used by the Emergency Operations Manager to activate the evacuation sequence.

Figure 78 graphically shows the sequence of events described in the previous section. SysML explicitly allows for the modeling of sequences and associated organizations with the sequence. As shown in **Figure 78**, step 2 involves the Levee District notifying the Bridge Operations that they will be closing the flood gates. The Bridge Operations Manager then notifies the other

IHNC Stakeholders (U.S. Coast Guard, USACE IHNC Lock Master, Railroad, and Maritime industry) that the gates are closed.

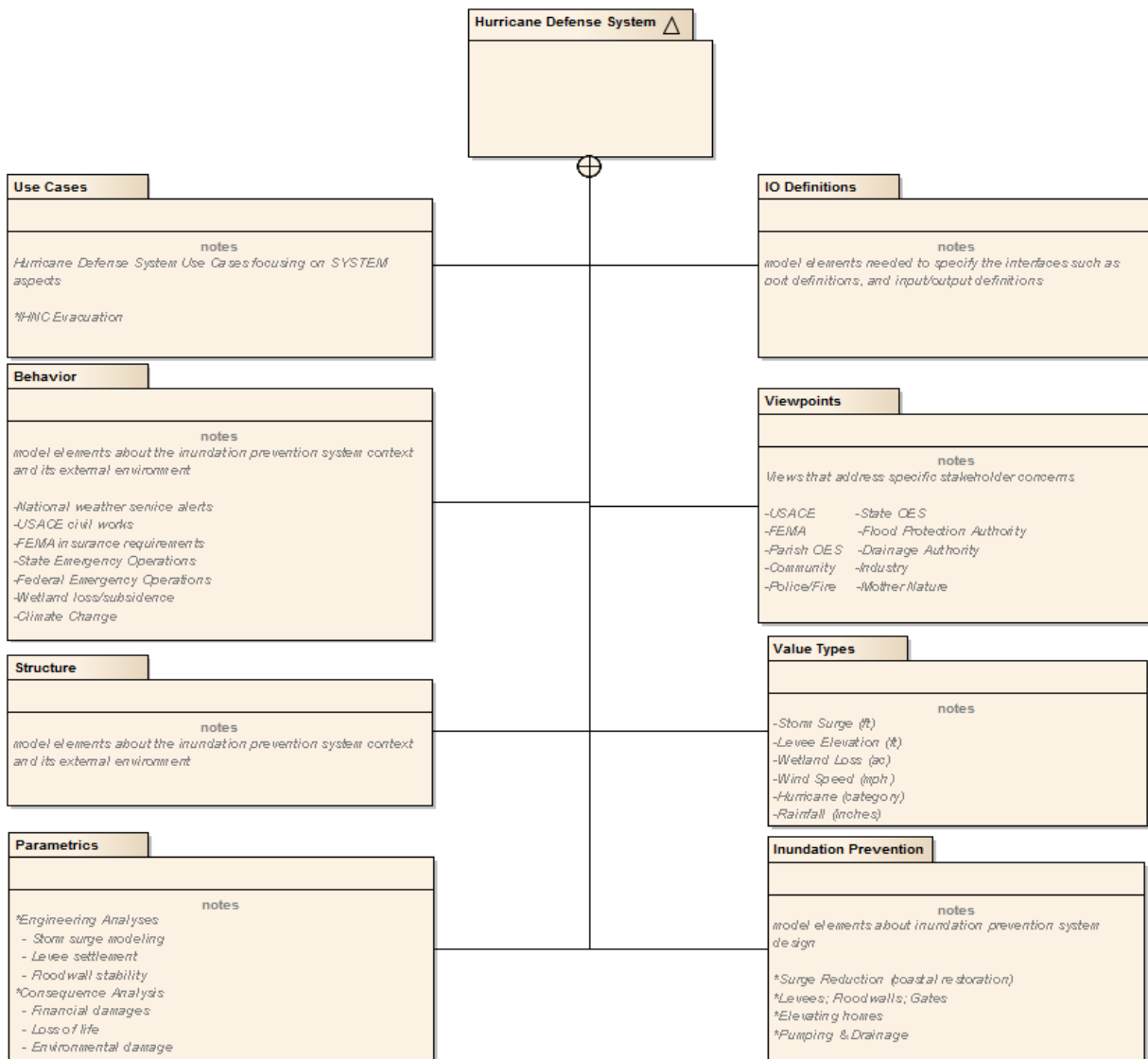


Figure 73: SysML Package Diagram for the Hurricane Surge Defense System.

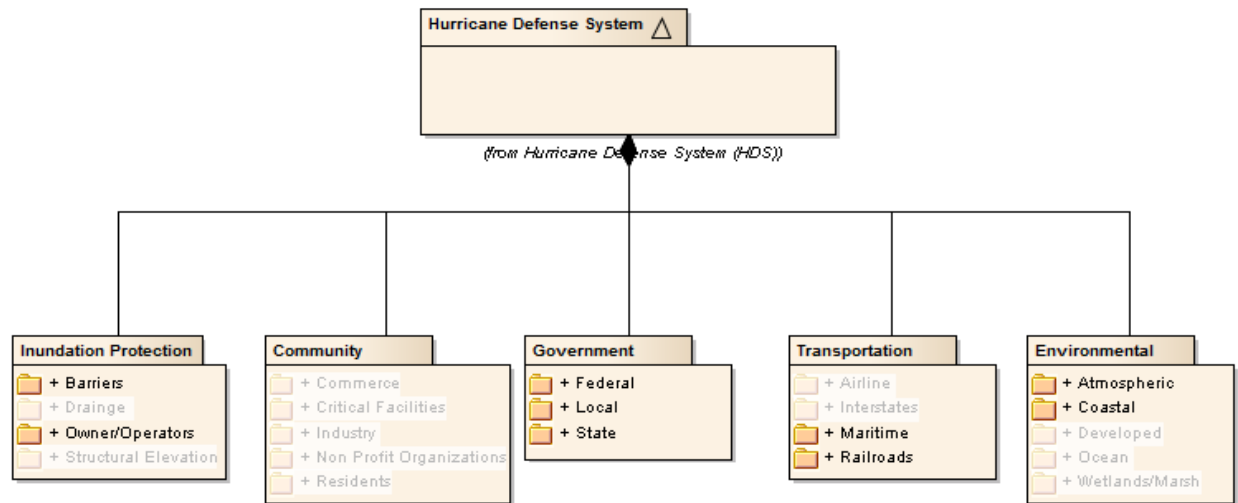


Figure 74: The Hurricane Surge Defense System is composed of other Domains/Systems.

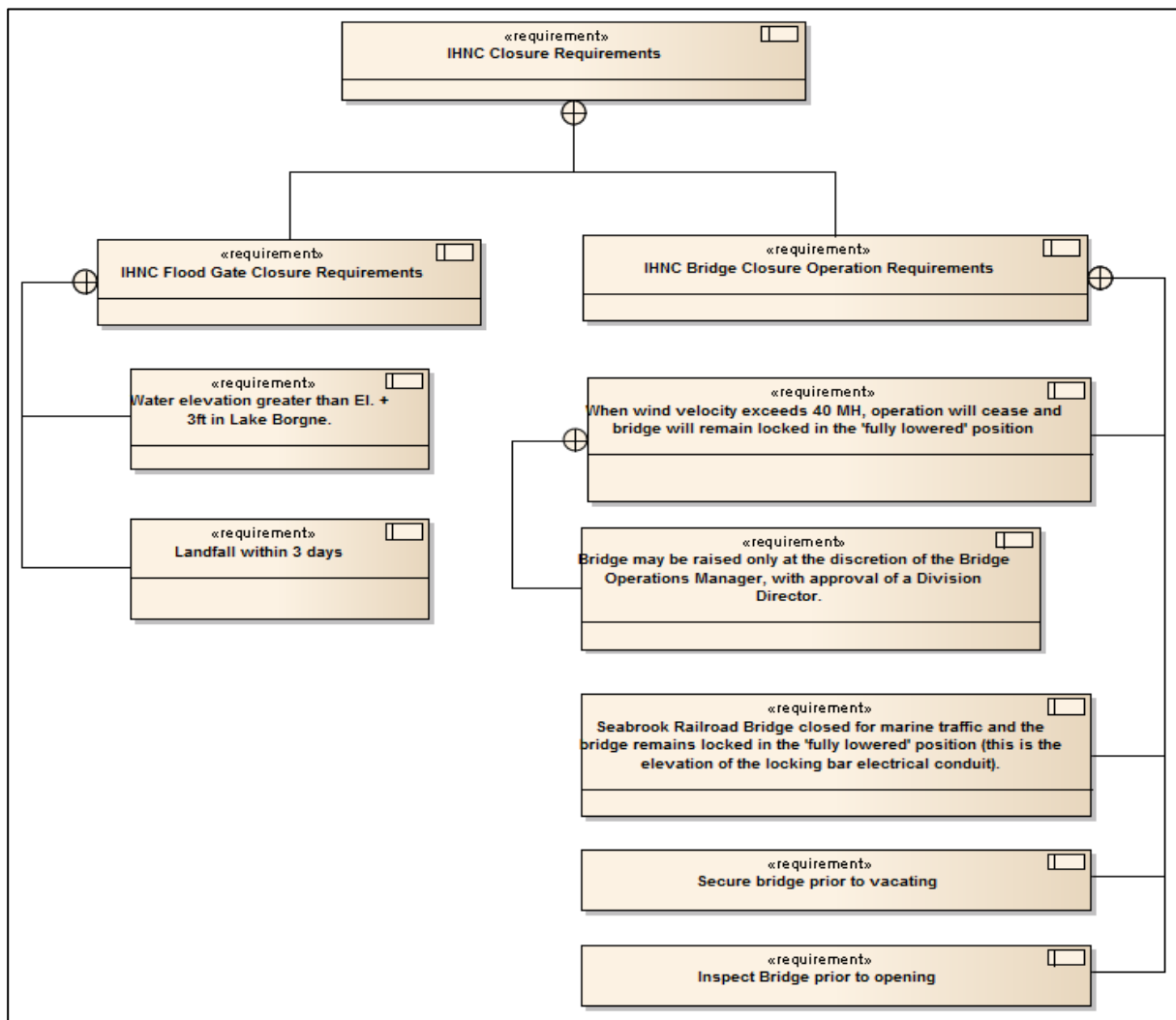


Figure 75: IHNC/GIWW closure requirements are comprised of Flood Gate closure and Bridge closure requirements.

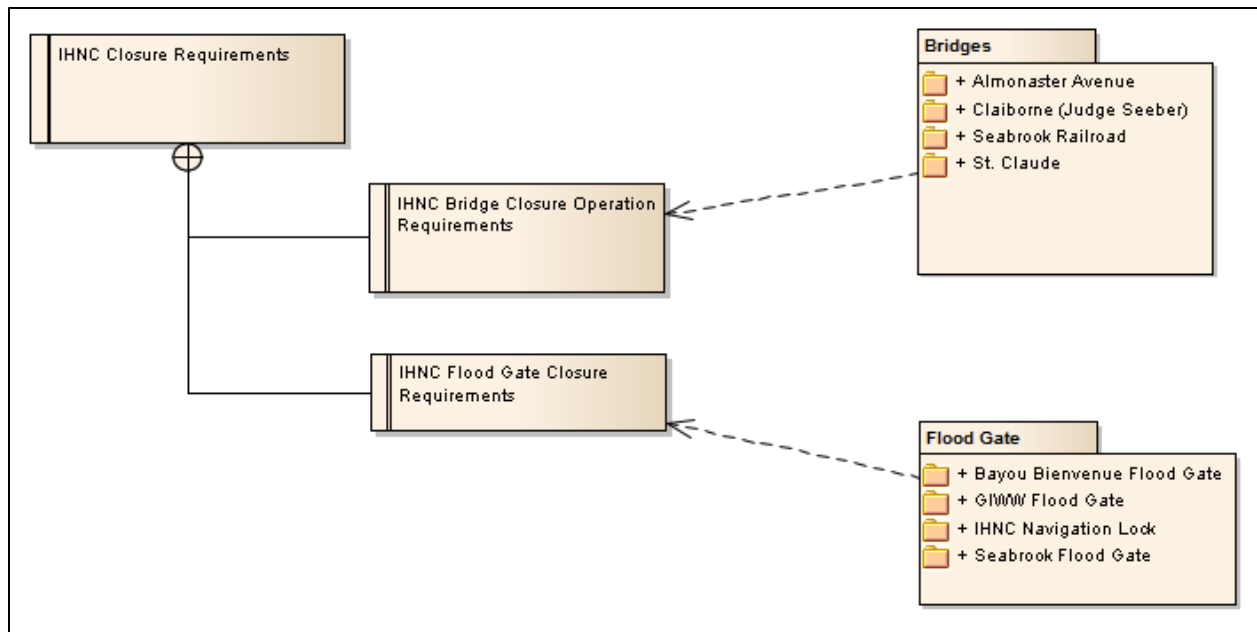


Figure 76: Assigning system elements to the system requirements.

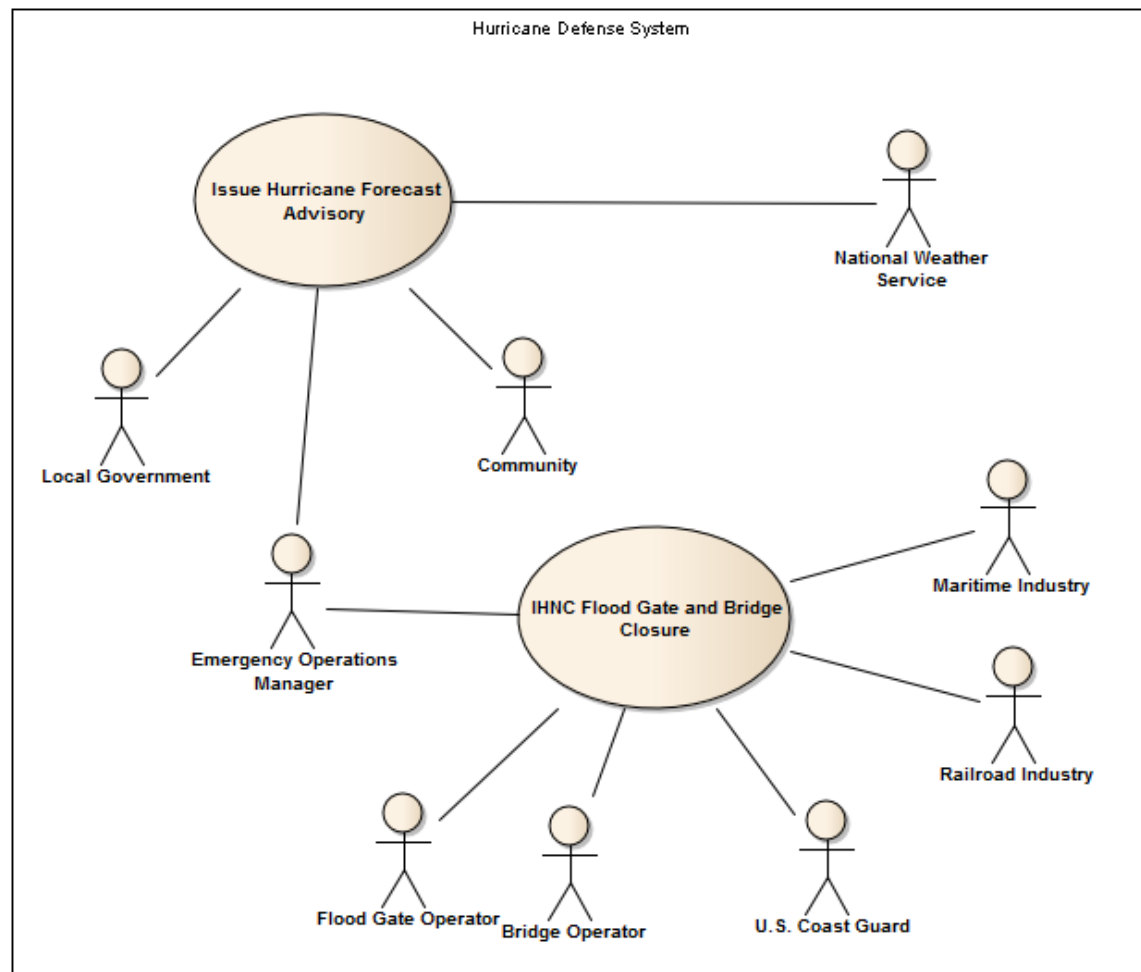


Figure 77: Uses Cases for the closure of the IHNC/GIWW flood gates and bridges.

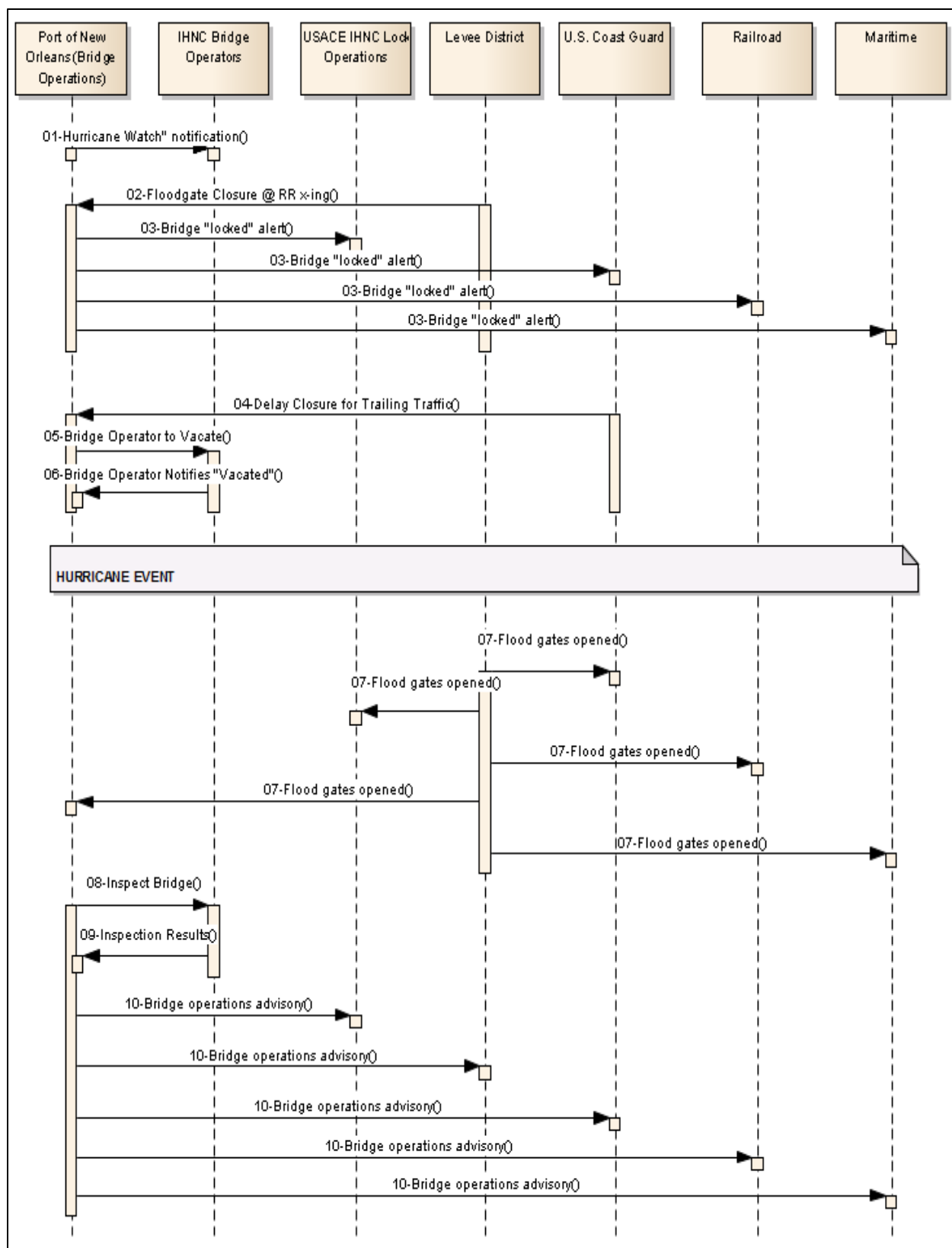


Figure 78: Sequence diagram showing interactions between Stakeholders during the IHNC/GIWW closure operations.

IHNC/GIWW Subsystem Results

The primary outputs from the SysML modeling of the IHNC/GIWW case study is the ability to capture and model the IHNC/GIWW Closure Operations system elements and their relationships and to summarize that information in an understandable format in order to demonstrate the application of the Systems Engineering modeling methods to better assess complicated HSDS operations. The tendency of these systems is to get extremely complex. For example, if we show the mapping of the bridge closure requirements to the bridges (**Figure 79**) and the flood gate closure requirements for the flood gates (**Figure 80**), things can quickly get cluttered and confusing.

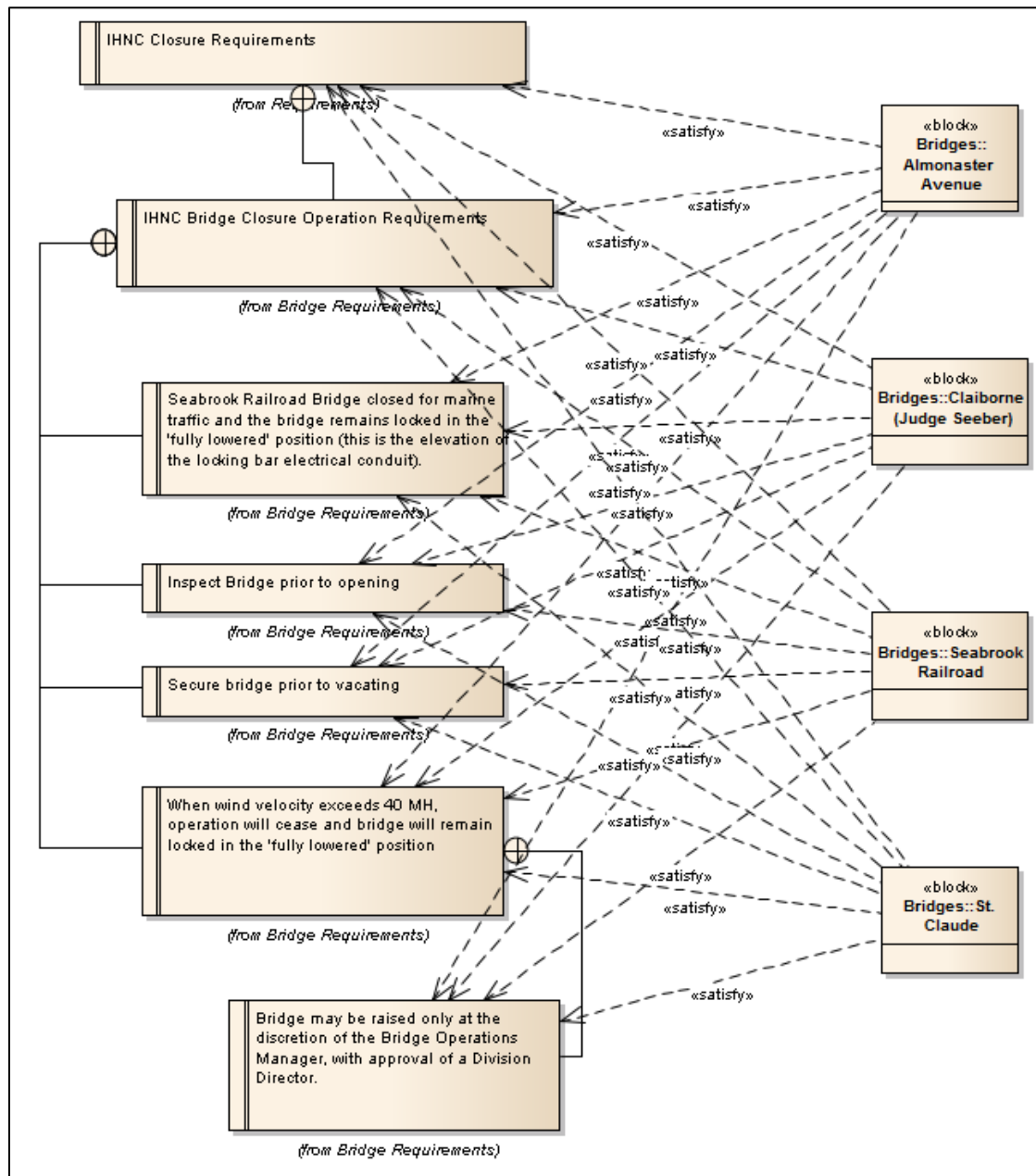


Figure 79: Requirement mapping for bridges in the IHNC/GIWW bridge closure requirements.

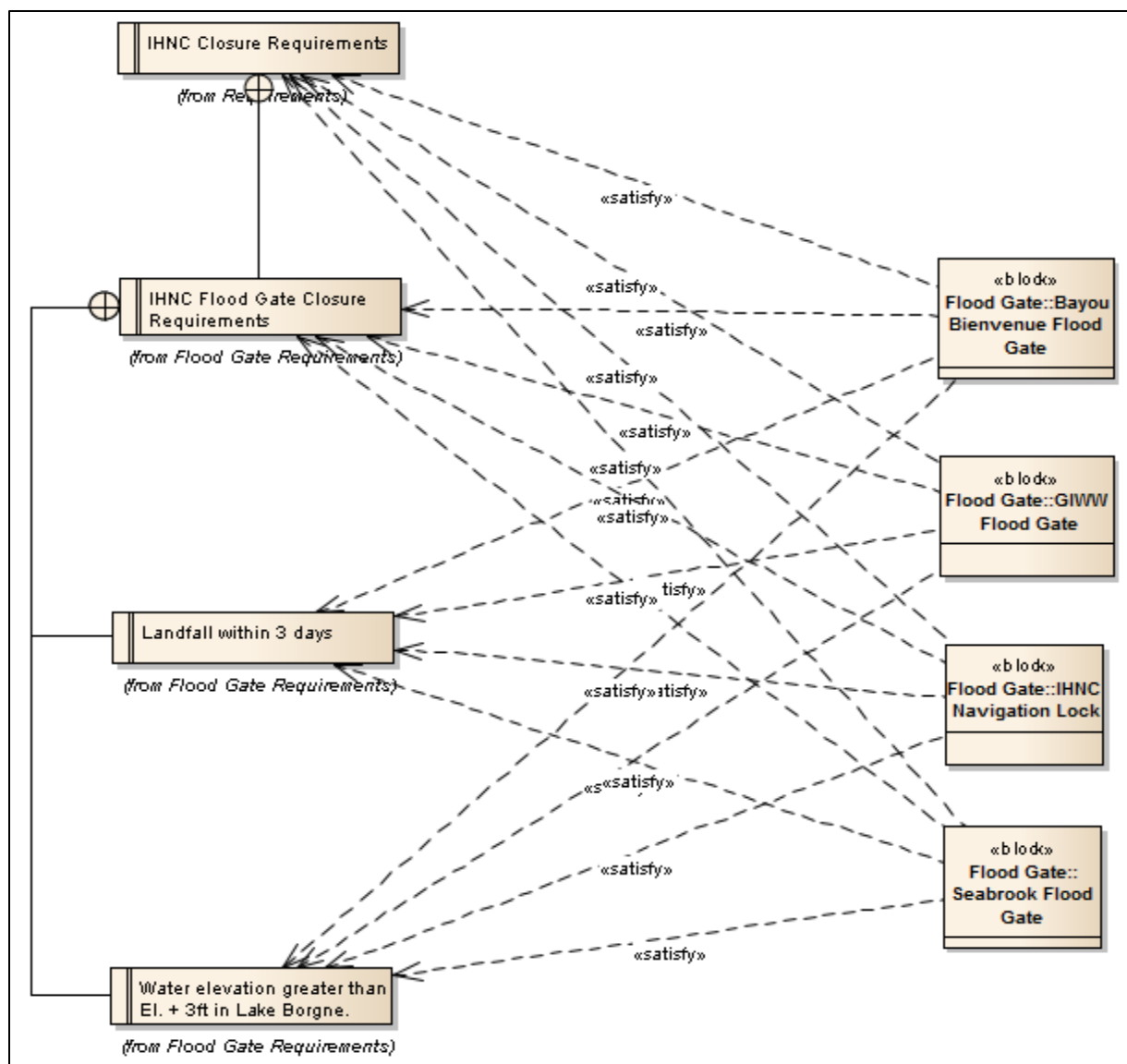


Figure 80: Requirement mapping for flood gates in the IHNC/GIWW flood gate closure requirements.

Instead, complex inter-relationships can be presented in a matrix format (**Figure 81**) where information can be better organized. **Figure 81**, for example, shows a summary of the IHNC closure requirements (left column) and the bridges and flood gates in the right columns. An arrow summarizes that the bridge/flood gate needs to satisfy the requirement on that row. You will note that the bridges show that they need to meet the bridges requirements and the flood gates need to meet the flood gate requirements. All structures need to meet the IHNC closure requirement (as the IHNC closure requirement is comprised of the bridge and flood gate requirements combined).

Beyond our scope, this matrix provides the basis for completing “gap analysis” (for example, either structures not having requirements or requirements not having structures) as well as “clash analysis” where conflicting requirements or structures are assigned.

In addition to the matrix, the SysML model also features the ability to trace elements by their relationships (**Figure 82**).

	Bridges::Almonaster Avenue	Bridges::Clalborne (Judge Seeber)	Bridges::Seabrook Railroad	Bridges::St. Claude	Flood Gate::Bayou Bienvenue Flood Gate	Flood Gate::GIWW Flood Gate	Flood Gate::IHNC Navigation Lock	Flood Gate::Seabrook Flood Gate
Bridge Requirements::Bridge may be raised only at the discretion of the Bridge Ope...	↑	↑	↑	↑				
Bridge Requirements::IHNC Bridge Closure Operation Requirements	↑	↑	↑	↑				
Bridge Requirements::Inspect Bridge prior to opening	↑	↑	↑	↑				
Bridge Requirements::Seabrook Railroad Bridge closed for marine traffic and the br...	↑	↑	↑	↑				
Bridge Requirements::Secure bridge prior to vacating	↑	↑	↑	↑				
Bridge Requirements::When wind velocity exceeds 40 MH, operation will cease and...	↑	↑	↑	↑				
Flood Gate Requirements::IHNC Flood Gate Closure Requirements					↑	↑	↑	↑
Flood Gate Requirements::Landfall within 3 days					↑	↑	↑	↑
Flood Gate Requirements::Water elevation greater than El. + 3ft in Lake Borgne.					↑	↑	↑	↑
Requirements::IHNC Closure Requirements	↑	↑	↑	↑	↑	↑	↑	↑

Figure 81: Requirement Matrix summarizing the relationship between the IHNC/GIWW System Elements (blocks) and IHNC/GIWW Evacuation Requirements.

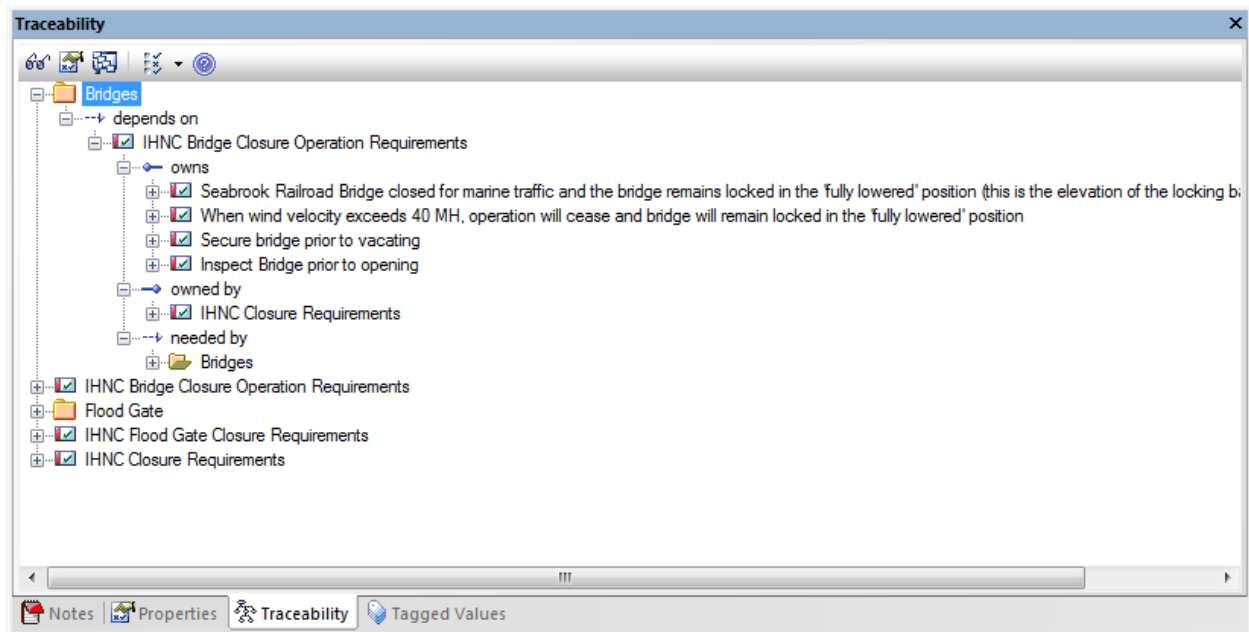


Figure 82: The SysML includes a "Traceability" function that tracks functional dependencies between elements. In this case, the IHNC/GIWW closure operations functionally depends on bridge closure operations and the associated requirements.

7) Conclusions and Recommendations

This study demonstrates the application of Systems Engineering and QMAS methods to holistically assess the Hurricane Surge Defense System (HSDS) for the greater New Orleans (GNO) Eastbank using the Multiple Lines of Defense Strategy. It provides an initial step towards a full-fledged method to model/manage the complete HSDS, which includes coastal and community lines of defense in addition to structural defenses, in an integrated fashion throughout its life-cycle.

Following Hurricane Katrina in 2005, all of the major investigations into the causes of the catastrophic flooding of GNO found that, in addition to specific design and engineering causes, the structural protection system lacked an overarching integrated systems approach to design and management. This has been termed the “system in name only” problem. Over \$15 billion (Resio, Irish et al. 2013) has been spent reconstructing the regional structural protection system to a 100-year hydraulically defined performance standard. These projects have addressed many of the specific design and engineering causes identified by levee investigations, but has the overriding systems engineering lesson been fully learned?

In general, our assessment found modest progress toward addressing the “system in name only” problem. Major upgrades to the structural protection have addressed many of the design vulnerabilities exploited by Hurricane Katrina. Importantly, floodgates and floodwalls at the IHNC/GIWW and the three Orleans Parish outfall canals in the levee system have closed major “gaps in the system”. Additionally, these improvements in structural protection have occurred concurrently with improvements in coastal protection and community resiliency. Thousands of homes, businesses, and government buildings have been elevated. Local Citizen’s Assisted Evacuation Plans and the new I-10 “Twin Spans” Bridge have increased evacuation capacity. Most recently, the 2012 State Master Plan links coastal management with flood risk reduction and, in the process, produced a massively coupled set of physical, ecosystem, and social computational models that provide a robust representation of the system in a policy and planning context.

However, the current HSDS lacks system-wide integration in many key aspects. While the stated level of protection is based on hydraulic engineering analyses, our study identified a number of behavioral factors that could reduce the actual level of protection to below the stated design criteria. In particular, major factors of concern included limited stakeholder education/input and operations, maintenance, and funding over the system’s life cycle. Additionally, we have found that the system objective is not fully defined and that the system lacks a holistic architecture which accounts for all the system elements scattered across the coastal, structural, and community lines of defense.

These results followed from our efforts to utilize the Multiple Lines of Defense Strategy (MLODS) as an architecture for the HSDS, and then to apply methods from systems engineering, the Systems Modeling Language (SysML), the Quality Management Assessment System (QMAS), and Geographical Information Systems (GIS) as tools to assess the system performance. We concluded that MLODS provides a robust system architecture that links coastal, structural, and community lines of defense. Most importantly it helps organize and inventory the parts, procedures, and processes that make up the system in a logical hierarchy that

spans spatial footprints and jurisdictional boundaries. Through the QMAS workshops and additional research we developed a hierarchy of system elements along with an initial “straw list” of system interactions, factors of concern, and scenarios of concern. We then presented and modeled this information using GIS and SysML.

During the course of this study, a series of three workshops were organized to engage system stakeholders (Assessment Team) to aid in the initial definition of the Hurricane Surge Defense System, to determine the Factors of Concern (FOCs), to develop a suite of test-cases (scenarios), and to ascertain if key lessons from Hurricane Katrina have been implemented.

Overall, four primary FOC themes were described by the Assessment Team:

- system interactions
- system integrity over its life-cycle
- jurisdiction and coordination
- long term system funding

Within these themes, the Assessment Team also described specific Factors of Concerns. Most notable, the IHNC/GIWW corridor was repeatedly described as the system’s “Achilles’ Heel”. Operators described how closure operations for the complex structure of floodwalls and gates must be initiated 48 - 72 hrs in advance of a hurricane landfall. Furthermore, assessors described a complicated closure sequence that must be coordinated with vessel evacuation and that impacts vehicular evacuation. Finally, they noted that floodwalls are not designed to sustain impacts from loose vessels and other large objects and that there remains considerable uncertainty about requirements for evacuating or securing vessels and other objects within the canals.

As an example of an FOC related to system integration between elements spanning spatial scales, the workshop participants described the pieces of a complex puzzle that links the state of the Chandeleur Islands, a major coastal line of defense, with Interstate-10, a major evacuation route and element of the community lines of defense. During the first workshop, assessors discussed choke points on I-10, including a section of the interstate just before the “Twin Span” bridges out of the city that is low-lying and outside the levee protected areas. Since it is low-lying and unprotected, this section is flooded when the surge heights reach around 7 ft. Sixty miles east of this system element, the Chandeleur Islands are a barrier island whose surge mitigation effects have been undermined by erosion and land loss. As discussed during the second workshop, hydrologic studies concluded that as the islands disintegrate, incoming storm surges peak higher and earlier. Despite their geographic distance, these two disparate system elements are linked via the dynamics of storm surge. If the Chandeleur Islands continue to deteriorate, then storm surges will flood the low point on I-10 sooner and longer, thus reducing evacuation capacity and delaying recovery.

Building from the workshops, an initial SysML model of the HSDS was developed. The model includes a high level system requirements diagram and hierarchy of system elements. Then SysML was used to depict the interaction between the deteriorating Chandeleur Islands and the unprotected low spot on Interstate 10. A second example modeled the IHNC closure sequence, a

complex sequence of operations involving four navigation gates located within the IHNC/GIWW: IHNC Navigation Lock, Seabrook Lock, GIWW gate, and Bayou Bienvenue gate.

Recently, Garret Graves, the former Chair of Louisiana's Coastal Protection and Restoration Authority, was quoted as stating that "New Orleans has the best protection system today that it's ever had" (Taflanidis, Kennedy et al. 2013) because the improvements to the structural protection system that have been implemented since Hurricane Katrina. While this assessment may be accurate if one looks at just the city's structural lines of defense, Lopez (Lopez 2009) points out that as recently a century ago the City had a much higher level of protection due to community and coastal lines of defense. These differing assessments, point to crucial unanswered questions on what makes the HSDS a "system." According to international systems engineering professional standards, a system is an "integrated set of elements to achieve a defined objective". In this regard, the HSDS possesses two of these pieces, an objective and individual elements, though some ambiguity remains with these pieces. However, we have found that the HSDS lacks a complete system integration and the lack of integration undermines the system's ability to fulfill its stated objective.

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Appendix A: Summary of Workshops

Our assessment of the GNO HSDS centers around three workshops based on the QMAS process. Each workshop included 10 – 20 participants with good representation of the different lines of defense. Table 20 gives a list of participants along with their individual areas of expertise. The table suggests that structural and coastal LODs were sufficiently represented, while community LODs could have been better represented.

This section summarizes the information and insights provided by the system assessors during the workshops. To the best extent possible, this is just a synopsis of their words with more indepth analysis presented in the main text of this report. Naturally, discussions of this type jump around between topics. In synthesizing these statements, we have grouped them into related topics with a logical flow. In some cases, this section also includes clarifying statements obtained during individual follow-ups. Other than these two qualifications, this section is a verbatim synopsis of the information provided by assessors during the workshops.

The sections below describe each of the three workshops in detail, but first here is a list of some of major points that were raised by assessors:

- “We can’t go back to what it was” which was followed by the “System has changed so much that we are forever dependent on structural protection”
- “Every flood protection structure shifts risk” and that there is “no risk neutral flood structure”
- “We bought into the systems approach, but money knows jurisdictional boundaries while water does not.”
- The stated level of protection is based on hydrology. “Does adding behavior reduce the level of protection?”
- Design-Build means “Built to a fixed budget for design and construction” so cost overruns during design come out construction.
- Local governments “will not be able to raise enough tax revenue to cover operations and maintenance”
- “Will \$15 billion spent on protection lead people to think New Orleans is safe?”
- “People look for reasons not to evacuate” and “We are obligated to realize” some people will not evacuate.
- “Evacuation authority is fragmented and exercise on a case-by-case basis.”
- “The INHC is the Achilles heel”
- When asked “What are our vulnerabilities?” an assessor with intimate knowledge of the HSDS replied “Besides the IHNC, operations”
- “Understanding the failure modes is only as good as our knowledge of the sub-surface.”
- “Pinnacle of the systems engineering approach is the shared vision of the system.”
- Limited availability of detailed information on the HSDS and flood risk for the region constrains the ability of the public, business interests, policy makers, and other stakeholders from making a fully risk informed decision on issues such as landuse and evacuation.

Table 20: List workshop participants along with agency and areas of expertise.

Name	Agency	Expertise	Workshop Attendance
Ezra Boyd PHD	LPBF	Hazard Geography, Risk Analysis	Facilitator
Rune Storesund	Storesund Consulting	Engineering, Risk Analysis	Facilitator
John Lopez PHD	LPBF	Engineering, Environmental Management	Facilitator
Bob Turner	SLFPAE	Engineering	1, 2, 3
Stradford Goins	Formerly SLFPA-E	Engineering	1, 2
Joe Suhayda PHD	LSU	Engineering	1, 2, 3
Officer Ron Tardo	LSP	Evacuation, Public Safety	1
Bruce Cuber	Red Cross	Disaster Response	1
Scott Eustis	GRN	Environmental Management	1, 2
Hal Needham	LSU	Climatology	1, 2
Elissa Miller	USACE	Engineering	1, 2
Mark Kulp	UNO	Geology, Environmental Science	1
HJ Bothsworth PE	Levees.org	Engineering	1, 2
Stevan Spencer	SLFPAE	Engineering	2, 3
Barry Keim	LSU Climate Center	Climatology	2
Robert Hoover	East Jefferson Levee District	Engineering	2
Earthea Nance	University of New Orleans	Engineering, Urban Planning	2
Roger Cowell	SLFPAE	GIS	2, 3
Bob Jacobsen PE	SLFPAE/BJPE	Engineering	2, 3
Steve Picou	LSU Ag Center	Water Management	2, 3
Grasshopper Mendoza	Horizon Water Committee	Water Management	2, 3
Sheila Grissett	SLFPAE	Public Information Specialist	2, 3
Dinah Maygarden	UNO	Environmental Science	2
Carlton Dufrechou	GNOEC	Environmental Management	1, 2, 3
Harold Daigle	CPRA	Environmental Management	2
KC King	Consultant	Systems Engineering	3

Summary of Workshop 1: Defining the system

Workshop #1 was held April 6, 2012 with ten assessors and the three authors serving as facilitators. The morning portion of the workshop largely consisted of presentations from the three facilitators that introduced the assessors to the project and its goals. Since systems engineering, systems modeling, and QMAS were all new topics to the assessors, the presentations focused on introducing these topics along with the goals of the project and purpose of the workshops. These introductory presentations were followed by an open ended discussion with assessors describing important issues related to the performance of the HSDS. An additional afternoon discussion focused on evacuation planning and procedures. Early during the morning discussion, an assessor asked “are the requirements right?” Specifically, the assessor questioned the validity of the storm surge height-return frequency relationships that formed the basis for determining the design heights of levees and floodwalls. He questioned the data and modeling behind the development of the 100 yr. storm surge heights (based on the USACE’s post-Katrina hydrologic analysis). Significant concern was expressed that not “starting from the beginning” would lead to a situation where inaccurate estimates of storm surge heights are used to guide system configurations. In this regard, many assessors agreed that the modeling and analysis tasks for determining surge risk levels are an element of the system.

Another important point to come out of the morning discussion was how adding behavior affects the actual level of performance of the Hurricane Surge Defense System (HSDS). It was pointed out that the structural components of the system (levees, floodwalls, and gates) are based on criteria derived from hydrologic analysis. In this sense, stating that the system provides 100 yr. level of protection just means that the crown elevations of the structures are at or above the estimated 100 yr. surge level. This assessment does not consider important behavioral aspects, such as operational procedures for gates and pump stations, maintenance procedures for earthen levees, and management procedures for local authorities. All of these behavioral factors influence the effective level of protection.

As a more specific example of how behavioral aspects of the system can comprise the level of safety provided by the structural perimeter protection, one assessor noted the risks posed by unsecured objects in the IHNC, specifically large navigation vessels that could break loose during a storm surge and impact with a levee. The assessor noted that during the design phase of levee improvements along the IHNC, the US Coast Guard mandated that all vessels evacuate the IHNC before a landfalling hurricane. For this reason, the USACE did not consider vessel impacts when designing the improved sections of the floodwall. However, after the fact, it was realized that the Coast Guard, nor any other agency, has the authority to enforce the vessel evacuation mandate. Since there is no authority enforcing vessel evacuation from the IHNC, it is probable, even likely, that a loose vessel could impact the IHNC floodwalls, resulting in a catastrophic failure. Because of this factor, the IHNC was described as one of the system’s “Achilles Heel.”

The newly constructed navigational floodgates in and around Lake Borgne are another example of where operational behavior could undermine performance. These complex engineering structures have been designed and constructed by the USACE, but once construction is complete

maintenance will be turned over to state and local agencies (the non-Federal sponsors). However, it is not clear that the state and local agencies have the technical expertise or funding to properly maintain this floodwall and floodgate complex. One assessor noted that “if we are unable to maintain them, they will not function as promised.”

Assessors also noted behavioral issues over the systems life cycle of improved segments of levees and floodwalls. With these system elements, design heights were derived based on current stage height calculations, with an intention to complete lifts in the future. The lifts compensate for settling of levee material and for changes in the design heights due to sea level rise and climate trends. The Louisiana Coastal Protection and Restoration Authority (CPRA) is currently the “non-Federal sponsor” of these projects and is responsible for maintenance once construction is completed. However, the authorization legislation does not include language that allows a Federal cost share for the lifts, and it is not clear that CPRA will have the funding to complete these lifts without a Federal cost share. In this regard, it was noted that some levee sections will need lifts as early as 2017, and assessors expressed doubt that CPRA or local agencies will acquire the needed funds by this date.

Life cycle considerations also arose when the discussion turned the topic of the lack of rust coating on the steel sheet piles used in newly constructed floodwalls around St. Bernard parish. In lieu of applying a rust prevention coating to the sheet piling, the USACE opted to add an extra 1/8 in. of “sacrificial steel” that supposedly can corrode while the sheet pile maintains integrity. According to the USACE, this option will allow the floodwall to function reliably for the next 50 years. In justifying this option, the USACE cited the June 1, 2011 deadline imposed by Congress and noted that coating the sheet piling would push the construction phase past this deadline. However, meeting the deadline in this manner created another life-cycle deadline: what happens after the 50 year lifespan of the uncoated sheet piling?

The assessors also voiced concerns regarding USACE policies on levee armoring, which rely extensively on grass and high performance geotextile mats (HPGM). Assessors conveyed that there are significant concerns regarding the performance of this armoring strategy. One assessor pointed out the HPGM will be susceptible to being torn by mowing equipment. He stated that as the levee materials settle unevenly, ruts will emerge in the grass surface and that when a mower passes over a rut the blades will tear into the ground, damaging both the grass and mat. He concluded that the “local levee districts will probably not have enough funding for the maintenance requirements. So in my opinion it will lead to poor performance of the levees over time.”

Furthermore, another assessor complained that the USACE was granted categorical exclusions for armoring certain levee sections, implying that sections that need armoring will not get it. Additionally, this categorical exclusion eliminated an investigation of scour on the flood side. As one example of why this is important, an assessor noted that 8 to 10 miles of a newly constructed floodwall in St. Bernard Parish consisted of a sandcore Earthen levee with a clay top into which sheet pile was driven with concrete floodwall poured over it. While the external appearance of a concrete and steel floodwall gives this new structure an appearance of high reliability, the hidden sand core along with the lack of armoring actually reduces the reliability considerably.

Additional issues raised during the morning session related to questions of overall systems integration. One assessor pointed out that pumps, levees, and canals should “all be under one umbrella.” Another assessor followed up that funding has to be regional so that function can be regional and that there is a need for better integration across system elements and local jurisdictions. Currently, the assessor noted, “money is spent based on jurisdictions, not based on system specifications.”

Another system interaction issue was noted in regards to many of the pump stations that convey water from the interior drainage network. While the pump stations are a crucial aspect of the hurricane defense system, many of them were designed using criteria based on rainfall events without consideration for hurricane or storm surge events. In fact, an assessor noted that many pumps were designed with sufficient capacity to handle a 10 yr. rainfall event, but they are not designed to handle the much heavier rainfall that occurs during a tropical system. Another assessor noted that the capacity of the new pump stations built at the end of the three outfall canals in Orleans parish equals the existing station capacity plus some consideration for the capital improvement plan. However, this constrains upgrades to the older pumps because of incompatibilities that can arise if the capacity to pump into the outfall canals is greater than the capacity to pump out of them, particularly given the low “safe water levels” in the outfall canals. Finally, on this topic an assessor lamented on the need to insure that the pump stations and interior drainage network be better integrated into the perimeter protection system.

On a related note, an assessor stated that unwatering operations following significant inundation of protected areas is not well coordinated between the local levee districts, the local drainage boards, and the USACE.

Another assessor noted the need for a thorough, independent investigation of the work completed by the USACE. In addition to the above mentioned sand core levee example, he noted that approximately 65 electrical service lines cross the Hayne Boulevard levee in New Orleans East, and that these protrusions through the levee may compromise systems performance. It was generally noted that TF Guardian just “piled on top of what was there.” While TF Hope devoted time and resources to the design phase, they generally assumed that TF Guardian did reliable work. Another general theme was that the Federal government funds projects piecemeal, and that the USACE is only completing independent review of some project designs. Finally, one assessor pointed out that the SLFPA-E has limited capacity for some independent assessments, including consultants looking at remediation of outfall canal floodwalls and structural analysis of floodgates along with surveyors checking elevation and benchmarks, while another assessor noted that the authority made a mistake by not hiring an engineering department.

Finally, at one point during the discussion, there were questions regarding technical aspects of the design used for a particular system element to which no one had the appropriate information nor knew how to access the information. At this point, it was suggested that fuller public access to design and technical information was one of the most important steps toward facilitating a systems approach to flood risk management.

Evacuation

The afternoon session of the first workshop focused on the evacuation line of defense, with a number of important factors of concern raised by the assessors. Early on assessors pointed out that the timing of the different evacuation phases is based on winds, but most storm surges increase water levels before tropical storm winds impact the coastline. This issue is particularly salient due to unprotected, low sections of key evacuation routes. One example includes the low spot (approximately 4 ft. above sea level) along I-10, a key evacuation out of the city going east, just before entering the newly build Twin Spans bridge over Lake Pontchartrain. Assessors also identified similar low spots along evacuation routes west of the city in and around the town of LaPlace, LA.

Relating to previous comments about the USACE's design process, an assessor pointed out that the current design process does not account for fast developing storms that can cause early flooding of low lying evacuation routes. It was also noted that some hurricanes, particularly ones that form in the Gulf of Mexico, develop and make landfall without sufficient time to implement the 50 hr evacuation plan. Historic hurricanes such as Audrey (1957) and Opal (1995) are examples, and it was pointed out that the NHC's forecasts for Hurricane Katrina provided less than 40 hrs for preparations and evacuations. In this regard, it was suggested that storm and storm surge prediction capabilities (such as the National Hurricane Center forecast products and ADCIRC predictive simulations) be considered an element of the system.

Timing was an important theme throughout the evacuation discussion. It was noted by an assessor involved in emergency planning and response that the tiered evacuation plan called for mandatory evacuations by parish with timing based on the parish's location relative to the coast. The tiered timeline was designed to prevent gridlock on major evacuation routes that would occur if coastal populations who have to travel through the urban areas evacuated at the same time of the urban populations. Implementing this plan requires coordination between towns/cities, parishes, and the state along with neighboring states. However, it was noted that for many residents the decision to evacuate is a personal, emotional decision, and that an individual's specific timing in deciding to evacuate does not always correspond with the official timeline. So, even in cases where there is sufficient coordination between different agencies, the actual evacuation sequence can vary and result in chokepoints.

From a system interactions perspective, an assessor pointed out that numerous floodgates cross evacuation routes and that a coordinated effort was required between local levee agencies and emergency managers implementing evacuation procedures. Gates, particularly those that cross over major evacuation routes, have to remain open during the entire evacuation period, but they also have to be closed before high winds (and possible storm surge) make operating the gates difficult and dangerous. Another assessor also pointed out that a number of small bridges along evacuation routes have not been widened and that these are traffic bottlenecks during a major evacuation.

At one point during the discussion, a facilitator asked whether omitting evacuation from the State Master Plan (SMP) was a problem. One assessor followed up by observing that the SMP, which is based on economic decision criteria, did not capture the risk reduction benefits of addressing

infrastructure improvements related to evacuation. The assessor argued that widening the narrow bridges or raising the low points on interstate both represent capital expenditures, but because these projects do not protect property they do not have any direct economic benefit. Benefits, such as increased public safety, were not considered in the SMP evaluation of proposed projects.

However, a different assessor replied to the question by stating that it is “not an issue, because the existing plan is good” with an additional assessor corroborating that assessment by stating that the plan we have “is the best we have to offer.” It was suggested that instead of major capital investments to improve evacuation infrastructure, what we need is “more education because people are stubborn” and tend to underplay the risks. Another assessor stated that many citizens are worried about the ability of authorities to protect property while they are evacuated. He pointed out that in his neighborhood, “only thugs and the NRA stayed” after the evacuation for Hurricane Katrina.

The facilitator followed up that question with another: will the \$15 billion spent on structural protection lead people to think that New Orleans is safe during a hurricane and that evacuating is unnecessary? In response, one assessor repeated that public education and risk awareness are important elements of the system. He also pointed out that for many individuals other factors beyond just objective risk assessments impact a person’s evacuation decision. These factors include economic constraints, social-cultural views toward authorities, and the stress of repeat evacuations. He also stated that even if the decision to evacuate is made, other factors can get in the way. As an example, he cited St. Rita's Nursing Home in St. Bernard parish, which decided to evacuate before Katrina but could not get adequate transportation for all its residents because its contracted transportation provider failed to fulfill its obligations as specified in the contract.

Finally, another assessor pointed out that officials need to be more precise when communicating hurricane risk and evacuation guidance. In particular, it is important to avoid false alarms that cost residents time, money, and stress and undermine the credibility of subsequent evacuations calls. At this point, a facilitator who has completed extensive research on evacuation issues for GNO, suggested that a robust flood protection system coupled with adequate understanding of the systems performance during tropical weather events will allow public officials to make more reliable risk assessments of specific storms and thus limit unnecessary evacuations and their associated impacts.

Summary of Workshop 2: System interactions, factors of concern

Workshop 2 was held on June 29, 2012 and it included 19 assessors along with the same three facilitators. Some, but not all, of the participants in workshop 2 had also participated in the first workshop. In contrast to the first workshop’s unstructured discussion format, the second workshop followed a more structured format. Like the first workshop, we began the day with brief introductory presentations, though they were much briefer than workshop #1. From there, the day consisted of three focused discussions on coastal lines of defense (morning) and structural lines of defense (afternoon). In addition to following a more structured agenda, Workshop #2 also benefited for the utilization of various worksheets for assessors to list system

elements along with considering factors of concern under various scenarios. Naturally, the discussion throughout the day covered many additional topics.

Community Lines of Defense

Early during the day, a facilitator observed that there are no elevation requirements for GNO in the SMP and asked whether this was a problem given that the SMP calls for 500 yr. perimeter protection for GNO. Echoing the previous workshop's discussion on making more data on the HSDS publically available, one assessor pointed out that there's no reliable and readily available source of data on the percent of homes that are and are not elevated. A different assessor then asserted that "what worries me most are slab on grade house" further stating that "people who make the most noise about flood protection live in slab on grade houses... someone needs to tell them how stupid that is." This discussion wrapped up with an assessor stating that Jefferson and Orleans parishes require 3 ft. elevation, which is based on advisory flood elevations. It was then suggested that these requirements created "institutional momentum" and the SMP was better to focus on areas that had not adopted any elevation requirements.

The topic of evacuation also arose during the second workshop, with assessors warning of the need to guard against false sense of security. On this topic, one assessor stated bluntly that "people look for reasons not to evacuate" and that "we are obliged to realize that not everyone will evacuate." This statement is significant because it contradicts the basic policy assumption that a full evacuation is attainable and that public shelters in GNO are not necessary. To help the public with the evacuation decision, it was suggested that the USACE should do more than just publish flood maps. They need to put out the models, assumptions, and uncertainties that go along with the maps. Finally, after comparing the odds of winning the lottery (1 in millions) to the odds of a 100 yr. storm (1 in 100), an assessor suggested a new lottery game where players are confronted with a hurricane threat.

Coastal Lines of Defense

The discussion of coastal lines of defense began by focusing on the Chandeleurs Islands. A facilitator presented the results of an analysis that looked at landloss there and how that impacted storm surge height and timing. The analysis concluded that a complete loss of the islands would both increase the peak surge height and shorten the evacuation window. He then prompted the assessors for their opinions on failure modes of the islands during fair weather and extreme weather conditions.

It was pointed out that the islands were created and sustained by accretion during fair-weather conditions, noting that the process works materials from inside the system via forces from outside the system. In terms of fair-weather failure modes, two were noted: subsidence and sea level rise. It was also noted that wave action during storms can degrade the island 2-3 ft. When prompted about failure modes of this system component, assessors pointed out two: loss of the shoal over the short term and risk to HSDS over long term.

At one point, an assessor asked if there is any benefit to structural reinforcement (rock shoreline stabilization) for the barrier island. A facilitator with extensive knowledge of the barrier island system stated that a breakwater at northern end could capture sand that would otherwise be lost, but that this benefit is countered by ecological concerns related to rock shoreline in a sandy habitat. Another assessor, active in coastal restoration projects, pointed out that structural reinforcement address erosion problems, but the barrier islands are suffering from landloss predominantly due to subsidence. He further stated that other shoreline stabilization projects that utilize large rocks have been deauthorized because of the expense of repeated lifts due to the rocks sinking from subsidence.

The session on the Chandeleurs eventually lead to a robust discussion of the implications of including Chandeleurs as part of the HSDS. It was pointed out that this would entail regular monitoring of the islands similar to the monitoring of the structural protection (levees and floodwalls), but a different assessor stated that inspections, maintenance, and certification of the barrier islands would not be the same as the process with levees. An assessor also pointed out that including the Chandeleur Islands in the HSDS obliges FEMA to pay for any storm damage. But, this prompted the question of who pays for maintenance? Additionally, an assessor asked how will ecological issues be balanced with flood protection? Finally, another assessor asked how the Chandeluers would be tied to flood insurance certification if they are part of the HSDS and particularly would the decay of the islands cause flood insurance premiums to rise?

The New Orleans East Landbridge was a second topic to be discussed during the morning session on coastal lines of defense. As is the landbridge, it currently reduces storm surge by up to 6 ft., but tidal passes through the land bridge allows surge to flow into Lake Pontchartrain at up to 200,000 cubic feet per second.

In terms of the structure of this line of defense, assessors noted that the landbridge was mostly brackish marsh with a remnant natural ridge (the Gentilly Ridge). A geologic fault underneath the marsh landbridge was also listed as a part. Additional parts of this system component include a railroad, US Highway 90 which was built atop the Gentilly Ridge, numerous lakes, canals and spoil Banks, the GIWW and associated bank erosion, two tidal passes connecting Lake Borgne and Lake Pontchartrain, numerous camps, and some industrial facilities. While portions of the landbridge are held in a mitigation bank, oil and gas activity continues in other parts, implying that wells, rigs, and pipelines are also parts of this LOD. Also, assessors noted numerous restoration projects, at various stages of implementation.

It was also noted that shoreline retreat on both the Lake Pontchartrain and Lake Borgne sides occur, at a rate of about 4 ft. per year. It was also noted that Hurricane Katrina induced erosion created a weak spot and that further erosion there could completely breach the landbridge and create a new tidal pass.

When prompted about fair weather factors of concern for the marsh landbridge, assessors noted concerns related to ongoing and possible future dredging for navigational purposes. The need to limit oil and gas activity was also a major concern. One assessor stated “don’t cut up the marsh if it is critical to flood defense.” Another assessor suggested that notification of permit applications should go out beyond the Department of Natural Resources to include

environmental, flood prevention and emergency management agencies. It was then stated that CPRA gets notifications of oil and gas permits if it located within a CPRA project area.

A different assessor expressed concerns about two potential USACE navigation projects: the IHNC Lock Replacement and possible dredging the GIWW. While the IHNC lock and GIWW are both configured for shallow draft navigation, he indicated that the expansion of the Panama Canal will likely create pressure to modify these for deep draft navigation. He pointed that the recently constructed Lake Borgne Surge Barrier had a sill depth of -16 ft., but that the lock replacement proposal calls for a sill depth of -34 ft. to accommodate deep draft vessels. Matching the depths of these structures along essentially the same navigation route would require expensive modifications to the Surge Barrier. Further, these modifications are likely to exacerbate the previously described problem related to potential presence of debris on surge barrier sill.

In terms of severe weather factors of concern, an assessor described two processes that increase surge risk for communities along the shores of Lake Pontchartrain. First there is “Filling the Lake” as the storm is offshore and winds push waters into Lake Pontchartrain, and then there is “Tilting the Lake” as the storm moves overland and wind directions change. He stated that the Barrier plan cannot stop tilting, but it would reduce filling. At this point, a different assessor described a control structure in Houston that can be open during events to allow quick drainage following overtopping. Assessors also noted storm induced erosion as a major factor of concern.

The discussion on Orleans East Landbridge ended with assessors listing the most vulnerable parts of the Landbridge:

- Katrina erosion spot
- GIWW
- Passes (some have enlarged)
- Grand Coin Pocket: small bay that is about to breach to the Rigolets
- Narrow part of ridge/landbridge along Highway 90
- Foot of I-10

Structural Lines of Defense

The afternoon sessions covered structural lines of defense with the first session covering the interior drainage component of the system. Particular emphasis was given to the metro Orleans and Jefferson drainage systems. Concerns with the interior drainage were illustrated when an assessor stated that the pump system only provided 18% of their full capacity during Hurricane Katrina. One assessor stated that a “pump station failure is a failure of the system.”

Early during the discussion, issues related to the dual pump system in metro Orleans were raised. One assessor stated that the city will always need multiple pump stations, adding that “pump stations in a flat city add energy for water to get to the lake.” He also noted that some drainage canals in Orleans parish require three pump stations. Assessors suggested that our system hierarchy for the interior drainage component include floodwalls and the double pumps in the metro Orleans outfall canals.

Assessors also described design capacity concerns with the older pump stations. These pump stations were built to mitigate rainfall flooding, and the design did not consider storm surge. It was stated that the USACE model for pump stations uses 10 yr. rainfall amounts (8 – 9 inches over 12 hrs) even though the system needs to handle a 100 yr. hurricane event, which could produce both more intense rainfall along with water due to overtopping.

It was also noted that design criteria was not uniform and that different pump stations have different design capacities. Likewise, assessors noted that the provision of safe houses and backflow prevention has not been universal.

A few assessors described problems with the pumps on the new floodgates at the end of three outfall canals in metro Orleans. One assessor stated the “hydraulic pumps are a joke... not the right type of pump for what they need.” These pumps were described as very inefficient and maintenance intensive. It was noted that the seals tend to blowout requiring frequent maintenance, requiring that some pumps be pulled out of service for maintenance. One assessor described this as a system failure because the S&WB would have to slow down pumping into the canals, so water from rain and overtopping would accumulate in the protected areas, resulting in flood damage. It was suggested that direct dry pumps would have been a much better option. One assessor summarized the situation as “we needed Audi’s but we got Pinto’s quickly.”

While it was noted that the USACE bi-weekly “Pump Tracker Report” provides updates on the status of the pumps, assessors seemed to agree that this behavioral (maintenance) aspect of this system component represented a substantiated factor of concern that have not been modeled in current risk assessments. An assessor then noted that it would be 2 – 3 yrs. for permanent structures to be built, while a different assessor observed that there has not been a full analysis of pumping options for the outfall canals.

On a related note, assessors described how the new pump stations along the Lakefront meant that the outfall canals now serve as retention areas. One assessor noted that the floodwalls are built to 12 ft., but that the current safe water levels vary between 5 and 8 ft. He then asked rhetorically what “If someone falls asleep at the switch,” meaning too much water is pumped into canals. Such a lapse in operations could lead to a floodwall failure. When asked if automatic controls fix this, he replied “even those fail. Better solution if to have walls that don’t fail.” He further noted that during fair weather, there is a possibility that low water could also cause walls to fall into the canal. Assessors also noted the subsidence problems associated with the floodwalls along the outfall canals.

After noting that storage is a part of the drainage component with Lake Pontchartrain as the major receiving basin, assessors discussed the role of interior retention. One assessor noted that while we have natural retention outside the perimeter levee system, with the levee system all water retention is engineered. She then asked “how do we connect what we are doing inside the levees with what is being done outside?” A facilitator responded that we “can’t go back in many cases,” citing the example of Bayou St. John. He stated that it will always be a pond (meaning cut off from Lake Pontchartrain), but it could potentially be modified to resemble a natural waterway. He also stated that while the storage capacity of the bayou is limited, it is still a good

tool for public outreach. In this regard, another assessor stated that the “system has changed so much, we are forever dependent on structural protection.”

Compartmentalization within the levee protected areas has been proposed as a way to limit flood damages, and this topic inevitably was discussed during this working session. One assessor pointed out that the interior topography already creates sump areas. However, it was also pointed out that interior polders (compartments) are contrary to a systems approach because it segments the protected areas and encourages parochialism. Additionally, this option was described as politically counter-productive because it creates winners and losers that would inevitably fall along social and economic lines. A facilitator also noted that compartmentalization would require major design changes with implications across the entire system. He further stated that the incident polders created by various natural and manmade ridges (most of which parallel the river) within the protected areas are oriented wrong and that compartmentalization would require guide levees going river to lake, similar to the Bonnet Carret spillway. In response, one assessor, a hydraulic engineer open to the compartmentalization option stated that “managing water means living with it” further elaborating that “maybe it’s better to save something instead of nothing... that is a political decision.” Another assessor followed by observing that “every flood structure shifts risk. There is no risk neutral structure.” It was also noted that major storm water retention areas are already built into the system, including the Central Wetlands Unit and IHNC/GIWW. Finally, another assessor suggested using undeveloped tracts for retention and storage.

Earthen Levees, Concrete Floodwalls and Hybrids

After discussing the interior drainage component of the system, the group next discussed the earthen levees, concrete floodwalls, and hybrid levee/floodwalls

In term of system parts, assessors listed grass, armoring, and HPGM (engineered grass) as crucial to preventing erosion during a storm. Foreshore protection was also noted, with one assessor mentioning a Jefferson parish augmentation project. Assessors also listed transitional armor. It was also noted that valves, pipes, sluice gates and similar elements are necessary to facilitate drainage through levees. Underlying geology and faults were listed, with particular emphasis given to differential settlement and the resultant need for periodic lifts. Finally, an assessor pointed out that foreshore protection is settling in many places.

When prompted about severe weather failure modes, assessors listed breaches due to overtopping and breach before overtopping due to unknown soil conditions. However, one assessor stated that generally while “earthen levees are robust,” the lack of flood-side wave erosion is a key vulnerability. It was noted that while most earthen levees now have wave erosion prevention on the protected side, these measures are still untested. One assessor asserted that focusing mechanisms, which are storm currents that focus energy into a particularly spot, are a major factor of concern. Finally, it was observed that levees along Lake Pontchartrain’s south shore are better now than before Katrina

Floodwalls are an alternative to earthen levees that require a smaller footprint. It was noted that L-walls, which are vessel impact resistant, are going up at Franklin Ave, while another stated that repaired floodwalls along the IHNC contained L-walls. Assessors also listed three parts to this system component: transitions (sloped pavement), relief wells on protected sides at areas where seepage is a problem (basically a controlled sand boil) and seepage berms. Another assessor pointed out that sheet piling and other temporary barriers are elements of the current “as-is” levee/floodwall component of the system. In regards to the floodwall repairs, one assessor stated that “lots of after the fact things are going on” that could create long term operations and maintenance problems.

Next, the discussion turned to the topic of hybrid levee/floodgate elements, with many assessors agreeing that most sections of the perimeter protection system are actually earthen/floodwall hybrids. Examples include the Hayne Boulevard levee in New Orleans East, which consists of I-wall anchored by an earthen levee and the upgraded MRGO levee, which consists of a T-wall on top of an earthen levee. One assessor expressed particular concern about a hybrid section of the perimeter protection in St. Bernard parish (which had also been mentioned in the first workshop). He described an eight mile stretch of MRGO levee that consists of a new T-wall driven into a relict sand core levee. He stated that under the original Lake Pontchartrain and Vicinity authorization the project was constructed as a sand core levee with a clay cap. Later, TF Guardian added clay on top, but then TF Hope removed the new clay and added a T-wall. He stated that if the clay that is left washes away, then the sand will quickly erode likely resulting in a T-wall failure. When asked if the T-Walls were designed to stand without sand and clay, he responded that the USACE claims that the T-wall will stand if the sand erodes away, but that they have not released the analysis behind this claim. He further stated that high hydrostatic forces exerted below the surface could result in seepage and piping. When asked if armoring should be added to this levee section, he responded that local levee authorities are looking at that option but the USACE can only fund (100% Federal) armoring on the protected side. Next, when asked if the failure mechanism for T-walls are the same as with I-walls, he responded that we do not know because T-walls did not fail during Katrina.

Large objects impacting floodwalls was listed as a factor of concern during the first workshop, and assessors elaborated on this concern during the second workshop. In particular, an assessor noted the number of potential unsecured objects along the banks (unprotected section) of the IHNC that could become water borne and impact a floodwall. In addition to the previously noted barges, the assessor also listed large pleasure crafts in the Marina at Seabrook, railroad cars not tied down or evacuated, and an RV park located along the IHNC, and a facility manufacturing building floating houses. However, other assessors asked that given that this is now a retention area, will water levels ever be high enough for this to be a problem? While no one provided a resounding “no” to this question, one assessor noted that the best approach requires closing the surge barrier gates and then pumping the water level in the IHNC/GIWW down to -1 ft. to maximize storage capacity and limit the risk of large water borne objects.

An assessor noted the importance of grass on the earthen levees. In particular, he noted that the levees are designed for limited overtopping, so grass is necessary to prevent erosion. However, he also stated that grass maintenance requires resources, including mowing, fertilizer, and herbicides, and that this will cause ongoing funding issues for local agencies. When asked if

hogs are a problem for earthen levees/T-wall hybrids, assessors answered “yes, and so are armadillos.”

Finally, the discussion of the structural perimeter protection turned to the topic of floodgates, a crucial element of this system component. It was observed that the system contains over 200 flood gates, but most of these are located on land where a road crosses a levee. For the GNO east bank HSDS, there are 6 floodgates located in navigable waterways along with the flood gates at the end of the three outfall canals in metro Orleans (these canals are not navigable however). One assessor stated “gates are a big vulnerability” due to different design criteria. For example, some gates have backup generators and safe houses, while others do not. It was also noted that all gates are going to local control, which creates concerns regarding operations and maintenance. Finally, assessors stated that operation of the gates will be compromised if navigational interests trump flood protection goals.

Assessors expressed particular concern over a floodgate where the Lake Borgne Surge Barrier crosses the GIWW. The floodgate includes a bypass gate made of a large, concrete barge that is sunken to close the gate. However, during its first test, the barge cracked due to debris at the bottom of the GIWW. More generally, assessors noted that up to 72 hrs. are required to lower the barge and close the surge barrier. A diver must visually inspect the sill to make sure the pad is free of obstruction before gate can be closed. Further, logs and other such debris on the sill can prevent a full closure which could allow water to flow underneath. It was suggested that local operators need bathymetry scanning equipment to facilitate quicker gate closings. Assessors also noted extreme operations and maintenance costs for this structure, which has been described as the “world’s largest surge barrier.”

Summary of workshop 3: Scenarios of concern

Workshop 3 was held on October 26 with 9 assessors participating. As the final workshop, it served as an opportunity to wrap-up the QMAS process by reviewing the compiled information and considering the factors of concerns in the context of specific use cases. The agenda started with a brief recap of the projects and its objectives, followed by two sessions focused on summarizing the system definition, factors of concern, and the stakeholders (all information from the first two workshops) and then two sessions that considered important scenario’s of concern. Of note, Hurricane Isaac had occurred just a few weeks prior to thisd workshop, and this event was mentioned many times during the discussion.

Project Recap and General Thoughts

The initial project recap session led to an early open discussion of both the project and the surge defense system in general. Like the earlier workshops, public access to information was a key point. Greater system exposure via making as much information as possible available online was a major theme of the discussion. One participant proclaimed that “the pinnacle of the system engineering approach is the shared vision of the system.” He further elaborated that it would be a collaborative interaction between the various stakeholders.

Pointing to the pitfalls of information overload, one assessor pointed out that the “geotechnical details of borings are boring” and that we “don’t want to create a landfill by dumping data.” However, he did note that there may be certain “hotspots” that are more important to system performance and the details of these locations could be useful to the public. In response, an assessor affirmed that systems engineering creates an architecture to intelligently manage information at the appropriate levels. Themes related to the availability of information on the system popped up at other times throughout the day.

System Definition and Factors of Concern

The first working session of the workshop reviewed the System Definition and Factors of Concern. During this session, we presented a draft list of “Factors of Concern” identified in the previous workshops and solicited comments and criticisms from the participants. A workshop facilitator noted that we are dealing with a unique system with “a lot of natural elements, as opposed to many engineered systems that are all fabricated elements.” After pointing out that we don’t have a pre-established architecture for the system, we discussed the need to develop a narrative (use case or scenario) that links the technical engineering and scientific data with the system’s stakeholders and their goals.

At this point, an assessor pointed out that stakeholders are generally considered any actors who gain value from system, and that among stakeholders there is a sub-group of critical stakeholders who look to the system for important values. Examples of critical stakeholders and important values include citizens who want to protect property and levee boards who are held accountable for performance. When asked who ends up with responsibility with what has been built, assessors responded that there is no complete organizational chart that specifies authorities over the system. One assessor noted that some stakeholders include specific groups within agencies. For example, the University of New Orleans has research groups dedicated to coastal defenses and other groups that study community defenses. It was suggested that we develop a map of jurisdictions for the system and then put it on Google Earth with links to agencies.

At this point, an assessor with extensive systems engineering experience pointed out that “there are different views of system architecture. The spatial view is one of those, but it is not the only view and it is not all encompassing.” It was then noted that SysML allow views from different stakeholder perspective along with aggregate views and system level views.

Along with this discussion, another assessor noted that the Corps has a massive database system for projects, called ProjectWise. He stated that the database has everything related to structural aspects of the perimeter protection, including 20,000 scanned .pdf files covering details such as certification documents for specific levee reaches. However, he lamented that the Corps doesn’t allow state and local stakeholders, specifically CPRA or SLFPA-E, access to the database. It was also pointed out that SLFPA-E has created a Levee Information Management System (LIMS) with extensive information on the New Orleans East levee system, including borings, specs, failure mechanisms, photos, and the year built. However, this data source is not available to the general public.

At this point, it was observed that these document centric databases are characteristic of a project perspective, and that what is needed is a systems perspective that models structure and behavior. While the other branches of the Department of Defense adopted systems engineering in the 1960's, the Corps is legally required to follow a project approach. An assessor then asked should we recommend that Congress mandate systems engineering for infrastructure systems designed and built by the Corps?

Changing direction somewhat, an assessor then stated that defining the system implies a boundary, but that there are other systems outside the boundary which impact the system. For example, some elements of the HSDS are also part of the Federal highway system but that system also includes many other elements outside the HSDS boundary. He then stated that we impose local flood protection on Federal agencies. Our system is part of larger systems and that ownership and responsibility is a problem, which he described as a "50 year problem." To this an assessor responded that if we keep just measuring structural components of the system (while ignoring coastal and community components), then we are going to get wet.

Stakeholders

Throughout the workshops, assessors have listed numerous stakeholders involved or linked with the surge defense system. Systems engineering practice specifies that identifying stakeholders is an important early step in the process. To better list our stakeholders, we presented a preliminary list of stakeholders and solicited participants to identify additional stakeholders. In general, the discussion jumped between listing stakeholder organizations and defining who constitutes a stakeholder. During this discussion, an assessor commented that "A raindrop [that falls within protected areas] has many masters," indicative of the many stakeholders involved.

Assessors initially listed key research organizations, including the Water Institute of the Gulf (TWIG) along with the University of North Carolina- Chapel Hill (UNC-CH) and Notre Dame University, both of which are active developers and users of ADCIRC. An assessor then suggested that we include a category of stakeholders for "Professional Organizations" such as INCOSE, ASCE, and the National Floodplain Management Association (NFMA). Additionally someone suggested a category for "Research Institutions." Assessors then observed that some organizations have multiple groups each with their own stake in the system. For example, Louisiana State University has distinct research groups for levee engineering, evacuation planning, and coastal restoration and management.

At this point, an assessor pointed out that there is a distinction between those organizations that are responsible for designing, building, maintaining, or operating the system and those stakeholders who are users that live within the protected area. Additionally, assessors discussed the nuances of stakeholders associated with the Federal government and their level of authority of system components. It was pointed out that some Federal agencies are policy cognizant, while others are system users, and others are participants in the system life cycle. Not surprisingly, many Federal agencies have multiple roles in the surge defense system.

At this point, an assessor posited that “some stakeholders have a ways to go.” As an example, he described recent construction at a local university that he was done without good design, resulting in poor drainage near an unelevated first floor. The assessor then asserted that we “cannot continue to design as if this is a city on a hill.”

Assessors then discussed residents that live outside the levee system, and many agreed that they are also stakeholders. As evidence of this, one assessor stated that “since Hurricane Isaac, we have a new definition between the haves and have-nots... and it has nothing to do with economics or class.... People inside protected areas are complaining about Entergy [the local electrical power supplier], while people outside the levee system are gutting their homes... there is a huge disconnect.” At this point, it was suggested that we need to breakdown the “Participant in Lifecycle” category of stakeholders. Elaborating on this point, it was pointed that “we know that a lot of people will have to move” because it is not possible to provide them with an adequate level of protection. It was then suggested that “a big part of what we are missing is political leadership... they are not making hard decisions... just complaining.”

At this point, a facilitator asked how can we better “align people with infrastructure?” He further stated that “generally speaking, cities rebuild themselves every 100 years, so we don’t need to force change, but guide it.” Along this point, an assessor stated that “in a few years, people will wish NORA [New Orleans Redevelopment Agency] created multiple family units,” a reference to a locally administered rebuilding and recovery effort which focuses on rebuilding single family homes. A different assessor pointed out that elements of the “system are being built where there isn’t money to operate them.”

To wrap-up the stakeholders discussion, an assessor pointed that “Societies create narratives to convey what a 100 yr. event means” and then described an analogy between personal automobile safety and storm surge protection:

Car analogy:	<u>Car</u>		<u>Surge Protection</u>
	Highway safety		Coastal Defense
	Bumper		Structural Defense
	Seatbelts, Safe driving		Community Defense

System Requirements & Factors of Concern

The second working session for the workshop aimed to further refine our list of crucial factors of concern. Following workshop #2, facilitators created a draft “Factors of Concern” summary based on the findings of both of the previous two workshops. This draft document was distributed to participants before the workshop and projected during the session. We then solicited comments and criticisms from the facilitators.

System Requirements

Early on, an assessor suggested that we need a bullet that states a clear functional objective. When we pointed out that we are assuming 100 yr. level of protection requirement for the system, he responded that “protection is a loaded term.” He suggested that we need a functional objective that is clear, not loaded, and not arbitrary. (Many experts maintain that the 100 yr. standard was chosen arbitrarily.) The assessor then elaborated that the “USACE was authorized to build a system that FEMA [specifically the NFIP] could draw on map to lower BFE’s [Based Flood Elevations].” He then stated that the “original authority is not the current authority.” Following up, a different assessor pointed out that extreme, outlier events do not follow a normal distribution and pointed to the “need to be actuarially realistic with the NFIP... These might be stated as risks that have not been mitigated.”

Next, the facilitators asked are there any other system requirements that have not been listed? One assessor stated that there is a “clamoring need for more protection since Isaac [revealed] changes in the system for other areas.”

Assessors also suggested that we add specific evacuation requirements. At which point, it was again noted that evacuation is not in the State Master Plan. An assessor then observed that “evacuation authority is fragmented and exercised case-by-case... [and that] Hurricane Isaac revealed problems.” A different assessor responded that “of all the things we are examining, evacuation is the best coordinated.” However, “Isaac showed there are still problems” responded another assessor. Finally, an assessor stated “at the end of the day, local governments tell people when to evacuate.”

Interior Drainage

Next, the group covered factors of concern related to interior drainage. Immediately, an assessor stated that “we keep finding more and more issues in the [outfall] canals.” She stated that we are shooting for an 8 ft. safe water level, but that is going to require “a lot more mitigation” work. She also stated that we are “looking at hardening gauges.” Citing water levels in the canals, another assessor pointed out that there is a “technology gap between what we know and what we need to know.”

This prompted one assessor to suggest that “everyone has to adopt continuous process improvement... [We] cannot get resilient systems without feedback loops.” In response, an assessor pointed out that there is “no city wide vision for monitoring water levels.” It was then suggested that we “use areas of concerns as defined by risk to suggest policies.”

Next, as assessor stated that the “limitations of the drainage system are not necessarily related to hurricanes.” He elaborated that most levee protected areas are still in the 100-yr. floodplain due to rainfall and that there is also the flood risk from river. Comparing this situation to a boat with one side that is 3 ft. higher than the other side, an assessor pointed out that “river levees are built to 800 yr. level,” a much greater level of protection than that provided against the surge threat.

Levees and Floodwalls

Following interior drainage, we went over the identified factors of concern related to levees and floodwalls. Early on, an assessor stated that the less reliable I-walls are present in the current system but not pervasive, and further stated that they generally are not part of the mainline defense. She stated that there are two areas with I-walls in the mainline, but that these are both undergoing mitigation.

Assessors then asked whether the perimeter system is redundant or resilient. It was pointed out that the Corps uses the term capacity instead of redundancy. An engineer then stated that there is no redundancy on the Eastbank, but that Westbank levees have redundancy. Assessors agreed that, as a principle, there is no 100% reliable backup system. An assessor then stated that “resiliency is an umbrella concept and redundancy is a practice for achieving resiliency.” He then elaborated that we need to answer “what is the nature of the strengths and weaknesses of the system?” That answer guides decisions on evacuation, home design, land use, flood fighting, and related issues.

To close out, an assessor suggested that we refine the wording on lifts, specifically pointing to the wording used in design documents. On this topic, an assessor stated that the funding situation has changed with locals responsible for the entire cost of future lifts along with armoring.

Pollution Controls

Before wrapping up the Factors of Concern, an assessor suggested that we create a category for pollution and environmental degradation, stating that “pollution controls are important to the system.” Another assessor supported this suggestion and stated that pollution is more than a health issue: “It becomes a recovery issue when roads are restricted” due to hazardous releases. He pointed to a major hazardous release during Hurricane Isaac that restricted traffic on the main highway on the eastbank of Plaquemines parish for a few weeks after the storm. Further support was provided when an assessor stated that a barge got loose in IHNC/GIWW during Isaac.

Scenario’s of Concern

The last session centered on scenario’s of concern. Two real life case studies – Tropical Storm Lee and Hurricane Isaac – were considered as examples of a frequent, low level scenario and a less frequent, moderate level scenario. A third, hypothetical scenario posited how the current, “as-is” system would perform given a storm like Hurricane Katrina – presumably an infrequent, major level scenario. As part of the session, we gave a brief presentation summarizing basic observations from these events, and then opened it up for a general discussion on what factor’s of concern actually materialized during these events.

Tropical Storm Lee, a mild storm that impacted the area in September 2011, was the first case study. As a mild storm, it did not place any major stresses on the system. The discussion of this

event was brief with only one assessor pointing out a failure of temporary flood barriers. He stated that there was a HESCO basket failure (collapse) at a repair location along the IHNC.

Hurricane Isaac was a Category 1 storm that moved slowly across southeast Louisiana in August 2012. Despite being a relatively mild wind storm, there was a major storm surge that caused catastrophic flood damage for locations outside the HSDS. Since it was a recent event with significant impacts (some areas were still in initial cleanup and recovery at the time of the workshop), this event was on everyone's minds.

Shortcomings in the evacuation procedures were preeminent in this discussion. Early on, it was pointed out that not everyone assumed the worst for this event, and that many people who should have evacuated did not. Pointing to Braithwaite, a small community in Plaquemines parish just outside the Federal levees on the eastbank that was devastated by Hurricane Isaac, an assessor stated that residents made their evacuation decisions based on their experiences during Hurricane Katrina. (While this community experienced some flooding, it was much less severe.) He stated that generally speaking if you have a large Category 1 storm, some locations in the region will experience localized surge heights at the 100-yr. level. In regards to this specific event, he stated that Isaac was a 1 to 2 yr. event if you consider pressure, and that considering wind it was a 1 in 5 yr. event. However, looking at surge, Braithwaite experienced a 100 yr. surge height. He further posited that someone gets a 100 yr. rainfall event every year.

At this point, an assessor asked "What are we learning?" Is it that these events are unpredictable and everyone should just assume the worst for every event? To illustrate this point, we observed that I-10 flooded in both directions (out of New Orleans), something that was unprecedented. In response, an assessor responded by suggesting we communicate to the public that "a catastrophic storm may have these characteristics... and list them all... then explain that we aren't smart enough to know which ones will pop up with a given event at a given time."

This then prompted an open discussion about the level of protection and what the language implies. An assessor stated that "when we are talking about a 100 yr. event, we are talking across the entire system. But, we can still have 100 yr. conditions somewhere much more frequently." He then elaborated that the analytical "techniques are not new... just applying them to surge is new." (For example, in the context of rainfall, the analysis is based on a time series of rainfall data from a gauge at a specified location and the 100 yr. rainfall amount is determined for that point location. In contrast to a site specific analysis, storm surge analyses are conducted over larger regions.)

"Hurricane Katrina Like Storm" with Current, "As-is" System

While the first two use cases considered less severe, but more frequent scenarios, the final use case sought to examine the system performance under a more severe, potentially catastrophic event. General speaking, this use case was described as a Hurricane Katrina like storm interacting the current system. However, assessors were also encouraged to vary the storm characteristics to consider similar potentially catastrophic hurricane scenarios.

To start this discussion, it was pointed out that the Hurricane Katrina levee failures created storage in the city. Without such failures in the system, the new system cannot utilize this area for surge storage, which then potentially impacts water levels outside the levee system. To emphasize this point, an assessor asserted that “with water, you are always just packing the buck.” He then talked about unintended consequences of structural defenses, and then pointed out that we are still reacting to past engineering decisions, some of which go back to 1800s. He then asserted that there are areas where people cannot live unless they are self-sufficient. Another assessor then suggested we list barriers to the ideal level of protection.

Then, someone asked what about a Hurricane Katrina like storm on a track similar to Hurricane Isaac? At this point, an assessor explained that Hurricane Katrina was a rare meteorological event, and that it was probably 100 yr. event though that was not known exactly. For comparison, he stated that a Category 5 storm somewhere in the Gulf of Mexico is probably 10 – 20 yr. event, in other words much more frequent than a 100 yr. event. He then stated that in terms of storm surge heights, Katrina was probably a 500 – 1000 yr. event. He then concluded his point by stating that the entire system can handle a surge from a 30 mph wind event, but that different parts of the system will perform differently during a 65 mph wind event.

He was then asked what sections are worse off? In response, he stated that on the eastbank levee reaches in St. Charles parish pose a serious concern. Generally speaking, levee reaches behind swamps perform worse because freeboard does not consider inundation of the swamp and how that impacts wave amplitudes and periods. The Corps designs levees based on an overtopping rate, not based on freeboard. A different assessor then pointed out that FEMA’s certification process is not the same as the Corps design process. FEMA ignores overtopping of levees, while the Corps allows a limited rate of overtopping along with rainfall.

Assessors then pointed out that erosion before overtopping is a big unknown, and also noted that levees are not designed to protect against erosion on protected side. A facilitator agreed that there is extreme uncertainty in erosion due to overtopping. As an example, someone pointed to St. Charles where overtopping with occur sooner because of no freeboard.

Changing course, an assessor then asserted that failures the system are not just failures of the structural elements. Failures can also include the electrical system, highways, supply line, no standards for homeowners, and for pollution releases. A different assessor then stated that the vulnerability we typically talk about is breach risk, but agreed that there are others.

Elaborating further, an assessor point out that surge barrier is designed to be overtopped at a rate of up to $0.1 \text{ ft}^3/\text{s}/\text{linear ft}$. But, we still have barges in IHNC, which represents a vulnerability even when the surge barrier performs adequately. He suggested that we can leave the Bayou Bienvenue Floodgate, which connects the CWU with the IHNC/GIWW retention area, open to increase the available storage area. He further suggested that those procedures be part of the comprehensive water management plan. He also stated in New Orleans East, the levees create a bottleneck at the surge barrier and that there is no armoring along the north bank of the GIWW there.

Next, the group discussed the possibility that the levee failures witnessed during Katrina could reoccur in a future storm. In St. Bernard parish, dramatic improvements in the levee system were noted. It was also noted that sections of the levees along IHNC had been improved, but that these reaches had not been completely rebuilt. Assessors also noted improvement with outfall canals in Orleans parish. While pump stations have been improved, many still needed repairs and upgrades to the buildings. However, one important factor in the Katrina breaches had not changed -- subsurface geotech issues. It was pointed out that some conditions might not be known and that these uncertainties “can come back and bite us in the future.” He concluded that “understanding of failure modes is only as good as knowledge of sub-surface.” Building on this line of thought, an assessor pondered that we have answers for issues that were revealed during Hurricane Katrina, but what about the other issues that did not come up during that storm?

An assessor with a background in civil engineering observed that the newer T-Walls have a much greater factor of safety, which at least particularly compensates for unknown sub-surface soil conditions. He then noted that some areas are still vulnerable due to seepage pathways that can lead to breaching. Another engineer then noted that subsidence, corrosion, and the bending moment are all concerns that could contribute to a catastrophic levee breach, like what occurred during Hurricane Katrina.

Next, a facilitator asked if we need to be concerned about river levees breaching during a surge event? An assessor familiar with current and pending levee construction projects stated that the Corps is currently raising river levees for river floods, but that some reaches still need to be improved for hurricane protection. Finally, an assessor pointed out that on the Southshore of Lake Pontchartrain, surge analysis since Katrina led to a lower design height than previous analysis. As a result, some levee heights were reduced post-K. He stated that we “went from 200 yr. protection to 100 yr. protection.”

Next, the discussion moved to the catastrophic conditions experienced throughout the region in the aftermath of Katrina. It was noted that limits to evacuation effectiveness remained a problem. While plans assume that a 100% evacuation rate will be achieved, in reality this has never occurred and likely never will occur. An assessor then noted that breaching was never contended with in emergency plans prior to Hurricane Katrina. He stated that the design process assumed a “small” amount of overtopping.

An assessor then asked if it is possible to do emergency repair to a breach? One person responded that access to the breach site with heavy equipment is a major constraint. Then it was noted that FEMA requires locals with non-certified levees to consider impacts of breaches, but they assume that Federal levees will not breach. This discussion then concluded with an assessor stating that we need to consider multitude of possibilities for breaches including erosion due to overtopping and erosion due to underseepage, which creates a huge crater and is harder to repair.

Narrative to Prove Concept

To wrap up the Scenario’s of Concern sessions, we prompted the participants to help us sketch an illustrative scenario and narrative. We want to develop a demonstrative use case that we

could model in SysML to illustrate the utility of this approach. A facilitator stated that “we are seeking a storyline that makes the model meaningful.”

At this point, an assessor suggested that our efforts be geared toward guiding change and adaptations. We need to triage the major issues, inform the stakeholders, and assist them in buying down risk. As an example, a different assessor mentioned the consequences of not elevating a home and this initiative could better help homeowners understand those consequences. He then noted that there are other classes of significant stakeholders for whom we can help guide decisions toward the options that best reduce flood risk.

We then asked what hypothetical scenario will best evaluate the “as-is” system, emphasizing that we are interested in the whole system spanning from the continental shelf to evacuation routes out of the city. An assessor suggested that we consider a storm that quickly intensifies to Category 5. Because of its quick intensification, there was no major evacuation effort. Still, this storm would produce surge heights that are greater than levee heights and the ensuing overtopping would result in considerable flooding of protected areas.

A facilitator then prompted the group to consider what if every house is elevated? Assessors responded that you still need a logistics system to get Meals Ready to Eat (MREs), water, medication, and other supplies to people’s homes which are surrounded by floodwaters. Assessors also pointed to the need for safe refuges within communities, describing it as “a within community evacuation plan.”

Another assessor then asked how do we avoid what he termed “the Bridge Scenario” again? (He was referring to the pedestrian evacuation across the Mississippi River following Hurricane Katrina to which Jefferson parish responded by blockading the bridge and possibly violating evacuee civil rights.) Assessors suggested that the region needs to encourage neighborhood groups that are tasked to identify and track elderly, children, and other persons with special needs and to also identify elevated buildings within the neighborhood that could be utilized as neighborhood shelters. One assessor pointed to the example of Cuba where neighborhood captains are identified and they have 3 – 5 people working under them to provide assistance at the neighborhood level. Finally, in this regard it was noted that state plans encourage personal preparedness, but that they do not mentioned neighborhood or community level planning and preparedness.

Building on this topic, an assessor noted that there are opportunities to increase the level of protection by utilizing infrastructure that is not part of the Federal system. He suggested that we “need to broaden our perspective and think outside of the box...[for example there is the] potential to open up river levees [to] allow water to flow across river... [we need to] allow water to move instead of stacking up.”

An assessor then suggested that we start by identifying a set of catastrophic scenarios and then extend our analysis interactively to cover more likely scenarios. A different assessor responded by suggesting we examine “what happens if a given line of defense fails?” She suggested that we start with the most important (levees) and then build scenarios that work down.

At this point, it was again noted that the:

“Achilles heel of the entire system is IHNC/GIWW and surge gate.... We need to change the way the Corps operates... We have a small window of time to close the gate... need 48 – 72 hrs... We need divers to check the sill because of possibility of debris... if debris is there, then it takes time to get equipment and manpower in place to move debris.... If not closed, the walls in the IHNC are still fragile... It’s impossible to forcibly remove movable items.”

We then asked if barge floodgate is left open, how long until IHNC fills up to dangerous water levels? The assessor responded that we would be forced to leave Seabrook gate open, which would increase surge levels in Lake Pontchartrain. After noting that is not in the written operations guidance, a different assessor described the issue as one of “the human factors that limit performance.” Which prompted the response that

“Back in 2006, when we talked about risk we were only talking about hydraulics... We need to complete the failure tree analysis, so that we can mitigate some of the vulnerabilities by having extra equipment on hand... [We should] assign probabilities of failure for each branch, then we can calculate risk for system.”

At this point, the discussion shifted toward the interactions between the flood defense system and the navigation system. An assessor stated that we “can’t leave the barge gate closed permanently, because of currents in the GIWW and how that would impact navigation.” He elaborated that we “originally thought they would use two sector gates, but because of deadlines issues [one sector gate was converted to] barge gate.” At that point, operational plans specified that “it was supposed to stay closed except during maintenance... Navigation complained and then environmental concerns arose to keeping gate closed. So the decision was made to adjust operations to keep it open.”

One assessor noted that our “system exists within a larger navigation system.” A different assessor described this as one of many jurisdictional issues and further pointed out that “no one can declare that barge traffic and flood protection work together.” Someone elaborated that this is “one of the things that happens when you have project management instead of system management.” Building on this point, someone later noted initially during the post-Hurricane Katrina rebuilding a life cycle view was adopted, but in the case of the IHNC and the surge barrier the timeline got in the way and the life cycle was compromised as a result.

From here, the discussion drifted to the Central Wetlands Unit in St. Bernard Parish and portions of Orleans parish east of the IHNC. It was stated that there is “no way to disconnect the CWU from system” and that there is a “positive gain from using CWU.” Someone else responded that we have been describing using the CWU as part of the flood protection system while “the CWU is boxed as an environmental issue.”

That’s when someone observed that “When the storm pushes the limit, the system issues come to the forefront.” Using the surge barrier as an example, someone pointed out that we “don’t know the trigger to close the surge barrier, we just have interim guidance.” Someone else elaborated that “even after we have an interim permanent, we have to wait another year the final permanent guidance.”

After it was noted that the GIWW is Regulated Navigation Area (RNA), someone noted that “if the Coast Guard doesn’t do its job, then our level of protection goes down.” At which point an assessor pointed to the difficulties of evacuating of vessels from the IHNC coincident with the general population evacuation. The vessels need the bridges to be raised, while the general population needs the bridges down.

At the end of session, the scenario of concern that emerged included the following sequence of events:

- 1) Rapid track change and storm intensification
- 2) No time to close surge barrier results in the IHNC/GIWW filling
- 3) No time to clear IHNC/GIWW so there are loose vessels
- 4) No time to evacuate and the population shelters in place
- 5) Surge exceeds design heights and/or vessels impacts cause breach.

In closing, someone stated that “we are really out here by ourselves... we need the support of the scientific community.”

Appendix B: Listing of Significant Observations from QMAS Workshops

Category	Comment
System Issues	The system lacks a clear functional objective Arbitrary construction timelines mandated from congress meant that life cycle considerations were sacrificed A system implies boundary, but elements extend outside boundary Residents outside Federal protection system Surge height-frequency relationship varies spatially, within events, and is non-stationary River levees breaching during surge Nearby infrastructure not considered part of system Post-K design and construction process driven by congressional deadlines No thorough, independent investigation of all projects completed by USACE

Category	Comment
System Requirements	"Are the requirements right?" Validity of storm surge height-frequency relationships based just on hydrologic modeling Risk analysis does not account for behavior, both maintenance and operations, that affects the level of protection FEMA & USACE requirements and standards are different Interaction between swamps/marsh & levees impacts freeboard calculation. USACE design based on acceptable overtopping rate, but FEMA uses freeboard for NFIP. Freeboard calculation does not consider inundation of swamp and how that impacts wave amplitude and periods. Impacts food risk for areas with swamps/marsh directly in front of levee, 100 yr. flooding from 85 yr. storm River levee requirements based on 800 yr. risk level Drainage and pump stations requirements based on 10 yr. risk level Need complete failure tree analysis No elevation requirements for GNO in SMP because it calls for 400-yr. structural protection Current 3 ft BFE's creates institutional momentum Need clear, realistic evacuation requirements

Category	Comment
Availability of Information on System	"Pinnacle of a system approach is a shared vision of the system" which requires crucial stakeholders have full access information on the system Public, independent experts need fuller access to design and technical information on system SLPFA-E and CPRA cannot access the Corps Project Wise database which has detailed information about the perimeter protection system Stakeholders: Actors who gain value from system; Critical Stakeholders: sub-group who require full info The SLPFA-E Levee Information System (LIMS) only covers New Orleans East Document centric databases are characteristic of a project perspective. What is needed is a system perspective that models structure and behavior What are the strengths & weaknesses of the system? The answer guides decisions on evacuation, settlement, elevation, landuse, emergency planning and response Need to complete the failure tree analysis

Category	Comment
Stakeholders	Organizations that design, build, maintain, and operate system Stakeholders who are users that live within the protected areas Residents who live outside the perimeter protection system

Category	Comment
Behavior: Operations, Maintenance , Jurisdictions, and Funding	"Money is spent based on jurisdictions not system specifications" Federal government funds projects piecemeal Independent review not standard State encourages personal preparedness, not community level planning. Need to encourage neighborhood groups to identify and track elderly, children, and other special needs residents Unique system with a lot of natural elements along with fabricated elements No complete organizational chart showing who has authority over system elements Funding and expertise for maintenance & operations lacking at the state & local level Grass maintenance on Earthen levees will be expensive, requires equipment, fertilizer, herbicides Concerns regarding floodgate maintenance and operations when turned over to local/state control

Category	Comment
Levee & Floodwall Issues	"Achilles heel of the entire system is IHNC/GIWW... We have a limited time to close the date.... We need divers to check sill because of debris... if debris is there it takes time to get equipment and manpower in place to move debris... if the barrier is not closed, the walls in the IHNC are still fragile" Most are hybrid floodwall & levee Unsecured objects in the IHNC/GIWW could impact levee, causing a breach. IHNC/GIWW is a Regulated Navigational Area, but if the Coast Guard does not enforce it then the level of protection is reduced. L-Walls are impact resistant, but not widely used Surge barrier design allows overtopping at rate of $0.1\text{ft}^3/\text{s}/\text{linear ft}$ Foreshore protecting, grass, armoring, and HPGM are all crucial to preventing erosion during a storm HPGM: Scars are common (particularly mower blades that dig in) and can cause erosion 8-mile section of T-wall with relict sand core levee in St. Bernard Parish I-walls still present, though not used widely in the perimeter protection system Lack of rust coating on sheet piles used in newly constructed floodwalls in St. Bernard Parish USACE granted "categorical exclusions" from armoring certain levee segments No investigation of scour on the flood side Temporary barriers prone to failure Erosion due to overtopping Erosion before overtopping is a big unknown Sub-surface geotechnical issues have not changed since Katrina Higher factor of safety in rebuild elements, particularly T-Walls Breach repairs could be better coordinated CPRA (local partner) lacks funding for lifts. Some will be required as early 2017.

Category	Comment
Floodgates	"Gates are a big vulnerability" Designs of different floodgates based on different standards and criteria New floodgates are very complex and state/local agencies are not properly trained Not all gates have backup generators and safe houses

Category	Comment
Interior Drainage	"A raindrop that falls [within the protected areas] has many masters" "Pump stations in a flat city add energy for water to drain to the lake" Different system within and across parishes. Five systems with study area: Jefferson Eastbank, Central New Orleans, New Orleans East, Lower 9th Ward, and St. Bernard Parish Failure of the drainage sub-system is a failure of the entire system Pumps and drainage are not well integrated into the perimeter system "No way to disconnect the CWU from the system" Unwatering not coordinated Central New Orleans will always require 2 - 3 pumps. Capacity concerns with older pump stations and new pump stations along single drainage path Designs of different pump stations based on different standards and criteria Hydraulic pumps at the new pump stations for the outfall canals are very inefficient and maintenance intensive Newly formed retention areas Natural retention outside levee system versus engineered retention inside levee system 100 yr. flood plain defined by rain, not surge Outfall canal levees remain a problem No full analysis of pumping options for the outfall canals
Category	Comment
Chandeleur Islands	Loss of islands increases surge height, quickens arrival, shortens evacuation window Created and sustained by accretion during fair weather Subsidence and sea level rise are Fair weather failure modes Loss of shoal, islands creates risk to HSDS Implications of Including Chandeleur Islands in Federal system: Regular monitoring, FEMA obligated to pay for damage, Ecological issues must be balances with flood protection, impacts flood insurance certification

Category	Comment
New Orleans	Brackish marsh with remnant Gentilly Ridge
East Land bridge	Shoreline retreat: ~ 4 ft/ yr. on both Lake Pontchartrain and Lake Borgne sides Katrina induced weak spot could breach Navigational dredging: Ongoing and possible future Need to limit oil and gas activities, but CPRA only reviews permits that fall within its project areas Potential USACE Navigational Projects: IHNC lock replacement, dredging GIWW for Panama Canal expansion "Filling the Lake" vs. "Tilting the lake": Barrier plan will reduce filling, but not impact tilting Most vulnerable parts of the land bridge: Katrina erosion spot, GIWW, Passes are eroding, Grain Coin Pocket (small bay that is about to breach to the Rigolets), Narrow part of ridge/land bridge along Hwy 90, Foot of I-10

Category	Comment
Evacuation	"a catastrophic storm has these characteristics... and you list them all.... Then explain that we aren't smart enough to know which ones will pop up with a given storm at a given time" Public education and risk awareness are important parts of the system Timing of the coordinated, regional plan is based on winds, not surge Storm and storm surge prediction capabilities need to be considered as part of the system No official, credible storm surge prediction capabilities SMP looks at property protection, not public health benefits of projects SMP neglects risk reduction benefits related to evacuation Because it just assess projects on property protection, it emphasizes capital improvements over health Unprotected low spots on evacuation routes Numerous floodgates cross evacuation routes. Closure must be coordinated. Levee design process does not account for fast moving storms and early flooding of evacuation routes Loss of Chandeluer Islands, Biloxi Marsh, and Orleans land bridge would reduce total evacuation capacity Not everyone assumed the worst for Isaac Coordination between agencies is difficult Even with excellent coordination, evacuation is still a personal decision based on perceived risk. Timing of individual decisions and actions can vary and cause chokepoints Warnings need to be precise and avoid false alarms "People look for reasons not to evacuate"; we are obligated to acknowledge

and plan for that

Need to avoid the "bridge scenario" again where post-storm pedestrian evacuation is forcefully blocked by neighboring parish authorities

Category	Comment
Pollution Controls	"Pollution controls are important to system" Hazardous releases cause direct damage and health impacts Hazardous materials containers can damage HSDS Hazardous cleanup operations cause traffic problems, hinders economic recovery

Appendix C: System requirements table.

Label	Objective	Source	Level
FEMA1	"To be recognized as providing a 1-percent-annual-chance level of flood protection on the modernized NFIP maps, called Digital Flood Insurance Rate Maps (DFIRMs), levee systems must meet and continue to meet the minimum design, operation, and maintenance standards (44 CFR Section 65.10).."	FEMA, "Meeting the Criteria for Accrediting Levee Systems on NFIP Flood Maps." (Factsheet)	Federal
USACE1	"One hundred year level of protection actually means reducing risk from a storm surge that has a 1% chance of being equaled or exceeded in any given year. The 1% chance is based on the combined chances of a storm of a certain size and intensity (pressure) following a certain track. Different combinations of size, intensity and track can result in a 100-year surge event."	USACE, "100-Year Level of Protection." (http://www.mvn.usace.army.mil/hps2/hps_100_year.asp)	Federal
LAMP1	Long-term solutions: "sustainable long term solution for coastal restoration and for comprehensive risk reduction from storm surge and waves."	CPRA. 2012 Master Plan Principles.	State
LAMP2	Seeking sustainability: "characterized by consistent levels of productivity and resilience (the ability to withstand naturally variable conditions and/or recover from disturbances)."	CPRA. 2012 Master Plan Principles.	State
LAMP3	Systems approach: "evaluated using a systems approach to flood risk reduction and restoration, whereby benefits of actions and trade-offs among potential solutions will be identified and considered."	CPRA. 2012 Master Plan Principles.	State
LAMP4	Clear expectations: "must seek to fashion a new landscape that will support viable natural and human communities into the future."	CPRA. 2012 Master Plan Principles.	State

LAMP5	Acknowledging residual risk: "acknowledges that flood risk reduction systems (both structural and nonstructural) and restored wetlands cannot eliminate all flooding risks, and that some degree of residual storm related risk will be inevitable in coastal Louisiana."	CPRA. 2012 Master Plan Principles.	State
LAMP6	Public's Role: "In addition to effective government action, success will require citizens to offer their ideas as planning proceeds and make informed decisions about living and working in south Louisiana."	CPRA. 2012 Master Plan Principles.	State
LAMP7	Providing for transitions: "As we seek to address this crisis, sensitivity and fairness must be shown to those whose homes, lands, livelihoods, and ways of life may be affected, in the near-term and long-term, by Master Plan projects or by continued rates of land loss and flooding."	CPRA. 2012 Master Plan Principles.	State
LAMP8	Participatory process: "The Master Plan will be developed and implemented with the participation of the many diverse interests that live, work, play, and own property in coastal Louisiana, along with national interests that have an interest in coastal Louisiana's landscape."	CPRA. 2012 Master Plan Principles.	State
LAMP9	Accounting for uncertainties: "consider how both financial and scientific/technical uncertainties influence the selection of projects."	CPRA. 2012 Master Plan Principles.	State
LAMP10	Adapting to changing circumstances: "The Master Plan will include an effective monitoring and evaluation process that seeks to reduce scientific and engineering uncertainty, assesses the success of the plan, and supports adaptive management."	CPRA. 2012 Master Plan Principles.	State

LAMP11	Efficient use of resources: "the need for efficient use of resources, such as funding, fresh water, and sediment."	CPRA. 2012 Master Plan Principles.	State
LAMP12	Sediment for restoration: "maximize use of sediment sources outside the system."	CPRA. 2012 Master Plan Principles.	State
LAMP13	A plan to guide local, state, and federal action while also ensuring consistency and sound resource management: "imperative that all government agencies act quickly and in accord with the principles of the Master Plan."	CPRA. 2012 Master Plan Principles.	State
LAMP14	Coastwide Objectives: Reduce storm surge (2.59), Harness natural processes (2.59)	CPRA, 2012 Master Plan Community Exercise: Coastwide Results	State
LAMP15	Flood Protection Service: 50-yr. Risk Reduction Levels (2.21), 100-yr. Risk Reduction Levels (2.09)	CPRA, 2012 Master Plan Community Exercise: Coastwide Results	State
LAMP16	Ecosystem Service: Storm surge & wave reduction by Wetlands (2.78), Availability of Fresh Water (2.49)	CPRA, 2012 Master Plan Community Exercise: Coastwide Results	State
NOMP1	Multiple-lines-of-defense strategy	New Orleans Mast Plan, Executive Summary	Local
NOMP2	A City Department of Environmental Affairs to coordinate strategy—including policy, mitigation, advocacy, liaison	New Orleans Mast Plan, Executive Summary	Local
NOMP3	A range of protection levels, from a 1-in-400-year event to a 1-in-1,000-year event, as appropriate to conditions	New Orleans Mast Plan, Executive Summary	Local
NOMP4	Public education about risk and federal funding to elevate, relocate and storm-proof buildings	New Orleans Mast Plan, Executive Summary	Local
NOMP5	Exploration of polder and canal systems to manage water	New Orleans Mast Plan, Executive Summary	Local
NOMP6	Natural drainage and stormwater-management strategies	New Orleans Mast Plan, Executive Summary	Local
NOMP7	A Climate Change Policy Group to plan for adaptation to climate change	New Orleans Mast Plan, Executive Summary	Local

JEFF1	Additional federal funding for 100-year flood protection	Jefferson Edge 2020 Strategic Implementation Plan: Flood Protection	Local
JEFF2	Reduced local share of the cost of 100-year protection	Jefferson Edge 2020 Strategic Implementation Plan: Flood Protection	Local
JEFF3	Alternate means of financing the local cost share component of 100-year protection	Jefferson Edge 2020 Strategic Implementation Plan: Flood Protection	Local
JEFF4	Federal authorization and funding of a “polder” system of compartmentalized flood protection.	Jefferson Edge 2020 Strategic Implementation Plan: Flood Protection	Local
JEFF5	Expanded Corps of Engineers oversight and reform	Jefferson Edge 2020 Strategic Implementation Plan: Flood Protection	Local
JEFF6	Renewed call for 500-year flood protection	Jefferson Edge 2020 Strategic Implementation Plan: Flood Protection	Local
JEFF7	Systematic public outreach campaign to keep the public informed about external flood control projects that are completed, underway, and planned	Jefferson Edge 2020 Strategic Implementation Plan: Flood Protection	Local
JEFF8	Additional funding through the SELA program to implement additional SELA-authorized projects	Jefferson Edge 2020 Strategic Implementation Plan: Flood Protection	Local
JEFF9	Federal funding for automated climber screens for the pump stations.	Jefferson Edge 2020 Strategic Implementation Plan: Flood Protection	Local
JEFF10	Obtain funding to improve the “local” drainage system in Jefferson	Jefferson Edge 2020 Strategic Implementation Plan: Flood Protection	Local
JEFF11	Authorize the Hoey’s Basin pump to the river project and work with Congress to secure funding	Jefferson Edge 2020 Strategic Implementation Plan: Flood Protection	Local
STB1	Reduce repetitive loss of structures due to flooding	Advertisement - Request for Proposals Integrated Water Resource Management Plan For St. Bernard Parish	Local

STB2	Increase the health, quality of life, and sustainable livelihoods of St. Bernard citizens	Advertisement - Request for Proposals Integrated Water Resource Management Plan For St. Bernard Parish	Local
STB3	Reduce the need for new large scale/conventional storm water infrastructure	Advertisement - Request for Proposals Integrated Water Resource Management Plan For St. Bernard Parish	Local
STB4	Provide new zoning and ordinance recommendations that address: Existing green space, LLT properties, Increased buffer areas for residential neighborhoods adjacent to industry, Integrated water resource management, Hurricane resilience	Advertisement - Request for Proposals Integrated Water Resource Management Plan For St. Bernard Parish	Local
STB5	Land use / water management implementation strategy	Advertisement - Request for Proposals Integrated Water Resource Management Plan For St. Bernard Parish	Local
STB6	Integrate with regional restoration and protection efforts	Advertisement - Request for Proposals Integrated Water Resource Management Plan For St. Bernard Parish	Local
STB7	Improve the environmental quality within the urban environment	Advertisement - Request for Proposals Integrated Water Resource Management Plan For St. Bernard Parish	Local
JONKMA N1	The optimal protection level for the central part of New Orleans is estimated to be in the order of magnitude of 1/5000 per year	S.N. Jonkman, M. Kok, M. van Ledden and J.K. Vrijling. "Risk-based design of flood defence systems:a preliminary analysis of the optimal protection level for the New Orleans metropolitan area."	Third Party
JONKMA N2	The optimal level of protection for New Orleans East dike ring and the southern dike ring are 1/1000 per year.	S.N. Jonkman, M. Kok, M. van Ledden and J.K. Vrijling. "Risk-based design of flood defence systems:a preliminary analysis of the optimal protection level for the New Orleans metropolitan area."	Third Party

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Appendix E: Contact Information for Relevant Agencies

Southeast Louisiana Flood Protection Authority-East

<http://www.slfpae.com/>

504-280-2411

2045 Lakeshore Dr., Room 422, New Orleans, La 70122

US Army Corps of Engineers, New Orleans District

<http://www.mvn.usace.army.mil/>

504-862-2201

7400 Leake Ave., New Orleans, LA 70118

Orleans Levee District

<http://www.orleanslevee.com/>

504-286-3100

6920 Franklin Avenue, New Orleans, Louisiana 70122

Louisiana Coastal Protection and Restoration Authority

<http://coastal.la.gov/>

225.342.7308

P.O. Box 44027, Baton Rouge, LA 70804-4027

East Jefferson Levee District

<http://www.ejld.com/>

504-733-0087

203 Plache Court, Harahan, Louisiana 70123

Louisiana Governor's Office of Homeland Security and Emergency Preparedness

<http://www.gohsep.la.gov/>

(225) 925-7500

7667 Independence Blvd., Baton Rouge, LA 70806

Lake Borgne Basin Levee District

<http://lbbld.com/>

(504) 682-5941

6136 E. St. Bernard Hwy., P.O. Box 216, Violet, La. 70092

Red Cross, Southeast Louisiana Chapter

<http://www.redcross.org/la/new-orleans>

(504) 620-3105

2640 Canal St., New Orleans, LA 70119

Pontchartrain Levee District

<http://www.leveedistrict.org/>

225-869-9721

4 Albert Street, P.O. Box 426, Litcher, LA 70071

Federal Emergency Management Agency, Region VI

<http://www.fema.gov/>

940-898-5399

FRC 800 North Loop 288, Denton, TX 76209-3698

Sewerage and Water Board of New Orleans

<https://www.swbno.org/>

504-529-2837

625 Saint Joseph Street Room 237, New Orleans, LA 70165

National Hurricane Center

<http://www.nhc.noaa.gov/>

(305) 229-4470

11691 S.W. 17th Street, Miami, Florida 33165-2149